MIDDLE YEARS PROGRAMME
YEAR 1 · MATHEMATICS

FIRST EDITION
2026

MYP

MATHEMATICS 1

Student Coursebook

A concept-driven inquiry course in number, proportion, algebra, shape and data.

Number systems · Place value and powers · Factors and primes
Fractions and decimals · Percentages · Patterns and equations
Lines, angles and constructions · Sets · Perimeter, area and volume · Data and graphs

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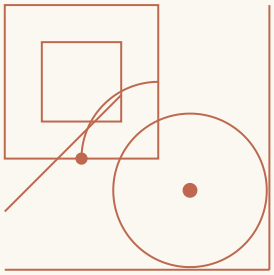
STUDENT COURSEBOOK
FIRST EDITION · 2026

MYP

Mathematics *1*

Student Coursebook

An inquiry-led introduction to number, place value, proportion, algebra, geometry, measurement, sets and data, built around the concepts, contexts and assessment criteria of the Middle Years Programme.



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MYP Mathematics 1 — Student Coursebook

FIRST EDITION · 2026

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Originality. All explanatory prose, examples, exercises, investigations, sample responses, diagrams, page layouts and cover artwork in this book are original to this edition. Public curriculum descriptions and the publicly available contents list of Oxford University Press’s *MYP Mathematics 1* were consulted only as reference points for topic coverage and progression; no Oxford text, exercise, illustration, layout or page design is reproduced or adapted.

Scope and sequence. The order of chapters is one defensible school sequence, not an official required sequence. MYP schools design their own units around conceptual understandings and local contexts. Teachers should check the sequence against their programme documentation and their students’ prior learning.

Data and contexts. Real-world scenarios use rounded, simplified or classroom-generated data. Figures are chosen to support the mathematics and may not reflect the most recent published statistics unless a source is named. Students should verify current data from a reliable source when a task requires it.

Assessment. Criterion tasks, rubrics and level descriptors in this book are teaching tools written for formative classroom use. They paraphrase and interpret the MYP assessment criteria; they are not official IB descriptors and do not replace the current subject guide.

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How to use this coursebook

This is a working book. Expect to write in the margins, test an idea, compare your method with a classmate's, and return to a question after your thinking has changed.

Year 1 mathematics is where many familiar ideas become more precise. You already know how numbers are written, how shapes are named and how data can be displayed. This course asks you to look underneath those familiar forms: why a digit's position changes its value, why equivalent fractions represent the same quantity, why an angle rule must hold for every triangle rather than merely one drawing.

The book is arranged in five parts. Part I establishes the structure of the number system and the rules that keep calculations reliable. Part II develops fractions, decimals and percentages as connected ways of describing parts of a whole. Part III introduces algebraic thinking through patterns and balance. Part IV brings together sets, shape and measurement. Part V examines data, graphs and the claims people make from them.

Every chapter follows an inquiry rhythm. A situation raises a question; definitions and representations give you the tools to investigate it; worked examples make the reasoning visible; practice makes the method dependable; an investigation, application and criterion task ask you to use the mathematics beyond the original example. The separate answer section confirms your reasoning rather than replacing it.

ANATOMY OF A CHAPTER

SIX MOVEMENTS, REPEATED IN EVERY CHAPTER

01 Chapter opener

A visual context and one central question that makes the chapter's mathematics worth developing.

02 MYP unit planner

Key and related concepts, global context, statement of inquiry, inquiry questions and criterion mapping.

03 Intentions and prerequisites

Learning intentions, success criteria, prior-knowledge check and the boundary of the Year 1 treatment.

04 Concept sequence

Definitions, representations, worked examples, common errors and graduated practice for each concept.

05 Investigate and apply

A conjecture-building investigation, a practical application and a mini-task mapped to MYP criteria.

06 Review and mixed practice

Summary, reflection, revision checklist and cumulative questions combining the chapter's ideas.

MATHEMATICAL HABITS THAT THIS COURSE DEVELOPS

→ **Represent.** Move between concrete materials, diagrams, tables, symbols and words.

→ **Justify.** Say why a method works, not only what answer it produces.

→ **Estimate.** Predict the size of an answer before calculating it.

→ **Reflect.** Name what you understand, what is still fragile, and what you will do next.

→ **Connect.** Ask how a fraction, decimal, percentage, ratio or diagram describe the same quantity.

HOW THE LABELS WORK

Definition boxes state precise meanings. **Key idea** boxes connect a method to a larger principle. **Common error** boxes show a mistake worth anticipating. **ATL** boxes make a learning skill explicit and give you a way to practise it. **Criterion** strips name the MYP criterion and strands developed by a section.

EXERCISE TYPES

F

FLUENCY

Short, deliberate practice to make a method accurate and efficient.

R

REASONING

Explain, compare, justify, find a counterexample or diagnose an error.

P

PROBLEM SOLVING

Multi-step and unfamiliar contexts that require selection and planning.

E

EXTENSION

Deeper structure, generalisation or a more open task.

ATL · USING THE ANSWER KEY WELL

Attempt the question fully before checking. If your answer differs, locate the first line where your reasoning diverges. Mark the correction in a different colour and write one sentence explaining why the alternative was wrong. That sentence is often more valuable than the correct number.

ACADEMIC HONESTY

Copying a solution without understanding it creates the appearance of progress and hides a gap. Use sample answers to compare *methods*, not to replace your attempt. In investigations, report your own data and reasoning, and acknowledge any source you use.

The MYP framework in this course

The Middle Years Programme asks students to learn mathematics as a connected, conceptual and applicable discipline. Concepts organise the content, contexts give it purpose, and criteria make the quality of thinking visible.

Aims of MYP mathematics

The MYP mathematics framework aims to help students enjoy mathematics, develop curiosity and confidence, build conceptual understanding, apply mathematics to authentic situations, communicate reasoning, think critically and creatively, and appreciate the international and historical dimensions of the subject. In Year 1, the emphasis is on strong foundations: fluent number sense, careful use of notation, spatial reasoning and the first deliberate steps from arithmetic towards algebra.

KEY CONCEPT 01

Form

The shape, structure and organisation of mathematical objects: how a decimal is arranged, how an angle is drawn, how an expression is built. Form helps us notice what is mathematically the same beneath different appearances.

KEY CONCEPT 02

Logic

The reasoning that connects claims to conclusions. Logic asks not only "Is it true?" but "How do we know, and for which cases?" It is the discipline behind classification, justification and generalisation.

KEY CONCEPT 03

Relationships

Connections between quantities, variables, representations and ideas. A percentage, a formula, an angle pair and a graph are different ways of expressing relationships.

Related concepts used in this course

CHANGE
How one quantity responds when another changes.

EQUIVALENCE
Different forms with the same value or meaning.

GENERALISATION
Extending a relationship to a whole class of cases.

JUSTIFICATION
Reasons and evidence that support a claim.

MEASUREMENT
Assigning numbers and units to attributes of objects.

MODELS
Simplified representations of a real situation.

PATTERNS
Regular structure that repeats or grows predictably.

QUANTITY
Number, magnitude, measure and unit.

REPRESENTATION
Words, symbols, tables, diagrams and graphs.

Global contexts

Each chapter is anchored in one of the six MYP global contexts. The context is not decoration: it shapes the questions asked, the data used and the kind of solution that counts as useful.

GLOBAL CONTEXT	HOW MATHEMATICS ENTERS THE INQUIRY
Identities and relationships	How categories, groups and belonging can be described and interrogated; how sets and classifications represent inclusion and difference.
Orientation in space and time	How number systems, calendars and measurement developed across cultures; how scale and position structure space.
Personal and cultural expression	How proportion, pattern and geometry shape music, design, craft and architecture.
Scientific and technical innovation	How accurate measurement, formulas and data displays support invention, engineering and scientific claims.
Fairness and development	How budgets, percentages, comparisons and data can support or distort fair decisions.
Globalization and sustainability	How quantities are compared across communities and environments, and how claims about resources are communicated.

Inquiry in a mathematics lesson

Inquiry does not mean that students are left to rediscover every definition without guidance. It means that a question genuinely drives the sequence of work, that students make and test conjectures, and that they can explain how a result was reached. The course uses a repeating pathway.

1
PROVOKE
Notice a situation, pattern or claim that is not yet explainable.

2
QUESTION
Turn the noticing into a mathematical question with quantities or objects.

3
REPRESENT
Choose diagrams, tables, symbols or materials that reveal structure.

4
GENERALISE
Move from examples to a rule, testing it against further cases.

5
APPLY
Use the rule in a context and judge whether the answer makes sense.

6
REFLECT
Evaluate the method, the evidence and the limits of the conclusion.

Assessment criteria in practice

MYP mathematics uses four criteria. They describe different qualities of mathematical work, not four separate syllabuses. A single rich task can provide evidence for more than one criterion.

A

KNOWING AND UNDERSTANDING

- Select appropriate mathematics for familiar and unfamiliar problems.
- Apply the selected mathematics successfully.
- Solve problems correctly across a range of contexts.
- In this course: definitions, place value, operations, fractions, percentages, angle rules, area and volume.

B

INVESTIGATING PATTERNS

- Apply problem-solving techniques to recognise patterns.
- Describe patterns as relationships or general rules consistent with findings.
- Verify whether a pattern works for other examples.
- In this course: sequences, factor structures, constructed figures, set patterns and measurement relationships.

C

COMMUNICATING

- Use appropriate mathematical language, notation and symbols.
- Use different forms of representation and move between them.
- Present complete, coherent and concise lines of reasoning.
- Organise information using a logical structure.
- In this course: explanations, diagrams, tables, graphs, units and justified working.

D

APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

- Identify relevant elements of authentic situations.
- Select and apply mathematical strategies.
- Justify the degree of accuracy of a solution.
- Justify whether the solution makes sense in context.
- In this course: budgets, recipes, designs, packaging, announcements and data claims.

Formative practice is not a formal assessment

Most questions in this book are practice. They are designed for learning, feedback and improvement, and they may be attempted more than once as understanding develops. A criterion task is different: it is planned to collect evidence against one or more criteria, and it has a defined audience, product and set of expectations. The label at the head of each criterion task states which criteria and strands are targeted, but it is not a formal IB grade.

WHAT A STRONG RESPONSE LOOKS LIKE

— **Accurate mathematics.** Numbers, units, labels and notation are correct.

- **Visible reasoning.** A reader can follow the decisions and see why each step follows.
- **Representation chosen for purpose.** A diagram is not decoration; it carries information.
- **Appropriate precision.** Rounding and accuracy are justified by the context.
- **Evaluation.** The answer is checked against the situation and its limitations are acknowledged.

LEVEL	GENERIC EVIDENCE IN A SHORT TASK
0	The response does not reach the standard described below.
1–2	Limited use of relevant mathematics; reasoning is incomplete or partly inaccurate.
3–4	Adequate use of relevant mathematics; some reasoning or representation is clear; application has gaps.
5–6	Accurate and mostly complete use of mathematics; coherent reasoning; suitable representation; context considered.
7–8	Accurate, complete and efficient; precise communication; thoughtful evaluation of accuracy and fit to context.

The generic bands above are classroom feedback tools written for this book. They are not official IB level descriptors. Departments should use the current subject guide's criterion descriptors and their own task-specific clarifications when reporting achievement.

Command words that shape the evidence

COMMAND	WHAT IT ASKS YOU TO DO	EXAMPLE FROM THIS COURSE
State	Give a concise answer without extended reasoning.	State the value of the digit 7 in 3,472.
Explain	Make the reasoning visible, including why a step is valid.	Explain why dividing numerator and denominator by 4 keeps a fraction's value.
Justify	Support a claim with evidence, an example or a rule.	Justify why angle x is 65° in the diagram.
Investigate	Explore cases systematically and describe what remains true.	Investigate the digit sums of multiples of 3.
Evaluate	Judge quality, accuracy or fitness for purpose.	Evaluate whether the graph fairly supports the headline.
Create	Design a product or model that meets stated constraints.	Create a net and cost estimate for a small box.

Course map and progression

The ten chapters form a cumulative course. Each chapter names the knowledge it assumes, the new structure it builds, and the Year 2 territory it deliberately does not yet enter.

PART	CHAPTER	CORE CONTENT	ASSESSMENT FOCUS	SUGGESTED TIME
I The grammar of number	1 · Number systems and place value	Place value, ordering, rounding, integers on the number line, number systems, expanded form	A, C	12 h
	2 · Powers, expanded form and order of operations	Powers, squares, cubes, roots, powers of 10, order of operations, estimation	A, B, C	14 h
	3 · Factors, multiples and primes	Factor pairs, divisibility, primes, prime factorisation, HCF and LCM	A, B, D	13 h
II Parts and proportions	4 · Fractions and decimals	Equivalence, ordering, the four operations, place value in decimals, links between fractions and decimals	A, B, C, D	16 h
	5 · Percentages and applications	Percent as per hundred, conversions, percentage of a quantity, increase and decrease, discounts and budgets	A, B, C, D	14 h
III Patterns and symbols	6 · Patterns, expressions and equations	Sequences, general rules, expressions, substitution, simplifying, one- and two-step equations, formulas	A, B, C, D	16 h
IV Shape, structure and measure	7 · Lines, angles, triangles and constructions	Geometric language, angle relationships, triangle properties, ruler-and-compass constructions	A, B, C	16 h
	8 · Sets and classification	Set notation, subsets, complement, union and intersection, two-set Venn diagrams	A, C, D	10 h
	9 · Perimeter, area, surface area and volume	Units, perimeter, area of common polygons, composite areas, nets and surface area, volume of prisms	A, B, D	15 h
V Data and decisions	10 · Data, tables and graphs	Data types, frequency tables, bar and line graphs, pie charts, summary statistics, misleading representations	A, B, C, D	14 h

Suggested hours total 140. They are a planning aid, not a mandated allocation. Schools should adapt the sequence, time and contexts to their own programme.

ENTRY ASSUMPTIONS

Chapter 1 assumes fluent counting, basic multiplication facts, familiarity with simple fractions and the ability to read a number line. Chapter 2 assumes multiplication facts and the concept of a square number from primary

school. Chapter 3 assumes multiplication and division facts. Chapter 4 assumes simple equivalent fractions and decimal notation to hundredths. Chapter 5 assumes the fraction and decimal ideas of Chapter 4. Chapter 6 assumes the four operations and simple number patterns. Chapter 7 assumes the vocabulary of points, lines, angles and common shapes. Chapter 8 assumes sorting and simple counting. Chapter 9 assumes multiplication, simple angle knowledge and measurement units. Chapter 10 assumes reading tables and bar graphs.

If a prerequisite check reveals gaps, use the review suggestions in the answer section before proceeding. Gaps are information, not a verdict: they identify the next useful thing to learn.

DELIBERATE YEAR 1 BOUNDARIES

- **Proportion.** Year 1 builds fraction, decimal and percentage fluency. Formal ratio notation, rate reasoning and proportional graphs belong to the next stage.
- **Algebra.** Year 1 uses expressions and linear equations with one unknown. Factorising algebraic expressions, simultaneous equations and gradients are not assumed.
- **Probability.** This Year 1 course develops data reasoning. Formal probability sample spaces and experiments are treated in the following course.
- **Geometry.** Constructions use ruler, compasses and protractor; formal congruence proofs and circle theorems are deferred.
- **Number.** Negative numbers are introduced and ordered. Extensive signed-number arithmetic and integer exponent laws are Year 2 work.

How the four criteria accumulate

THROUGH THE YEAR	CRITERION A	CRITERION B	CRITERION C	CRITERION D
Terms 1–2	Number structure and operations	Noticing sequences and factor patterns	Precise notation and diagrams	Short classroom contexts
Terms 2–3	Fractions, percentages, elementary algebra	Generalising a rule and testing it	Moving between representations	Budgets, recipes and design decisions
Terms 3–4	Geometry, measurement, sets and data	Investigating relationships in figures and data	Coherent multi-step reports	Evaluating accuracy and fitness for purpose
End of year	Selecting methods across topics	Justifying and extending a generalisation	Organising a complete mathematical argument	Modelling a situation and evaluating the solution

Notation, working and habits

Clear notation is part of the mathematics. It allows another person to follow your reasoning, and it helps you notice when a claim is not yet precise.

Notation you will use

$=$	is equal to
$\neq$	is not equal to
$\approx$	is approximately equal to
$<$, $>$	is less than, is greater than
$\leq$, $\geq$	is less than or equal to, is greater than or equal to
2^4	a power: base 2, exponent 4
$\sqrt{49}$	the positive square root of 49
$\frac{3}{4}$	three quarters: 3 parts out of 4 equal parts
35%	35 per hundred
x	an unknown number or a variable
$2x$	2 multiplied by x, not the two-digit number 2x
$\in$	is an element of
$\cup$, $\cap$	union, intersection
A'	the complement of set A
$\angle ABC$	the angle at vertex B, with sides BA and BC
AB	the length of segment AB; a line may be written AB with arrows in a diagram

Five habits that protect accuracy

HABIT 01

Align equal signs

Write each line of working below the previous one and keep the $=$ signs in a vertical line. A chain of equal expressions should be read left to right.

HABIT 02

Carry units

Write the unit in the calculation or state it unambiguously in the answer. Convert to a common unit before adding, subtracting or comparing.

HABIT 03

Estimate first

Use rounded values to predict the size of the answer. If the exact answer is wildly different, re-read the question or recheck the operation.

HABIT 04

Show the check

Substitute an answer back into an equation. Recombine parts after a division. Count squares after an area calculation. A check is evidence, not extra decoration.

HABIT 05

Name the rounding

If a value is rounded, say to what accuracy and why. A sensible context may require whole people, whole cents or a measurement to the nearest millimetre.

Reading a mathematics question

1	2	3	4	5
GIVEN	**ASKED**	**STRUCTURE**	**METHOD**	**CHECK**
What quantities, units, labels and constraints are stated?	What exactly must be found, explained or judged?	What kind of object is involved: number, fraction, expression, angle, shape, set or data set?	Which definition, rule or representation connects the given information to the answer?	Is the answer plausible, correctly labelled and stated to a sensible accuracy?

ATL · COMMUNICATION

When you explain a method to a partner, use three moves: name the rule, show the calculation, and state why the result answers the question. If a listener cannot follow the middle step, the gap is usually in the reasoning rather than the arithmetic.

COMMON ERROR · CALCULATOR DEPENDENCE

Unless a question allows a calculator, practise a written or mental strategy. In Year 1, understanding place value and estimation matters more than producing a decimal quickly. When a calculator is allowed, record the expression you entered and still judge whether the display is reasonable.

Academic honesty and collaboration

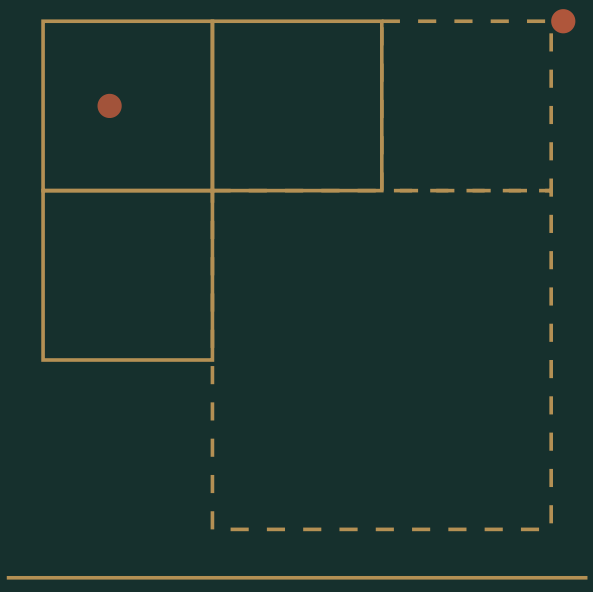
Discussing a problem with classmates is valuable, but the work you submit should make your own reasoning visible. If a classmate's idea helps you, say what you learned from the discussion and then write the solution in your own words. In an investigation, record the data you collected, the method you used and any assumptions you made. Do not present a classmate's conjecture or result as your own.

PART I · THE GRAMMAR OF NUMBER

I

Number, structure and proof

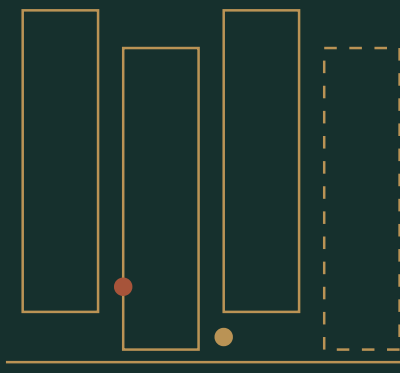
Numbers are not merely symbols to be operated on. Their positions, factors and powers form a system with rules that make calculation reliable. This part examines that system: how place value works, how powers compress repeated multiplication, and how divisibility reveals the hidden structure of whole numbers.



01 Number systems and place value

02 Powers, expanded form and order of operations

03 Factors, multiples and primes



1

NUMBER, STRUCTURE AND PROOF

Number systems and place value

Ten digits are enough to write any number we can name. Their power comes from position: the same symbol can mean seven, seventy or seven thousandths.

KEY CONCEPT · FORM

RELATED · QUANTITY

RELATED · REPRESENTATION

CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

How does a digit's position change its value, and how can we describe that structure so precisely that comparisons and calculations become reliable?

MYP UNIT PLANNER

Framing the inquiry

UNIT 1 · APPROX. 12 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Form. A number's written form is not a single symbol but an organised system of positions. Each position carries a value ten times the position to its right.

Quantity. Place value lets a small set of digits represent any magnitude, while rounding records an appropriate level of precision.

Representation. A number may appear as words, digits, an expanded sum, a position on a number line or a rounded claim.

GLOBAL CONTEXT AND EXPLORATION

Orientation in space and time. Exploration: how different cultures have recorded quantity, and why the invention of a place-value system with zero changed what numbers could do.

Place-value systems are technologies of representation. They make large quantities manageable, comparisons quick and written calculation possible. The history of zero is not an optional curiosity: it explains why our written system can distinguish 305 from 35 and 3.05 without inventing new symbols for every value.

STATEMENT OF INQUIRY

The position of a digit in a base-ten system encodes its value, so a small set of symbols can represent any magnitude when place and zero are used consistently.

FACTUAL

What is the value of a digit in a given position? How does multiplying or dividing by 10 change a number's digits?

CONCEPTUAL

Why is zero essential in a place-value system, and how can one number have several equivalent representations?

DEBATABLE

When a number is reported as "about 2,000", is the result more useful than the exact count?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Read, compare, round and rewrite integers and decimals; select sensible placements.
B · Investigating patterns	i, ii, iii	Investigate how counting possibilities grow with the number of digits; state and test a rule.
C · Communicating	i, ii, iii, iv	Move between words, digits, expanded form, number lines and rounded statements.
D · Applying mathematics	i, ii, iii, iv, v	Choose an appropriate accuracy for a public noticeboard and justify its effect on meaning.



ATL · CRITICAL THINKING IN ACTION

Write the claim "adding a zero to the end of a number multiplies it by 10". Test it on 4, 4.5 and 0.40. Find a case where adding a zero leaves the number unchanged, then revise the claim so it is mathematically accurate. Use your revised rule in Practice 1.2 question 7.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** how natural numbers, whole numbers and integers are related.
- **Know** that the value of a digit depends on its place in a base-ten number.
- **Be able to** write whole numbers and decimals in expanded form and return to standard form.
- **Be able to** compare and order integers and decimals using place value and a number line.
- **Be able to** round to a stated place and choose a reasonable level of accuracy in context.
- **Be able to** estimate a result and explain whether the estimate is likely to be too high or too low.

SUCCESS CRITERIA

- ☐ I can classify a number as natural, whole, integer or a rational number written as a fraction or decimal.
- ☐ I can state the value of any digit in a whole number or decimal and explain the role of a zero.
- ☐ I can write a number in expanded form and reconstruct it accurately.
- ☐ I can order positive and negative numbers and decimals and justify the order.
- ☐ I can round to the nearest stated unit or place and avoid an over-precise answer.
- ☐ I can estimate, compare the estimate with an exact calculation, and explain the difference.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Write “four thousand and sixty” using digits.
2. Which is greater, 0.5 or 0.45?
3. Calculate 8×7.
4. What is 100 more than $2{,}950$?
5. Write $\frac{7}{10}$ as a decimal.
6. Put $-2,\ 0,\ 3$ and -5 in order from least to greatest.

Answers are in the Chapter 1 answer section. If more than two are incorrect, review place value to hundreds and simple fractions before continuing.

BEFORE YEAR 1
Counting, number bonds, place value to thousands and simple fractions of a whole.

YEAR 1 · THIS CHAPTER
Number families; decimal place value; integer ordering; expanded form; rounding and estimation.

LATER
Signed-number operations, powers and standard index form; scientific notation; absolute error.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Museum labels report collection size and years in different forms.	What does each digit in a displayed number actually contribute?	Base-ten place value, zero as placeholder, precision.	Count the possible numbers of different lengths.	Expanded form, number lines and justified rounding.	When does extra precision help, and when does it mislead?

1.1 Number systems and the number line

CRITERION A · I ◆ NUMBER FAMILIES ◆ ORDERING

Mathematicians use different number sets for different purposes. Distinguishing them is not a vocabulary game: the classification tells you which operations are always possible and which values count as solutions.

DEFINITIONS

The **natural numbers** are the counting numbers $1, 2, 3, 4, \ldots$. The **whole numbers** are the natural numbers together with zero: $0, 1, 2, 3, \ldots$. The **integers** are the whole numbers and their negatives: $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$. A **rational number** can be written as a fraction $\frac{a}{b}$, where a and b are integers and $b \neq 0$.

Every integer is also rational because it can be written with denominator 1: $7 = \frac{7}{1}$ and $-3 = \frac{-3}{1}$. Terminating decimals such as 0.25 are rational too, because $0.25 = \frac{25}{100}$. The families are nested: natural numbers sit inside the whole numbers, which sit inside the integers.

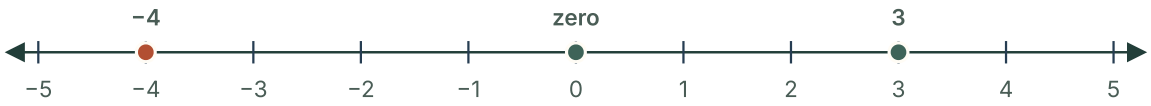


Figure 1.1 On a horizontal number line, values increase from left to right. Negative numbers lie to the left of zero; positive numbers lie to the right. The distance from zero matters, but direction determines the sign.

A number line is more than a display device. It gives a shared order to every integer, fraction and decimal. To compare two values, locate both: the one further to the right is greater. This remains true for negative numbers, where a larger digit means a smaller value: -8 is further left than -3, so $-8 < -3$.

KEY IDEA

The symbols $<$ and $>$ describe position, not magnitude alone. "$-5 < -2$" is correct because -5 lies to the left of -2, even though 5 is greater than 2. Reversing the symbol reverses the claim.

WORKED EXAMPLE 1.1.1 CLASSIFYING WITH EVIDENCE

PROBLEM

For each number, state whether it is natural, whole, an integer or rational. A number may belong to more than one family.

$0, \quad 7, \quad -3, \quad 2.5, \quad \frac{3}{4}$

1. 0 is not natural, because counting begins at 1. It is whole, an integer and rational: $0 = \frac{0}{1}$.
2. 7 is natural, whole, an integer and rational: $7 = \frac{7}{1}$.
3. -3 is an integer and rational: $-3 = \frac{-3}{1}$. It is not natural or whole because it is negative.
4. 2.5 is rational because $2.5 = \frac{5}{2}$. It is not an integer.
5. $\frac{3}{4}$ is rational because both 3 and 4 are integers and the denominator is not zero.

ANSWER 0**: whole, integer, rational;** 7**: natural, whole, integer, rational;** -3**: integer, rational;** 2.5 **and** $\frac{3}{4}$**: rational.**

WORKED EXAMPLE 1.1.2 A CONTEXT WITH NEGATIVE VALUES

PROBLEM

The temperature in a mountain town is recorded hourly: $-6°\text{C}$, $-3°\text{C}$, $0°\text{C}$, $4°\text{C}$ and $-1°\text{C}$. Order the temperatures from coldest to warmest.

1. "Coldest" means the least number on the number line. Negative values are less than zero, and a larger digit after the negative sign makes a colder temperature.
2. Compare the negatives: $-6 < -3 < -1$.
3. Zero is warmer than every negative value, and 4 is warmer still.

ANSWER $-6°\text{C}$, $-3°\text{C}$, $-1°\text{C}$, $0°\text{C}$, $4°\text{C}$**.**

COMMON ERROR

Reading the absolute size of the numerals instead of their positions: "$-8 > -3$ because 8 is bigger than 3". Mark both values on a number line. The further-left number is smaller, so $-8 < -3$.

ATL · CRITICAL THINKING

Skill in action. A classmate says, "Every fraction is less than 1." Find two fractions that make the statement false, one greater than 1 and one equal to 1. Then rewrite the statement so that it is true for all fractions you can name. The revision should refer to the numerator and denominator, not to a list of exceptions.

PRACTICE 1.1 · NUMBER FAMILIES AND ORDER

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Show enough working or explanation that your classification or order can be checked.

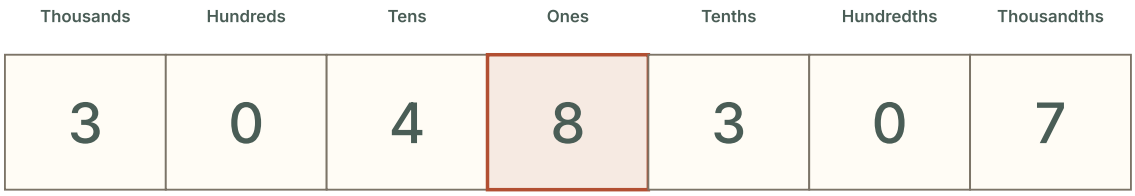
1. State whether each number is natural, whole, an integer or rational. Some answers will name more than one family: (a) 12 (b) 0 (c) -4 (d) $\frac{5}{8}$ (e) 1.2 F
2. Write all the integers from -3 to 2 inclusive. F
3. Replace each $\square$ with $<$, $>$ or $=$: (a) $-7 \square -2$ (b) $0 \square -1$ (c) $-5 \square 5$. F
4. Order these temperatures from coldest to warmest: $3°\text{C}$, $-2°\text{C}$, $-7°\text{C}$, $0°\text{C}$, $5°\text{C}$. F

5. Maya says, “Zero is a natural number because we can count zero items.” Explain whether Maya is using the mathematical definition of natural numbers correctly. R
6. Give an example to show that this statement is false: “If a number is written as a fraction, it must be less than 1.” R
7. A lift’s buttons show floors -2, -1, 0, 1 and 3. A person enters at floor 1 and travels to the lowest floor, then to floor 3. Describe the two changes as movements along a number line and state the final floor relative to the starting floor. P
8. Construct a number line from -5 to 5. Mark and label -4, 0 and $2\frac{1}{2}$. Explain why $2\frac{1}{2}$ lies between 2 and 3 even though it is not an integer. E

1.2 Place value and the role of zero

CRITERION A · I, II ◆ BASE TEN ◆ ZERO AS PLACEHOLDER

Our number system is called base ten because each position is worth ten times the position to its right. The digit 4 can stand for 4 ones, 4 tens, 4 hundreds or 4 tenths; the position supplies the value.



The highlighted 8 means 8 ones, not 8 thousands.

Place-value columns for 3,048.307. The decimal point marks the boundary between whole-number places and fractional places.

Figure 1.2 Each place is ten times the one on its right. A decimal point separates whole-number positions from positions representing tenths, hundredths and smaller parts.

Moving one place to the left multiplies the value of a digit by 10. Moving one place to the right divides it by 10. This is why multiplying a number by 10 does not require inventing new digits: every digit shifts one place left. A zero appears at the end of a whole number because there are no units in that position, not because zero is being “added”.

> DEFINITION
>
> A **place value** is the value of a digit determined by its position. In 3,048.307, the digit 4 is in the tens place, so its value is $4 \times 10 = 40$. The digit 7 is in the thousandths place, so its value is $7 \times 0.001 = 0.007$.

Zero has two related jobs. As a number, it marks a position on the number line between the positive and negative integers. As a digit, it holds an empty place so that the other digits keep their intended values. Without the zero in 3,048, the digits would read 348 and mean a completely different quantity. Without the zero in 3.05, the digit 5 would shift into the tenths place and produce 3.5.

WORKED EXAMPLE 1.2.1 READING PLACE VALUE

PROBLEM

For the number 6,502.94, state the value of each underlined digit: (a) the 5, (b) the 4, (c) the 9.

1. (a) The 5 is in the hundreds place. Its value is $5 \times 100 = 500$.
2. (b) The 4 is the second digit after the decimal point, in the hundredths place. Its value is $4 \times 0.01 = 0.04$.
3. (c) The 9 is the first digit after the decimal point, in the tenths place. Its value is $9 \times 0.1 = 0.9$.

ANSWER **(a) 500 (b) 0.04 (c) 0.9**

Check. The three values are not the same even though the digits are all single digits. Read the position first, then the digit.

WORKED EXAMPLE 1.2.2 SHIFTING BY POWERS OF TEN

PROBLEM

Calculate (a) 4.06×100, (b) $37.5 \div 10$, (c) $0.8 \times 1{,}000$.

1. (a) Multiplying by 100 moves every digit two places left: $4.06 \times 100 = 406$. The decimal point is not "moved" as a trick; the place values have changed.
2. (b) Dividing by 10 moves every digit one place right: $37.5 \div 10 = 3.75$.
3. (c) $0.8 \times 1{,}000 = 800$. Eight tenths of a thousand is eight hundred.

ANSWER **(a) 406 (b) 3.75 (c) 800**

GUIDED PRACTICE · COPY AND COMPLETE

In 28.407, the digit 4 is in the _____ place, so its value is _____. The digit 7 is in the _____ place, so its value is _____. If the number is multiplied by 100, the new number is _____. Compare your completed sentence with a partner: did you name the place before calculating the value?

COMMON ERROR

Believing that adding a zero at the end always multiplies a number by 10. Adding a zero to the whole number 4 gives 40, which is ten times as large. Adding a zero after the decimal part of 4.5 gives 4.50, which is the same number. The rule depends on where the zero is inserted, not merely that one more zero appears.

PRACTICE 1.2 · PLACE VALUE AND ZERO

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. State the value of each underlined digit: (a) 6,$\underline{7}$42 (b) 23.$\underline{5}$8 (c) 0.$\underline{2}$04 (d) 8,10$\underline{6}$. F
2. Write the number represented by $4{,}000 + 600 + 30 + 2$. F
3. Calculate: (a) 36.2×10 (b) 5.48×100 (c) $920 \div 10$ (d) $7.3 \div 100$. F
4. In the number 2.06, what does the zero do? Write what the number would become if the zero were removed and explain the change in value. R
5. Which is larger, 3.5 or 3.05? Explain your answer using place value rather than the number of digits. R

6. A warehouse receives 24 boxes containing 100 pencils each. Write the total number of pencils using place value and state the value of the digit 2 in your answer. P
7. Test the claim "adding a zero to the right of a number multiplies it by 10" on 4, 0.4 and 4.50. State a corrected rule and give one counterexample. E
8. Using the digits 5, 0, 8 and 2 exactly once each, make the greatest and least four-digit numbers possible. Explain why the least number cannot begin with 0. E

1.3 Expanded form and equivalent representations

CRITERION A · II, III ◆ EQUIVALENCE ◆ REPRESENTATION

Expanded form unpacks a number into the values of its digits. It is a diagnostic tool: it shows immediately whether you understand a digit's place and whether a zero is holding a position.

DEFINITION

Expanded form writes a number as the sum of each digit multiplied by its place value. For example, $5{,}207 = 5 \times 1{,}000 + 2 \times 100 + 7 \times 1$. A zero digit contributes zero, but writing it can make the structure visible: $5 \times 1{,}000 + 2 \times 100 + 0 \times 10 + 7 \times 1$.

Expanded form is not a different number; it is the same number written to reveal its parts. That makes it useful for checking calculations, explaining the meaning of a digit and preparing for powers of ten. Decimals expand in the same way: $8.43 = 8 \times 1 + 4 \times 0.1 + 3 \times 0.01$. The decimal point is not a digit; it marks where the ones place ends.

KEY IDEA

Expanded form makes place value visible. If you can expand a number, you can often reconstruct it, compare it or check a multiplication by 10 without relying on a remembered trick.

WORKED EXAMPLE 1.3.1 — WHOLE NUMBER AND DECIMAL

PROBLEM

Write $70{,}604.509$ in expanded form.

1. List the positions from left to right: ten-thousands, thousands, hundreds, tens, ones, tenths, hundredths, thousandths.
2. The digits are 7, 0, 6, 0, 4, 5, 0, 9. Multiply each digit by its place value.
3. $70{,}604.509 = 7 \times 10{,}000 + 0 \times 1{,}000 + 6 \times 100 + 0 \times 10 + 4 \times 1 + 5 \times 0.1 + 0 \times 0.01 + 9 \times 0.001$.

ANSWER $7 \times 10{,}000 + 6 \times 100 + 4 \times 1 + 5 \times 0.1 + 9 \times 0.001$**, with the zero terms understood.**

Check. The largest place is ten-thousands, so the standard number should have five digits before the decimal point. It does: 70,604.

WORKED EXAMPLE 1.3.2 RECONSTRUCTING A NUMBER

PROBLEM

Write the standard number represented by $3 \times 100 + 9 \times 1 + 2 \times 0.1 + 6 \times 0.01$.

1. The largest place is hundreds, so the number has three whole-number places. There are no tens: the tens digit is 0.
2. The whole-number part is $300 + 9 = 309$, but the zero must be written to keep the 9 in the ones place.
3. The decimal part is $0.2 + 0.06 = 0.26$. Combine: 309.26.

ANSWER 309.26

COMMON ERROR

Counting the number of terms instead of using the named places. In 0.409, the 4 is in the tenths place, not the hundredths place, because the first position after the decimal point is tenths. Write the place-word above each digit before expanding.

PRACTICE 1.3 · EXPANDED FORM SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Write in expanded form: (a) 2,506 (b) 30.78 (c) 0.409. F
2. Write as a standard number: (a) $4 \times 100 + 9 \times 10 + 6 \times 1$ (b) $6 \times 1 + 5 \times 0.1 + 8 \times 0.01$ (c) $2 \times 1{,}000 + 7 \times 10 + 3 \times 0.001$. F
3. Fill in the missing value: (a) $8{,}000 + \square + 40 + 5 = 8{,}745$ (b) $0.6 + 0.07 + \square = 0.679$. F
4. Explain why writing $2 \times 0.1 + 5 \times 0.01$ describes 0.25, while $2 \times 0.01 + 5 \times 0.001$ describes 0.025. R
5. Two students expand 407.3 differently. Student A writes $400 + 7 + 0.3$. Student B writes $4 \times 100 + 0 \times 10 + 7 \times 1 + 3 \times 0.1$. Whose form shows the tens place more clearly? Does either form change the value? R
6. A label gives a length as $5 \times 10 + 3 \times 1 + 4 \times 0.1$ centimetres. Write the length in standard form and explain what physical quantity each term contributes. P
7. Create a four-digit number with a 6 in the hundreds place and a 0 in the tens place. Write it in standard and expanded form, then make a different number by swapping the 0 and the 6. Explain how the values differ. E
8. Explain why the expanded form of 1,000 can be written with a single non-zero term, while its place-value chart still contains four positions. E

1.4 Comparing, ordering and rounding

CRITERION A · II, III ◆ ORDER ◆ PRECISION

To compare decimals, compare the places that matter, not the number of digits. To round, choose a place, examine the digit immediately to its right, and decide whether the chosen digit stays the same or increases by one.

For 3.6 and 3.58, align the decimal points and compare place by place. Both have 3 ones. In the tenths place, 6 is greater than 5, so $3.6 > 3.58$. Writing 3.60 can help: it uses the same value and

shows that the hundredths place is zero. For negative decimals, the same left-to-right comparison applies, but a value further left is smaller.

ROUNDING RULE

To round to a given place value, locate that place. Look at the single digit immediately to its right. If that digit is 5 or more, increase the chosen digit by 1. If it is less than 5, leave the chosen digit unchanged. Replace the digits to the right with zeros (for whole numbers) or drop them (for decimals), keeping enough zeros to show the intended precision.

WORKED EXAMPLE 1.4.1 COMPARING DECIMALS

PROBLEM

Insert $<$, $>$ or $=$: (a) $0.7 \square 0.70$ (b) $2.68 \square 2.7$ (c) $-4.2 \square -4.25$.

1. (a) 0.70 has the same value as 0.7 because the extra zero in the hundredths place contributes zero. So $0.7 = 0.70$.
2. (b) Compare the ones: both 2. In the tenths place, $6 < 7$, so $2.68 < 2.7$. The number of digits is irrelevant.
3. (c) Both are negative, so the one further left is smaller. -4.25 is further left than -4.2, so $-4.2 > -4.25$.

ANSWER **(a)** $=$ **(b)** $<$ **(c)** $>$

WORKED EXAMPLE 1.4.2 ROUNDING WITH A CARRY

PROBLEM

Round 39.95 to the nearest tenth.

1. The tenths digit is 9. The digit immediately to its right is 5 in the hundredths place.
2. Because 5 means "round up", increase the 9 by one. Increasing 9 tenths by one tenth produces 10 tenths, which is one whole.
3. The whole-number part increases from 39 to 40, and the tenths digit becomes 0. Keep the zero to show the accuracy: 40.0.

ANSWER 40.0 **(to the nearest tenth)**

Check on a number line. 39.95 lies exactly halfway between 39.9 and 40.0. The convention of rounding a 5 up sends it to 40.0.

GUIDED PRACTICE · ROUNDING WITH PURPOSE

Round 4,682 to the nearest hundred. The hundreds digit is ____. The digit to its right is ____, so the hundreds digit _____. The rounded value is _____. Now round the same number to the nearest ten. Explain why the two rounded values are different even though the original number is the same.

COMMON ERROR

Rounding digits one after another, as in $4.96 \to 5.0 \to 5$. For "nearest tenth", only the hundredths digit matters. The value is 5.0, not 5, because the zero records that the answer is precise to the nearest tenth. A rounded answer should not silently lose the place value it claims.

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Insert $<$, $>$ or $=$: (a) $0.5 \square 0.50$ (b) $2.7 \square 2.68$ (c) $-3 \square -5$. F
2. Order from least to greatest: $0.7,\ 0.07,\ 0.77,\ 0.707$. F
3. Round $8{,}647$ to the nearest: (a) hundred (b) ten. F
4. Round 3.746 to the nearest hundredth and 0.897 to the nearest tenth. F
5. Round 59.96 to the nearest tenth. Explain why the whole-number part changes. R
6. A sign says a trail is 4.6 km long. Another says it is 4.56 km. Explain how both could be correct if one is rounded to the nearest tenth. R
7. List three decimal numbers that round to 6.5 when rounded to the nearest tenth. Are there other possibilities? Explain the range. E
8. A class reports its fundraising total as "about £2,000". The exact total is £1,948. Is the rounded claim reasonable? What extra information would you want before publishing it? P

1.5 Estimation and sensible accuracy

CRITERION D · I, IV ◆ ESTIMATION ◆ REASONABLENESS

An estimate is not a guessed answer. It is a deliberately simplified calculation that tells you the approximate size of a result and helps you judge whether an exact answer is plausible.

One useful strategy is to round the numbers to friendly values before calculating. For $4{,}912 + 3{,}186$, round to $5{,}000 + 3{,}000 = 8{,}000$. The exact total is $8{,}098$, so the estimate has the right magnitude. For 47×22, $50 \times 20 = 1{,}000$ gives a quick benchmark; the exact answer is $1{,}034$.

> **KEY IDEA**
>
> Keep track of whether your rounding made the estimate too high or too low. If both factors were rounded up, a product estimate is likely too high. If both were rounded down, a product estimate is likely too low. For sums, mixed adjustments can partially cancel.

WORKED EXAMPLE 1.5.1 — ESTIMATING A PRODUCT

PROBLEM

Estimate 19.8×4.1, then calculate the exact product and compare.

1. Round 19.8 to 20 and 4.1 to 4, both convenient values close to the originals.
2. $20 \times 4 = 80$. The first factor was rounded up and the second was rounded down, so the two errors act in opposite directions.
3. The exact product is $19.8 \times 4.1 = 81.18$. The estimate is close but slightly low, because decreasing 4.1 to 4 removed more than increasing 19.8 to 20 added.

ANSWER **Estimated product $= 80$; exact product $= 81.18$.**

Alternative estimate. Use $20 \times 4 = 80$ or $20 \times 4.1 = 82$. Both are useful; the purpose of the estimate determines which is preferable.

WORKED EXAMPLE 1.5.2 CHOOSING ACCURACY FOR AN AUDIENCE

PROBLEM

A school collects 3,846 books for a book drive. A newsletter says, "Almost 4,000 books were collected." Is this a reasonable statement? Which exact value should the school publish if donors need proof?

1. 3,846 is 154 less than 4,000, so "almost 4,000" is reasonable for a short promotional sentence.
2. The statement rounds to the nearest thousand and does not claim an exact count.
3. For an accountability report, the exact value 3,846 is better because it can be checked against the count and gives donors precise evidence.

ANSWER **Yes for the newsletter; publish 3,846 in a report that requires verification.**

Also acceptable. A more informative rounded claim such as "about 3,850 books" retains more precision than "almost 4,000".

COMMON ERROR

Reporting a calculator display as if every digit were meaningful. If the inputs are approximate measurements, a long decimal answer suggests accuracy that the data do not support. Round the result to a sensible place and state the unit.

PRACTICE 1.5 · ESTIMATE AND JUDGE ACCURACY

SUPPORT 1–3 · CORE 1–5 · EXTENSION 6–8

1. Estimate $692 + 3{,}141$ by rounding each number to the nearest hundred. F
2. Estimate 58×31 by rounding each factor to the nearest ten. F
3. Estimate 9.8×4.2 and state whether your estimate is likely to be above or below the exact product. F
4. Estimate $604 \div 29$ using $600 \div 30$. Then compute the exact quotient and compare. F
5. A shop records 397 visitors on Friday, 412 on Saturday and 289 on Sunday. Estimate the weekend total using rounding, then calculate it exactly. P
6. A runner completes 9.95 km on Monday and 10.08 km on Tuesday. A poster says she ran "about 20 km". Is this reasonable? What rounding did the poster use? R
7. Explain why the statement "The library has 1,247,936 books" might be inappropriate in a short public noticeboard, even if the figure is exact. Suggest a rounded alternative and state its accuracy. E
8. A calculation displays 2.673913043 on a calculator. The inputs are lengths measured to the nearest centimetre. Write a sensible final answer and explain your rounding choice. E

Investigation 1 · How many numbers can a given number of digits write?

CRITERION B · I, II, III

Ten digits are available: $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$. A number cannot begin with 0. Investigate how many different positive whole numbers can be written using exactly n digits when repetition is allowed.

Number of digits n	First digit choices	Remaining digit choices	Total count
1	9	—	9
2	9	10	90
3	9	10×10	900
4	9	$10 \times 10 \times 10$	?

1. Complete the table for 4 digits. Verify the count for 2-digit numbers by listing the smallest and largest examples and checking with a systematic count.
2. Describe the pattern in the "total count" column as the number of digits increases by one.
3. State a rule for the number of n-digit positive whole numbers. Test the rule for $n = 5$.
4. Explain why the first digit has 9 choices but each later digit has 10 choices.
5. How many n-digit positive whole numbers can be written in base 5 using the digits $0, 1, 2, 3, 4$, if a number cannot begin with 0? Record at least three cases, form a conjecture and test it.

Record your counts in a table, write the rule in words, and verify it with a case you did not use to discover it. Your conclusion should distinguish what you noticed from what you proved.

Real-life application · Numbers for a community noticeboard

A coastal town is preparing a one-page annual noticeboard. These are the draft figures supplied by different departments:

Item	Draft figure	What readers need
Resident population	128,460	A sense of scale, comparable with nearby towns
Visitors during the year	1,047,315	A memorable headline figure
Recycling collected	3,680 tonnes	An exact figure for a grant application
Year the library opened	1937	Historical context

Your task. The noticeboard has room for four numbers only. Work through the prompts below and keep a record of each decision.

1. Write each of the four figures in the form you recommend, and justify the precision chosen for each.
2. Round the population and the visitor figure to the same precision, then write the comparison a reader would see. Does the comparison remain fair?
3. Define "visitor" so the visitor figure can be interpreted. State what the definition must clarify.
4. Identify one figure for which rounding would be inappropriate, and the figure most sensitive to rounding. Explain each choice.

REFLECTION

Which figure is most sensitive to rounding, and why?

Criterion-mapped mini-task

A MUSEUM LABEL FOR THE HALL OF NUMBERS

A · KNOWING AND UNDERSTANDING
C · COMMUNICATING
D · APPLYING MATHEMATICS

Context. A small museum is creating a label about its collection. The label will be read by visitors who are 10 to 14 years old and by adults who want a quick overview. The curator gives you a fact sheet with the following information:

Fact	Exact value
Objects in the collection	48,672
Objects on display	6,509
Year the collection began	1894
Annual visitors	214,830
Floor area of the galleries	7,408.5 square metres

Product. Design a label of no more than 70 words that uses each fact in an appropriate form. Include at least one expanded form, one rounded value and one comparison. Add a short "how we rounded" note for the curator.

Requirements.

1. State the place value of one named digit in each of the whole-number facts.
2. Write the number of objects on display in expanded form.
3. Round at least two figures for the public-facing label and justify each choice in your curator's note.
4. Explain how rounding affects a reader's impression of the collection's size.
5. Identify one risk in using an exact figure where the data source may not be exact, and one risk in rounding too heavily.

CRITERIA AND STRANDS A i, ii, iii: accurate place value and rounding; C i, ii, iv: precise language and complete justification; D i, ii, iii, iv, v: select and apply representations for an audience, justify accuracy and evaluate fitness for purpose.

BEFORE YOU SUBMIT

Check that every number has a unit where relevant, that your label actually fits the word limit, and that your rounded figures are traceable to the exact values. A reader should be able to explain why each rounded number was chosen.

Chapter review

SUMMARY

- **Number families.** Natural numbers begin at 1; whole numbers include 0; integers include negatives; rational numbers can be written as a fraction of two integers with a non-zero denominator.
- **Order.** On a horizontal number line, values increase from left to right. Negative numbers reverse the usual reading of digit size.
- **Place value.** Each place is ten times the place to its right. Zero holds an empty place and prevents other digits from shifting.
- **Expanded form.** A number is a sum of digits multiplied by their place values. Equivalent forms have the same value.
- **Rounding.** Identify the target place, inspect the next digit, and apply the 5-or-more rule. Keep enough zeros to show the intended precision.
- **Estimation.** Round to friendly values, calculate, and compare the estimate with an exact result. Accuracy should fit the use and the data.

Place-value shift: one place left = multiply by 10; one place right = divide by 10.

Expanded form: digit × place value for every non-zero digit (and optionally for zero digits).

REVISION CHECKLIST

- ☐ I can explain the difference between a whole number and an integer.
- ☐ I can give the value of any digit in a whole number or decimal.
- ☐ I can write a number in expanded form and reconstruct it.
- ☐ I can compare two decimals without being misled by digit count.
- ☐ I can order positive and negative values on a number line.
- ☐ I can round to a specified place, including a case with a carry.
- ☐ I can estimate a result and state whether the estimate is likely above or below.
- ☐ I can choose an appropriate precision for a stated audience.

KEY VOCABULARY

place value
The value assigned by a digit's position.

integer
A whole number or its negative; includes zero.

expanded form
A number written as the sum of its place-value parts.

rounding
Replacing a number by a nearby value at a chosen accuracy.

estimate
An approximate value obtained from a simplified calculation.

precision
The fineness of a stated measurement or rounded value.

rational number
A number expressible as a fraction of integers.

zero
The number between positive and negative integers; also a place-holding digit.

REFLECTION AND SELF-ASSESSMENT

Choose one number from today and explain what its digits would mean if the decimal point moved two places. What changes, and what stays the same?

Rate your confidence from 1 to 4 on each success criterion. For any criterion rated 1 or 2, write the specific exercise you will revisit and the help you need. Then explain in two sentences why place value is essential to comparing quantities fairly.

END-OF-CHAPTER PRACTICE

Mixed practice 1

Do not use a calculator unless a question states otherwise. Show a method or explanation for questions that ask for one.

FLUENCY Build accuracy

1. Write the value of the digit 8 in (a) 3,812 (b) 20.83 (c) 0.081.
2. Write 5,040.207 in expanded form.
3. Write $7 \times 1{,}000 + 3 \times 10 + 6 \times 0.01$ as a decimal.
4. Round 64,782 to the nearest thousand, hundred and ten.
5. Round 0.496 to the nearest hundredth, and 12.95 to the nearest tenth.
6. Calculate: (a) 0.64×100 (b) $8.2 \div 10$ (c) $3.5 \times 1{,}000$.
7. Order from least to greatest: $-2.5,\ 0.25,\ -0.5,\ 2.05$.
8. Insert $<$, $>$ or $=$: (a) $0.08 \square 0.8$ (b) $1.30 \square 1.3$ (c) $-6 \square -2$.

REASONING AND PROBLEM SOLVING Connect ideas

9. A number rounds to 4,700 when rounded to the nearest hundred. Give three possible exact values and explain the range of possibilities.
10. Two students compare 6.09 and 6.1. One says 6.09 is larger because it has more digits. Explain the correct comparison.
11. A temperature changes from -3°C to 4°C. Use a number line to explain which temperature is colder and by how many degrees the values differ.
12. A shop's stock list gives 12,480 items. The manager says "about 12,000". Is this an underestimate or an overestimate? By how many items? Suggest an alternative rounded statement that is more informative.
13. The digit 0 appears in 4,507 but not in 457. Explain the different value of the 5 in each number using place value.
14. Estimate $3{,}912 + 4{,}087 + 2{,}998$ using numbers rounded to the nearest thousand. Is the estimate above or below the exact total? Calculate the exact total and check.
15. A number is written as $6 \times 100 + 4 \times 10 + \square \times 1$. It is even and lies between 646 and 650. Find the missing digit and explain why there is only one possibility.
16. Explain why "2.60" and "2.6" have the same value, but the first may still be a more appropriate way to report a measurement.
17. Write a number between 3.5 and 3.6 that has three decimal places. Then write a second number in the same interval that is closer to 3.5. Justify the order.
18. Construct a four-digit whole number whose digits have a sum of 12, whose thousands digit is 3, and whose tens digit is twice its ones digit. Find more than one possible answer if possible.

CRITERION-STYLE Apply and evaluate

19. **Community garden receipts (Criteria A, C, D).** A garden receives 2,495 visitors in one month. It rounds this to "about 2,500" in a funding bid. A volunteer gives the exact figure in a report. Explain when each statement is appropriate, calculate the difference between the two values, and suggest one additional figure that would make the reporting more informative.
20. **Archive number line (Criterion C).** An archive stores boxes numbered from -40 to 60. Boxes above 0 are in the new wing; boxes below 0 are in the basement. Create a clear number-line diagram or written guide showing where boxes -12, 0 and 45 are stored. Explain how a visitor should interpret the number zero.

Check your progress

ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

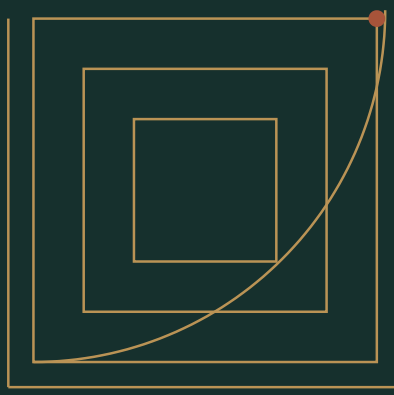
- **More digits does not mean a larger decimal.** Compare the same place values: 0.8 is greater than 0.08, although 08 uses more digits.
- **Zero is not always "nothing".** In 3.05 it holds the tenths place and keeps the 5 in the hundredths; removing it changes the value.
- **Rounding must preserve the intended precision.** 4,700 rounded to the nearest hundred is not the same statement as 4,700 rounded to the nearest ten.
- **An estimate needs a direction.** Saying "about 80" is incomplete if you cannot say whether the exact value is likely above or below it.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can classify a number as natural, whole, integer or rational.			
I can state the value of any digit and explain the role of a zero.			
I can write a number in expanded form and reconstruct it.			
I can order positive and negative numbers and decimals.			
I can round to a stated place and avoid over-precision.			
I can estimate and explain why my estimate is above or below the exact value.			

WHERE THIS LEADS

Chapter 2 replaces repeated multiplication with powers, and shows how expanded form becomes a compact language for very large and very small numbers. The order of operations then settles what a written expression such as $3 + 2 \times 5^2$ actually means. **Chapter 3** asks which numbers can be built from others through factors and multiples — a question that depends on the same place-value structure you have just used.

Review the answer key only after attempting each question.



2

PATTERN, COMPRESSION, PRECISION

Powers, expanded form and order of operations

A power is a compressed instruction: it records how many copies of a factor to multiply. A shared order of operations keeps that compression unambiguous.

KEY CONCEPT · FORM

RELATED · SIMPLIFICATION

RELATED · QUANTITY

CONTEXT · SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

When does a shorter notation make a calculation clearer, and what rules must everyone share before a compact expression can be read correctly?

MYP UNIT PLANNER

Framing the inquiry

UNIT 2 · APPROX. 14 HOURS
ATL FOCUS · TRANSFER

KEY AND RELATED CONCEPTS

Form. A power has a precise form: a base and an exponent. Changing either part changes the number, and expanded form and index form are two ways of writing the same quantity.

Simplification. Index notation shortens repeated multiplication, while an agreed order of operations makes a long expression readable in one consistent way.

Quantity. Powers of ten and powers of two let us describe and compare very large quantities such as storage capacity, distance and population.

GLOBAL CONTEXT AND EXPLORATION

Scientific and technical innovation. Exploration: how compact notation and agreed calculation rules allow people and machines to process quantities far beyond what can be written digit by digit.

A computer stores information in binary, so its capacities grow by powers of two, while storage is often advertised in powers of ten. The difference is small at first and large at scale. The same compact notation that describes a memory chip also describes the distance to a star.

STATEMENT OF INQUIRY

Compressing repeated multiplication into a power, and agreeing on the order in which operations apply, allows complex calculations to be recorded compactly and reproduced reliably.

FACTUAL

What does 5^3 mean? Which operation is performed first in $6 + 2 \times 3$? What is the value of 10^4?

CONCEPTUAL

Why does an exponent of 1 leave the base unchanged? How can 2^{10} and 10^3 be compared fairly? Why is a shared order of operations necessary?

DEBATABLE

Should a calculator be trusted to decide the meaning of an expression, or should the person writing it make the meaning visible?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Evaluate powers, roots and expressions using the agreed order.
B · Investigating patterns	i, ii, iii	Investigate doubling from paper folding; generalise thickness after n folds and test its limits.
C · Communicating	i, ii, iii, iv	Translate between repeated multiplication, index form, expanded form and standard form.
D · Applying mathematics	i, ii, iii, iv	Choose storage capacity and interpret advertised versus binary measures with justified assumptions.

ATL · TRANSFER IN ACTION

A classmate claims that $(a + b)^2 = a^2 + b^2$. Test the claim with $a = 3$, $b = 4$, then with $a = 2$, $b = 5$. Record both results and decide whether one counterexample is enough to reject a general claim. Use your conclusion in Practice 2.5 question 7.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** a power as repeated multiplication and identify its base and exponent.
- **Know** the place values represented by powers of ten and use them in expanded form.
- **Be able to** recognise square and cube numbers and find square roots of perfect squares.
- **Be able to** apply the order of operations, including brackets and indices.
- **Be able to** substitute values into expressions and formulas containing powers.
- **Be able to** compare quantities written as different powers and justify the comparison.

SUCCESS CRITERIA

- ☐ I can write repeated multiplication using an exponent and expand a power correctly.
- ☐ I can evaluate powers of whole numbers, including squares and cubes.
- ☐ I can write a whole number in expanded form using powers of ten and return to standard form.
- ☐ I can find square roots of perfect squares and explain what the root means.
- ☐ I can evaluate expressions using brackets, indices, multiplication, division, addition and subtraction in the agreed order.
- ☐ I can substitute a value into an expression with a power and explain the meaning of the result.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate 6×7 and 9×9.
2. Write $4{,}000 + 500 + 30 + 6$ as a single number.
3. Calculate $12 + 3 \times 4$.
4. Write $3 \times 3 \times 3$ as a product and evaluate it.
5. What is the area of a square with side 8 cm?
6. Insert brackets so that $2 + 4 \times 3$ has the value 18.

Answers are in the Chapter 2 answer section. If more than two are incorrect, review multiplication facts and the meaning of brackets before continuing.

BEFORE YEAR 1
Multiplication tables, place value, simple brackets and areas of squares and rectangles.

YEAR 1 · THIS CHAPTER
Index notation, powers of ten, squares, cubes and roots, order of operations, substitution.

LATER
Negative and fractional indices, standard form, algebraic expansion and sequences.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Storage, distance and population are reported in compact units.	How can repeated multiplication be recorded without losing meaning?	Base and exponent, powers of ten, inverse roots.	Model the thickness of folded paper as a power of two.	Move between expanded, index and standard forms.	When does compression clarify a quantity, and when does it hide it?

2.1 From repeated multiplication to powers

CRITERION A · I ◆ INDEX NOTATION ◆ BASE AND EXPONENT

Writing $3 \times 3 \times 3 \times 3$ is correct but inefficient, especially when a factor repeats many times. Index notation records the repetition in two small pieces of information: which factor is repeated, and how many copies there are.

DEFINITIONS

In the expression 3^4, the number 3 is the **base** and the number 4 is the **exponent** (or index). The whole expression is called a **power**. It means $3 \times 3 \times 3 \times 3$, so $3^4 = 81$.

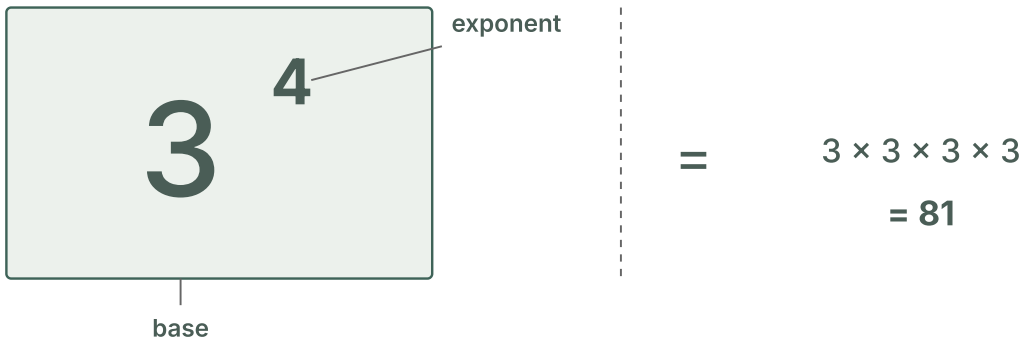


Figure 2.1 The base 3 is the repeated factor. The exponent 4 counts how many copies are multiplied. The value 81 is the result, not part of the notation.

The exponent is written above and to the right of the base. It is not a multiplier. The expression 3^4 does not mean 3×4; it means four factors of 3. Reading the notation aloud as "three to the fourth power" helps keep that distinction visible.

WORKED EXAMPLE 2.1.1 EXPANDING AND EVALUATING

PROBLEM

Write each power as repeated multiplication and evaluate it: (a) 2^5, (b) 5^2, (c) 10^4, (d) 4^3.

1. (a) $2^5 = 2 \times 2 \times 2 \times 2 \times 2$. Multiply in pairs: $2 \times 2 = 4$, $4 \times 2 = 8$, $8 \times 2 = 16$, $16 \times 2 = 32$.
2. (b) $5^2 = 5 \times 5 = 25$. A power with exponent 2 is called a square.
3. (c) $10^4 = 10 \times 10 \times 10 \times 10 = 10{,}000$. The exponent counts four zeros in the result.
4. (d) $4^3 = 4 \times 4 \times 4 = 16 \times 4 = 64$. A power with exponent 3 is called a cube.

ANSWER **(a) 32 (b) 25 (c) 10,000 (d) 64**

Check. The number of factors in the expanded product must equal the exponent. Counting factors is a quick way to catch an expansion that is too short or too long.

WORKED EXAMPLE 2.1.2 FROM EXPANDED FORM TO INDEX FORM

PROBLEM

Write each product as a power and evaluate it: (a) $6 \times 6 \times 6 \times 6$, (b) 9×9, (c) $7 \times 7 \times 7 \times 7 \times 7 \times 7$.

1. (a) The factor 6 appears four times, so the power is 6^4. Then $6^2 = 36$ and $6^4 = 36 \times 36 = 1{,}296$.
2. (b) The factor 9 appears twice, so the power is $9^2 = 81$.
3. (c) The factor 7 appears six times, so the power is 7^6. Evaluate in stages: $7^2 = 49$, $7^3 = 343$, $7^6 = 343 \times 343 = 117{,}649$.

ANSWER **(a)** $6^4 = 1{,}296$ **(b)** $9^2 = 81$ **(c)** $7^6 = 117{,}649$

Two special cases follow from the definition. A power with exponent 1 has one factor, so $7^1 = 7$. A power with exponent 0 is defined to be 1 for any non-zero base: $10^0 = 1$, $5^0 = 1$. The second rule is not obvious from "repeated multiplication"; it is chosen so that the pattern of dividing by the base continues consistently. You will see that pattern formalised when negative indices are introduced later.

COMMON ERROR

Reading the exponent as a multiplier: $3^4 \neq 3 \times 4$. The exponent tells you how many times to use the base as a factor. Likewise, $5^2 \neq 5 \times 2$; it is $5 \times 5 = 25$.

GUIDED PRACTICE · COPY AND COMPLETE

In 8^3, the base is _____ and the exponent is _____. Expanded, the power is _____, and its value is _____. If the exponent were 1 instead of 3, the value would be _____. Explain why the exponent cannot be read as "multiply the base by 3".

ATL · CRITICAL THINKING

Skill in action. Sort these expressions into two groups without evaluating all of them: 2^6, 6^2, 3^4, 4^3, 10^3. Which comparisons require calculation, and which can be decided by reasoning about the factors? Record your decision rule, then test it.

PRACTICE 2.1 · POWERS AND INDEX NOTATION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Show the expanded product at least once for each new power.

1. Write each product as a power: (a) $4 \times 4 \times 4$ (b) 7×7 (c) $10 \times 10 \times 10 \times 10 \times 10$ (d) $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$. F
2. Evaluate: (a) 2^3 (b) 3^2 (c) 5^3 (d) 10^3 (e) 2^6 (f) 4^3. F
3. Find the missing exponent: (a) $3^{\square} = 81$ (b) $2^{\square} = 64$ (c) $10^{\square} = 1{,}000$ (d) $5^{\square} = 125$. F
4. Explain the difference between 4^2 and 4×2. Include the value of each expression. R
5. A square tile has side length 9 cm. Write its area using a power and evaluate it. State the unit. P
6. Compare 2^4 and 4^2. Are they equal? Compare 2^3 and 3^2. What does this suggest about the claim "swapping base and exponent never changes the value"? R
7. A rumour starts with one person. In round 1, that person tells 3 people. In round 2, each of those 3 people tells 3 new people, and the pattern continues. Write the number of new people told in round 4 as a power, evaluate it, and find the total number of new people told across rounds 1 to 4. P

8. Without calculating the full values, decide which is larger: 2^{10} or 10^3. Explain your reasoning using factors or benchmark powers, then check by calculation. E

2.2 Powers of ten and expanded form

CRITERION A · II, III ◆ PLACE VALUE ◆ EQUIVALENT FORMS

Place value is built from powers of ten. The ones place is 10^0, the tens place is 10^1, the hundreds place is 10^2, and so on. Expanded form makes that structure explicit.

PLACE VALUES WRITTEN AS POWERS OF TEN

Place	Millions	Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones
Power of ten	10^6	10^5	10^4	10^3	10^2	10^1	10^0
Value	1,000,000	100,000	10,000	1,000	100	10	1

The pattern continues in both directions: each step to the left multiplies by 10, and each step to the right divides by 10. This is why the exponent of 10 matches the number of places the digit is to the left of the ones place. It also explains why the number of zeros in 10^n is n when n is a positive whole number.

WORKED EXAMPLE 2.2.1 — EXPANDED FORM WITH POWERS

PROBLEM

Write 5,207 in expanded form using powers of ten.

1. Read the digits by place: 5 thousands, 2 hundreds, 0 tens and 7 ones.
2. Replace each place with its power of ten: $5 \times 10^3 + 2 \times 10^2 + 0 \times 10^1 + 7 \times 10^0$.
3. The zero term contributes nothing, but it preserves the place structure. It may be omitted once the positions are clear.

ANSWER $5 \times 10^3 + 2 \times 10^2 + 0 \times 10^1 + 7 \times 10^0$**, which simplifies to** $5{,}000 + 200 + 0 + 7$**.**

Check. Evaluate each term and add: $5{,}000 + 200 + 7 = 5{,}207$.

WORKED EXAMPLE 2.2.2 — FROM POWERS TO STANDARD FORM

PROBLEM

Write $6 \times 10^4 + 0 \times 10^3 + 3 \times 10^2 + 9 \times 10^1 + 0 \times 10^0$ in standard form. State the value of the digit 3.

1. Evaluate each non-zero term: $6 \times 10^4 = 60{,}000$, $3 \times 10^2 = 300$, $9 \times 10^1 = 90$.
2. The zero terms leave empty places for the thousands and ones.
3. Place the digits in order: 6 ten-thousands, 0 thousands, 3 hundreds, 9 tens, 0 ones, giving 60,390.
4. The digit 3 is in the hundreds place, so its value is 300.

ANSWER **60,390; the 3 is worth 300.**

Expanded form is longer than standard form, so why use it? It exposes the value of every digit and makes the effect of multiplying or dividing by 10 visible. It is also the form in which many

algorithms and computer representations are built.

> **COMMON ERROR**
>
> Treating 10^4 as 40. The exponent is not a multiplier: $10^4 = 10 \times 10 \times 10 \times 10 = 10{,}000$. A useful check is to count the zeros: 10^4 has four zeros.

PRACTICE 2.2 · POWERS OF TEN AND EXPANDED FORM

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Evaluate: (a) 10^2 (b) 10^4 (c) 10^6 (d) 10^0. F
2. Write 8,304 in expanded form using powers of ten. F
3. Write $4 \times 10^3 + 7 \times 10^2 + 0 \times 10^1 + 6 \times 10^0$ in standard form. F
4. Write 90,050 in expanded form, then state the value of the digit 5. F
5. A number is written as $2 \times 10^4 + 5 \times 10^1$. Explain why the standard form contains zeros, and write the number. R
6. A school has 12,480 students. Write this number in expanded form using powers of ten and explain which digit contributes the most to the total. P
7. Compare 3×10^3 and 3×10^2. How many times larger is the first expression? Explain using place value. R
8. Write a five-digit number whose expanded form contains exactly one zero term and whose digits add to 18. Explain how you constructed it and whether more than one answer is possible. E
9. Use the pattern $10^3 = 1{,}000$, $10^2 = 100$, $10^1 = 10$ and repeated division by 10 to explain why $10^0 = 1$. Then explain why the same reasoning supports $10^1 = 10$ as the ones place's immediate neighbour. R

2.3 Squares, cubes and roots

CRITERION A · I, III ◆ SQUARE NUMBERS ◆ INVERSE OPERATIONS

A square number can be arranged as a square array of dots or tiles. A cube number can be arranged as a cube. These pictures explain both the names and the link between a power and its root.

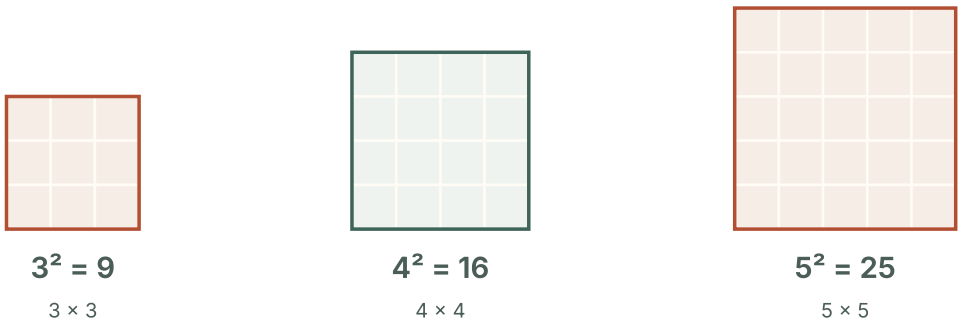


Figure 2.2 The side length of each array is the base. The number of cells inside is the square number. A square with side n contains n^2 cells.

DEFINITIONS

A **square number** is the result of multiplying a whole number by itself: $1, 4, 9, 16, \ldots$. The **square root** of a number is the positive value that produces it when squared. We write $\sqrt{49} = 7$ because $7^2 = 49$. A **cube number** is the result of multiplying a whole number by itself twice: $1, 8, 27, 64, \ldots$. The **cube root** of 64 is 4 because $4^3 = 64$.

USEFUL SQUARE AND CUBE NUMBERS

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2	1	4	9	16	25	36	49	64	81	100	121	144
n^3	1	8	27	64	125	216	343	512	729	1,000	1,331	1,728

Squaring and taking a square root are inverse operations. If you square 9 you get 81, and if you take the square root of 81 you return to 9. The same relationship holds for cubes and cube roots. The symbol $\sqrt{\ }$ asks "which positive number, when squared, gives this value?"

WORKED EXAMPLE 2.3.1 POWERS AND ROOTS

PROBLEM

Evaluate: (a) 12^2, (b) 7^3, (c) $\sqrt{81}$, (d) $\sqrt{144}$, (e) $\sqrt[3]{64}$.

1. (a) $12^2 = 12 \times 12 = 144$.
2. (b) $7^3 = 7 \times 7 \times 7 = 49 \times 7 = 343$.
3. (c) $9^2 = 81$, so $\sqrt{81} = 9$.
4. (d) $12^2 = 144$, so $\sqrt{144} = 12$.
5. (e) $4^3 = 64$, so $\sqrt[3]{64} = 4$.

ANSWER **(a) 144 (b) 343 (c) 9 (d) 12 (e) 4**

Check. Reverse each root by squaring or cubing the result. If the original number returns, the root is correct.

WORKED EXAMPLE 2.3.2 ROOTS IN GEOMETRY

PROBLEM

(a) A square has area $121\ \text{cm}^2$. Find its side length. (b) A cube has volume $125\ \text{cm}^3$. Find its edge length. (c) If the edge of a cube doubles, what happens to its volume?

1. (a) The area of a square is s^2. Since $11^2 = 121$, the side length is $\sqrt{121} = 11$ cm.
2. (b) The volume of a cube is s^3. Since $5^3 = 125$, the edge length is $\sqrt[3]{125} = 5$ cm.
3. (c) Doubling the edge replaces s by $2s$. The new volume is $(2s)^3 = 2^3 \times s^3 = 8s^3$, so the volume becomes eight times as large.

ANSWER **(a) 11 cm (b) 5 cm (c) the volume is multiplied by 8.**

COMMON ERROR

Distributing a square root over a sum: $\sqrt{36 + 64} \neq \sqrt{36} + \sqrt{64}$. The left side is $\sqrt{100} = 10$, while the right side is $6 + 8 = 14$. A root applies to the whole value underneath it, not to each part separately.

GUIDED PRACTICE · COPY AND COMPLETE

Since $_____^2 = 169$, we have $\sqrt{169} = _____$. Since $_____^3 = 216$, we have $\sqrt[3]{216} = _____$. A square with area $196\ \text{cm}^2$ has side length $_____$ cm. Explain why the side length is a root rather than a power.

PRACTICE 2.3 · SQUARES, CUBES AND ROOTS SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. List the first twelve square numbers. F
2. Evaluate: (a) 13^2 (b) 15^2 (c) 6^3 (d) 10^3. F
3. Evaluate: (a) $\sqrt{49}$ (b) $\sqrt{100}$ (c) $\sqrt{169}$ (d) $\sqrt[3]{8}$ (e) $\sqrt[3]{27}$ (f) $\sqrt[3]{125}$. F
4. A square mosaic uses 144 identical tiles. How many tiles are along each side? P
5. Find the positive value of x: (a) $x^2 = 64$ (b) $x^3 = 216$ (c) $x^2 = 225$. F
6. Compare $\sqrt{81} + \sqrt{16}$ with $\sqrt{81 + 16}$. Are they equal? Explain the difference. R
7. Between which two consecutive whole numbers does $\sqrt{50}$ lie? Explain using the square numbers on either side, then state which whole number it is closer to. E
8. A cube has volume $343\ \text{cm}^3$. Find its edge length. If the edge length is doubled, calculate the new volume and explain how your answer connects to 2^3. E

2.4 Order of operations

CRITERION A · II ◆ BRACKETS AND INDICES ◆ CONVENTION

An expression such as $2 + 3 \times 4$ could produce 20 or 14 depending on the order you choose. Mathematics avoids that ambiguity by fixing an order that everyone uses.

ORDER OF OPERATIONS

Priority	Operations	How to apply it
1	Brackets and other grouping symbols	Work from the innermost bracket outwards. A fraction bar groups everything above and below it.
2	Indices and roots	Evaluate powers and roots before multiplication, division, addition or subtraction.
3	Multiplication and division	Same priority: work from left to right.
4	Addition and subtraction	Same priority: work from left to right.

Brackets are the writer's tool for changing the order. Without brackets, $2 + 3 \times 4 = 2 + 12 = 14$. With brackets around the sum, $(2 + 3) \times 4 = 5 \times 4 = 20$. Multiplication and division share one priority level, and so do addition and subtraction; within a level, move left to right.

WORKED EXAMPLE 2.4.1 — APPLYING THE PRIORITY ORDER

PROBLEM

Evaluate: (a) $7 + 2 \times 3^2$, (b) $(7 + 2) \times 3^2$, (c) $24 \div 4 \times 2$, (d) $24 \div (4 \times 2)$.

1. (a) Evaluate the index first: $3^2 = 9$. Then multiply: $2 \times 9 = 18$. Finally add: $7 + 18 = 25$.
2. (b) Evaluate the bracket first: $7 + 2 = 9$. Then the index: $3^2 = 9$. Then multiply: $9 \times 9 = 81$.
3. (c) Multiplication and division have equal priority, so work left to right: $24 \div 4 = 6$, then $6 \times 2 = 12$.
4. (d) The bracket groups the product: $4 \times 2 = 8$, then $24 \div 8 = 3$.

ANSWER **(a) 25 (b) 81 (c) 12 (d) 3**

Check. The only difference between (a) and (b) is the bracket, yet the values differ by a factor of more than three. Brackets are not decoration.

WORKED EXAMPLE 2.4.2 — NESTED BRACKETS AND A FRACTION BAR

PROBLEM

Evaluate: (a) $(18 - 4 \times 2)^2 \div 5$, (b) $\dfrac{36 - 6}{2 + 4}$.

1. (a) Inside the bracket, multiplication comes before subtraction: $4 \times 2 = 8$, so $18 - 8 = 10$. Then square: $10^2 = 100$. Then divide by 5: $100 \div 5 = 20$.
2. (b) The fraction bar groups the whole numerator and the whole denominator. Numerator: $36 - 6 = 30$. Denominator: $2 + 4 = 6$. Then divide: $30 \div 6 = 5$.

ANSWER **(a) 20 (b) 5**

A fraction bar acts like a pair of brackets around the numerator and another pair around the denominator. This is why $\frac{36-6}{2+4} = 5$, not $36 - 6 \div 2 + 4$. Writing the fraction with a horizontal bar makes the grouping visible; entering the expression into a calculator without brackets often does not.

COMMON ERROR

Assuming multiplication always comes before division. They have the same priority, so $24 \div 4 \times 2$ is 12, not 3. The correct order within a priority level is left to right. A second common error is evaluating addition before multiplication: in $7 + 2 \times 3^2$, the addition waits until the multiplication is complete.

PRACTICE 2.4 · ORDER OF OPERATIONS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Evaluate $8 + 2 \times 5$. F
2. Evaluate $20 - 3 \times 4$. F
3. Evaluate: (a) $6 + 4^2$ (b) $(6 + 4)^2$ (c) $2^3 + 3^2$. F
4. Evaluate: (a) $24 \div 4 \times 2$ (b) $24 \div (4 \times 2)$ (c) $36 \div 6 + 3$. F
5. Evaluate $5 \times 2^3 - 3^2$, showing each stage of your working. F
6. Insert brackets to make each statement true: (a) $3 + 5 \times 2 = 16$ (b) $12 - 4 \times 2 = 16$ (c) $18 \div 3 + 6 = 2$. R
7. A student calculates $12 + 6 \div 3$ as $18 \div 3 = 6$. Identify the mistake, give the correct value, and explain which operation should be performed first. R
8. Use the digits 2, 3 and 4 once each, together with any operations and brackets, to make as many different values as you can. Record at least four expressions and their values; one expression must contain an index. E

2.5 Substitution and expressions with powers

CRITERION A · III ◆ FORMULAS ◆ STRUCTURE

A letter in an expression stands for a number. Substitution means replacing the letter with a value and evaluating the resulting expression using the order of operations. The position of the letter inside a power matters.

WORKED EXAMPLE 2.5.1

SAME SUBSTITUTION, DIFFERENT STRUCTURE

PROBLEM

Evaluate $3x^2$ and $(3x)^2$ when $x = 4$. Explain why the results differ.

1. In $3x^2$, the exponent applies only to x: $3x^2 = 3 \times (4^2) = 3 \times 16 = 48$.
2. In $(3x)^2$, the brackets make $3x$ the base: $(3 \times 4)^2 = 12^2 = 144$.
3. The bracket changes the base of the power. Without it, only x is squared. The second result is three times the first because $(3x)^2 = 3^2 \times x^2 = 9x^2$.

ANSWER $3x^2 = 48$ $(3x)^2 = 144$

WORKED EXAMPLE 2.5.2 EVALUATING FORMULAS

PROBLEM

(a) The area of a square is $A = s^2$. Find A when $s = 7$ cm. (b) The volume of a cube is $V = s^3$. Find V when $s = 4$ cm. (c) A model for the distance fallen by an object is $d = 5t^2$, where d is in metres and t is in seconds. Find d when $t = 3$.

1. (a) Substitute $s = 7$: $A = 7^2 = 49$ cm^2.
2. (b) Substitute $s = 4$: $V = 4^3 = 64$ cm^3.
3. (c) Substitute $t = 3$: $d = 5 \times 3^2 = 5 \times 9 = 45$ m. The index is evaluated before the multiplication by 5.

ANSWER **(a)** 49 cm^2 **(b)** 64 cm^3 **(c) 45 m**

Substitution is more than a calculation routine. It is a test of structure: two expressions that look similar may behave differently when a value is inserted. Comparing $3x^2$ with $(3x)^2$, or $a^2 + b^2$ with $(a + b)^2$, reveals where the exponent actually applies.

COMMON ERROR

Applying the exponent to every factor in a product: $(3x)^2 \neq 3x^2$. The first expression squares the whole product $3x$; the second squares only x. Write the substitution in brackets before calculating.

GUIDED PRACTICE · COPY AND COMPLETE

For $x = 5$, $2x^2 =$ _____ while $(2x)^2 =$ _____. For $x = 2$, $x^2 + 3^2 =$ _____ while $(x + 3)^2 =$ _____. In one sentence, explain what the brackets change about the base of the power.

PRACTICE 2.5 · SUBSTITUTION AND POWERS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Evaluate $2n^2$ for (a) $n = 3$, (b) $n = 5$, (c) $n = 10$. F
2. Evaluate $(2n)^2$ for (a) $n = 3$, (b) $n = 5$, (c) $n = 10$. Compare each result with the corresponding result in question 1. F
3. For $a = 3$ and $b = 4$, evaluate $a^2 + b^2$ and $(a + b)^2$. Are they equal? F
4. A square has side length $s = 12$ cm. Use $A = s^2$ to find its area, with units. P
5. Use $V = s^3$ to find the volume of a cube with edge (a) 3 cm, (b) 5 cm. F
6. The distance fallen model is $d = 5t^2$. Find d when (a) $t = 2$, (b) $t = 4$. Explain why doubling t does not double d. R
7. Insert one pair of brackets to make each statement true: (a) $2x^2 = 36$ when $x = 3$ (b) $a + b^2 = 49$ when $a = 2$ and $b = 5$. Explain how the brackets change the base of the power. R
8. A square has area 81 cm^2. Find its side length. If the side length is tripled, find the new area and explain how 3^2 appears in your answer. E

Real-life application · The school data store

A media club stores video projects on a server with three drives. The labels use decimal units: 1 TB = 1,000 GB and 1 GB = 1,000 MB.

Drive	Label	Capacity in GB	Power form
A	4 TB	4,000	4×10^3
B	4 TB	4,000	4×10^3
C	2 TB	2,000	2×10^3
Total	**10 TB**	**10,000**	10^4

Two constraints shape the club's planning. The operating system and recovery files reserve 15% of the total capacity. Each finished project needs 5 GB for the master file and 1 GB for its backup.

Your task. Work through the prompts below and show every unit.

1. Write a single calculation that finds the number of complete projects the server can hold after the reserve is removed. Show the order of operations and state the space left over.
2. Explain why the answer must be rounded down rather than to the nearest whole number.
3. A computer reports the 10 TB total as 9.09 TiB. One tebibyte is 2^{40} bytes, while one terabyte is 10^{12} bytes. Explain why the same capacity can carry two different numbers.
4. Decide which number the club should quote to (a) a technician ordering a replacement drive and (b) a family reading a poster. Explain what "honest" should mean when units are chosen.

Investigation 2 · The doubling sheet

CRITERION B · I, II, III

A sheet of paper is about 0.1 mm thick. Each fold doubles the number of layers, so the height of the folded stack doubles. Assume that every fold is perfect and that the paper keeps the same thickness.

Number of folds n	Layers 2^n	Thickness in mm	Thickness as a power
0	1	0.1	0.1×2^0
1	2	0.2	0.1×2^1
2	4	0.4	0.1×2^2
3	8	?	0.1×2^3
4	16	1.6	0.1×2^4
5	32	3.2	0.1×2^5
6	64	?	0.1×2^6
7	128	12.8	0.1×2^7
8	256	25.6	0.1×2^8
9	512	?	0.1×2^9
10	1,024	102.4	0.1×2^{10}

1. Complete the three missing thickness values. Check each one by doubling the row above it.
2. How many times thicker is the stack after 10 folds than after 0 folds? Relate your answer to a power of 2.
3. Write a rule for the thickness T millimetres after n folds.
4. Find the least number of folds for which the stack is thicker than 1 m. Show the values on either side of your answer.
5. A classroom is about 3 m high. Which fold count first produces a stack taller than the room? Compare this with your answer to question 4.
6. People often say that paper cannot be folded more than about eight times. Does the table prove that claim false? Distinguish what the model predicts from what a physical sheet allows.
7. State two assumptions of the model and one way to improve the estimate of the sheet's thickness.
8. Extension: a different sheet triples in thickness with each fold, so $T = 0.1 \times 3^n$. Make a table for $n = 0$ to 5 and compare it with the doubling stack at $n = 5$. Which grows faster, and by what factor after 10 folds?

Present your rule in words and symbols. A convincing conclusion should separate what the numbers predict from what can actually be done with a sheet of paper.

Criterion-mapped mini-task

A POWER PRIMER FOR THE SCIENCE LAB

A · KNOWING AND UNDERSTANDING
C · COMMUNICATING
D · APPLYING MATHEMATICS

Context. The science department is producing a one-page reference card for Year 1 students. It will be pinned beside the microscopes. The card must explain powers, squares, cubes, roots and the order of operations for a reader who has not yet studied this chapter.

Product. Design the card using no more than 180 words of explanation, plus diagrams and one fully worked example. The card should be readable from about one metre away.

Requirements.

1. Define *power*, *base* and *exponent*, using $5^3 = 125$ as your example.
2. Explain 5^1 and 5^0 with a pattern that makes clear why $5^0 = 1$ rather than 0.
3. Show $4^2 = 16$ as a square of 16 unit squares. Explain why "square" is a suitable name. Add one sentence and a sketch or description explaining why 4^3 is called a cube.
4. Work through $4 + 3 \times (8 - 2)^2 \div 6 = 22$, with one line per operation and a label naming the rule used at each step.
5. Explain the difference between $3x^2$ and $(3x)^2$ when $x = 2$. State both values.
6. Include a science example: a culture of 500 bacteria doubles every hour. Write an expression for the number after h hours, find the number after 5 hours, and explain why it is not 500×5.
7. Ask a classmate to use your card to evaluate $2 + 5 \times 3^2$. Record whether they succeed, then revise one part of the card in response.
8. Evaluate your card: state one strength, one limitation and one change you would make for a younger audience.

CRITERIA AND STRANDS A i, ii, iii: accurate definitions and calculations; C i, ii, iii, iv: precise language, notation, diagrams and audience awareness; D i, ii, iii, iv: select and apply strategies, justify steps and evaluate the fitness of the card for its purpose.

BEFORE YOU SUBMIT

Check every number on the card independently. A reader should be able to follow the worked example without consulting the chapter. Ask your test reader to explain one step back to you; if they cannot, the card needs clearer language rather than more words.

Chapter review

SUMMARY

- **Index notation.** In a^n, the base a is the repeated factor and the exponent n counts the factors. The value is not $a \times n$.
- **Special cases.** $a^1 = a$ for every base, and $a^0 = 1$ for every non-zero base. The zero rule continues the pattern of dividing by a.
- **Powers of ten.** Each place is ten times the place to its right, so expanded form can be written with powers of ten. 10^n has n zeros for whole-number n.
- **Squares and cubes.** A square number is $n \times n$; a cube number is $n \times n \times n$. Square roots and cube roots reverse these operations.
- **Order of operations.** Work through brackets, then indices, then multiplication and division from left to right, then addition and subtraction from left to right.
- **Substitution.** Replace each variable with its value, keep brackets around the replacement, then evaluate using the order of operations.

Power: $a^n = a \times a \times \cdots \times a$ with n factors.

Special cases: $a^1 = a$; $a^0 = 1$ for $a \neq 0$.

Roots: $\sqrt{a^2} = a$ for $a \geq 0$; $\sqrt[3]{a^3} = a$.

Priority: brackets → indices → $\times$, $\div$ (left to right) → $+$, $-$.

REVISION CHECKLIST

- ☐ I can write a repeated product as a power and evaluate it.
- ☐ I can explain why $a^0 = 1$ and $a^1 = a$.
- ☐ I can write a whole number in expanded form using powers of ten.
- ☐ I can find squares, cubes, square roots and cube roots.
- ☐ I can apply the order of operations without skipping a step.
- ☐ I can evaluate an expression after substituting a value, including brackets.
- ☐ I can explain the difference between $3x^2$ and $(3x)^2$.
- ☐ I can describe how a doubling or tripling model grows and state its limits.

KEY VOCABULARY

power
An expression such as a^n, or its value.

base
The factor that is repeated in a power.

exponent
The number of times the base is used as a factor; also called the index.

square number
A number of the form $n \times n$, or n^2.

cube number
A number of the form $n \times n \times n$, or n^3.

square root
The positive value whose square is the given number.

cube root
The value whose cube is the given number.

order of operations
The agreed sequence for evaluating an expression with several operations.

REFLECTION AND SELF-ASSESSMENT

Choose one calculation from this chapter that you first answered incorrectly. Explain what rule you had misapplied and how you now recognise the situation.

Rate your confidence from 1 to 4 on each success criterion. For any criterion rated 1 or 2, name the exercise you will revisit and the specific help you need. Then explain in two sentences why a shared order of operations is necessary for expressions to communicate one meaning.

END-OF-CHAPTER PRACTICE

Mixed practice 2

Do not use a calculator unless a question states otherwise. Show the order of operations for questions with more than one operation.

FLUENCY Build accuracy

1. Evaluate: (a) 3^4 (b) 2^7 (c) 10^6 (d) 6^3 (e) 11^2 (f) 4^4.
2. Write as a single power: (a) $9 \times 9 \times 9$ (b) 12×12 (c) $10 \times 10 \times 10 \times 10 \times 10$ (d) $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$.
3. Write 60,304 in expanded form using powers of ten.
4. Write $8 \times 10^4 + 2 \times 10^2 + 5 \times 10^0$ in standard form.
5. Evaluate: (a) $\sqrt{81}$ (b) $\sqrt{196}$ (c) $\sqrt[3]{27}$ (d) $\sqrt[3]{1{,}000}$.
6. Calculate: (a) $6 + 2 \times 5$ (b) $(6 + 2) \times 5$ (c) $3^2 \times 4 - 10$ (d) $48 \div (4 + 4)$ (e) $5 + 12 \div 3$ (f) $(7 - 2)^2$.
7. Substitute $x = 3$ and $y = 2$: (a) $x^2 + y^2$ (b) $(x + y)^2$ (c) $2x^3$ (d) $(2x)^2$.
8. Insert one pair of brackets to make each statement true: (a) $2 + 3 \times 4 = 20$ (b) $24 - 8 \div 4 = 4$ (c) $3 \times 2^2 = 36$ (d) $10 - 4^2 = 36$.

REASONING AND PROBLEM SOLVING Connect ideas

9. Explain why $2^5 = 32$ but $5^2 = 25$. Which is larger, and why does the position of the exponent matter?
10. A student writes $7^2 = 14$. Identify the error and give the correct value with a one-sentence explanation.
11. Find the missing whole number: (a) $2^n = 64$ (b) $10^n = 100{,}000$ (c) $3^n = 81$ (d) $n^3 = 125$.
12. Explain why $10^0 = 1$, using the pattern $10^3 = 1{,}000$, $10^2 = 100$, $10^1 = 10$.
13. Evaluate $10^3 + 10^2 + 10^1 + 10^0$. Describe the pattern in the digits of the answer and explain why it occurs.
14. A square rug has area 169 cm^2. Find its side length and perimeter. If the side length is doubled, find the new area and perimeter. Explain why the area changes by a factor of 2^2 while the perimeter changes by a factor of 2.
15. A cube has edge 2 cm. Find its volume. Repeat for a cube with edge 4 cm. How many times larger is the second volume? Explain using 2^3.
16. Using the numbers 4, 3 and 2 in that order, insert operations and brackets to make (a) 14 and (b) 20. Check each result with the order of operations.
17. A rumour spreads in rounds. One student tells 3 friends in round 1. In round 2, each of those 3 tells 3 new people, and the pattern continues. (a) How many new people hear it in round 4? (b) How many people in total have heard it after 4 rounds, including the original student? (c) Explain why the answer to (b) is not 3^5.
18. Explain why $\sqrt{36 + 64} \neq \sqrt{36} + \sqrt{64}$. Evaluate both expressions and state the rule that the example disproves.

CRITERION-STYLE Apply and evaluate

19. **Bacterial culture (Criteria A, C, D).** A laboratory culture starts with 400 bacteria and doubles every hour. (a) Write an expression for the number of bacteria after h hours. (b) Calculate the number after 3, 5 and 8 hours. (c) Explain why the number after 3 hours is not 400×3. (d) The laboratory reports "about 100,000 bacteria" after 8 hours. Is this a sensible approximation? Justify your answer using the exact value. (e) A second culture starts with 800 bacteria and also doubles every hour. Will the first culture ever catch up? Explain.
20. **A fair storage label (Criteria C, D).** A 2 TB drive is advertised. A computer reports its capacity as 1.82 TiB. (a) Show that 2 TB $= 2 \times 10^{12}$ bytes. (b) Use 1 TiB $= 2^{40}$ bytes to verify the computer's report. (c) Explain why the capacity appears smaller and whether any storage is actually missing. (d) The drive already holds 250 GB of school files. Calculate the percentage of the advertised capacity in use. (e) Write a label of no more than 30 words that explains the two numbers fairly to a school office manager.

Check your progress

ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

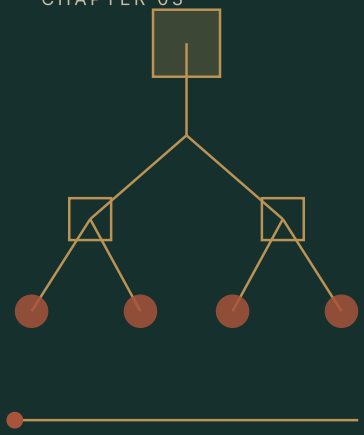
- **An exponent is not a multiplier.** $5^2 = 5 \times 5 = 25$, not $5 \times 2 = 10$. The exponent counts factors.
- **Brackets decide the base.** $3x^2$ squares only x; $(3x)^2$ squares the whole product. For $x = 4$, the values are 48 and 144.
- **Multiplication and division share a rank.** Work from left to right: $24 \div 4 \times 2 = 12$, not $24 \div 8 = 3$.
- **Roots do not distribute over sums.** $\sqrt{36 + 64} = 10$, while $\sqrt{36} + \sqrt{64} = 14$.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can write and evaluate powers, including the cases a^1 and a^0.			
I can write a number in expanded form using powers of ten.			
I can find squares, cubes and the corresponding roots.			
I can apply the order of operations accurately and in the correct sequence.			
I can substitute values into expressions with powers while respecting brackets.			
I can explain how a doubling or tripling model grows and where it stops describing reality.			

WHERE THIS LEADS

Chapter 3 asks which numbers can be built from others. Factors, multiples and primes expose the multiplicative structure that powers describe in shorthand, and prime factorisation will let you write a number such as 360 as $2^3 \times 3^2 \times 5$. The same index notation returns whenever a factor repeats, from divisibility tests to the area and volume formulas in Part IV.

Review the answer key only after attempting each question.



3

STRUCTURE BENEATH THE SURFACE

Factors, multiples and primes

Every whole number is built from smaller whole numbers. Learning to see that structure turns division, fair sharing and repeating schedules into questions with exact answers.

KEY CONCEPT · RELATIONSHIPS

RELATED · SYSTEM

RELATED · REPRESENTATION

CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

When is a number best understood by what it is made of, and when does the way we group things decide whether a division is fair?

MYP UNIT PLANNER

Framing the inquiry

UNIT 3 · APPROX. 13 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Relationships. Factors and multiples describe how whole numbers are connected: one number divides another exactly, or two numbers share a common structure.

System. The primes are building blocks: every whole number greater than 1 is assembled from them in exactly one way, apart from order.

Representation. Factor trees, index notation and Venn diagrams display the same factorisation in different forms.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Exploration: how number structure shapes fair division, shared timetables and the allocation of resources.

A school cannot make identical mixed teams from 84, 60 and 48 students unless the number of teams divides all three. Two bells ring together only at common multiples. Arithmetic decides what is possible before anyone negotiates.

STATEMENT OF INQUIRY

Whole numbers are built from factors in a structured way, and that structure makes it possible to divide quantities fairly, schedule repeating events and communicate precisely.

FACTUAL

What is a factor of 36? What is a multiple of 7? Which numbers below 100 are prime?

CONCEPTUAL

Why is 1 neither prime nor composite? Why is every prime factorisation unique? How are HCF and LCM related?

DEBATABLE

Is a number with many factors more useful than a prime number, or does usefulness depend entirely on the problem?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Find factors and multiples, apply divisibility tests, identify primes, calculate HCF and LCM.
B · Investigating patterns	i, ii, iii	Investigate how many factors a number has; generalise why squares have an odd number of factors.
C · Communicating	i, ii, iii, iv	Represent factorisations with trees, index notation and diagrams, using correct vocabulary.
D · Applying mathematics	i, ii, iii, iv	Solve grouping and scheduling problems, justify assumptions and evaluate the fairness of a plan.

ATL · CRITICAL THINKING IN ACTION

A classmate says: “Every number with a 3 in it is divisible by 3.” Test the claim with 13, 23, 36 and 132. Decide which examples support it, which refute it, and what a better divisibility test would look at. Record the counterexample you would use to settle the question in a class discussion.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** the relationship between factors and multiples, and list factor pairs systematically.
- **Know** the divisibility tests for 2, 3, 4, 5, 6, 8, 9 and 10 and explain how they work.
- **Be able to** classify numbers as prime, composite or neither, and test whether a number is prime.
- **Be able to** write any composite number as a product of primes using a factor tree or division.
- **Be able to** find the highest common factor and lowest common multiple of two or three numbers.
- **Be able to** choose and justify HCF or LCM in sharing and scheduling contexts.

SUCCESS CRITERIA

- ☐ I can list all factors of a number without missing any, using factor pairs.
- ☐ I can explain the difference between a factor and a multiple with examples.
- ☐ I can decide whether a number is divisible by 2, 3, 4, 5, 6, 8, 9 or 10 and give evidence.
- ☐ I can identify prime numbers and explain why 1 is excluded.
- ☐ I can write a prime factorisation in index form and reconstruct the number from it.
- ☐ I can calculate HCF and LCM and explain which one a context requires.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate 7×8 and 9×6.
2. Divide 96 by 8.
3. List the first five multiples of 4.
4. Write 100 as a product of two equal factors.
5. Which is larger: $3 \times 3 \times 3$ or 4×4?
6. Is 56 in the 7 times table? Explain how you know.

Answers are in the Chapter 3 answer section. If more than two are incorrect, review multiplication and division facts before continuing.

BEFORE YEAR 1
Multiplication tables, division facts and simple fractions of amounts.

YEAR 1 · THIS CHAPTER
Factor pairs, divisibility, primes, prime factorisation, HCF and LCM.

LATER
Laws of indices, algebraic factorisation, number theory and cryptography.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Groups, timetables and shared resources must divide exactly.	What smaller numbers build a given whole number?	Factors, multiples, primes and common structure.	Count the factors of the numbers 1 to 30 and explain the pattern.	Represent factorisations as trees, index products and diagrams.	When does the largest common group produce the fairest result?

3.1 Factors, multiples and factor pairs

CRITERION A · I ◆ RELATIONSHIPS ◆ EXACT DIVISION

Some divisions leave a remainder and some do not. When a division is exact, the divisor and the quotient are both factors of the number, and the number is a multiple of each of them.

DEFINITIONS

A **factor** of a whole number divides it exactly, with no remainder. Because $36 = 4 \times 9$, both 4 and 9 are factors of 36. A **multiple** of a number is the result of multiplying it by a whole number: 36 is a multiple of 4 because $36 = 4 \times 9$. A **factor pair** is a pair of whole numbers whose product is the given number.

The two ideas are reverses of each other. If 7 is a factor of 84, then 84 is a multiple of 7. The equation $84 = 7 \times 12$ shows both facts at once: 7 and 12 are factors, and 84 is a multiple of each. A single equation can therefore answer two different questions.

FACTOR PAIRS OF 36, FOUND IN ORDER

First factor	1	2	3	4	6
Partner	36	18	12	9	6
Product	36	36	36	36	36

Listing factors in pairs is safer than listing them in a single stream. Start with 1 and its partner, then 2 and its partner, and continue until the two factors meet. For 36 the pair 6×6 is the meeting point: any larger first factor would pair with a number already listed. The factors are $1, 2, 3, 4, 6, 9, 12, 18, 36$ — nine in total.

WORKED EXAMPLE 3.1.1 — A SYSTEMATIC FACTOR LIST

PROBLEM

List all factors of 48. How many are there?

1. Begin with 1: $1 \times 48 = 48$, so 1 and 48 are a pair.
2. Try 2: $2 \times 24 = 48$. Try 3: $3 \times 16 = 48$. Try 4: $4 \times 12 = 48$.
3. Try 5: 48 is not a multiple of 5. Try 6: $6 \times 8 = 48$.
4. Try 7: no. The next whole number is 8, whose partner 6 has already appeared. The pairs would now repeat, so the list is complete.
5. Collect the factors in order: $1, 2, 3, 4, 6, 8, 12, 16, 24, 48$.

ANSWER **Ten factors: 1, 2, 3, 4, 6, 8, 12, 16, 24 and 48.**

Check. Pair each factor with its partner: 1×48, 2×24, 3×16, 4×12, 6×8. Every factor appears once and every product is 48.

WORKED EXAMPLE 3.1.2 FACTORS IN A SEATING PROBLEM

PROBLEM

A hall has 24 identical chairs. They must be arranged in equal rows with no chairs left over. Which numbers of rows are possible, and how many chairs would each row contain?

1. Each possible number of rows must be a factor of 24.
2. Factor pairs of 24: 1×24, 2×12, 3×8, 4×6.
3. A pair gives two arrangements, because either number can describe the rows. For example, 4 rows of 6 chairs, or 6 rows of 4 chairs.

ANSWER **1, 2, 3, 4, 6, 8, 12 or 24 rows. The partner factor gives the number of chairs in each row.**

Check. Rows multiplied by chairs per row must return 24. If the organiser wants more than 5 rows but fewer than 10, the possibilities are 6 rows of 4 or 8 rows of 3.

COMMON ERROR

Confusing factors with multiples. Factors of 12 are 1, 2, 3, 4, 6 and 12; multiples of 12 begin 12, 24, 36, 48. A number has finitely many factors but infinitely many multiples. A second common error is counting the square factor twice: 36 has nine factors, not ten, because the pair 6×6 contributes one factor, 6.

GUIDED PRACTICE · COPY AND COMPLETE

The factor pairs of 40 are $1 \times$ ____, $2 \times$ ____, $4 \times$ ____ and $5 \times$ ____. So 40 has ____ factors. The number 40 is a multiple of 8 because ____.

PRACTICE 3.1 · FACTORS AND MULTIPLES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

List factors in pairs so that none are missed.

1. List all factors of (a) 20, (b) 28, (c) 45. State how many factors each number has. F
2. Write all factor pairs of (a) 30, (b) 42, (c) 64. F
3. Write the first five multiples of (a) 8, (b) 13. F
4. Decide whether each statement is true or false, and correct any false statement: (a) 6 is a factor of 42; (b) 42 is a factor of 6; (c) 1 is a factor of 37; (d) 16 is a multiple of 4. R
5. A box holds 36 eggs. Eggs are packed in cartons of 6, 10 or 4. Which carton sizes use every egg with none left over? Give the number of cartons in each case. P
6. Two factors of a number are 8 and 9. Write the number and list all of its factors. P
7. Explain why 6 is a factor of 36 and 36 is a multiple of 6, using a single multiplication equation. Then explain why 5 is not a factor of 36. R
8. Find every number between 20 and 30 that has exactly four factors. Describe the order in which you searched, and explain why checking only numbers of a certain kind made the search quicker. E

3.2 Divisibility tests

CRITERION A · I, II ◆ PLACE VALUE ◆ JUSTIFICATION

Dividing every number by every possible factor would be slow. Divisibility tests use the digits of a number to decide whether a division is exact, without carrying out the division.

A test is a shortcut with a reason. The test for 2 works because 10 is even, so every place from the tens upwards is even. The test for 4 looks only at the last two digits because 100 is a multiple of 4, and every higher place is a multiple of 100.

DIVISIBILITY TESTS FOR WHOLE NUMBERS

Divisible by	Test	Reason
2	The last digit is 0, 2, 4, 6 or 8.	Every higher place is a multiple of 10.
5	The last digit is 0 or 5.	Every higher place is a multiple of 10.
10	The last digit is 0.	A multiple of 10 ends in 0.
3	The digit sum is a multiple of 3.	Every place is 1 more than a multiple of 3.
9	The digit sum is a multiple of 9.	Every place is 1 more than a multiple of 9.
4	The last two digits form a multiple of 4.	100 is a multiple of 4.
8	The last three digits form a multiple of 8.	1,000 is a multiple of 8.
6	The number passes the tests for 2 and for 3.	$6 = 2 \times 3$, with no shared factor.

The digit-sum tests for 3 and 9 are consequences of place value. In 4,728, the digit 4 is worth $4{,}000 = 4 \times 999 + 4$, the 7 is worth $700 = 7 \times 99 + 7$, and so on. All the multiples of 9 disappear when we ask about divisibility by 3 or 9, leaving the sum of the digits. This is why the test works for any size of number.

WORKED EXAMPLE 3.2.1 — TESTING ONE NUMBER EIGHT WAYS

PROBLEM

Which of 2, 3, 4, 5, 6, 8, 9 and 10 divide 4,728 exactly?

1. Ends in 8: divisible by 2.
2. Digit sum 21, a multiple of 3, so divisible by 3. Not by 9, since 21 is not a multiple of 9.
3. Last two digits 28, and $28 = 4 \times 7$: divisible by 4.
4. Ends in 8, not 0 or 5: not divisible by 5 or 10.
5. Last three digits 728, and $728 = 8 \times 91$: divisible by 8.
6. Divisible by 2 and by 3: divisible by 6.

ANSWER **2, 3, 4, 6 and 8 divide 4,728 exactly; 5, 9 and 10 do not.**

Check. $4{,}728 \div 6 = 788$, and $6 \times 788 = 4{,}728$.

WORKED EXAMPLE 3.2.2 — A FAILED TEST AND A REMAINDER

PROBLEM

Without dividing, decide whether 3,156 is divisible by 8 and by 9. Then state the remainder when 3,156 is divided by 8.

1. For 8, examine the last three digits: 156. Since $8 \times 19 = 152$ and $8 \times 20 = 160$, 156 is not a multiple of 8. So 3,156 is not divisible by 8.
2. For 9, add the digits: $3 + 1 + 5 + 6 = 15$. The nearest multiples of 9 are 9 and 18, so 15 is not a multiple of 9. So 3,156 is not divisible by 9.
3. To find the remainder on division by 8, only the last three digits matter: $156 = 8 \times 19 + 4$. The remainder is 4.

ANSWER **Not divisible by 8 (remainder 4) and not divisible by 9.**

Check. $3{,}156 \div 8 = 394.5$, and $8 \times 394 = 3{,}152$, leaving 4. The digit sum 15 also shows that 3,156 is divisible by 3 but not by 9.

COMMON ERROR

Assuming that a number with digit sum divisible by 3 must also be divisible by 9. The digit sum 12 passes the test for 3 but not for 9, so 84 is divisible by 3 but not by 9. A second error is using only the 3 test for divisibility by 6: the number must also be even. For example, 513 has digit sum 9 but is odd, so it is not divisible by 6.

GUIDED PRACTICE · COPY AND COMPLETE

For 2,514: the last digit is _____, so the number _____ divisible by 2. The digit sum is _____, so it _____ divisible by 3 and _____ divisible by 9. The last two digits are 14, so it _____ divisible by 4. Therefore it _____ divisible by 6.

PRACTICE 3.2 · DIVISIBILITY TESTS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. For each number, state which of 2, 3, 5, 9 and 10 divide it exactly: (a) 84 (b) 95 (c) 120 (d) 216 (e) 315. F
2. Use the tests for 4 and 8 to decide, for each number, whether it is divisible by 4 and whether it is divisible by 8: (a) 1,432 (b) 2,516 (c) 7,208 (d) 5,310. F
3. Use digit sums to test divisibility by 3 and by 9: (a) 4,281 (b) 7,542 (c) 2,007 (d) 8,136. F
4. Find every possible digit for the box: (a) 3□4 is divisible by 3 (b) 52□ is divisible by 9 (c) 7□2 is divisible by 4 (d) 4□0 is divisible by 6. P
5. Explain why every number ending in 0 is divisible by both 2 and 5, but is not necessarily divisible by 4. Give an example and a non-example. R
6. A student claims that 513 is divisible by 6 because its digit sum is 9. Identify the missing part of the test, decide whether the claim is correct, and explain your reasoning. R
7. A number is divisible by 3 and by 4. Must it be divisible by 12? Test 36 and 48. Then investigate whether a number divisible by 8 and by 6 must be divisible by 48, using 24 as a counterexample. R
8. Using the digits 4, 5 and 6 exactly once each, write the largest three-digit number that is divisible by 4. Show the test you applied and explain why the other arrangements fail. E

3.3 Prime and composite numbers

CRITERION A · I, III ◆ CLASSIFICATION ◆ PROOF BY TESTING

Once a number is written as a product of smaller factors, those factors may themselves break down further. Numbers that cannot be broken down at all are the atoms of arithmetic.

> DEFINITIONS
>
> A **prime number** is a whole number greater than 1 with exactly two factors: itself and 1. A **composite number** is a whole number greater than 1 with more than two factors. The number 1 has only one factor, so it is neither prime nor composite.

The number 2 is prime because its only factors are 1 and 2, even though it is even. It is the only even prime, since every other even number has 2 as a factor as well as 1 and itself. The number 1 is excluded by definition because including it would allow a number to be written as a product of primes in infinitely many ways — for example $6 = 2 \times 3 = 1 \times 2 \times 3 = 1 \times 1 \times 2 \times 3$ — and the structure would lose its precision.

THE 25 PRIME NUMBERS BELOW 100

Range	Primes	Count
1–20	2, 3, 5, 7, 11, 13, 17, 19	8
21–40	23, 29, 31, 37	4
41–60	41, 43, 47, 53, 59	5
61–80	61, 67, 71, 73, 79	5
81–100	83, 89, 97	3

To decide whether a number n is prime, it is enough to test the primes up to $\sqrt{n}$. If any of them divides n, the number is composite. If none does, n is prime. The reason is that factors always occur in pairs: a factor below the square root is matched by a partner above it. If no prime factor exists below the square root, no factor pair can exist at all.

WORKED EXAMPLE 3.3.1 — CLASSIFY AND EXPLAIN

PROBLEM

Classify each number as prime or composite: (a) 21, (b) 29, (c) 51, (d) 91, (e) 97.

1. (a) $21 = 3 \times 7$, so 21 is composite.
2. (b) 29 is not divisible by 2, 3 or 5, and $5^2 = 25 < 29 < 7^2 = 49$, so no factor below $\sqrt{29}$ exists. It is prime.
3. (c) $51 = 3 \times 17$ because $5 + 1 = 6$. It is composite.
4. (d) 91 is not divisible by 2, 3 or 5, but $91 = 7 \times 13$. It is composite — a useful reminder that a number can look prime and still fail.
5. (e) Test the primes up to $\sqrt{97} \approx 9.8$: 2, 3, 5 and 7 all fail. So 97 is prime.

ANSWER **(a) composite (b) prime (c) composite (d) composite (e) prime**

Check. A composite number has at least one prime factor no larger than its square root. Finding such a factor is a complete proof of compositeness; testing every integer is unnecessary.

WORKED EXAMPLE 3.3.2 — IS 149 PRIME?

PROBLEM

Show that 149 is prime. Then explain why testing primes beyond 12 is unnecessary.

1. $\sqrt{149}$ lies between 12 and 13, because $12^2 = 144$ and $13^2 = 169$.
2. Test the primes up to 12: 2 (149 is odd); 3 (digit sum 14); 5 (does not end in 0 or 5); 7 ($7 \times 21 = 147$); 11 ($11 \times 13 = 143$ and $11 \times 14 = 154$).
3. None divides 149, so 149 has no factor pair with a factor below its square root.
4. If 149 had a factor greater than 12, its partner would be less than 12, and we have already ruled those out.

ANSWER **149 is prime.**

Check. The nearest composite candidates illustrate why the boundary matters: 143 has the factor 11, and 147 has the factor 7. Both lie below 13^2, and both would be found by tests up to 12.

COMMON ERROR

Calling 1 a prime number. It has only one factor, not two, so it is neither prime nor composite. Another error is assuming that 2 is not prime because it is even, or that a number is prime as soon as it is not divisible by 2, 3 or 5. For example, 91 passes those three tests but equals 7×13.

PRACTICE 3.3 · PRIMES AND COMPOSITES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. List all prime numbers between 20 and 40. F
2. Which of these are composite: 33, 43, 57, 61, 87? Give the factors that prove it. F
3. Decide whether each number is prime: (a) 121 (b) 127 (c) 131 (d) 133. Show the primes you tested. P
4. Explain why 2 is prime even though it is even, and why 1 is neither prime nor composite. R
5. A student says 87 is prime because it is not divisible by 2, 3 or 5. Identify the error and give the correct classification. R

6. Two prime numbers that differ by 2 are called twin primes. Find a twin-prime pair between 50 and 70, and state the sum of the pair. P

7. Goldbach's conjecture states that every even number greater than 2 can be written as the sum of two primes. (a) Write 28, 40 and 100 as sums of two primes. (b) Does checking these three numbers prove the conjecture? Explain. R

8. Every prime greater than 3 has a remainder of 1 or 5 when divided by 6. Test this claim with 5, 7, 11, 13, 17 and 19, then explain why the other remainders are impossible for a prime. E

3.4 Prime factorisation

CRITERION A · II ◆ FACTOR TREES ◆ INDEX NOTATION

A composite number can be broken into factors repeatedly until every factor is prime. The result is the number's prime factorisation, and it is unique apart from the order in which the factors are written.

A factor tree records the breaking-down process visually. Start with the number, split it into a factor pair, and continue splitting any composite branch until only primes remain. The primes at the ends of the branches multiply to the original number. The division method reaches the same result by repeatedly dividing by the smallest prime that works.

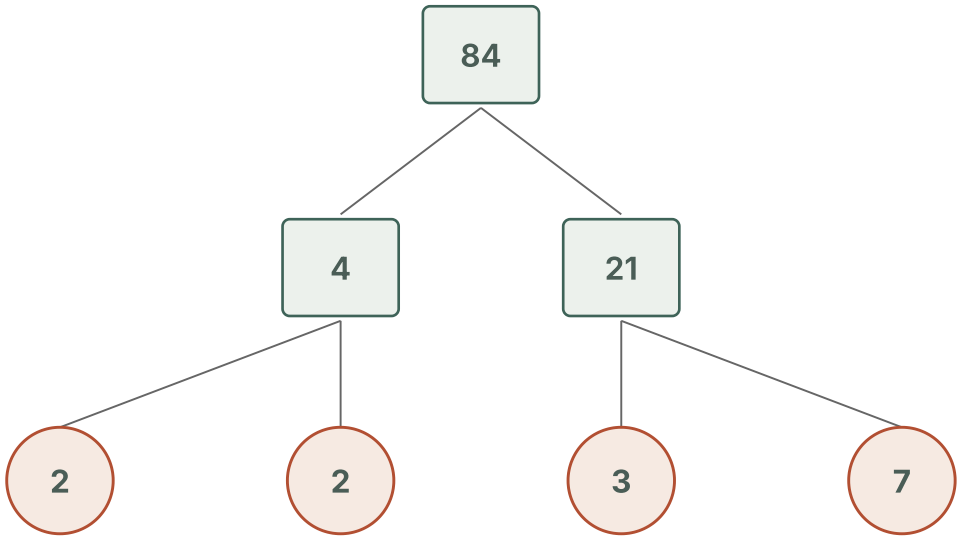


Figure 3.1 A factor tree for 84. The composite branches are split until only primes remain, giving $84 = 2 \times 2 \times 3 \times 7$.

WORKED EXAMPLE 3.4.1 — FACTOR TREE AND INDEX FORM

PROBLEM

Write 84 as a product of prime factors, using index notation.

1. Split 84 into 4×21. Both factors are composite, so each needs further splitting.
2. $4 = 2 \times 2$, and $21 = 3 \times 7$. Every branch now ends in a prime.
3. The prime factors are $2, 2, 3, 7$, so $84 = 2 \times 2 \times 3 \times 7$.
4. Collect the repeated 2s into a power: $84 = 2^2 \times 3 \times 7$.

ANSWER $84 = 2^2 \times 3 \times 7$.

Different tree, same primes. Starting with $84 = 6 \times 14$ gives 2×3 and 2×7, and the same four primes. Changing the first split changes the picture but not the factorisation.

WORKED EXAMPLE 3.4.2 — THE DIVISION METHOD

PROBLEM

Use repeated division to write 360 as a product of primes in index form.

1. 360 is even, so divide by 2: $360 \div 2 = 180$.
2. Divide by 2 again: $180 \div 2 = 90$, then $90 \div 2 = 45$. Now 45 is odd.
3. Divide by 3: $45 \div 3 = 15$, then $15 \div 3 = 5$.
4. Divide by 5: $5 \div 5 = 1$. The divisions stop when the quotient reaches 1.
5. The divisors used were 2, 2, 2, 3, 3 and 5, so $360 = 2^3 \times 3^2 \times 5$.

ANSWER $360 = 2^3 \times 3^2 \times 5$.

Check. $2^3 \times 3^2 \times 5 = 8 \times 9 \times 5 = 72 \times 5 = 360$.

WORKED EXAMPLE 3.4.3 — RECONSTRUCTING THE NUMBER

PROBLEM

A number has prime factorisation $2^3 \times 3^2 \times 5$. Find the number without using a calculator.

1. Evaluate the powers: $2^3 = 8$, $3^2 = 9$ and $5 = 5$.
2. Multiply in a convenient order: $8 \times 5 = 40$, then $40 \times 9 = 360$.

ANSWER **360.**

Check. Dividing 360 repeatedly by 2, 2, 2, 3, 3 and 5 returns exactly the exponents in the factorisation.

COMMON ERROR

Stopping the factor tree before every branch is prime. Writing $84 = 4 \times 21$ is a factorisation, but not a prime factorisation, because 4 and 21 are composite. Another error is losing an exponent: $2 \times 2 \times 2 \times 3 \times 5$ is $2^3 \times 3 \times 5$, not $2^3 \times 3^2 \times 5$. Count each repeated prime carefully.

GUIDED PRACTICE · COPY AND COMPLETE

$48 = 4 \times 12 = 2 \times$ ____ $\times 3 \times$ ____ $\times 2$, so $48 =$ ____$^4 \times 3$. Check by evaluating: ____ $\times 3 = 48$.

PRACTICE 3.4 · PRIME FACTORISATION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Complete each factorisation: (a) $60 = 2 \times 30 = 2 \times 2 \times ____ \times 5$ (b) $90 = 2 \times 45 = 2 \times ____ \times ____ \times 5$ (c) $126 = 2 \times 63 = 2 \times ____ \times ____ \times 7$. F
2. Write the prime factorisation of each number in index form: (a) 36 (b) 54 (c) 72 (d) 96 (e) 150 (f) 225. F
3. Write each product in index form and evaluate it: (a) $2 \times 2 \times 2 \times 3 \times 3 \times 5$ (b) $3 \times 3 \times 5 \times 7$ (c) $2 \times 5 \times 5 \times 5 \times 11$. F
4. Find the number represented by each prime factorisation: (a) $2^2 \times 7$ (b) $3^3 \times 5$ (c) $2 \times 5^2 \times 7$ (d) $2^4 \times 3^2$. F
5. A student writes $84 = 4 \times 21$ and calls it a prime factorisation. Explain why this is incomplete and correct the answer. R
6. Is $2^3 \times 3^2 = 72$ a square number? Explain how you know from the exponents. Find the smallest whole number by which 72 must be multiplied to make a square. R
7. Write 1,000,000 as a product of primes in index form. Begin from $1{,}000{,}000 = 10^6$, and explain how the indices arise. P
8. Two numbers have prime factorisations $2^2 \times 3 \times 5$ and 2×3^2. Find both numbers, state which is larger, and identify the largest common factor of the two numbers. E

3.5 Highest common factor and lowest common multiple

CRITERION A · III ◆ COMMON STRUCTURE ◆ OPTIMISATION

Two numbers often need to be divided into the same-sized groups, or two repeating events need to coincide. The first question uses the highest common factor; the second uses the lowest common multiple.

> **DEFINITIONS**
>
> The **highest common factor** (HCF) of two whole numbers is the largest whole number that divides both exactly. The **lowest common multiple** (LCM) is the smallest non-zero whole number that is a multiple of both. Every pair of whole numbers has both, because 1 divides both and the product is a common multiple.

For small numbers, listing works well. For 24 and 36, the factors of 24 are $1, 2, 3, 4, 6, 8, 12, 24$ and the factors of 36 are $1, 2, 3, 4, 6, 9, 12, 18, 36$. The common factors are $1, 2, 3, 4, 6, 12$, so the HCF is 12. The multiples of 24 begin $24, 48, 72, \ldots$ and those of 36 begin $36, 72, 108, \ldots$, so the LCM is 72.

Prime factorisation gives a faster method for larger numbers. Write both numbers as products of primes and align the factors:

FINDING HCF AND LCM FROM PRIME FACTORS OF 24 AND 36

Number	Prime factorisation	HCF uses	LCM uses
24	$2^3 \times 3$	$2^2 \times 3$	2^3
36	$2^2 \times 3^2$	$2^2 \times 3$	3^2
Result		$2^2 \times 3 = 12$	$2^3 \times 3^2 = 72$

The HCF takes the *lower* power of each shared prime; the LCM takes the *higher* power of every prime that appears in either number. A useful check for two numbers is that HCF $\times$ LCM equals the product of the numbers: $12 \times 72 = 864$ and $24 \times 36 = 864$. This identity holds for pairs, but not in general for three numbers.

WORKED EXAMPLE 3.5.1 HCF AND LCM OF 24 AND 36

PROBLEM

Find the HCF and LCM of 24 and 36 using prime factors. Verify the result with the product check.

1. $24 = 2^3 \times 3$ and $36 = 2^2 \times 3^2$.
2. Shared primes are 2 and 3. For the HCF take the lower power of each: $2^2 \times 3 = 12$.
3. For the LCM take the higher power of every prime in either number: $2^3 \times 3^2 = 8 \times 9 = 72$.
4. Check: $12 \times 72 = 864$ and $24 \times 36 = 864$.

ANSWER **HCF = 12, LCM = 72.**

Listing check. The common factors of 24 and 36 end at 12, and the first multiple the two lists share is 72. Both methods agree.

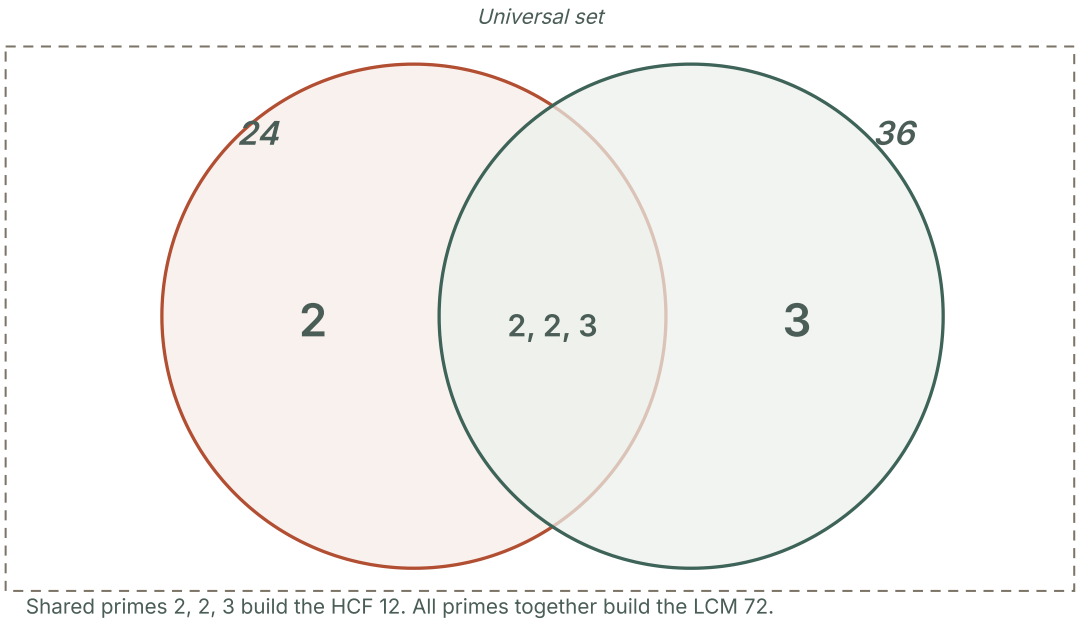


Figure 3.2 Prime factors of 24 and 36 arranged by sharing. The overlap contains the shared factors used in the HCF; the whole diagram contains the factors used in the LCM.

WORKED EXAMPLE 3.5.2 THE LARGEST EQUAL GROUPS

PROBLEM

A school has 84 Year 7 students and 60 Year 8 students. They are placed into mixed groups of the same size, with no student left out. What is the largest possible group size, and how many groups are there?

1. The group size must divide 84 and 60 exactly, so it is a common factor. The largest is the HCF.
2. $84 = 2^2 \times 3 \times 7$ and $60 = 2^2 \times 3 \times 5$. The HCF is $2^2 \times 3 = 12$.
3. Each group has 12 students: $84 \div 12 = 7$ Year 7 groups' worth and $60 \div 12 = 5$ Year 8 groups' worth.
4. There are $7 + 5 = 12$ groups, each containing 7 Year 7 students and 5 Year 8 students.

ANSWER **12 students per group; 12 groups, each with 7 Year 7 and 5 Year 8 students.**

Check. $12 \times (7 + 5) = 12 \times 12 = 144$, which equals $84 + 60$. No student is unplaced and every group is the same size.

WORKED EXAMPLE 3.5.3 WHEN DO TWO BELLS RING TOGETHER?

PROBLEM

One bell rings every 12 minutes and another every 18 minutes. They ring together at 9:00. When do they next ring together?

1. The times they ring together must be multiples of both 12 and 18, so we need the LCM.
2. $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$, so the LCM is $2^2 \times 3^2 = 36$.
3. They coincide every 36 minutes. Starting at 9:00, the next time is 9:36.

ANSWER **9:36, then 10:12, 10:48 and so on.**

Check. In 36 minutes the first bell rings 3 times and the second rings twice. The next simultaneous ring after 36 minutes is at 72 minutes, which is also a multiple of both intervals.

COMMON ERROR

Choosing the product of two numbers whenever a common multiple is needed. For 12 and 15, the product 180 is a common multiple, but 60 is smaller: $60 = 4 \times 15 = 5 \times 12$. The LCM uses the highest power of each prime, not every factor multiplied together. A second error is using the LCM when the problem asks for the largest equal group; that question needs the HCF.

GUIDED PRACTICE · COPY AND COMPLETE

$18 = 2 \times 3^2$ and $30 = 2 \times 3 \times 5$. The HCF is $2 \times 3 =$ ____, and the LCM is $2 \times 3^2 \times 5 =$ ____. The product $18 \times 30 = 540$, and the product of the HCF and LCM is ____, so the check _____.

PRACTICE 3.5 · HCF AND LCM

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Find the HCF: (a) 12 and 18 (b) 20 and 30 (c) 16 and 40 (d) 45 and 60. F
2. Find the LCM: (a) 6 and 8 (b) 10 and 15 (c) 12 and 18 (d) 9 and 12. F
3. Find the HCF and LCM of 36 and 48. Verify that HCF × LCM equals 36×48. F
4. Find the HCF and LCM of 18 and 30 using prime factors. F

5. Find the HCF and LCM of 8, 12 and 18. Check that the LCM is a multiple of each number and the HCF divides each number. P
6. Two ribbons are 48 cm and 60 cm long. They are cut into equal pieces of the longest possible length, with no ribbon wasted. Find the piece length and the total number of pieces. P
7. Two bus routes leave the station every 15 minutes and every 20 minutes. A bus from each route leaves together at 8:00. When do they next leave together? Explain why you used the LCM rather than the HCF. R
8. Explain why the HCF of two numbers cannot be larger than either number, and why the LCM cannot be smaller than either number. Then find two different pairs of numbers with HCF 6 and LCM 36. E

Real-life application · The fairness relay

A primary school is holding a relay festival for three year groups. Every student must take part exactly once. The numbers available are 84 in Year 4, 60 in Year 5 and 48 in Year 6.

Plan	Teams	Team size	How the teams are formed
A · mixed	12	16	7 Year 4, 5 Year 5 and 4 Year 6 students in every team
B · single-year	16	12	7 Year 4 teams, 5 Year 5 teams and 4 Year 6 teams
Students used			**192 in each plan:** $12 \times 16 = (7 + 5 + 4) \times 12$

Both plans use every student, but they answer different questions. For Plan A, the number of identical mixed teams must divide 84, 60 and 48, so the largest possible number of teams is their HCF. Since $84 = 2^2 \times 3 \times 7$, $60 = 2^2 \times 3 \times 5$ and $48 = 2^4 \times 3$, the shared factor is $2^2 \times 3 = 12$ teams, and $192 \div 12 = 16$ students per team. For Plan B, the team size must divide each year group, so the largest team size is again the HCF, 12, giving $84 \div 12 = 7$, $60 \div 12 = 5$ and $48 \div 12 = 4$ single-year teams.

Your task. Work through the prompts below and refer to the numbers in the table.

1. Decide which plan is fairer: 12 mixed teams of 16 students, or 16 single-year teams of 12 students. State the criterion you used, and write a recommendation to the organiser that names one disadvantage of the plan you choose.
2. The festival has two activity stations. One changes activity every 20 minutes, the other every 30 minutes. The festival begins at 9:00 and lunch is at 12:30. Find every time before lunch when both stations change at the same moment, and explain how the LCM produces those times.
3. Does “fair” mean equal team size, equal numbers from each year, or equal proportions? Explain how the arithmetic constrains what the organiser can promise.
4. The HCF sets the limit in both plans. When might smaller teams — and therefore more of them — be the fairer choice? Check that your proposed team size divides each year group exactly.

Investigation 3 · The factor table

CRITERION B · I, II, III

The table shows how many factors each number from 1 to 20 has. A factor is counted once, even when it is paired with itself.

n	1	2	3	4	5	6	7	8	9	10
Factors	1	2	2	3	2	4	2	4	3	4
n	11	12	13	14	15	16	17	18	19	20
Factors	2	6	2	4	4	5	2	6	2	6

1. Extend the table to 30, recording the number of factors for each number.
2. Which numbers have exactly two factors? What name is given to this set of numbers?
3. Which numbers have an odd number of factors? Describe them and test your description with 36 and 49.
4. Explain why most numbers have an even number of factors. Use the idea of factor pairs, and identify the special case that produces an odd count.
5. Generalise: how many factors does p^2 have when p is prime? How many does p^3 have? How many does $p \times q$ have when p and q are different primes? Test your rule on 25, 27 and 15.
6. Which number below 30 has the most factors? State how many factors it has and list them in pairs.
7. Does a larger number always have more factors? Compare 24 with 25, and 28 with 29. Use your comparisons to answer the question.
8. Extension: a number has exactly three factors. Explain why it must be the square of a prime, and list every such number below 100.

Record your table and your conjectures separately. A conjecture is a claim that fits the evidence; a generalisation needs a reason that explains why the claim must hold for every case, not just the ones you tested.

Criterion-mapped mini-task

THE NUMBER MUSEUM

A · KNOWING AND UNDERSTANDING
C · COMMUNICATING
D · APPLYING MATHEMATICS

Context. Your class is opening a "number museum" for younger students. Each exhibit explains the hidden structure of one whole number between 60 and 100. Visitors will walk past the exhibits and should leave able to see that a number is not just a label but a structure built from factors.

Product. An A3 display card for one composite number of your choice, with a short visitor guide. The card must be accurate, readable from one metre and self-explanatory without a presenter.

Requirements.

1. State your number and list all of its factor pairs systematically. State how many factors it has.
2. Show divisibility evidence for 2, 3, 4, 5, 6, 9 and 10, saying clearly which tests pass and which fail.
3. Draw a factor tree and write the prime factorisation in index form. Show that the product returns your number.
4. Choose a second exhibit number from a classmate. Find the HCF and LCM of the two numbers using prime factors, and explain in two sentences which one answers the question "largest equal groups?"
5. Create one real context for your number: an arrangement of chairs, a repeating timetable, or a fair sharing problem. Solve it with working visible.
6. Add a "surprising fact" panel: a square, prime, highly divisible or otherwise unusual feature of your number, with evidence.
7. Write three visitor questions on a separate card, with answers on the back. At least one question must ask the visitor to use the prime factorisation.
8. Ask a younger student to read the card. Record one thing they understood and one thing they misread, then revise the card.

CRITERIA AND STRANDS A i, ii, iii: accurate factors, tests, factorisation, HCF and LCM; C i, ii, iii, iv: clear notation, labelled diagrams and language suited to the audience; D i, ii, iii, iv: apply number structure to a real context, justify method choice and evaluate the exhibit's clarity.

BEFORE YOU SUBMIT

Ask someone to find all factors of your number using only the card. If they miss one, the display is not yet systematic. Check every product in the factor tree independently, then check the HCF and LCM by a second method. An exhibit is finished when a visitor can use it without you.

Chapter review

SUMMARY

- **Factors and multiples.** A factor divides a number exactly; a multiple is the product of the number and a whole number. They are two directions of one equation.
- **Factor pairs.** List factors in pairs from 1 and stop when the pair meets the square root. No factor can be missed.
- **Divisibility.** Last digits detect 2, 5 and 10; digit sums detect 3 and 9; the last two or three digits detect 4 and 8; 6 needs both 2 and 3.
- **Primes.** A prime has exactly two factors. A composite has more than two. The number 1 is neither. Test primes only up to $\sqrt{n}$.
- **Prime factorisation.** Every composite number is a product of primes in exactly one way, apart from order. Trees and repeated division both find it.
- **HCF and LCM.** The HCF is the largest shared factor; the LCM the smallest shared multiple. From primes, use lowest shared powers for the HCF and highest powers for the LCM.

Factor test: a is a factor of N when $N = a \times k$ for a whole number k.

Digit sum: $3 \mid N$ or $9 \mid N$ exactly when the digit sum is a multiple of 3 or 9.

Primality: test primes $p \leq \sqrt{N}$; if none divides N, then N is prime.

HCF and LCM: HCF uses the lowest shared powers; LCM uses the highest powers. For two numbers, HCF × LCM = product.

REVISION CHECKLIST

- ☐ I can list all factors of a number using pairs.
- ☐ I can distinguish a factor from a multiple.
- ☐ I can apply divisibility tests for 2, 3, 4, 5, 6, 8, 9 and 10.
- ☐ I can classify numbers as prime, composite or neither, with reasons.
- ☐ I can test whether a number is prime using primes up to its square root.
- ☐ I can write a prime factorisation using a tree or division and express it in index form.
- ☐ I can calculate HCF and LCM using lists and prime factors.
- ☐ I can choose HCF or LCM for a context and justify the choice.

KEY VOCABULARY

factor
A whole number that divides another exactly.

multiple
The product of a number and a whole number.

factor pair
Two whole numbers whose product is the given number.

divisible
Leaving no remainder.

prime
A whole number greater than 1 with exactly two factors.

composite
A whole number greater than 1 with more than two factors.

prime factorisation
A number written as a product of primes.

HCF
The largest factor shared by two or more numbers.

LCM
The smallest non-zero multiple shared by two or more numbers.

index form
A repeated product written with a base and exponent.

REFLECTION AND SELF-ASSESSMENT

Choose one number whose structure surprised you. Explain what its factors or prime factorisation revealed that its size did not.

Rate your confidence from 1 to 4 on each success criterion. For any rating of 1 or 2, name the exercise you will revisit and the help you need. Then explain in two sentences how you decide between the HCF and the LCM.

END-OF-CHAPTER PRACTICE

Mixed practice 3

Do not use a calculator. Show the test, pair or factorisation that supports each answer.

FLUENCY Build accuracy

1. List all factors of 54.
2. Write the first six multiples of 9.
3. Write all factor pairs of 48.
4. State which of 2, 3, 4, 5, 9 and 10 divide 5,742. Give evidence for each decision.
5. List the prime numbers between 40 and 60.
6. Classify each number as prime or composite, giving a factor or a complete test: 87, 89, 117, 149, 221.
7. Write the prime factorisation in index form: (a) 84 (b) 120 (c) 196 (d) 450.
8. Evaluate: (a) $2^3 \times 3^2$ (b) 5×7^2 (c) $2^2 \times 3 \times 5^2$ (d) $3^3 \times 5^2$.
9. Find the HCF and LCM of 28 and 42.
10. Find the HCF and LCM of 45 and 75.
11. Find the HCF and LCM of 18 and 24.
12. Find the HCF and LCM of 12, 16 and 20.

REASONING AND PROBLEM SOLVING Connect ideas

13. A student says the LCM of 12 and 15 is 180 because $12 \times 15 = 180$. Explain the error, give the correct LCM, and state why the product method sometimes gives the right answer.
14. Explain why 1 is neither prime nor composite, and why 2 is prime. Your answer should refer to the number of factors each has.
15. Two numbers have HCF 8 and LCM 48. Find two different pairs of numbers with these values and check each pair.
16. The prime factorisation of 90 is $2 \times 3^2 \times 5$. Find the smallest whole number by which 90 must be multiplied to make a perfect square, and verify the result.
17. A floor is 240 cm by 360 cm. It is covered exactly by square tiles of the largest possible whole-centimetre side length. Find the tile side length and the number of tiles needed.
18. Three lights flash every 8 seconds, 12 seconds and 18 seconds. They flash together at 10:00:00. When do they next flash together? Give your answer in minutes and seconds.
19. **The fruit bags (Criteria A, C, D).** A canteen has 96 apples, 72 bananas and 60 oranges. (a) Find the largest number of identical fruit bags that can be made with no fruit left over. (b) State the contents of each bag. (c) A late delivery adds 90 apples, so the canteen now has 186 apples. Find the largest number of identical bags possible now, and state the new contents. (d) Explain why the answer changed so much when only one quantity changed.
20. **The station displays (Criteria C, D).** Two station displays flash every 45 seconds and every 60 seconds. (a) Find the LCM of 45 and 60 and interpret it in seconds and minutes. (b) The displays flash together at 07:00. List the next three times they flash together. (c) How many times do they flash together between 07:00 and 08:00 inclusive? (d) A third display flashes every 90 seconds. Explain whether this changes the times at which all three flash together, using prime factors.

Check your progress

ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

- **Factors and multiples run in opposite directions.** 6 is a factor of 36; 36 is a multiple of 6. A number has finitely many factors but infinitely many multiples.
- **Stop factor lists at the square root.** Once the pair meets or passes $\sqrt{n}$, every later factor would repeat a pair already found.
- **1 is neither prime nor composite.** It has one factor, not the two required for a prime.
- **A prime factorisation contains only primes.** $84 = 4 \times 21$ is incomplete; $84 = 2^2 \times 3 \times 7$ is finished.
- **HCF and LCM answer different questions.** The HCF is at most the smaller number; the LCM is at least the larger. Use the HCF for largest equal groups, the LCM for events that coincide.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can list all factors of a number systematically.			
I can explain the difference between a factor and a multiple.			
I can apply divisibility tests and justify the result.			
I can identify primes and test whether a number is prime.			
I can write and use a prime factorisation in index form.			
I can find HCF and LCM and choose the right one for a context.			

WHERE THIS LEADS

Chapter 4 moves from whole-number division to parts of a whole. Factors will reappear whenever a fraction is simplified: dividing numerator and denominator by a common factor does not change the value. The same idea of equivalent structure, expressed through factors, becomes the foundation for comparing fractions and for working with decimals.

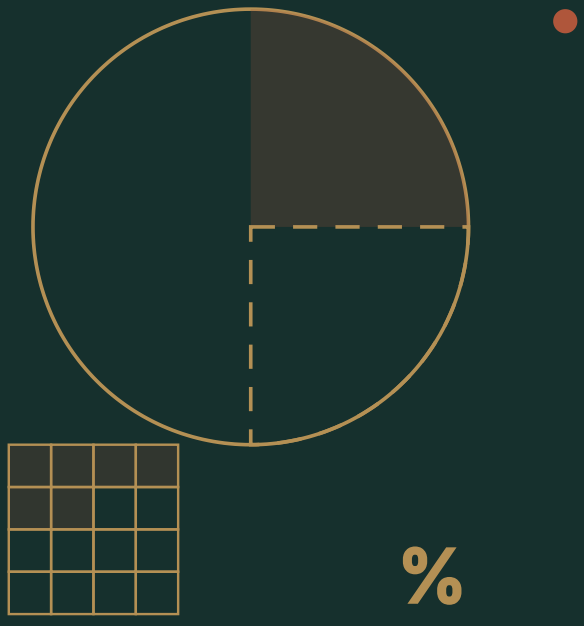
Review the answer key only after attempting each question.

PART II · PARTS AND PROPORTIONS

II

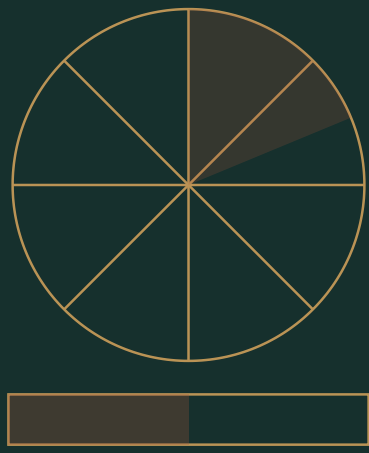
Fractions, decimals and percentages

A fraction, a decimal and a percentage can all describe the same portion of a whole. Choosing among them is a communication decision as well as a calculation decision. This part connects the three forms, builds reliable methods with each, and uses them to make sense of prices, budgets and everyday comparisons.



04 Fractions and decimals

05 Percentages and applications



4

ONE QUANTITY, SEVERAL FORMS

Fractions and decimals

A fraction and a decimal can name the same point on a number line. Learning to move between forms makes it easier to compare quantities, carry out calculations and judge how precise an answer really is.

KEY CONCEPT · RELATIONSHIPS

RELATED · EQUIVALENCE

RELATED · REPRESENTATION

CONTEXT · SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

When two numbers look different but describe the same amount, how do we choose the form that is easiest to compare, calculate with or trust?

MYP UNIT PLANNER

Framing the inquiry

UNIT 4 · APPROX. 16 HOURS
ATL FOCUS · TRANSFER

KEY AND RELATED CONCEPTS

Relationships. A fraction records a relationship between part and whole: the denominator sets the part size, the numerator counts parts.

Equivalence. Different fractions and decimals can name the same number, so a calculation can be rewritten in a convenient form.

Representation. A number line, area diagram, fraction or decimal makes different features visible; the form should serve the task.

GLOBAL CONTEXT AND EXPLORATION

Scientific and technical innovation. Exploration: precision and measurement — how the way a quantity is recorded affects comparison, calculation and trust in a result.

A 3/8 inch drill bit and a 0.375 inch drill bit are the same size. A laboratory that records 1/3 litre as 0.33 litre has changed the quantity slightly. In measurement, the form of a number carries information about precision.

STATEMENT OF INQUIRY

The same quantity can be represented in equivalent forms, and choosing an appropriate form improves the precision of comparison, calculation and communication.

FACTUAL

What is an equivalent fraction? What does each digit after a decimal point represent? How can a fraction be written as a decimal?

CONCEPTUAL

Why do different fractions name the same number? How does the form of a number influence the calculation you choose?

DEBATABLE

Is a fraction always a more exact record of a quantity than a decimal?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Generate equivalent fractions, order fractions, calculate with fractions and decimals, round decimals.
B · Investigating patterns	i, ii	Investigate which fractions produce terminating decimals and generalise from the prime factors of denominators.
C · Communicating	i, ii, iii	Explain equivalence and method choice, use number lines and diagrams, record results with suitable notation.
D · Applying mathematics	i, ii, iii, iv	Scale recipes, compare measurements, justify the choice of fraction or decimal and evaluate the effect of rounding.

ATL · TRANSFER IN ACTION

A recipe serves 6 and must be rewritten for 4 and for 15. Choose fractions or decimals for each scaling step, compare the records for the awkward quantity 4/9 cup, and decide which form a cook needs and which a supplier needs.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** equivalent fractions and write a fraction in simplest form.
- **Be able to** compare and order fractions using common denominators, benchmarks or decimals.
- **Be able to** add and subtract fractions, including mixed numbers.
- **Be able to** multiply and divide fractions and interpret both operations.
- **Know** decimal place value to thousandths, and be able to order, round and calculate with decimals.
- **Be able to** convert between fractions and decimals and explain which decimals terminate.

SUCCESS CRITERIA

- ☐ I can generate equivalent fractions and simplify using a common factor.
- ☐ I can compare and order fractions with unlike denominators and justify my method.
- ☐ I can add and subtract fractions and mixed numbers, regrouping when needed.
- ☐ I can multiply and divide fractions, using the reciprocal correctly.
- ☐ I can read, order, round and calculate with decimals to thousandths.
- ☐ I can convert between fractions and decimals and explain which decimals terminate.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Shade $\frac{3}{4}$ of a rectangle divided into 8 equal parts.
2. Write 0.5 and 0.25 as fractions in simplest form.
3. Find $\frac{3}{4}$ of 20.
4. Which is larger: $\frac{1}{2}$ or $\frac{3}{8}$?
5. Calculate $\frac{1}{4} + \frac{1}{4}$.
6. Write “seven tenths” as a decimal and as a fraction.

Answers are in the Chapter 4 answer section. If more than two are incorrect, review simple equivalent fractions and decimal tenths before continuing.

BEFORE YEAR 1
Halves, quarters and tenths; simple fractions of shapes and amounts; decimal notation to hundredths.

YEAR 1 · THIS CHAPTER
Equivalence, ordering, the four operations with fractions, decimal place value and fraction–decimal links.

LATER
Percentages, ratio and proportion, rational expressions and algebraic fractions.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Recipes, measurements and shared quantities rarely arrive as whole numbers.	How can different-looking numbers describe exactly the same amount?	Equivalence, representation and precision.	Which fractions produce terminating decimals, and how can denominators predict it?	Number lines, fraction diagrams and decimal notation.	When is an exact fraction more useful than a rounded decimal?

4.1 Equivalent fractions and simplest form

CRITERION A · I ◆ EQUIVALENCE ◆ SIMPLIFYING

Two fractions can look different and describe the same quantity. The same amount of pizza can be cut into halves, quarters or eighths; the number of pieces changes, but the amount eaten need not.

Multiplying the numerator and the denominator by the same non-zero whole number produces an equivalent fraction. Each part is cut into the same number of smaller parts, so both the number of parts taken and the total number of parts are multiplied by that number. The relationship between the part and the whole is unchanged.

DEFINITIONS

Equivalent fractions are fractions that name the same number. For any non-zero whole number k, $\frac{a}{b} = \frac{a \times k}{b \times k}$. A fraction is in **simplest form** when the numerator and denominator have no common factor greater than 1.

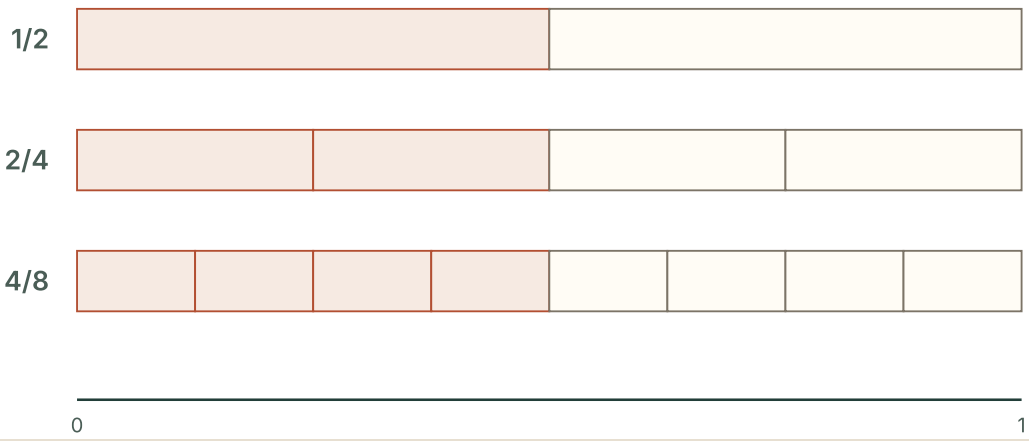


Figure 4.1 Cutting every part into more pieces multiplies both the numerator and the denominator by the same number, so the shaded length does not change.

Division reverses the process. If the numerator and denominator share a factor, divide both by it. Doing this once with the highest common factor gives simplest form immediately; doing it repeatedly with smaller factors reaches the same place. From Chapter 3, the highest common factor of 42 and 56 is 14, so $\frac{42}{56} = \frac{42 \div 14}{56 \div 14} = \frac{3}{4}$.

WORKED EXAMPLE 4.1.1 — SIMPLIFYING ONCE, WITH THE HCF

PROBLEM

Write $\frac{42}{56}$ in simplest form.

1. Find the HCF of 42 and 56: $42 = 2 \times 3 \times 7$ and $56 = 2^3 \times 7$, so the HCF is $2 \times 7 = 14$.
2. Divide both terms by 14: $42 \div 14 = 3$ and $56 \div 14 = 4$.
3. The numbers 3 and 4 share no factor greater than 1, so the fraction cannot be simplified further.

ANSWER $\frac{42}{56} = \frac{3}{4}$.

Check. Reverse the division by multiplying both terms of $\frac{3}{4}$ by 14: $\frac{3\times14}{4\times14} = \frac{42}{56}$. Simplifying and multiplying are inverse operations.

WORKED EXAMPLE 4.1.2 — RESTORING A MISSING TERM

PROBLEM

Find the missing numbers: (a) $\frac{7}{12} = \frac{\square}{60}$, (b) $\frac{21}{28} = \frac{3}{\square}$.

1. (a) The denominator has been multiplied by 5, because $12 \times 5 = 60$. Multiply the numerator by the same number: $7 \times 5 = 35$.
2. (b) The numerator has been divided by 7, because $21 \div 7 = 3$. Divide the denominator by 7: $28 \div 7 = 4$.

ANSWER **(a) 35 (b) 4.**

Check. $\frac{7}{12} = \frac{35}{60}$ and $\frac{3}{4} = \frac{21}{28}$. Both original and new fractions simplify to the same value.

COMMON ERROR

Adding the same number to the numerator and denominator. $\frac{7}{12}$ is not equal to $\frac{8}{13}$, because adding 1 adds one whole part to the denominator but only one unit to the numerator; the parts are no longer the same size relationship. A second error is stopping too early: $\frac{18}{24} = \frac{9}{12}$ is correct but incomplete, since both terms still share the factor 3, giving $\frac{3}{4}$.

GUIDED PRACTICE · COPY AND COMPLETE

$\frac{24}{36}$: the HCF of 24 and 36 is ____, so the fraction simplifies to $\frac{\quad}{3}$. Also $\frac{5}{8} = \frac{\quad}{40}$, because the denominator is multiplied by ____. And $\frac{3}{7} = \frac{15}{\quad}$, because the numerator is multiplied by ____.

PRACTICE 4.1 · EQUIVALENT FRACTIONS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Show the multiplier or divisor, not only the answer.

1. Write two fractions equivalent to (a) $\frac{2}{5}$, (b) $\frac{3}{4}$, (c) $\frac{7}{10}$. F
2. Fill in the missing numbers: (a) $\frac{3}{4} = \frac{\square}{20}$, (b) $\frac{5}{6} = \frac{25}{\square}$, (c) $\frac{14}{21} = \frac{2}{\square}$, (d) $\frac{8}{12} = \frac{\square}{3}$. F
3. Write in simplest form: (a) $\frac{18}{24}$, (b) $\frac{25}{40}$, (c) $\frac{36}{54}$, (d) $\frac{45}{60}$. F
4. Which fractions are equivalent to $\frac{3}{5}$: $\frac{9}{15}$, $\frac{12}{18}$, $\frac{15}{25}$, $\frac{18}{30}$? Explain how you decided without drawing a diagram. R

5. A class has 27 students, and 18 of them walk to school. Write the fraction who walk in simplest form, and state what the simplified fraction means in the context. P
6. A student writes $\frac{12}{18} = \frac{13}{19}$ "because I added 1 to both parts". Explain why this is incorrect, and give the correct simplest form. R
7. Explain, using the idea of cutting each part into smaller pieces, why $\frac{3}{4} = \frac{9}{12}$. Your explanation should say what happens to the number of pieces and to the size of each piece. R
8. Is there a fraction equivalent to $\frac{3}{7}$ with denominator 70? With denominator 50? Justify both answers using factors. Then write the fraction equivalent to $\frac{4}{5}$ whose numerator is 100. E

4.2 Comparing and ordering fractions

CRITERION A · II ◆ ORDER ◆ COMMON DENOMINATORS

Comparing fractions with the same denominator is easy: the numerators count parts of the same size. When denominators differ, the parts differ too, so both fractions must be expressed in a common unit before their sizes can be compared.

There are several reliable methods. A common denominator uses the lowest common multiple, keeping the numbers as small as possible. Using the product of the denominators also works, but often creates larger numbers to simplify later. A benchmark such as $\frac{1}{2}$ or 1 settles some comparisons faster than any calculation, and converting to decimals is useful when a calculator or a decimal context is already present.

METHODS FOR COMPARING TWO FRACTIONS

What you notice	What to do	Example
Same denominator	Compare the numerators.	$\frac{5}{8} > \frac{3}{8}$
Same numerator	The fraction with the larger denominator is smaller, because the whole is cut into more, smaller parts.	$\frac{3}{10} < \frac{3}{4}$
One fraction is above or below a benchmark	Compare each with $\frac{1}{2}$ or with 1.	$\frac{2}{5} < \frac{1}{2} < \frac{5}{8}$
Different numerators and denominators	Rewrite with a common denominator, or convert to decimals.	$\frac{5}{8} = \frac{15}{24}$, $\frac{7}{12} = \frac{14}{24}$

WORKED EXAMPLE 4.2.1 — COMPARING UNLIKE FRACTIONS

PROBLEM

Which is larger, $\frac{5}{8}$ or $\frac{7}{12}$?

1. The LCM of 8 and 12 is 24, so use 24 as the common denominator.
2. $\frac{5}{8} = \frac{5\times3}{8\times3} = \frac{15}{24}$ and $\frac{7}{12} = \frac{7\times2}{12\times2} = \frac{14}{24}$.
3. The denominators are now equal, so compare numerators: $15 > 14$.

ANSWER $\frac{5}{8} > \frac{7}{12}$.

Check with decimals. $\frac{5}{8} = 0.625$ and $\frac{7}{12} \approx 0.583$, which confirms the order. The decimal values also show how close the two fractions are: they differ by only $\frac{1}{24} \approx 0.042$.

WORKED EXAMPLE 4.2.2 ORDERING THREE FRACTIONS

PROBLEM

Write $\frac{3}{5}$, $\frac{5}{8}$ and $\frac{7}{12}$ in ascending order.

1. The LCM of 5, 8 and 12 is 120, so use 120 as the common denominator.
2. $\frac{3}{5} = \frac{72}{120}$, $\frac{5}{8} = \frac{75}{120}$ and $\frac{7}{12} = \frac{70}{120}$.
3. Compare numerators: $70 < 72 < 75$.

ANSWER $\frac{7}{12} < \frac{3}{5} < \frac{5}{8}$.

Check. As decimals the three values are approximately 0.583, 0.6 and 0.625. All three lie between $\frac{1}{2}$ and 1, which is consistent with the number line placement.

COMMON ERROR

Comparing denominators as if they were whole numbers. The fraction $\frac{1}{8}$ is smaller than $\frac{1}{3}$, even though $8 > 3$, because eighths are smaller parts than thirds. A second error is cross-multiplying and then reading the result the wrong way round: for $\frac{5}{8}$ and $\frac{7}{12}$, $5 \times 12 = 60$ and $7 \times 8 = 56$, so $\frac{5}{8}$ is larger. Always state which fraction the larger product belongs to.

GUIDED PRACTICE · COPY AND COMPLETE

To compare $\frac{4}{7}$ and $\frac{5}{9}$, use the common denominator ____. Then $\frac{4}{7} = \frac{\quad}{63}$ and $\frac{5}{9} = \frac{\quad}{63}$. Since ____ $>$ ____, the larger fraction is ____.

PRACTICE 4.2 · COMPARING AND ORDERING

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Insert $<$, $>$ or $=$: (a) $\frac{3}{5} \square \frac{7}{12}$, (b) $\frac{5}{8} \square \frac{2}{3}$, (c) $\frac{7}{10} \square \frac{11}{15}$, (d) $\frac{3}{4} \square \frac{5}{6}$. F
2. Write in ascending order: $\frac{2}{3}$, $\frac{5}{8}$, $\frac{7}{12}$, $\frac{3}{5}$. F
3. Write in descending order: $\frac{7}{8}$, $\frac{5}{6}$, $\frac{11}{12}$, $\frac{3}{4}$. F
4. Place $\frac{1}{4}$, $\frac{5}{8}$, $\frac{2}{3}$ and $\frac{7}{10}$ in order on a number line between 0 and 1. Describe the position of each relative to $\frac{1}{2}$. P
5. Find a fraction between $\frac{2}{5}$ and $\frac{1}{2}$. Explain how a common denominator helped you find it, and check that your fraction really lies between the two. P
6. A student says $\frac{5}{9} < \frac{4}{7}$ because $9 > 7$. The conclusion happens to be correct, but the reasoning is not. Explain why the size of the denominator alone cannot decide, and justify the comparison correctly. R
7. Which is larger, $\frac{7}{12}$ or $\frac{9}{16}$? Use two different methods and state which method was more efficient for these numbers. R
8. Compare $\frac{99}{100}$ and $\frac{100}{101}$. Both are close to 1. Explain which is closer to 1 and why, without converting them to decimals. E

4.3 Adding and subtracting fractions

CRITERION A · III ◆ COMMON UNITS ◆ MIXED NUMBERS

Addition counts parts. The denominator names the size of each part and the numerator counts how many there are, so parts must be the same size before the counts can be combined.

When denominators are equal, add or subtract the numerators and keep the denominator: $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$. When denominators differ, rewrite both fractions with a common denominator. The lowest common multiple of the denominators is usually the most efficient choice, because it keeps the numbers small and the final simplification easy.

WORKED EXAMPLE 4.3.1 — ADDING UNLIKE FRACTIONS

PROBLEM

Calculate $\frac{3}{4} + \frac{5}{6}$.

1. The LCM of 4 and 6 is 12, so use twelfths.
2. $\frac{3}{4} = \frac{9}{12}$ and $\frac{5}{6} = \frac{10}{12}$.
3. Add the numerators: $\frac{9}{12} + \frac{10}{12} = \frac{19}{12}$.
4. Write as a mixed number: $\frac{19}{12} = 1\frac{7}{12}$.

ANSWER $\frac{19}{12} = 1\frac{7}{12}$.

Check. $\frac{3}{4} = 0.75$ and $\frac{5}{6} \approx 0.833$, so the sum is about 1.583. And $1\frac{7}{12} = 1.583\ldots$, which agrees. The answer is reasonable because both fractions are less than 1 but together exceed 1.

WORKED EXAMPLE 4.3.2 — SUBTRACTING MIXED NUMBERS

PROBLEM

Calculate $5\frac{1}{3} - 2\frac{3}{4}$.

1. Write both as improper fractions: $5\frac{1}{3} = \frac{16}{3}$ and $2\frac{3}{4} = \frac{11}{4}$.
2. The LCM of 3 and 4 is 12: $\frac{16}{3} = \frac{64}{12}$ and $\frac{11}{4} = \frac{33}{12}$.
3. Subtract: $\frac{64}{12} - \frac{33}{12} = \frac{31}{12}$.
4. Write as a mixed number: $\frac{31}{12} = 2\frac{7}{12}$.

ANSWER $2\frac{7}{12}$.

Second method. Regroup first: $5\frac{1}{3} = 5\frac{4}{12} = 4\frac{16}{12}$ and $2\frac{3}{4} = 2\frac{9}{12}$. Then $4\frac{16}{12} - 2\frac{9}{12} = 2\frac{7}{12}$. Regrouping borrows one whole as twelve twelfths when the fraction being subtracted is larger.

COMMON ERROR

Adding the denominators: $\frac{2}{3} + \frac{1}{4}$ is not $\frac{3}{7}$. The denominator records the size of the parts, not a quantity to be added. A second error appears in mixed-number subtraction when the fractional parts are subtracted in the wrong order, as in $5\frac{1}{3} - 2\frac{3}{4}$ giving $3\frac{2}{12}$; regrouping or converting to improper fractions prevents this.

GUIDED PRACTICE · COPY AND COMPLETE

$\frac{2}{3}+\frac{1}{4}=\frac{__}{12}+\frac{__}{12}=\frac{__}{12}$. For $3\frac{1}{2}-1\frac{2}{5}$, use tenths: $3\frac{1}{2}=3\frac{__}{10}$ and $1\frac{2}{5}=1\frac{__}{10}$, so the difference is _____.

PRACTICE 4.3 · ADDITION AND SUBTRACTION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Calculate: (a) $\frac{1}{5}+\frac{3}{5}$, (b) $\frac{2}{7}+\frac{3}{7}$, (c) $\frac{5}{8}+\frac{1}{8}$, (d) $\frac{7}{12}+\frac{5}{12}$. F
2. Calculate: (a) $\frac{1}{2}+\frac{1}{3}$, (b) $\frac{3}{4}+\frac{1}{6}$, (c) $\frac{5}{6}+\frac{3}{8}$, (d) $\frac{2}{5}+\frac{7}{10}$. F
3. Calculate: (a) $\frac{7}{8}-\frac{3}{8}$, (b) $\frac{5}{6}-\frac{1}{2}$, (c) $\frac{3}{4}-\frac{2}{5}$, (d) $1-\frac{5}{9}$. F
4. Calculate: (a) $2\frac{1}{4}+1\frac{1}{3}$, (b) $4\frac{5}{6}-2\frac{1}{2}$, (c) $5\frac{1}{3}-2\frac{3}{4}$, (d) $3\frac{1}{5}+2\frac{3}{10}$. P
5. A recipe uses $\frac{2}{3}$ cup of milk and $\frac{3}{4}$ cup of water. How much liquid is used in total? Give the answer as a mixed number. P
6. A student calculates $\frac{3}{5}+\frac{2}{3}=\frac{5}{8}$. Explain the error, show the correct method, and state why the answer must be greater than 1. R
7. Find the missing fraction: (a) $\frac{1}{2}+\square=\frac{5}{6}$, (b) $\square-\frac{2}{5}=\frac{1}{10}$. R
8. A painter uses $\frac{1}{3}$ of a can of paint on Monday and $\frac{1}{4}$ of the can on Tuesday. What fraction of the can remains? Show your working, then check that the three fractions add to 1. E

4.4 Multiplying and dividing fractions

CRITERION A · III ◆ AREA MODELS ◆ RECIPROCALS

Multiplying by a fraction can be read as taking a fraction of something. A fraction of a fraction is a smaller fraction of the whole, and an area diagram shows why the numerators and denominators multiply.

To multiply fractions, multiply the numerators and multiply the denominators: $\frac{a}{b}\times\frac{c}{d}=\frac{ac}{bd}$. The diagram below shows $\frac{2}{3}\times\frac{3}{5}$. A square is divided into 3 rows and 5 columns, making 15 equal parts. Three of the five columns represent $\frac{3}{5}$, and two of the three rows represent $\frac{2}{3}$. Their overlap contains $2\times3=6$ parts out of 15.

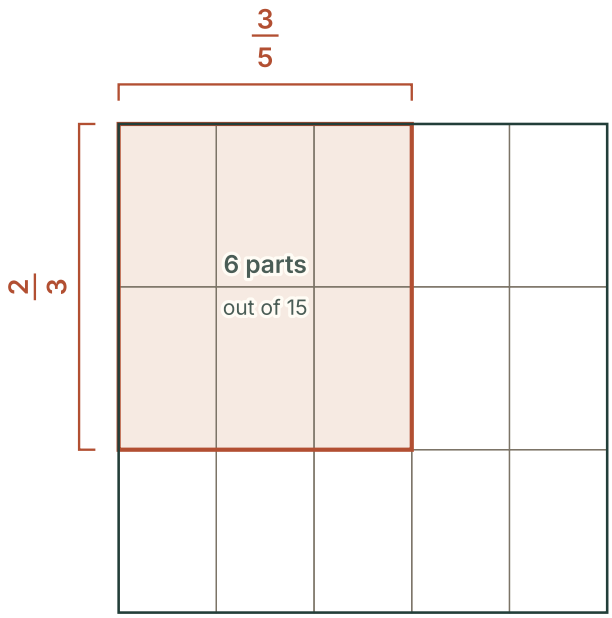


Figure 4.2 An area model for $\frac{2}{3} \times \frac{3}{5}$. The overlap is 6 parts out of 15, so the product is $\frac{6}{15} = \frac{2}{5}$.

Simplifying before multiplying keeps the numbers small. In $\frac{2}{3} \times \frac{3}{5}$, the 3 in the numerator and the 3 in the denominator share a factor, so the product is $\frac{2}{5}$ directly. Division uses the reciprocal: dividing by $\frac{c}{d}$ is the same as multiplying by $\frac{d}{c}$, because the question "how many $\frac{c}{d}$ are in the dividend?" is answered by scaling both quantities by $\frac{d}{c}$.

WORKED EXAMPLE 4.4.1 — MULTIPLYING WITH AN AREA MODEL

PROBLEM

Calculate $\frac{2}{3} \times \frac{3}{5}$ and explain the result using Figure 4.2.

1. Multiply numerators and denominators: $\frac{2 \times 3}{3 \times 5} = \frac{6}{15}$.
2. The diagram shows 15 equal parts in the whole and 6 parts in the overlap.
3. Simplify by dividing both terms by 3: $\frac{6}{15} = \frac{2}{5}$.

ANSWER $\frac{2}{5}$.

Check. $\frac{3}{5} = 0.6$, and $\frac{2}{3}$ of 0.6 is 0.4. Also $\frac{2}{5} = 0.4$, so the product is consistent. The result is smaller than either factor, as expected when both factors are less than 1.

WORKED EXAMPLE 4.4.2 DIVIDING BY A FRACTION

PROBLEM

Calculate $\frac{3}{4} \div \frac{3}{8}$.

1. Interpret the division: how many groups of $\frac{3}{8}$ fit into $\frac{3}{4}$?
2. Write with a common denominator: $\frac{3}{4} = \frac{6}{8}$. So the question becomes how many $\frac{3}{8}$ are in $\frac{6}{8}$.
3. There are 2 groups, because $6 \div 3 = 2$.

ANSWER **2.**

Rule check. Multiply by the reciprocal: $\frac{3}{4} \times \frac{8}{3} = \frac{24}{12} = 2$. The answer is greater than $\frac{3}{4}$ because the divisor is less than 1; there are more than one group of $\frac{3}{8}$ in $\frac{3}{4}$.

WORKED EXAMPLE 4.4.3 MIXED NUMBERS IN BOTH OPERATIONS

PROBLEM

Calculate (a) $1\frac{1}{2} \times \frac{2}{3}$, (b) $2\frac{1}{2} \div \frac{3}{4}$.

1. (a) Change the mixed number to an improper fraction: $1\frac{1}{2} = \frac{3}{2}$. Then $\frac{3}{2} \times \frac{2}{3} = \frac{6}{6} = 1$.
2. (b) $2\frac{1}{2} = \frac{5}{2}$. Multiply by the reciprocal of $\frac{3}{4}$: $\frac{5}{2} \times \frac{4}{3} = \frac{20}{6} = \frac{10}{3}$.
3. Write $\frac{10}{3}$ as $3\frac{1}{3}$.

ANSWER **(a)** 1 **(b)** $3\frac{1}{3}$.

Check. (a) As decimals, $1.5 \times 0.666\ldots = 1$. (b) $3\frac{1}{3}$ groups of $\frac{3}{4}$ make $3\frac{1}{3} \times \frac{3}{4} = \frac{10}{3} \times \frac{3}{4} = \frac{5}{2} = 2\frac{1}{2}$.

COMMON ERROR

Using the multiplication rule for addition: $\frac{1}{2} + \frac{1}{3}$ is not $\frac{1}{6}$. A second error is forgetting to invert the second fraction in a division, or inverting the first one instead: $\frac{3}{4} \div \frac{3}{8} = \frac{3}{4} \times \frac{8}{3}$, not $\frac{4}{3} \times \frac{3}{8}$. A third error is expecting division always to make a number smaller; dividing by a fraction less than 1 makes it larger.

GUIDED PRACTICE · COPY AND COMPLETE

$\frac{3}{5} \times \frac{5}{9}$: the 5s cancel, giving $\frac{\quad}{\quad}$. For $\frac{5}{8} \div \frac{5}{12}$, multiply by the reciprocal: $\frac{5}{8} \times \frac{\quad}{\quad} = \frac{\quad}{\quad} = ____$.

PRACTICE 4.4 · MULTIPLICATION AND DIVISION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Calculate: (a) $\frac{1}{2} \times \frac{3}{5}$, (b) $\frac{2}{3} \times \frac{3}{4}$, (c) $\frac{5}{6} \times \frac{3}{10}$, (d) $\frac{7}{8} \times \frac{4}{7}$. F
2. Calculate: (a) $\frac{1}{2} \div \frac{1}{4}$, (b) $\frac{3}{4} \div \frac{3}{8}$, (c) $\frac{5}{6} \div \frac{2}{3}$, (d) $\frac{4}{5} \div 8$. F
3. Calculate: (a) $1\frac{1}{2} \times \frac{2}{3}$, (b) $2\frac{1}{4} \times 1\frac{1}{3}$, (c) $3\frac{1}{3} \div \frac{5}{6}$, (d) $2\frac{1}{2} \div 1\frac{1}{4}$. F
4. Find (a) $\frac{2}{3}$ of 45, (b) $\frac{3}{8}$ of 64 m, (c) $\frac{5}{6}$ of 2 hours. Give the last answer in hours and minutes. P
5. Without calculating both values exactly, decide which is larger: $\frac{3}{4} \times \frac{5}{6}$ or $\frac{3}{4} \div \frac{5}{6}$. Explain your reasoning, then calculate both to check. R
6. A student writes $\frac{2}{3} \div \frac{4}{5} = \frac{8}{15}$. Identify the error, give the correct answer, and explain what the reciprocal is and why it is used. R

7. A recipe uses $\frac{3}{4}$ kg of flour per batch. (a) How many whole batches can be made from $4\frac{1}{2}$ kg? (b) How much flour is needed for $\frac{2}{3}$ of a batch? P

8. Explain why dividing by a fraction less than 1 increases a number, using $\frac{3}{4} \div \frac{1}{8} = 6$. Then calculate $\left(\frac{2}{3} \div \frac{4}{9}\right) \times \frac{3}{8}$, showing each stage. E

4.5 Decimals: place value, ordering and calculation

CRITERION A · II, III ◆ PLACE VALUE ◆ ROUNDING

Decimal notation extends place value to the right of the units digit. Each position is one tenth of the position before it, so the value of a digit depends on where it stands, exactly as it does for whole numbers.

In 3.472, the digit 4 is worth 4 tenths, the 7 is worth 7 hundredths and the 2 is worth 2 thousandths. The number can be written in expanded form as $3 + 0.4 + 0.07 + 0.002$, or as the fraction $\frac{3472}{1000}$. Reading the decimal aloud — "three and four hundred seventy-two thousandths" — confirms the last place used.

PLACE VALUE IN 3.472

Place	Ones	Tenths	Hundredths	Thousandths
Digit	3	4	7	2
Value	3	0.4	0.07	0.002
Fraction	$\frac{3}{1}$	$\frac{4}{10}$	$\frac{7}{100}$	$\frac{2}{1000}$

To order decimals, align the decimal points and compare digits place by place from the left. Writing extra zeros at the end does not change a decimal, so 0.7 can be written 0.700; this makes the comparison with 0.680 straightforward. To round, locate the required place and look at the single digit immediately to its right: 5 or more rounds the digit up, 4 or less leaves it unchanged.

WORKED EXAMPLE 4.5.1 ORDERING DECIMALS

PROBLEM

Write 0.7, 0.68, 0.708 and 0.8 in ascending order.

1. Give every number three decimal places: 0.700, 0.680, 0.708 and 0.800.
2. Compare the tenths first: 0.680 has 6 tenths and is smallest; 0.800 has 8 tenths and is largest.
3. For 0.700 and 0.708, the tenths and hundredths agree, so compare thousandths: 0 < 8.

ANSWER **0.68, 0.7, 0.708, 0.8.**

Check on a number line. All four lie between 0.6 and 0.8. The two values 0.7 and 0.708 lie very close together, so the thousandths digit is the only way to separate them.

WORKED EXAMPLE 4.5.2 ROUNDING TO A REQUIRED PLACE

PROBLEM

Round 4.675 to one decimal place and to two decimal places. Then round 12.996 to two decimal places.

1. To one decimal place, look at the hundredths digit of 4.675: it is 7, so round the tenths digit 6 up to 7. The result is 4.7.
2. To two decimal places, look at the thousandths digit: it is 5, so round the hundredths digit 7 up to 8. The tenths digit stays the same, giving 4.68.
3. For 12.996 to two decimal places, the thousandths digit is 6, so the hundredths 9 rounds up; this carries through 9 and 9, giving 13.00.

ANSWER **4.7, 4.68 and 13.00.**

Check. 4.675 lies between 4.67 and 4.68, and closer to 4.68. 12.996 lies between 12.99 and 13.00, and closer to 13.00. Rounding never moves a number further than half of the smallest place retained.

WORKED EXAMPLE 4.5.3 CALCULATING WITH DECIMALS

PROBLEM

Calculate (a) $3.7 + 2.45$, (b) $9.2 - 4.65$, (c) 0.6×0.7, (d) $3.6 \div 0.4$.

1. (a) Align the decimal points and write 3.70: $3.70 + 2.45 = 6.15$.
2. (b) Write 9.20, then subtract: $9.20 - 4.65 = 4.55$.
3. (c) $6 \times 7 = 42$, and tenths multiplied by tenths gives hundredths, so $0.6 \times 0.7 = 0.42$.
4. (d) $3.6 \div 0.4$ asks how many 0.4 are in 3.6. Both are tenths: $36 \div 4 = 9$.

ANSWER **(a) 6.15 (b) 4.55 (c) 0.42 (d) 9.**

Check. (a) $6.15 - 2.45 = 3.7$. (b) $4.55 + 4.65 = 9.2$. (c) $0.42 \div 0.6 = 0.7$. (d) $9 \times 0.4 = 3.6$. Each answer reverses correctly.

COMMON ERROR

Thinking a longer decimal is larger: 0.25 is not greater than 0.3, because 0.25 has 2 tenths while 0.3 has 3 tenths. A second error applies whole-number multiplication habits to decimals and writes $0.6 \times 0.7 = 4.2$; the correct product is 0.42, because the factors are smaller than 1 and the place value shifts two positions. A third error is misaligning digits when adding, for example writing 3.7 above 2.45 without lining up the decimal points.

GUIDED PRACTICE · COPY AND COMPLETE

Order 0.5, 0.45 and 0.504 by writing them as 0.500, 0.450 and 0.504: the smallest is _____ and the largest is _____. Round 6.347 to two decimal places: the digit in the thousandths place is ___, so the answer is _____. And $1.2 \times 0.5 =$ _____, because $12 \times 5 = 60$ and the place value moves two positions.

PRACTICE 4.5 · DECIMALS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. State the value of the stated digit: (a) 5.274, digit 7 (b) 5.274, digit 2 (c) 0.708, digit 7 (d) 12.05, digit 5. F

2. Write in expanded form: (a) 4.36, (b) 0.708, (c) 12.05. F
3. Write in ascending order: 0.6, 0.56, 0.605, 0.65. F
4. Round: (a) 3.846 to two decimal places and to one decimal place; (b) 0.995 to two decimal places; (c) 12.349 to one decimal place. F
5. Calculate: (a) $4.6 + 3.75$, (b) $9.2 - 4.65$, (c) 0.6×0.7, (d) $3.6 \div 0.4$. P
6. A student says 0.45 is larger than 0.5 "because 45 is larger than 5", and another writes $0.3 \times 0.4 = 1.2$. Identify both errors and give the correct answers with an explanation. R
7. Three measurements are recorded as 2.4 cm, 2.38 cm and 2.405 cm. Order them and state which measurement is the most precise. Explain what the extra digit tells you. P
8. Explain why 0.5 and 0.50 name the same number, and why 0.5 and 0.05 do not. Use place value in your answer. R

4.6 Fractions, decimals and the same number

CRITERION A · I, II ◆ CONVERSION ◆ TERMINATING DECIMALS

Every fraction is a division, and every terminating decimal is a fraction with denominator 10, 100, 1000 or another power of ten. Converting in either direction therefore uses ideas already met in this chapter.

To write a fraction as a decimal, divide the numerator by the denominator. If the denominator is a factor of 10, 100 or 1000, multiplying both terms to reach that denominator is often quicker. To write a decimal as a fraction, use the place value of its final digit: 0.36 means 36 hundredths, and 0.125 means 125 thousandths. Then simplify if possible.

Some fractions produce decimals that stop, and others produce decimals that repeat forever. In simplest form, a fraction terminates exactly when its denominator has no prime factor other than 2 or 5. This follows from the structure of Chapter 3: a denominator built only from 2s and 5s can be multiplied into a power of 10, while a denominator containing 3, 7 or 11 cannot.

WORKED EXAMPLE 4.6.1 — FROM FRACTION TO DECIMAL

PROBLEM

Write $\frac{7}{8}$ as a decimal and explain whether it terminates.

1. Divide 7 by 8: $7.000 \div 8 = 0.875$.
2. Alternatively, multiply both terms by 125 so the denominator becomes 1000: $\frac{7}{8} = \frac{875}{1000} = 0.875$.
3. The denominator $8 = 2^3$ has only the prime factor 2, so the decimal terminates.

ANSWER $\frac{7}{8} = 0.875$.

Check. $8 \times 0.875 = 8 \times 0.8 + 8 \times 0.07 + 8 \times 0.005 = 6.4 + 0.56 + 0.04 = 7$. The multiplication returns the numerator exactly.

WORKED EXAMPLE 4.6.2 — FROM DECIMAL TO FRACTION

PROBLEM

Write 0.64 and 1.25 as fractions in simplest form.

1. 0.64 means 64 hundredths: $\frac{64}{100}$.
2. The HCF of 64 and 100 is 4, so $\frac{64}{100} = \frac{16}{25}$.
3. 1.25 means $1\frac{25}{100}$, which simplifies to $1\frac{1}{4} = \frac{5}{4}$.

ANSWER **0.64 = $\frac{16}{25}$ and 1.25 = $\frac{5}{4}$.**

Check. $\frac{16}{25} = \frac{64}{100} = 0.64$, and $\frac{5}{4} = 1.25$. Converting back confirms both answers.

WORKED EXAMPLE 4.6.3 — CHOOSING A FORM TO COMPARE

PROBLEM

Which is larger, $\frac{5}{8}$ or 0.62?

1. Convert the fraction to a decimal: $\frac{5}{8} = 0.625$.
2. Compare 0.625 with 0.620 by place value: the hundredths are equal, and the thousandths are $5 > 0$.
3. So $\frac{5}{8} > 0.62$, by 0.005.

ANSWER $\frac{5}{8}$ **is larger.**

Alternative. Convert 0.62 to a fraction: $0.62 = \frac{62}{100} = \frac{31}{50}$. With common denominator 400, $\frac{5}{8} = \frac{250}{400}$ and $\frac{31}{50} = \frac{248}{400}$, confirming the decimal conclusion.

COMMON ERROR

Writing 0.3 as $\frac{1}{3}$. The decimal 0.3 is exactly three tenths, $\frac{3}{10}$, while $\frac{1}{3} = 0.333\ldots$ repeats without stopping. A second error is treating $\frac{1}{3}$ as exactly 0.33; 0.33 is a rounded value, and three of them make 0.99, not 1. A third error is writing 0.75 as $\frac{75}{10}$; the final digit is in the hundredths place, so the denominator is 100.

GUIDED PRACTICE · COPY AND COMPLETE

$\frac{9}{20}$: multiply both terms by 5 to reach hundredths, giving $\frac{\quad}{100} = 0.____$. Conversely, $0.45 = \frac{45}{100} = \frac{\quad}{\quad}$ in simplest form. And $\frac{2}{3} = 0.____$ with the digit 6 repeating, because $2 \div 3$ does not terminate.

PRACTICE 4.6 · FRACTIONS AND DECIMALS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Write as decimals: (a) $\frac{3}{5}$, (b) $\frac{7}{20}$, (c) $\frac{9}{8}$, (d) $\frac{5}{16}$. F
2. Write as fractions in simplest form: (a) 0.7, (b) 0.36, (c) 0.125, (d) 2.4. F
3. Decide which fractions terminate, using the prime factors of the denominator: (a) $\frac{3}{8}$, (b) $\frac{5}{6}$, (c) $\frac{7}{40}$, (d) $\frac{2}{15}$. F
4. Write in ascending order: 0.7, $\frac{2}{3}$, 0.67, $\frac{3}{4}$. P
5. A student claims that $\frac{1}{3} = 0.33$ exactly and that $0.3 = \frac{1}{3}$. Identify both errors, give the correct values, and explain why 0.33 is only an approximation. R

6. A drill bit is labelled $\frac{3}{8}$ inch; another is labelled 0.4 inch. Which is wider? Convert to compare, and state the difference in inches. P
7. Explain why $\frac{1}{3}$ repeats but $\frac{3}{8}$ terminates. Use the prime factors of the denominators in simplest form, and give one more example of each kind. R
8. Extension: some people write $0.999\ldots = 1$. Starting from $\frac{1}{3} = 0.333\ldots$, multiply both sides by 3 and explain what the result suggests. Then explain why the two numbers are equal rather than merely close. E

Real-life application · The scaling kitchen

A community kitchen prepares a sauce for 6 people. The recipe is recorded in cups. For a weekend event it must be scaled to 4 people and to 15 people, and the quantities must also be passed to a supplier who works in millilitres.

Ingredient	For 6	For 4 $(\times\frac{2}{3})$	For 10 $(\times\frac{5}{3})$	For 15 $(\times\frac{5}{2})$
Flour	$2\frac{1}{4}$ cups	$1\frac{1}{2}$ cups	$3\frac{3}{4}$ cups	$5\frac{5}{8}$ cups
Milk	$\frac{3}{4}$ cup	$\frac{1}{2}$ cup	$1\frac{1}{4}$ cups	$1\frac{7}{8}$ cups
Oil	$\frac{2}{3}$ cup	$\frac{4}{9}$ cup	$1\frac{1}{9}$ cups	$1\frac{2}{3}$ cups
Sugar	$\frac{1}{2}$ cup	$\frac{1}{3}$ cup	$\frac{5}{6}$ cup	$1\frac{1}{4}$ cups

Each row is one multiplication. For 15 people, the flour is $2\frac{1}{4} \times \frac{5}{2} = \frac{9}{4} \times \frac{5}{2} = \frac{45}{8} = 5\frac{5}{8}$ cups, and the oil is $\frac{2}{3} \times \frac{5}{2} = \frac{5}{3} = 1\frac{2}{3}$ cups. The fractions keep every quantity exact, which matters when the scaled recipe is scaled again.

Your task. The kitchen owns measuring cups of $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$ and 1 cup. Work through the prompts below and keep every quantity in its stated unit.

1. For the 4-person oil, $\frac{4}{9}$ cup cannot be measured exactly with these cups. Choose the closest single measure, calculate the error in cups, and decide whether the error is acceptable if a tolerance of 10 mL is allowed. Use 1 cup = 240 mL.
2. A cook converts $5\frac{5}{8}$ cups of flour to 5.6 cups for a supplier form. The exact decimal is 5.625. Calculate the difference in cups and in millilitres, and state whether the rounded record falls within a 10 mL tolerance. Then explain why the supplier may still prefer the round number.
3. A cook wants the 6-person recipe multiplied by $\frac{5}{2}$. Instead, they double the recipe and then add half of the original quantity again. Show with fractions that the result is the same, and explain which method is less likely to introduce a rounding error. Check your claim on the sugar.
4. Which record is more useful to the cook, and which is more useful to the supplier? Explain what information each form preserves and what each one hides.
5. The recipe uses fractions because the quantities were created by division. When does a measured decimal become the more honest record?

Investigation 4 · Which fractions stop?

CRITERION B · I, II

The tables show the decimal for $\frac{1}{n}$ for values of n from 2 to 12. An overline records the repeating block: $0.\overline{3}$ means 0.333… and $0.\overline{142857}$ means the block repeats without end.

DENOMINATORS 2 TO 7

n	2	3	4	5	6	7
$\frac{1}{n}$	0.5	$0.\overline{3}$	0.25	0.2	$0.1\overline{6}$	$0.\overline{142857}$

DENOMINATORS 8 TO 12

n	8	9	10	11	12
$\frac{1}{n}$	0.125	$0.\overline{1}$	0.1	$0.\overline{09}$	$0.08\overline{3}$

1. Sort the denominators 2 to 12 into two sets: those whose decimal terminates and those whose decimal repeats.
2. Write each denominator in the terminating set as a product of prime factors. What do they have in common?
3. Write each denominator in the repeating set as a product of prime factors. Which prime factors appear that were absent from the terminating set?
4. State a rule that predicts whether a fraction terminates, and test it on $\frac{1}{16}$, $\frac{1}{20}$, $\frac{1}{25}$, $\frac{1}{30}$ and $\frac{1}{40}$.
5. $\frac{1}{6}$ has a denominator containing a 2, yet it repeats. Explain why the presence of a 2 is not enough.
6. Simplify $\frac{3}{12}$ before converting it. Does it terminate? Explain why the original denominator 12 is misleading if you forget to simplify first.
7. Find one fraction with a denominator between 13 and 40 whose decimal terminates, and one whose decimal repeats. Give evidence for each.
8. Extension: the repeating block of $\frac{1}{7}$ has six digits and that of $\frac{1}{11}$ has two. Find a unit fraction whose repeating block has exactly three digits. Compare with a classmate and describe how you searched.

Record your sorted sets and your rule separately. A rule is only a generalisation when you can explain why it must hold, not merely that it worked for the first eleven cases.

Criterion-mapped mini-task

THE CONVERSION CARD

A · KNOWING AND UNDERSTANDING
C · COMMUNICATING
D · APPLYING MATHEMATICS

Context. A school science club sells measurement kits to younger classes. Each kit includes a conversion card showing five quantities in fraction, decimal and rounded form, so that students can move between representations without a calculator.

Product. A double-sided conversion card for one kit, plus a one-page design note explaining the choices. The card must be correct, readable at arm's length, and useful to someone who has not studied this chapter.

Requirements.

1. Choose five quantities between 0 and 2. At least two must be given as fractions in simplest form and at least two as decimals.
2. Show the exact equivalence for each quantity, with working. Include at least one mixed number and one decimal with thousandths.
3. Order the five quantities, show them on a number line, and mark the position of each.
4. Round each quantity to two decimal places. State which roundings are exact and which lose information.
5. Use your quantities in one addition and one subtraction, showing the method and giving the answer in both fraction and decimal form where possible.
6. Classify each fraction as terminating or repeating, and justify the classification using the prime factors of its denominator.
7. Add a "use this form" panel: one situation where the fraction is the better record, one where the decimal is better, and one risk of choosing the wrong form.
8. Ask a classmate to reproduce every conversion using only your card. Record one error they made, identify whether the card or the reader caused it, and revise the card.

CRITERIA AND STRANDS A i, ii, iii: accurate equivalence, ordering and calculation; C i, ii, iii, iv: clear notation, number line and language suited to the audience; D i, ii, iii, iv: apply conversion to measurement, justify form choice and evaluate the effect of rounding.

BEFORE YOU SUBMIT

Check each conversion in the opposite direction. A card that says $\frac{7}{8} = 0.875$ should also support $0.875 = \frac{7}{8}$. Verify the number line by converting every quantity to the same form, then ask someone who has not seen your working to use the card. If they hesitate, the card needs a clearer label, not a longer explanation.

Chapter review

SUMMARY

- **Equivalent fractions.** Multiplying or dividing both terms by the same non-zero number gives another name for the same value. Simplest form uses the HCF once.
- **Comparing.** Equal denominators allow direct comparison of numerators; unlike denominators need a common denominator, a benchmark or decimals.
- **Adding and subtracting.** Rewrite in the same-sized parts, combine the numerators, then simplify. Regroup when a mixed-number subtraction borrows a whole.
- **Multiplying and dividing.** Multiply across; divide by multiplying by the reciprocal. Dividing by a fraction below 1 increases the number.
- **Decimals.** Each place is one tenth of the place to its left. Align decimal points to compare or add; round using the next digit.
- **Conversions.** A fraction is a division; a decimal is a fraction with denominator a power of ten. In simplest form, only 2s and 5s terminate.

Equivalence: $\frac{a}{b} = \frac{a \times k}{b \times k}$, for any whole number $k \neq 0$.

Addition: $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$, then simplify.

Multiplication and division: $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$ and $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$.

Terminating decimals: in simplest form, the denominator has no prime factor except 2 and 5.

REVISION CHECKLIST

- ☐ I can generate equivalent fractions and simplify using the HCF.
- ☐ I can restore a missing numerator or denominator.
- ☐ I can compare and order fractions using two different methods.
- ☐ I can add and subtract fractions and mixed numbers.
- ☐ I can multiply and divide fractions and mixed numbers.
- ☐ I can read, order, round and calculate with decimals.
- ☐ I can convert between fractions and decimals and classify a decimal as terminating or repeating.
- ☐ I can choose between a fraction and a decimal for a context and justify the choice.

KEY VOCABULARY

numerator
Parts taken from the whole.

denominator
Equal parts in the whole.

equivalent fractions
Different names, same value.

simplest form
No common factor remains.

common denominator
A shared multiple of denominators.

mixed number
A whole number and a fraction.

reciprocal
Numerator and denominator swapped.

terminating decimal
Stops after finite digits.

repeating decimal
A digit block repeats forever.

place value
Value set by a digit's position.

REFLECTION AND SELF-ASSESSMENT

Choose a quantity you met this week that was recorded as a fraction or a decimal. Explain why that form was chosen and what another form would have made clearer.

Rate your confidence on each success criterion, name one exercise to revisit, and explain in two sentences how you choose between a fraction and a decimal.

END-OF-CHAPTER PRACTICE

Mixed practice 4

Do not use a calculator. Show the equivalent fraction or the place-value step that supports each answer.

FLUENCY Build accuracy

1. Write in simplest form: (a) $\frac{16}{24}$, (b) $\frac{30}{45}$, (c) $\frac{28}{42}$, (d) $\frac{54}{72}$.
2. Fill in the missing numbers: (a) $\frac{2}{3} = \frac{\square}{15}$, (b) $\frac{7}{8} = \frac{56}{\square}$, (c) $\frac{18}{24} = \frac{3}{\square}$, (d) $\frac{5}{9} = \frac{45}{\square}$.
3. Insert $<$, $>$ or $=$: (a) $\frac{3}{8} \square \frac{2}{5}$, (b) $\frac{7}{12} \square \frac{4}{7}$, (c) $\frac{5}{6} \square \frac{8}{9}$, (d) $\frac{9}{10} \square \frac{11}{12}$.
4. Write in ascending order: $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{6}$, $\frac{7}{12}$.
5. Calculate: (a) $\frac{5}{6} + \frac{3}{8}$, (b) $\frac{2}{3} + \frac{4}{9}$, (c) $\frac{7}{10} + \frac{1}{2}$.
6. Calculate: (a) $\frac{5}{6} - \frac{3}{4}$, (b) $\frac{3}{4} - \frac{5}{12}$, (c) $1 - \frac{7}{15}$.
7. Calculate: (a) $3\frac{2}{5} + 1\frac{3}{4}$, (b) $4\frac{1}{6} - 2\frac{5}{6}$.
8. Calculate: (a) $\frac{4}{9} \times \frac{3}{8}$, (b) $\frac{5}{6} \times \frac{9}{10}$, (c) $2\frac{1}{3} \times \frac{3}{7}$.
9. Calculate: (a) $\frac{3}{5} \div \frac{9}{10}$, (b) $\frac{4}{7} \div 2$, (c) $2\frac{1}{2} \div \frac{5}{8}$.
10. Write as decimals: (a) $\frac{7}{8}$, (b) $\frac{11}{20}$, (c) $\frac{5}{16}$, (d) $\frac{4}{3}$.
11. Write as fractions in simplest form: (a) 0.45, (b) 0.375, (c) 1.6, (d) 0.08.
12. Round: (a) 5.278 to two decimal places and to one decimal place; (b) 0.996 to two decimal places; (c) 17.45 to one decimal place.
13. Calculate: (a) $6.4 + 2.75$, (b) $10 - 3.6$, (c) 0.8×0.7, (d) $4.5 \div 0.9$.

REASONING AND PROBLEM SOLVING Connect ideas

14. A student calculates $\frac{2}{3} + \frac{1}{5} = \frac{3}{8}$. Identify the error, give the correct answer, and explain why the correct answer is greater than $\frac{2}{3}$.
15. Explain why $\frac{3}{3} \times \frac{4}{5}$ is equal to $\frac{4}{5}$. Use a diagram or a sentence about cutting a whole into pieces, then state why multiplying by $\frac{0}{0}$ is not allowed.
16. Explain why $\frac{3}{4} \div \frac{4}{5}$ is the same as $\frac{3}{4} \times \frac{5}{4}$, and calculate the result. Then explain why the answer is larger than $\frac{3}{4}$.
17. Which is larger, $\frac{5}{7}$ or $\frac{7}{10}$? Use two different methods and say which you would choose under time pressure.
18. A bottle contains $\frac{5}{8}$ litre of juice. A second bottle contains 0.6 litre. Which contains more, and by how much? Give the difference as a fraction and as a decimal.
19. **The water tank (Criteria A, C, D).** A tank is $\frac{3}{4}$ full. It uses $\frac{2}{5}$ of its full capacity during the day. (a) What fraction of the capacity remains? (b) Write the remaining fraction as a decimal. (c) If the full capacity is 2,400 litres, how many litres remain? (d) Explain whether the answer to (a) would change if the tank started half full.
20. **Three offcuts (Criteria C, D).** A workshop has three strips of metal measuring 0.35 m, $\frac{3}{8}$ m and 0.4 m. (a) Write the lengths in the same form and order them. (b) Find their total length in metres. (c) The longest strip is cut into pieces of $\frac{1}{8}$ m. How many pieces are made, and what length is left over? (d) Explain how rounding each length to one decimal place before adding would change the total, and whether that matters for the workshop.

Check your progress

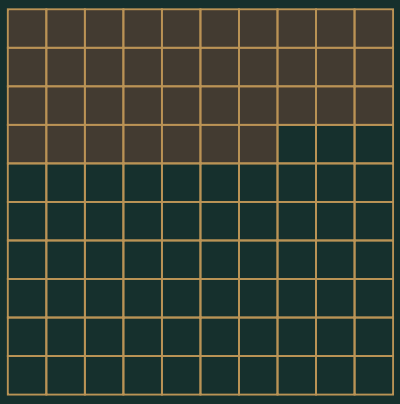
ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

- **Equivalent fractions use multiplication or division, never addition.** Adding the same number to both terms changes the relationship between the parts and the whole.
- **Denominators are part sizes, not quantities to add.** $\frac{2}{3} + \frac{1}{4}$ needs twelfths, not a denominator of 7.
- **Division by a fraction uses the reciprocal.** Invert the divisor, not the dividend, and check whether the answer should be larger or smaller than the starting number.
- **A longer decimal is not a larger decimal.** Line up place values; 0.25 is less than 0.3, and 0.5 equals 0.50.
- **Terminating depends on the simplest denominator.** Only prime factors 2 and 5 allow a decimal to stop; simplify before deciding.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can generate equivalent fractions and simplify using a common factor.			
I can compare and order fractions and justify my method.			
I can add and subtract fractions and mixed numbers.			
I can multiply and divide fractions using the reciprocal.			
I can read, order, round and calculate with decimals.			
I can convert between fractions and decimals and explain termination.			

WHERE THIS LEADS

Chapter 5 turns fractions with denominator 100 into percentages. Everything learned here applies directly: a percentage is an equivalent fraction with a fixed denominator, and a percentage change is a multiplication by a fraction. Comparing discounts, increasing a price and reading a test score all depend on the equivalence and calculation skills in this chapter.



per centum

5

A COMMON SCALE

Percentages and applications

A percentage compares a quantity with 100. Once a price, a budget or a change is written on that scale, very different quantities can be judged side by side — provided we ask what the percentage is of.

KEY CONCEPT · CHANGE

RELATED · QUANTITY

RELATED · PROPORTION

CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

When a shop advertises “20% off”, what exactly is being reduced, and how can a buyer tell whether the offer is genuinely good?

MYP UNIT PLANNER

Framing the inquiry

UNIT 5 · APPROX. 14 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Change. A percentage describes a change or a part relative to a base; the same absolute change is a different percentage when the base changes.

Quantity. A percentage is not an amount until it is applied to something. 20% of £50 and 20% of £80 are the same proportion but not the same money.

Proportion. Percentages compare a part with a whole scaled to 100, linking fractions, decimals and ratio.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Exploration: financial decisions and access — how percentage claims in prices, taxes and budgets shape what people can afford, and whether an offer is fair.

A 30% discount on a £20 item saves £6; the same 30% on a £200 item saves £60. The percentage is identical, but the effect on a household is not. Reading the base is part of reading the offer honestly.

STATEMENT OF INQUIRY

Percentages express change and comparison on a common scale of 100, and interpreting the base and the multiplier determines whether a financial claim is understood or misleading.

FACTUAL

What does per cent mean? How do you convert between percentages, fractions and decimals? How do you find a percentage of a quantity?

CONCEPTUAL

How can the same percentage produce different amounts, and why do successive changes not add?

DEBATABLE

Is a larger percentage discount always the better deal? Should a budget be judged by its percentages or by its amounts?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Convert between forms, calculate a percentage of a quantity, increase and decrease amounts, calculate discounts and taxes.
B · Investigating patterns	i, ii	Investigate successive percentage changes, record multipliers and generalise an equivalent single change.
C · Communicating	i, ii, iii	Explain the choice of base and method, represent percentages with grids and bars, justify a financial claim.
D · Applying mathematics	i, ii, iii, iv	Price a stall, compare offers, build a budget, evaluate rounding and recommend a course of action.

ATL · CRITICAL THINKING IN ACTION

A stall advertises “20% off, then an extra 10% at the till”. A rival advertises “30% off”. Decide which claim is better before calculating every price, then test your reasoning on £40 and £250.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** per cent as a proportion of 100 and convert between percentages, fractions and decimals.
- **Be able to** find a percentage of a quantity using benchmarks and multipliers.
- **Be able to** increase and decrease quantities by a percentage and explain the multiplier.
- **Know** how discounts, sale prices, taxes and budgets use percentages, and be able to calculate each.
- **Be able to** compare offers and justify a financial decision with evidence.
- **Investigate** what happens when two percentage changes are applied one after the other.

SUCCESS CRITERIA

- ☐ I can convert between percentages, fractions and decimals.
- ☐ I can find a percentage of a quantity and choose an efficient method.
- ☐ I can increase or decrease a quantity by a percentage using a multiplier.
- ☐ I can calculate a discount, a sale price and a price including tax.
- ☐ I can build and interpret a simple budget using percentages.
- ☐ I can compare two offers and explain which is better and why.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Write $\frac{3}{4}$ as a decimal.
2. Find $\frac{1}{10}$ of 80.
3. Calculate 0.2×150.
4. Write $\frac{45}{100}$ in simplest form.
5. Increase 50 by 10.
6. Which is larger: 0.6 or 0.06?

Answers are in the Chapter 5 answer section. If more than two are incorrect, review fraction–decimal conversion and finding a fraction of an amount before continuing.

BEFORE YEAR 1
Fractions and decimals; finding a fraction of an amount; simple money calculations and comparisons.

YEAR 1 · THIS CHAPTER
Percent as per hundred, conversions, percentage of a quantity, increase and decrease, discounts, tax and budgets.

LATER
Ratio and proportion, repeated and compound change, interest, and percentages in statistics.

INQUIRY PATHWAY FOR THIS CHAPTER

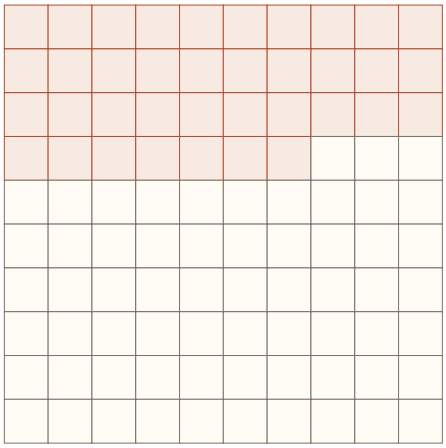
1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Offers, taxes and household budgets are written as percentages of something.	What is the percentage of, and does the claim tell the whole story?	Change, quantity and proportion.	Do two successive discounts add up to one larger discount?	Hundred grids, bar models, multipliers and clear financial statements.	When is a percentage the honest way to describe a change?

5.1 Per cent: a proportion of 100

CRITERION A · I ◆ CHANGE ◆ REPRESENTATION

Per cent means “per hundred”. 37% is 37 parts in every 100, so it can be written as $\frac{37}{100}$ or 0.37. A percentage fixes the denominator at 100, which makes proportions easy to compare.

The word comes from the Latin *per centum*, “by the hundred”. A percentage is therefore a fraction whose denominator is understood to be 100. The symbol % is a compact way of writing that denominator. The number in front of the symbol can be a whole number, a decimal or even greater than 100.



37%

Figure 5.1 Thirty-seven of the hundred squares are shaded. The same region is 37%, $\frac{37}{100}$ or 0.37.

DEFINITIONS

Per cent means “out of 100”. A **percentage** is a quantity expressed as a proportion of 100. The **base** is the whole amount that the percentage is taken of.

To convert a percentage to a fraction, divide by 100 and simplify. To convert a percentage to a decimal, divide by 100. To reverse either step, multiply by 100 and attach the % symbol.

Because the denominator is always 100, percentages can be compared at a glance, while fractions with different denominators cannot. The cost is that the base becomes invisible: “40%” tells us the proportion but not 40% of what. The next section makes that base explicit.

FOUR WAYS TO NAME THE SAME PROPORTION

Percentage	Fraction in simplest form	Decimal	Meaning
45%	$\frac{45}{100} = \frac{9}{20}$	0.45	45 parts in every 100
7.5%	$\frac{75}{1000} = \frac{3}{40}$	0.075	7 and a half parts in every 100
0.5%	$\frac{5}{1000} = \frac{1}{200}$	0.005	half of one part in every 100
120%	$\frac{120}{100} = \frac{6}{5}$	1.2	the whole and 20 parts more

WORKED EXAMPLE 5.1.1 — PER CENT TO FRACTION AND DECIMAL

PROBLEM

Write (a) 45%, (b) 7.5% and (c) 0.5% as fractions in simplest form and as decimals.

1. (a) $45\% = \frac{45}{100} = \frac{9}{20}$. Divide by 100 for the decimal: $45 \div 100 = 0.45$.
2. (b) $7.5\% = \frac{7.5}{100} = \frac{75}{1000} = \frac{3}{40}$. As a decimal, $7.5 \div 100 = 0.075$.
3. (c) $0.5\% = \frac{0.5}{100} = \frac{5}{1000} = \frac{1}{200}$. As a decimal, $0.5 \div 100 = 0.005$.

ANSWER **(a) $\frac{9}{20}$, 0.45 (b) $\frac{3}{40}$, 0.075 (c) $\frac{1}{200}$, 0.005.**

Check. 0.5% is half of 1%, and 1% of a quantity is one hundredth, so 0.5% must be half of one hundredth: 0.005. The answer is small, but it is not zero.

WORKED EXAMPLE 5.1.2 — FRACTION AND DECIMAL TO PER CENT

PROBLEM

Write (a) $\frac{3}{8}$, (b) 0.64 and (c) 1.05 as percentages.

1. (a) $\frac{3}{8} = 3 \div 8 = 0.375$. Multiply by 100: $0.375 \times 100 = 37.5$, so $\frac{3}{8} = 37.5\%$.
2. (b) $0.64 \times 100 = 64$, so $0.64 = 64\%$.
3. (c) $1.05 \times 100 = 105$, so $1.05 = 105\%$. The value is greater than the whole, which is why the percentage exceeds 100.

ANSWER **(a) 37.5% (b) 64% (c) 105%.**

Check. 37.5% is a little more than $\frac{1}{3}$, and $\frac{3}{8}$ is a little more than $\frac{1}{3}$. 1.05 is just over 1, so 105% is just over 100%.

COMMON ERROR

Moving the decimal point the wrong distance. 7.5% is 0.075, not 0.75 and not 7.5: dividing by 100 moves the digits two places, not one. A second error is treating a small percentage as zero: 0.5% is 0.005, which is small but not nothing. A third is reading 120% as "20 more than the whole" and writing 1.02; 120% is 1.2.

GUIDED PRACTICE · COPY AND COMPLETE

$62\% = \frac{62}{100} = \frac{\quad}{50} = 0.___$. $4\% = \frac{\quad}{1} = 0.___$. $12.5\% = \frac{12.5}{100} = \frac{\quad}{1} = 0.____$. And $0.09 = ___\%$, because $0.09 \times 100 = ___$.

PRACTICE 5.1 · PERCENT, FRACTIONS AND DECIMALS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Show the division or multiplication by 100 at each step.

1. Write each percentage as a fraction in simplest form and as a decimal: (a) 25%, (b) 60%, (c) 8%, (d) 150%. F
2. Write each fraction as a percentage: (a) $\frac{1}{4}$, (b) $\frac{7}{20}$, (c) $\frac{9}{8}$, (d) $\frac{3}{5}$. F
3. Write each decimal as a percentage: (a) 0.7, (b) 0.09, (c) 1.35, (d) 0.005. F
4. Copy and complete each set of equivalent forms: (a) $35\% = \frac{\square}{100} = 0.\square$; (b) $\frac{3}{8} = 0.\square = \square\%$; (c) $0.06 = \square\% = \frac{\square}{100}$; (d) $250\% = \frac{\square}{100} = \square$. F
5. Which is larger? (a) 0.4 or 39%; (b) $\frac{5}{8}$ or 62%; (c) 1.2 or 115%. Explain each choice. R

6. A student writes "$0.5\% = 0.5$ because 0.5 is a half." Explain the error, then give the correct decimal and fraction. R
7. A test has 40 marks. Sam scores 85%. Write the score as a fraction and as a decimal, then find the number of marks gained. P
8. Extension: a price is increased by 100%, then by another 100%. Is the final price 200% of the original? Test with £50, write the overall multiplier, and explain the result. E

5.2 Finding a percentage of a quantity

CRITERION A · II ◆ QUANTITY ◆ MULTIPLIERS

A percentage becomes an amount when it is applied to a base. 15% of £80 is not £15; it is fifteen hundredths of £80. The base is the quantity named after "of".

There are three dependable methods. The first converts the percentage to a decimal and multiplies: $15\% = 0.15$, so $0.15 \times 80 = 12$. The second uses a fraction: $15\% = \frac{3}{20}$, and $\frac{3}{20} \times 80 = 12$. The third builds the percentage from easy parts: 10% of 80 is 8, 5% is 4, and $8 + 4 = 12$. All three use the same underlying proportion.

BENCHMARK PERCENTAGES AND THE MOVES THEY ALLOW

Percentage	Equivalent move	Example on a base of 200
50%	Halve	100
25%	Quarter	50
10%	Divide by 10	20
5%	Half of 10%	10
1%	Divide by 100	2
75%	Three quarters	150
20%	One fifth	40

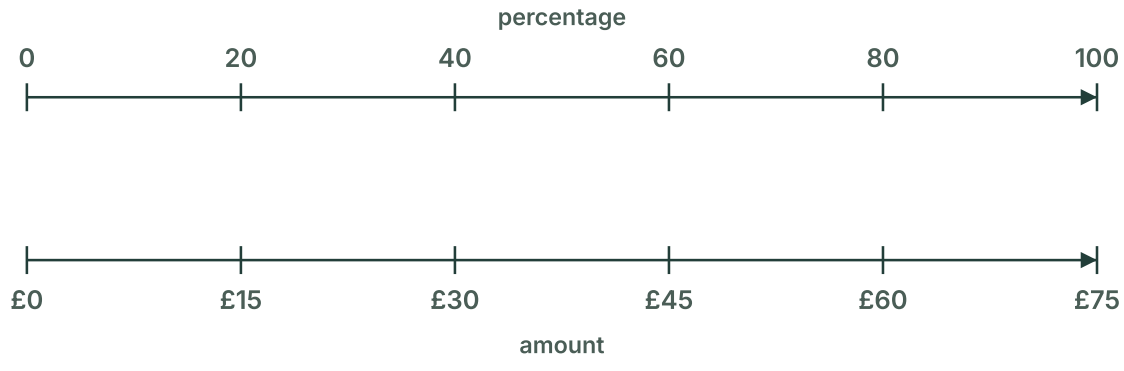


Figure 5.2 Two scales for the same proportion. A 20% share of £75 is £15, because £75 is the whole and 20% is one fifth of it.

WORKED EXAMPLE 5.2.1 — USING A DECIMAL MULTIPLIER

PROBLEM

Find 15% of 80 kg.

1. Write the percentage as a decimal: $15\% = 0.15$.
2. Multiply by the base: $0.15 \times 80 = 12$.
3. Attach the unit: 12 kg. Check with easy parts: 10% of 80 is 8 and 5% is 4, so 15% is $8 + 4 = 12$.

ANSWER **15% of 80 kg is 12 kg.**

Second method. $15\% = \frac{15}{100} = \frac{3}{20}$, and $\frac{3}{20} \times 80 = 3 \times 4 = 12$. The fraction method is efficient when the fraction is simple.

WORKED EXAMPLE 5.2.2 — DECIMAL PERCENTAGES AND MORE THAN 100%

PROBLEM

(a) Find 6.5% of £240. (b) Find 120% of 45.

1. (a) $6.5\% = 0.065$. Multiply: $0.065 \times 240 = 15.6$. Check with parts: 10% is 24, 5% is 12, 1% is 2.4 and 0.5% is 1.2. So 5% + 1% + 0.5% gives $12 + 2.4 + 1.2 = 15.6$.
2. (b) $120\% = 1.2$. Multiply: $1.2 \times 45 = 54$. Equivalently, 100% of 45 is 45 and 20% is 9, so the total is $45 + 9 = 54$.

ANSWER **(a) £15.60 (b) 54.**

Check. 6.5% is a little more than $\frac{1}{16}$ of 240, and 15.6 is close to that estimate. 120% must be larger than 45, and 54 is larger by 20% of 45.

COMMON ERROR

Using the percentage as if it were the amount: "15% of 80 is 15" or "£80 plus 15% is £95". A percentage has no size until it is multiplied by its base. A second error is misplacing the decimal point when the percentage has a decimal part: $6.5\% = 0.065$, not 0.65. A third is assuming a larger percentage always means a larger amount; 25% of £100 is less than 30% of £90.

GUIDED PRACTICE · COPY AND COMPLETE

35% of 60: 10% is ___, 30% is ___, and 5% is ___; so 35% is ___. 2.5% of 400: 1% is ___, so 2.5% is ___. 140% of 25: 100% is ___ and 40% is ___, so the total is ___.

PRACTICE 5.2 · A PERCENTAGE OF A QUANTITY

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Choose a benchmark, a fraction or a multiplier, and say which you used for at least two parts.

1. Find: (a) 10% of 90, (b) 25% of 60, (c) 50% of 7, (d) 1% of 450. F
2. Find: (a) 15% of 80, (b) 35% of 60, (c) 6.5% of 240, (d) 120% of 45. F
3. Find: (a) 2.5% of 400, (b) 0.5% of 900, (c) 140% of 25, (d) 250% of 12. F
4. A jacket costs £80 and is discounted by 15%. Find the discount in pounds. P
5. A school has 650 students, and 48% of them are in Key Stage 3. How many students is that? P
6. Which is larger: 30% of 80 or 25% of 100? By how much? R

7. A student calculates 6.5% of 240 as 156. Explain the error, give the correct answer, and describe how an estimate would have caught it. R
8. Extension: "15% of 80 equals 80% of 15." Explain why both products are 12, then find another pair of a percentage and a quantity with the same product. E

5.3 Increasing and decreasing by a percentage

CRITERION A · III ◆ CHANGE ◆ MULTIPLIERS

A percentage change is measured from the original amount. A 15% increase on £80 is £12, because 15% is taken of £80. The original amount is the base, not the new amount.

An increase of $p\%$ adds $p\%$ of the original, so the new amount is $100\% + p\%$ of the original. A decrease subtracts $p\%$, leaving $100\% - p\%$. Written as multipliers, an increase of $p\%$ multiplies by $1 + \frac{p}{100}$, and a decrease multiplies by $1 - \frac{p}{100}$.

Increase by $p\%$: $\text{new} = \text{original} \times \left(1 + \frac{p}{100}\right)$.

Decrease by $p\%$: $\text{new} = \text{original} \times \left(1 - \frac{p}{100}\right)$.

Percentage change: $\frac{\text{change}}{\text{original}} \times 100\%$.

MULTIPLIERS FOR COMMON CHANGES

Change	Multiplier	Effect on a base of £200
Increase by 5%	1.05	£210
Increase by 15%	1.15	£230
Increase by 100%	2	£400
Decrease by 10%	0.9	£180
Decrease by 35%	0.65	£130
Decrease by 100%	0	£0

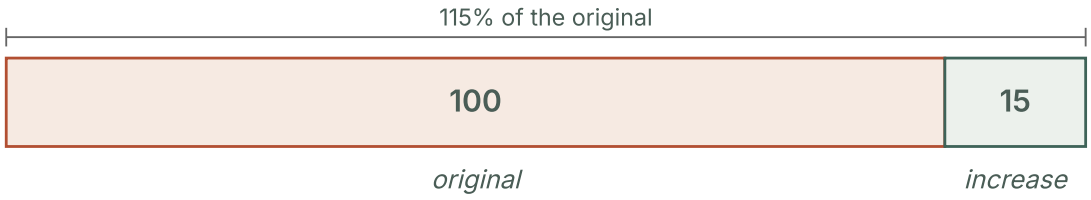


Figure 5.3 A 15% increase adds 15 parts to every 100. The new amount is 115% of the original, which is the multiplier 1.15.

WORKED EXAMPLE 5.3.1 INCREASE THEN DECREASE

PROBLEM

(a) Increase £80 by 15%. (b) Decrease 240 kg by 35%.

1. (a) The increase is $0.15 \times 80 = 12$. Add it: $80 + 12 = 92$. With a multiplier: $1.15 \times 80 = 92$.
2. (b) The decrease is $0.35 \times 240 = 84$. Subtract it: $240 - 84 = 156$. With a multiplier: $0.65 \times 240 = 156$.

ANSWER **(a) £92 (b) 156 kg.**

Check. A decrease of 35% leaves 65%, and 65% of 240 is a little less than two thirds of 240, which is 160. The answer 156 is reasonable.

WORKED EXAMPLE 5.3.2 WORKING BACKWARDS AND MEASURING CHANGE

PROBLEM

(a) After a 15% discount, a jacket costs £68. Find its original price. (b) A town's population rises from 80,000 to 92,000. Find the percentage increase.

1. (a) The sale price is 85% of the original, so $0.85 \times \text{original} = 68$. Divide: $68 \div 0.85 = 80$. Check: 15% of £80 is £12, and $80 - 12 = 68$.
2. (b) The increase is $92{,}000 - 80{,}000 = 12{,}000$. Divide by the original: $\frac{12{,}000}{80{,}000} = 0.15$. As a percentage, $0.15 \times 100\% = 15\%$.

ANSWER **(a) £80 (b) a 15% increase.**

Check. In (b), 10% of 80,000 is 8,000 and 5% is 4,000, so a 15% increase is 12,000. The new population 92,000 confirms the calculation.

COMMON ERROR

Adding the percentage rate as an amount: £80 increased by 15% is £92, not £95. Another error is dividing by the new amount when finding the original: after a 15% discount, £68 is 85% of the original, so divide by 0.85, not by 1.15. Finally, a 20% increase followed by a 20% decrease does not return to the start; the second percentage is taken of a larger base.

GUIDED PRACTICE · COPY AND COMPLETE

Increase 60 by 25%: the increase is ___, so the new amount is ___. Decrease 90 by 10%: the multiplier is ___, so the result is ___. After a 25% decrease an item costs £60, so the original price is 60 ÷ ___ = ___.

PRACTICE 5.3 · INCREASE AND DECREASE

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Write the multiplier before calculating, and state whether the answer should be larger or smaller than the original.

1. Increase: (a) 40 by 10%, (b) 70 by 20%, (c) 250 by 4%, (d) 18 by 50%. F
2. Decrease: (a) 80 by 25%, (b) 140 by 30%, (c) 64 by 12.5%, (d) 500 by 1%. F
3. A bicycle priced at £240 rises in price by 15%. Find the new price. P
4. A loaf of dough has mass 1,200 g. It loses 8% of its mass during baking. Find the mass of the baked loaf. P

5. A town's population falls from 45,000 to 42,750. Find the percentage decrease. P
6. After a 20% increase, a ticket costs £54. Find the original price. P
7. A student says: "A 20% increase followed by a 20% decrease returns to the start, because $+20 - 20 = 0$." Test the claim with £100, find the final amount, and explain the error. R
8. Extension: a price is increased by 10%, then reduced by 10%. Is the final price more or less than the original? By what percentage? Show the multipliers and explain why the order does not change the result. E

5.4 Discounts, taxes and budgets

CRITERION A · III · D · I–III ◆ PROPORTION ◆ FINANCIAL DECISIONS

Shops, taxes and household budgets all use percentages of a base. A discount is a decrease from the marked price; a tax is an increase on a price before tax; a budget is a whole against which each expense is a percentage.

DEFINITIONS

The **marked price** is the price before a reduction. A **discount** is the amount taken off; the **sale price** is what remains. **VAT** (value added tax) is a tax charged as a percentage of a price before tax. A **budget** is a plan for spending a fixed income, and each expense can be expressed as a percentage of that income.

These applications use the same multipliers as Section 5.3, with the base made explicit. A discount of 18% multiplies the marked price by 0.82. Adding 20% tax multiplies the price before tax by 1.20. A budget of £1,800 with 35% for rent allocates $0.35 \times 1{,}800 = £630$. In every case, identify the base before calculating.

WORKED EXAMPLE 5.4.1 DISCOUNT AND SALE PRICE

PROBLEM

A coat is marked at £250 and reduced by 18%. Find the discount and the sale price.

1. The discount is 18% of the marked price: $0.18 \times 250 = 45$.
2. Subtract the discount: $250 - 45 = 205$. With a multiplier: $0.82 \times 250 = 205$.

ANSWER **The discount is £45 and the sale price is £205.**

Check. 10% of £250 is £25 and 8% is £20, so the discount is £45. The sale price is £205, which is less than the marked price but more than 80% of it (£200), as expected for an 18% reduction.

WORKED EXAMPLE 5.4.2 ADDING TAX TO A PRICE

PROBLEM

A business buys a printer for £320 before VAT. VAT is charged at 20%. Find the VAT and the total price.

1. VAT is 20% of the price before tax: $0.20 \times 320 = 64$.
2. Add the tax: $320 + 64 = 384$. With a multiplier: $1.20 \times 320 = 384$.

ANSWER **VAT is £64 and the total price is £384.**

Check. £64 is one fifth of £320, and £384 is one fifth more than £320. The total is 120% of the pre-tax price, not 20%.

WORKED EXAMPLE 5.4.3 A MONTHLY BUDGET

PROBLEM

A household has a monthly income of £1,800. It spends 35% on rent, 22% on food, 12% on transport and saves 8%. Find each amount and the amount remaining.

1. Rent: $0.35 \times 1{,}800 = 630$. Food: $0.22 \times 1{,}800 = 396$. Transport: $0.12 \times 1{,}800 = 216$. Savings: $0.08 \times 1{,}800 = 144$.
2. The percentages used total $35 + 22 + 12 + 8 = 77\%$, so the remaining share is $100 - 77 = 23\%$.
3. Remaining amount: $0.23 \times 1{,}800 = 414$. Check by adding: $630 + 396 + 216 + 144 = 1{,}386$, and $1{,}800 - 1{,}386 = 414$.

ANSWER **Rent £630, food £396, transport £216, savings £144, remaining £414 (23%).**

Check. The four amounts plus the remainder must equal the income: $630 + 396 + 216 + 144 + 414 = 1{,}800$. The percentages must sum to 100%.

COMMON ERROR

Taking a discount off the sale price instead of the marked price, or adding the discount to the price. Another error is treating a percentage of a budget as an amount: 23% remaining is not £23. When adding tax, the base is the price before tax, not the price after a discount unless the problem states that rule. Always write what the percentage is of before you calculate.

GUIDED PRACTICE · COPY AND COMPLETE

30% off £80: the discount is ___, so the sale price is ___. A 5% service charge on £60: the charge is ___, so the total is ___. A budget of £1,200 with 45% spent: the amount spent is ___ and the amount left is ___, which is ___% of the budget.

PRACTICE 5.4 · DISCOUNTS, TAX AND BUDGETS

SUPPORT 1–4 · CORE 1–8 · EXTENSION 9–10

For each question, name the base before you calculate.

1. Find the discount and the sale price: (a) £80 with 25% off, (b) £150 with 12% off, (c) £36 with 5% off. F
2. Find the total price after tax: (a) £60 plus 20% VAT, (b) £240 plus 5% tax, (c) £18.50 plus 20% VAT. F
3. A family has a monthly income of £2,400. It spends 30% on rent, 15% on transport and 25% on food. (a) Find each amount. (b) What percentage and amount remains? P

4. A shopper says: “I saved £15 in a 20% sale, so the original price was £90.” Is the reasoning correct? Find the original price and explain the error. R
5. Two shops sell the same £60 game. Shop A offers 25% off. Shop B offers £12 off. Which is cheaper, and by how much? R
6. A café adds a 12.5% service charge to a £48 bill. Find the total. P
7. A student budgets 40% of £250 for materials and then spends £120. What percentage of the budget did they spend, and by how much were they over or under? R
8. A price of £80 is reduced by 20%, and then 20% tax is added to the reduced price. Is the final price £80? Show the multipliers and explain. R
9. A household spends £450 of a £1,500 monthly income on rent. What percentage is that? The household plans to reduce rent to 25% of income by moving. How much would it save each month? P
10. Extension: a shop offers “buy one, get one half price” on two items of equal price. What percentage is saved on the pair? Explain why the saving is not 50%. E

Real-life application · The community market stall

A youth group runs a stall at a community market. It buys three products, sets prices, offers a discount and pays fixed costs. Every decision is a percentage decision, but the group must also live with the whole-pound and whole-item constraints of a real stall.

COSTS, MARKED PRICES AND DISCOUNTS

Product	Cost each	Marked price	Discount	Sale price	Profit each
Honey	£6.00	£9.00	10%		
Soap	£2.50	£4.00	25%		
Jam	£3.20	£5.00	0%		

Your task. The group expects to sell 20 jars of honey, 36 bars of soap and 25 jars of jam.

1. Complete the sale price and profit columns. Profit is the sale price minus the cost of the item.
2. Find the total revenue, the total cost of the goods and the gross profit. Fixed costs are £45 for the stall and £18 for transport. Find the net profit.
3. Find each product's share of the revenue as a percentage, correct to one decimal place. Check whether the three percentages add to 100%; explain any difference.
4. The group expects 15% more customers next time and plans to buy 15% more of each item. Find the new quantities. Items must be whole, so round up. Find the new total cost of goods.
5. Fixed costs rise by 8%. Find the new fixed costs and the net profit if revenue and the cost of goods stay at their original values.
6. A supplier offers 10% off the cost of honey if the group buys 30 jars. Compare the cost of 30 jars at the offer price with the cost of the 23 jars the group actually needs. Should the group accept the offer? Explain the risk.
7. **Consider.** The group could advertise "15% off everything" instead of the individual discounts. Estimate the effect on revenue if the same number of items are sold, and explain why the group might prefer a discount on only some products.
8. **Reflect.** Which figure matters more to the group: gross profit or net profit? Which matters more to a customer: the percentage discount or the final price? Use the numbers from the stall to support your answer.

Investigation 5 · Do two discounts add up?

CRITERION B · I, II

A shop advertises “20% off, then an extra 10% off at the till”. A rival advertises “30% off”. A shopper assumes the two offers are the same because $20 + 10 = 30$. The table tests that assumption on a £250 item.

TWO ROUTES TO A LOWER PRICE

Step	Offer A: 20%, then 10%	Offer B: 30%
Start	£250	£250
After the first reduction	£200	£175
After the second reduction	£180	—
Total reduction	£70	£75

1. Check the table by calculating each reduction. Which offer is cheaper, and by how much?
2. Reverse the order: take 10% off first, then 20% off the reduced price. Does the final price change? Record both multipliers.
3. Write the two multipliers as decimals and multiply them. What single multiplier is equivalent to the pair? What single percentage discount does it represent?
4. Repeat the comparison for a £40 item and a £80 item. Is the difference between the offers always the same amount? Is it always the same percentage of the original?
5. A price is increased by 20% and then decreased by 20%. Record the multipliers, find the final price for a £100 start, and explain why it is not £100.
6. Compare three successive 10% discounts with a single 30% discount on £200. Which is cheaper, and by how much? What single discount is equivalent to the three reductions?
7. Write a rule for the single discount equivalent to successive discounts of $a\%$ and $b\%$. Test it on 20% and 10%, and on 25% and 20%.
8. Explain why the order of two percentage changes does not affect the final amount, but the order of “add £10” and “take 10%” does.
9. Extension: find two successive whole-number discounts, other than 20% and 10%, whose combined effect is exactly 28% off. Show that your pair works.

Record each step as a multiplier, not only as an amount. A rule is a generalisation only when it explains why the result must hold for every starting price, not merely when it works for £250.

Criterion-mapped mini-task

A · Knowing and understanding
C · Communicating
D · Applying mathematics

Context. A community centre is publishing a one-page guide to help families judge school-equipment offers. The guide must explain how a percentage discount, a tax and a budget interact, using one worked example that a reader can follow without a calculator.

Product. A single-sided "fair-price card" for one item, plus a short design note explaining the choices. The card must be accurate, readable at a glance, and honest about what a percentage does not tell the buyer.

Requirements.

1. Choose an item with a marked price between £40 and £400. State the price and the base of every percentage you use.
2. Compare two offers: one percentage discount and one fixed amount off. Calculate both sale prices and state which is cheaper, by how much and as a percentage of the marked price.
3. Add a tax scenario at 20%. Show the tax and the total, and explain why the total is 120% of the pre-tax price.
4. Include one percentage increase and one percentage decrease, each with a multiplier and a checking estimate.
5. Build a monthly budget of £2,000 for a family. Show the item as a percentage of income, alongside at least three other categories that sum to no more than 100%.
6. Explain one situation in which the larger percentage discount is not the better deal. Use calculated amounts, not opinion alone.
7. Include a "check the claim" panel with two common percentage errors, each shown with a wrong and a corrected calculation.
8. Ask someone who has not studied the chapter to use the card. Record one misunderstanding, decide whether the card or the reader caused it, and revise the card.

CRITERIA AND STRANDS A i, ii, iii: accurate conversion, calculation and percentage change; C i, ii, iii, iv: clear layout, correct notation and language suited to the audience; D i, ii, iii, iv: apply percentages to a real purchase, evaluate the offers and justify a recommendation.

BEFORE YOU SUBMIT

Check every percentage against its base. A 20% discount on £80 is £16, not £20, and a 20% tax added afterwards multiplies £64 by 1.2, not by 0.2. Make sure the budget percentages sum to 100% or less, and state what happens to the remainder. Ask a reader to find the final price using only your card; if they hesitate, the card needs a clearer base label, not more text.

Chapter review

SUMMARY

- **Per cent.** "Per hundred" fixes the denominator at 100. Divide by 100 to reach a decimal, multiply by 100 to return to a percentage.
- **A percentage of a quantity.** Multiply the decimal or fraction by the base. Benchmark parts such as 10%, 5% and 1% make mental calculation reliable.
- **Increase and decrease.** Use the multipliers $1 + \frac{p}{100}$ and $1 - \frac{p}{100}$. The original amount is the base, including when working backwards.
- **Discounts and tax.** A discount is a decrease from the marked price; tax is an increase on the price before tax. Name the base before calculating.
- **Budgets.** Each expense is a percentage of the income, and the percentages must sum to 100% or leave a stated remainder.
- **Successive changes.** Multipliers combine by multiplication, so two discounts do not add. 20% then 10% is equivalent to 28% off, not 30%.

Conversion: $p\% = \frac{p}{100} = p \div 100$.

Percentage of a quantity: $\text{part} = \frac{p}{100} \times \text{base}$.

Change: $\text{new} = \text{original} \times \left(1 \pm \frac{p}{100}\right)$.

Successive changes: overall multiplier $= m_1 \times m_2$.

REVISION CHECKLIST

- ☐ I can convert between percentages, fractions and decimals.
- ☐ I can find a percentage of a quantity using two methods.
- ☐ I can increase and decrease a quantity with a multiplier.
- ☐ I can find an original amount after a percentage change.
- ☐ I can calculate discounts, sale prices and prices including tax.
- ☐ I can build and interpret a budget in percentages and amounts.
- ☐ I can compare offers and justify the better one.
- ☐ I can explain why successive percentages do not add.

KEY VOCABULARY

per cent
Out of one hundred.

percentage
A proportion expressed out of 100.

base
The whole amount a percentage is of.

multiplier
The decimal used to apply a change.

discount
An amount taken off a price.

marked price
The price before a reduction.

sale price
The price after a discount.

VAT
A tax added to a price before tax.

budget
A plan for spending an income.

successive change
One percentage change after another.

REFLECTION AND SELF-ASSESSMENT

Find a discount or sale claim you have seen. Identify the base, the percentage and the actual saving, then explain what the advertisement does not say.

Rate your confidence on each success criterion, name one exercise to revisit, and explain in two sentences how you check that a percentage has been applied to the correct base.

END-OF-CHAPTER PRACTICE

Mixed practice 5

Do not use a calculator. Write the base and the multiplier for every change.

FLUENCY Build accuracy

1. Write each percentage as a fraction in simplest form and as a decimal: (a) 35%, (b) 8%, (c) 150%, (d) 0.5%.
2. Write each fraction as a percentage: (a) $\frac{3}{5}$, (b) $\frac{7}{25}$, (c) $\frac{11}{10}$, (d) $\frac{5}{8}$.
3. Write each decimal as a percentage: (a) 0.42, (b) 0.08, (c) 1.6, (d) 0.125.
4. Find 12% of 250.
5. Find 45% of 80.
6. Find 7.5% of 400.
7. Find 130% of 60.
8. Increase 150 by 12%.
9. Decrease 84 by 25%.
10. After a 20% increase, a price is £360. Find the original price.
11. A £140 jacket is reduced by 15%. Find the sale price.
12. A £85 service has 20% VAT added. Find the total.
13. A budget of £2,000 has 65% spent. Find the amount spent and the amount left.

REASONING AND PROBLEM SOLVING Connect ideas

14. "15% of 80 equals 80% of 15." Show that both are 12, and explain why the two calculations must agree.
15. Which is larger: 18% of 250 or 25% of 180? Do not calculate both fully until you have made an estimate; then check.
16. A price is reduced by 10% and then increased by 10%. Is the final price the same as the original? Show the multipliers and explain.
17. Shop A sells a £80 item with 20% off. Shop B sells the same item for £68 with 10% off. Which is cheaper, and by how much?
18. A household earns £2,500 a month and spends £800 on rent. (a) What percentage of income is rent? (b) Rent rises to £900. Find the new percentage and the percentage increase in rent.
19. **The stall's second market (Criteria A, C, D).** A stall takes £1,200. The stall fee is 8% of takings, transport is 4% and materials are 55%. (a) Find each cost and the profit. (b) The organisers offer a 10% discount on all prices and predict that sales will rise by 25%, so takings become £1,200 × 0.9 × 1.25. The fee and transport are still 8% and 4% of the new takings, but materials rise by 25% because more items are made. Find the new costs and profit. (c) Was the discount a good decision? Explain using both profits.
20. **The jacket rail (Criteria C, D).** A jacket is marked at £180. It is reduced by 25%, and then an extra 10% is taken off at the till. (a) Find the final price. (b) Find the single percentage discount equivalent to the two reductions. (c) A customer says the total discount is 35%; explain why this is wrong and by how much it overstates the saving.

Check your progress

ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

- **Per cent means per hundred.** Dividing by 100 moves the decimal point two places; 7.5% is 0.075, not 0.75.
- **A percentage needs a base.** 20% of £80 is £16; the percentage alone is not an amount.
- **An increase of 15% is a multiplier of 1.15.** It is not £15 added to the price.
- **To find the original, divide by the multiplier.** A price of £68 after 15% off started at £68 ÷ 0.85 = £80.
- **Successive percentages multiply, they do not add.** 20% off then 10% off is 28% off overall, not 30%.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can convert between percentages, fractions and decimals.			
I can find a percentage of a quantity.			
I can increase or decrease a quantity using a multiplier.			
I can calculate discounts, sale prices and tax.			
I can build and interpret a budget.			
I can compare offers and explain why successive percentages do not add.			

WHERE THIS LEADS

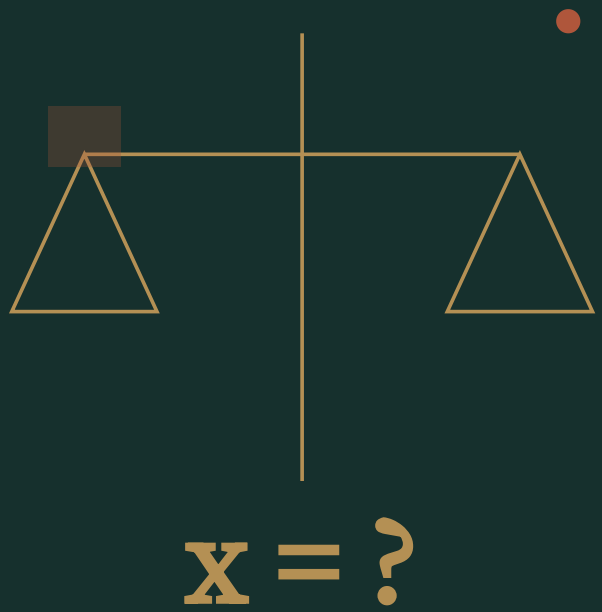
Chapter 6 replaces particular numbers with letters. A rule such as “multiply by 1.15” is a pattern that can be written once and applied to any starting amount; an expression such as $1.15x$ describes every 15% increase at the same time. The percentage multipliers in this chapter become the constants in the formulas and equations of algebra.

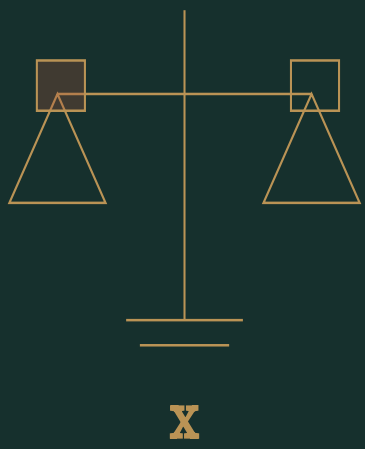
PART III · PATTERNS AND SYMBOLS

III

Algebra as compressed reasoning

Algebra begins with a simple but powerful move: replace a changing number with a letter so that one statement can describe many cases. This part develops that language from visual patterns, through expressions and formulas, to equations that can be solved and checked.





6

RULES THAT REPEAT

Patterns, expressions and equations

Algebra begins by noticing what stays the same while something else changes. A pattern can be shown in shapes, written in a table, described in words and compressed into a rule — and an equation lets us work backwards from a result to the unknown that produced it.

KEY CONCEPT · RELATIONSHIPS

RELATED · PATTERN

RELATED · EQUIVALENCE

CONTEXT · SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

A tiled border grows by the same number of tiles at every step. How can you predict the hundredth stage without drawing it, and how can you find which stage uses exactly 100 tiles?

MYP UNIT PLANNER

Framing the inquiry

UNIT 6 · APPROX. 16 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Relationships. A pattern relates a position to a value. An equation states that two expressions name the same number, so a solution can be checked by substitution.

Pattern. Regular change can be described, continued and predicted. The step and the starting value together determine the rule.

Equivalence. Expressions may look different and still be equal for every value of the variable. Collecting like terms and expanding rewrite an expression without changing its value.

GLOBAL CONTEXT AND EXPLORATION

Scientific and technical innovation. Exploration: how systems and patterns work — using rules to model growth, cost and design in construction.

A square pond of side n metres needs a one-slab border counted by $4n + 4$. The rule lets a builder order slabs without drawing the plan; an equation reverses the count.

STATEMENT OF INQUIRY

Patterns follow algebraic rules, and equivalent expressions and equations let us predict later stages and recover unknown values.

FACTUAL

What is a variable, a term and a coefficient? How do you substitute a value? What separates an expression from an equation?

CONCEPTUAL

How can different-looking expressions always be equal, and what does it mean to solve an equation?

DEBATABLE

Is a rule guessed from several terms trustworthy? When is a formula more useful than a table of values?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Continue and generalise sequences, write and simplify expressions, substitute values, solve one- and two-step equations.
B · Investigating patterns	i, ii	Investigate a matchstick chain, record counts in a table, justify the general rule and test it against a claim.
C · Communicating	i, ii, iii	Move between diagram, table, words and symbols; explain each step in a solution; justify that two expressions are equivalent.
D · Applying mathematics	i, ii, iii, iv	Model a courtyard border and a service charge, solve design equations, evaluate a budget and recommend a size.

ATL · CRITICAL THINKING IN ACTION

Two students describe the sequence 4, 7, 10, 13. One writes “add 3 each time”; the other writes $3n + 1$. Explain what each description can and cannot predict, then use both to find the 40th term.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** how a sequence is built from a position and a rule, and distinguish term-to-term from position-to-term descriptions.
- **Be able to** write, interpret and substitute into algebraic expressions.
- **Be able to** collect like terms and expand a single bracket.
- **Be able to** form and solve one- and two-step equations, and check the solution.
- **Know** how a formula links quantities and how to use one in a context.
- **Investigate** a growing matchstick pattern and justify a general rule.

SUCCESS CRITERIA

- ☐ I can continue a sequence and explain the rule in words.
- ☐ I can find a position-to-term rule and use it for a distant term.
- ☐ I can write and evaluate expressions, including with negative values.
- ☐ I can simplify expressions by collecting like terms and expanding brackets.
- ☐ I can solve linear equations and check my answer by substitution.
- ☐ I can form an equation from a context and interpret the solution.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate 6×7.
2. Calculate $20 - 3 \times 4$.
3. Write $5 + 5 + 5 + 5$ as a multiplication.
4. Find the next two terms of 2, 5, 8, 11, …
5. Calculate $3 \times 4 + 2$.
6. Solve $6 + \square = 15$.

Answers are in the Chapter 6 answer section. If more than two are incorrect, review multiplication facts, order of operations and simple missing-number sentences before continuing.

BEFORE YEAR 1
Number patterns, missing-number sentences, multiplication facts and the order of operations.

YEAR 1 · THIS CHAPTER
Sequences and rules, expressions, substitution, simplifying, expanding, equations and formulas.

LATER
Linear graphs, simultaneous equations, inequalities, quadratic expressions and algebraic proof.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Borders, chains and service charges grow by a regular rule.	How can one rule describe every stage, and how can we reverse it?	Relationships, pattern and equivalence.	How many matches make a chain of n squares?	Diagrams, tables, words, expressions and balanced equations.	When does a pattern count as explained rather than merely continued?

6.1 Patterns and rules

CRITERION A · I ◆ PATTERN ◆ RELATIONSHIPS

A sequence is a list of numbers in order. Each number is a term, and its place in the list is its position. Finding the relationship between position and term turns a list into a rule that can predict values you have not written down.

In the sequence 5, 8, 11, 14, ..., each term is 3 more than the one before. The **term-to-term rule** is "add 3". It tells you how to continue the list, but not how to jump straight to the 100th term without adding 3 ninety-nine times. A **position-to-term rule** does that job: it connects the position n directly to the term.

POSITION AND TERM FOR THE SEQUENCE 5, 8, 11, 14

Position n	1	2	3	4	5
Term	5	8	11	14	17
$3n$	3	6	9	12	15

Compare the term with $3n$. Every term is 2 more than $3n$, so the rule is $\text{term} = 3n + 2$. The multiplier 3 is the constant difference between consecutive terms; the +2 fixes the starting value. Check the rule at $n = 1$: $3(1) + 2 = 5$, the first term. At $n = 5$: $3(5) + 2 = 17$, as the table shows.



Stage 1
3 tiles

Stage 2
5 tiles

Stage 3
7 tiles

Figure 6.1 A row grows by two tiles each stage. Stage n has $2n + 1$ tiles: the first stage has 3, and each later stage adds 2.

DEFINITIONS

A **sequence** is an ordered list of numbers. Each number is a **term**, and its index is its **position**. A **term-to-term rule** describes how to get from one term to the next. A **position-to-term rule** calculates any term directly from its position n.

WORKED EXAMPLE 6.1.1 — FROM A LIST TO A RULE

PROBLEM

The sequence 4, 9, 14, 19, ... continues with the same step. (a) Write the term-to-term rule and the next two terms. (b) Find a position-to-term rule. (c) Use it to find the 12th term.

1. (a) $9 - 4 = 5$, $14 - 9 = 5$, so the term-to-term rule is "add 5". The next terms are $19 + 5 = 24$ and $24 + 5 = 29$.
2. (b) The difference is 5, so compare the terms with $5n$: $5, 10, 15, 20$. Each term is 1 less than $5n$, so the rule is $\text{term} = 5n - 1$.
3. (c) At $n = 12$, $5(12) - 1 = 60 - 1 = 59$.

ANSWER **(a) add 5; 24, 29 (b)** $5n - 1$ **(c) 59.**

Check. The rule gives $5(1) - 1 = 4$ and $5(4) - 1 = 19$, matching the first and fourth terms. A rule must reproduce every term it claims to describe.

WORKED EXAMPLE 6.1.2 — A GROWING TILE PATTERN

PROBLEM

Stage 1 of a pattern uses 3 tiles, stage 2 uses 5 tiles and stage 3 uses 7 tiles. (a) Write a position-to-term rule for the number of tiles. (b) How many tiles are needed for stage 10? (c) Which stage uses 41 tiles?

1. (a) Each stage adds 2 tiles, so the rule has the form $2n + c$. Stage 1 has 3 tiles, and $2(1) + c = 3$, so $c = 1$. The rule is $2n + 1$.
2. (b) At $n = 10$, $2(10) + 1 = 21$ tiles.
3. (c) Solve $2n + 1 = 41$. Subtract 1: $2n = 40$. Divide by 2: $n = 20$. Stage 20 uses 41 tiles.

ANSWER **(a)** $2n + 1$ **(b) 21 tiles (c) stage 20.**

Check. Stage 3 should use $2(3) + 1 = 7$ tiles, which matches Figure 6.1. Solving the rule backwards is how a pattern predicts an earlier or unknown stage.

COMMON ERROR

Using the difference as the whole rule. For 5, 8, 11, ... the rule is $3n + 2$, not $3n$: $3n$ alone gives 3, 6, 9, which is two less than each term. A second error is using $n = 0$ as the first position. The first term has position 1, so substitution must start there. A third is assuming every sequence is linear: 1, 4, 9, 16 has differences 3, 5, 7, not a constant step.

GUIDED PRACTICE · COPY AND COMPLETE

For the sequence 5, 9, 13, 17, ..., the difference is ____, so the multiplier is ____. Compare with $4n$: 4, 8, 12, 16. Each term is ____ more, so the rule is ____. The 10th term is $4(10) + \square = \square$. To find which position gives 45, solve $4n + 1 = 45$: $4n = \square$, so $n = \square$.

PRACTICE 6.1 · SEQUENCES AND GENERAL RULES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Show the difference or multiplier you used before writing a rule.

1. Write the next three terms and describe the term-to-term rule: (a) 6, 11, 16, 21, ... (b) 48, 42, 36, 30, ... (c) 3, 6, 12, 24, ... (d) 1, 4, 9, 16, ... F

2. Complete a table of positions 1 to 5 for the rule $4n + 2$, then find the 10th term. F
3. A pattern starts with 4 tiles and adds 3 tiles at each stage. Write a rule for stage n, then find the number of tiles in stage 12. F
4. A chain of squares uses 4 matches for 1 square, 7 matches for 2 squares and 10 matches for 3 squares. How many matches are needed for 15 squares? Which stage uses 61 matches? P
5. The sequence 8, 11, 14, 17 has position-to-term rule $3n + 5$. Show that the rule works for the first three terms, then find the 10th term. F
6. A student says the sequence 2, 5, 8, 11 has rule $2n$ "because it starts at 2". Explain both mistakes in this reasoning and give the correct rule. R
7. A taxi charges £3 plus £2 for each kilometre. Write an expression for a journey of d kilometres, then find the fare for 12 km and for 25 km. P
8. Extension: a sequence has rule $5n - 3$. Which position gives the term 72? Is 100 a term of this sequence? Justify your answer. E

6.2 Expressions and substitution

CRITERION A · II ◆ REPRESENTATION ◆ EQUIVALENCE

An expression combines numbers, variables and operations without an equals sign. It records a calculation that has not yet been carried out. Substituting a value for the variable turns the expression into a number.

In $3x + 5$, the letter x is a **variable**: it stands for a number that may change. The 3 is the **coefficient** of x, and the 5 is a **constant**, because it does not change with x. The expression has two **terms**, $3x$ and 5, separated by the plus sign. In $7 - 2y$ the terms are 7 and $-2y$; the coefficient of y is -2, not 2, because the subtraction belongs to the term.

FROM WORDS TO SYMBOLS

Words	Expression	Why
5 more than a number n	$n + 5$	start with n, add 5
3 less than a number n	$n - 3$	start with n, subtract 3
4 times a number n	$4n$	$4 \times n$, written without the sign
A number n divided by 5	$\frac{n}{5}$	the whole of n is divided
2 more than 3 times n	$3n + 2$	multiply first, then add

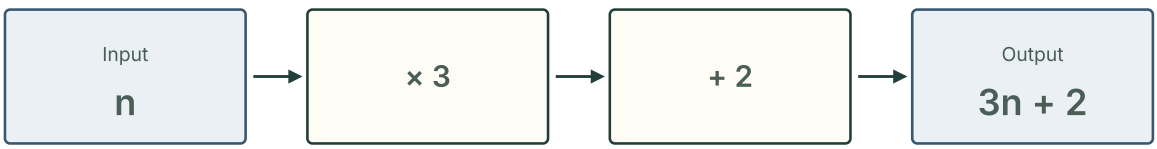


Figure 6.2 A function machine for the expression $3n + 2$. The order of the operations matters: multiply before adding.

To **substitute**, replace the variable with the given number and follow the order of operations. Brackets first, then powers, then multiplication and division, then addition and subtraction. Writing the multiplication sign in temporarily prevents mistakes: $4x$ with $x = 5$ becomes $4 \times 5 = 20$.

WORKED EXAMPLE 6.2.1 — WRITING EXPRESSIONS FROM WORDS

PROBLEM

Write an expression for each statement. (a) 7 more than n. (b) 3 times a number minus 4. (c) Half a number plus 5. (d) The cost in pounds of n pens at 85p each plus £2 delivery.

1. (a) Start with n and add 7: $n + 7$.
2. (b) Multiply the number by 3, then subtract 4: $3n - 4$. The 4 is subtracted from the product, not from n alone.
3. (c) Half of n is $\frac{n}{2}$, then add 5: $\frac{n}{2} + 5$.
4. (d) Each pen costs £0.85, so n pens cost $0.85n$ pounds. Add the delivery: $0.85n + 2$.

ANSWER **(a)** $n + 7$ **(b)** $3n - 4$ **(c)** $\frac{n}{2} + 5$ **(d)** $0.85n + 2$.

Check. Test (b) with $n = 5$: $3(5) - 4 = 11$. "3 times 5 minus 4" is also 11, so the expression matches the words.

WORKED EXAMPLE 6.2.2 — SUBSTITUTING VALUES

PROBLEM

Evaluate each expression. (a) $3x + 4$ when $x = 6$. (b) $5a^2$ when $a = 2$. (c) $4(y - 3)$ when $y = 10$. (d) $10 - 3b$ when $b = 4$.

1. (a) $3(6) + 4 = 18 + 4 = 22$.
2. (b) Square first: $a^2 = 4$, then $5 \times 4 = 20$. It is not $(5 \times 2)^2 = 100$.
3. (c) Brackets first: $10 - 3 = 7$, then $4 \times 7 = 28$.
4. (d) $3(4) = 12$, then $10 - 12 = -2$. The result can be negative.

ANSWER **(a) 22** **(b) 20** **(c) 28** **(d)** -2.

Check. For (c), substitute into the expanded form $4y - 12$: $4(10) - 12 = 28$. The two forms agree, which is evidence that $4(y - 3) = 4y - 12$.

COMMON ERROR

Reading $3x$ as $3 + x$. The coefficient multiplies: if $x = 5$, then $3x = 15$, not 8. A second error is squaring too late: $2a^2$ with $a = 3$ is $2 \times 9 = 18$, not $(2 \times 3)^2 = 36$. A third is ignoring brackets: $4(y - 3)$ with $y = 10$ is $4 \times 7 = 28$, not $4 \times 10 - 3 = 37$.

GUIDED PRACTICE · COPY AND COMPLETE THE TABLE

Expression	$x = 2$	$x = 3$	$x = 5$
$4x$			
$x + 7$			
$3x - 2$			
x^2			

PRACTICE 6.2 · EXPRESSIONS AND SUBSTITUTION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Show the substituted calculation before writing the final value.

1. Write an expression for each: (a) 6 more than x; (b) 8 less than x; (c) 5 times x; (d) one quarter of x; (e) double x plus 3. F
2. For each expression, state the number of terms, the coefficient of the variable and the constant: (a) $5x + 2$; (b) $7 - 3y$; (c) $4a + 2b - 1$. F
3. Substitute $x = 4$: (a) $3x$; (b) $x + 9$; (c) $5x - 2$; (d) x^2. Then substitute $x = 10$: (e) $2(x + 1)$; (f) $\frac{x}{2} + 3$. F
4. Substitute $a = 3$ and $b = 5$: (a) $2a + b$; (b) $3ab$; (c) $b^2 - a^2$; (d) $2(a + b)$. F
5. Substitute $y = -2$: (a) $y + 7$; (b) $4y$; (c) $3 - 2y$; (d) y^2. P
6. A student says that when $x = 5$, $2x + 1 = 8$, because "$2x$ means $2 + x$". Explain the error and give the correct value. R
7. A phone plan costs £8 per month plus £0.10 per minute of calls. Write an expression for the monthly cost of m minutes, then find the cost for 60 minutes and for 200 minutes. P
8. Extension: a rectangle has length $2x + 1$ and width x. Write an expression for its perimeter in terms of x, simplify it, and find the perimeter when $x = 4$. E

6.3 Simplifying and expanding

CRITERION A · II ◆ SIMPLIFICATION ◆ EQUIVALENCE

An expression can often be written in a shorter form without changing its value. Collecting like terms and expanding brackets are two ways of rewriting an expression so that its structure is easier to use.

Like terms contain exactly the same variable combination. $3x$ and $5x$ are like terms, so $3x + 5x = 8x$: three lots of x plus five lots of x is eight lots of x. $4x$ and $3y$ are not like terms, because x and y may stand for different numbers; $4x + 3y$ cannot be shortened. Constants are like terms with each other: $7 + 2 = 9$.

DEFINITIONS

Like terms have the same variable and the same power. To **simplify** an expression, collect like terms by adding or subtracting their coefficients. To **expand** a bracket, multiply every term inside by the factor outside: $a(b + c) = ab + ac$.

WORKED EXAMPLE 6.3.1 COLLECTING LIKE TERMS

PROBLEM

Simplify: (a) $5a + 3a - 2a$; (b) $4x + 2 + 3x - 5$; (c) $6m + 2n - m + 3n$.

1. (a) All three terms are like terms in a: $5 + 3 - 2 = 6$, so the result is $6a$.
2. (b) Collect the x-terms: $4x + 3x = 7x$. Collect the constants: $2 - 5 = -3$. The result is $7x - 3$.
3. (c) Collect m-terms: $6m - m = 5m$. Collect n-terms: $2n + 3n = 5n$. The result is $5m + 5n$; it cannot be simplified further.

ANSWER **(a)** $6a$ **(b)** $7x - 3$ **(c)** $5m + 5n$.

Check. Test (b) with $x = 2$: the original is $8 + 2 + 6 - 5 = 11$, and $7(2) - 3 = 11$. A correct simplification gives the same value for every x.

WORKED EXAMPLE 6.3.2 EXPANDING A BRACKET

PROBLEM

Expand: (a) $4(x + 3)$; (b) $3(2y - 5)$; (c) $2(3n + 1) + 5n$.

1. (a) Multiply both terms by 4: $4 \times x = 4x$ and $4 \times 3 = 12$. So $4(x + 3) = 4x + 12$.
2. (b) Multiply both terms by 3: $3 \times 2y = 6y$ and $3 \times 5 = 15$. The subtraction stays: $6y - 15$.
3. (c) Expand first: $2(3n + 1) = 6n + 2$. Then collect like terms: $6n + 5n = 11n$. The result is $11n + 2$.

ANSWER **(a)** $4x + 12$ **(b)** $6y - 15$ **(c)** $11n + 2$.

Check. For (a), substitute $x = 2$: $4(5) = 20$ and $4(2) + 12 = 20$. For (c), substitute $n = 1$: $2(4) + 5 = 13$ and $11(1) + 2 = 13$.

COMMON ERROR

Multiplying only the first term in a bracket: $3(x + 4) = 3x + 4$ is wrong because the 4 is also multiplied by 3, giving $3x + 12$. Combining unlike terms: $5a + 2b \neq 7ab$, since a and b are different variables. Dropping the variable: $7x - 3x = 4x$, not 4. And $4 + 3x$ stays as it is; 4 is a constant and $3x$ is not.

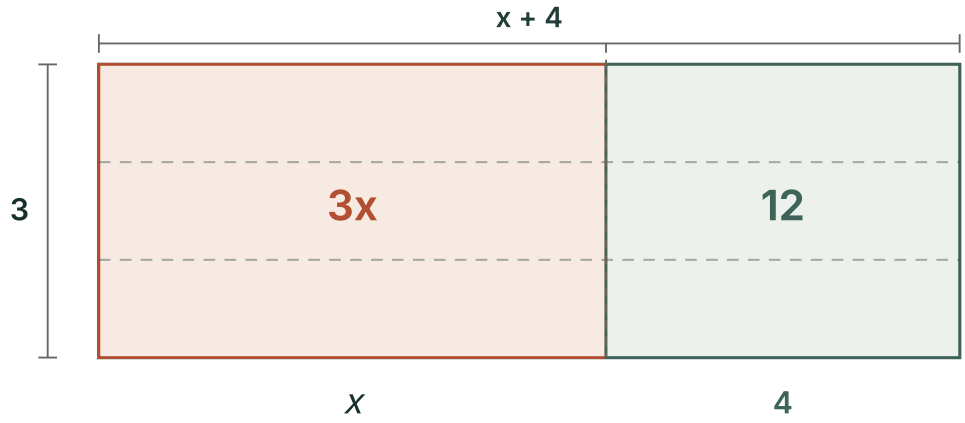


Figure 6.3 The rectangle is 3 rows high and $(x + 4)$ wide. Its area is $3(x + 4)$, and counting the two parts gives $3x + 12$.

GUIDED PRACTICE · COPY AND COMPLETE

$6x + 2x = \square x$. $9y - 4y = \square y$. $3(a + 5) = \square a + \square$. $2(4n - 1) = \square n - \square$. $5p + 2p - 3 = \square p - \square$. $4(x + 2) + 3x = \square x + 8$.

PRACTICE 6.3 · SIMPLIFY AND EXPAND

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Underline like terms before collecting them.

1. Simplify: (a) $4a + 3a$; (b) $9x - 2x$; (c) $6m + m$; (d) $8y - 8y$. F
2. Simplify: (a) $5x + 2 + 3x$; (b) $7a + 4 - 2a$; (c) $3n + 5 - n - 2$; (d) $6p + 2q - p + q$. F
3. Expand: (a) $2(x + 5)$; (b) $5(y - 3)$; (c) $3(2m + 1)$; (d) $4(3 - n)$. F
4. Expand and simplify: (a) $3(x + 2) + 4x$; (b) $2(3n - 1) + 5$; (c) $5(2a + 1) + 2a$. F
5. Which expressions are equivalent to $4(x + 2)$: $4x + 2$, $4x + 8$ or $8 + 4x$? Explain. R
6. A student writes $3(x + 4) = 3x + 4$ and $5a + 2b = 7ab$. Explain both errors and write the correct results. R
7. A rectangle has length $x + 5$ cm and width 3 cm. Write and simplify an expression for its perimeter and for its area. Find both when $x = 4$. P
8. Extension: simplify $2(3x + 1) + 3(x + 2)$, then evaluate the result when $x = 5$. Show that the original expression gives the same value. E

6.4 Equations and formulas

CRITERION A · III ◆ EQUIVALENCE ◆ MODEL

An equation says that two expressions have the same value. Solving an equation means finding the value of the variable that makes the statement true. A formula is an equation that links quantities which have a meaning in a context.

The equation $3x + 5 = 20$ can be read as a balance: the left-hand side and the right-hand side weigh the same. Whatever is done to one side must be done to the other, or the balance is destroyed. To isolate x, undo the operations in reverse order: first subtract 5, then divide by 3.

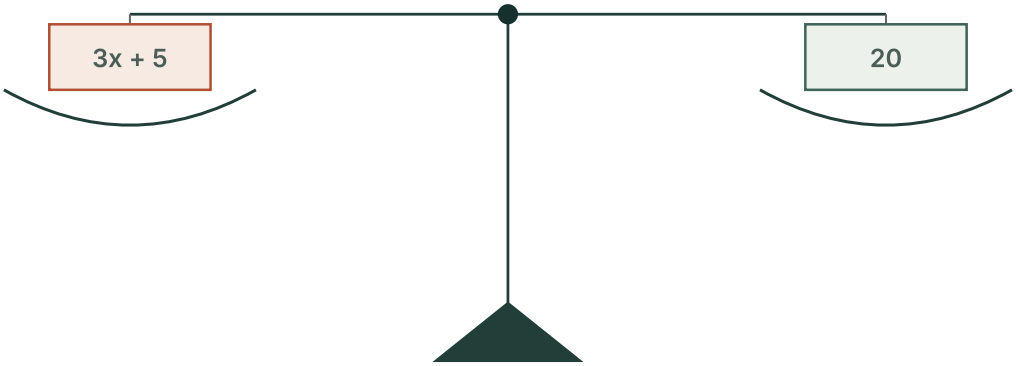


Figure 6.4 A balance model for $3x + 5 = 20$. The solution is the value that keeps the pans level.

DEFINITIONS

An **equation** states that two expressions are equal. A **solution** is a value of the variable that makes the equation true. To **solve** is to find all such values. A **formula** is an equation that relates quantities, such as $P = 2(l + w)$ for the perimeter of a rectangle.

WORKED EXAMPLE 6.4.1 — ONE-STEP EQUATIONS

PROBLEM

Solve and check: (a) $x + 9 = 14$; (b) $5x = 45$; (c) $\frac{n}{4} = 7$.

1. (a) Subtract 9 from both sides: $x + 9 - 9 = 14 - 9$, so $x = 5$.
2. (b) Divide both sides by 5: $\frac{5x}{5} = \frac{45}{5}$, so $x = 9$.
3. (c) Multiply both sides by 4: $4 \times \frac{n}{4} = 4 \times 7$, so $n = 28$.

ANSWER **(a)** $x = 5$ **(b)** $x = 9$ **(c)** $n = 28$.

Check. Substitute back: $5 + 9 = 14$, $5(9) = 45$, and $\frac{28}{4} = 7$. A solution that does not check is not a solution.

WORKED EXAMPLE 6.4.2 — TWO-STEP EQUATIONS

PROBLEM

Solve and check: (a) $4x - 7 = 25$; (b) $3(x + 2) = 21$.

1. (a) Add 7 to both sides: $4x = 32$. Divide by 4: $x = 8$.
2. (b) Divide both sides by 3: $x + 2 = 7$. Subtract 2: $x = 5$. Alternatively, expand first: $3x + 6 = 21$, then $3x = 15$, so $x = 5$.

ANSWER **(a)** $x = 8$ **(b)** $x = 5$.

Check. (a) $4(8) - 7 = 32 - 7 = 25$. (b) $3(5 + 2) = 3 \times 7 = 21$. Both solutions restore the original equality.

WORKED EXAMPLE 6.4.3 FORMING AND USING A FORMULA

PROBLEM

A rectangle has perimeter 34 cm and length 10 cm. The perimeter formula is $P = 2(l + w)$. Find the width.

1. Substitute the known values: $34 = 2(10 + w)$.
2. Divide both sides by 2: $17 = 10 + w$.
3. Subtract 10: $w = 7$. The width is 7 cm.
4. Check: $2(10 + 7) = 2 \times 17 = 34$ cm.

ANSWER **The width is 7 cm.**

Alternative. Expand: $34 = 20 + 2w$, so $2w = 14$ and $w = 7$. Both routes use the same balance idea.

COMMON ERROR

Doing an operation to one side only. In $x + 9 = 14$, subtracting 9 from the left but not the right gives $x = 14$, which fails the check. A second error is stopping too early: $3x = 15$ gives $x = 5$, not $x = 15$. A third is undoing operations in the wrong order; for $4x - 7 = 25$, add 7 before dividing by 4, because the multiplication by 4 happened first to x.

GUIDED PRACTICE · SOLVE AND CHECK

$x + 6 = 13$, so $x = \square$. $4x = 32$, so $x = \square$. $2x - 5 = 11$: add 5 to get $2x = \square$, then $x = \square$. $3(x + 1) = 18$: divide by 3 to get $x + 1 = \square$, then $x = \square$. Check the last answer by substitution: $3(\square + 1) = \square$.

PRACTICE 6.4 · EQUATIONS AND FORMULAS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Show each operation on both sides, then check every solution.

1. Solve: (a) $x + 8 = 15$; (b) $y - 4 = 9$; (c) $6m = 42$; (d) $\frac{n}{5} = 6$. F
2. Solve: (a) $2x + 3 = 13$; (b) $5x - 4 = 26$; (c) $3y + 7 = 28$; (d) $4n - 9 = 15$. F
3. Solve: (a) $2(x + 3) = 16$; (b) $5(x - 2) = 25$; (c) $3(2x + 1) = 27$. F
4. I think of a number, multiply it by 3 and add 7. The result is 25. Form an equation and solve it. P
5. A rectangle has perimeter 40 cm and length 12 cm. Use $P = 2(l + w)$ to find the width. P
6. A student solves $3x + 4 = 19$ by adding 4 to both sides, getting $3x = 23$ and $x = \frac{23}{3}$. Explain the error and solve correctly. R
7. A taxi charges £3 plus £2 per kilometre. The fare is £23. Form and solve an equation to find the distance travelled. P
8. Extension: a rectangle has length $x + 3$ cm and width x cm. Its perimeter is 26 cm. Form an equation, simplify it and find the dimensions. E

Investigation 6 · The matchstick chain

CRITERION B · I, II

Squares are made from matchsticks and joined in a straight chain. Each new square shares one side with the square before it. The first three stages are shown in Figure 6.5.

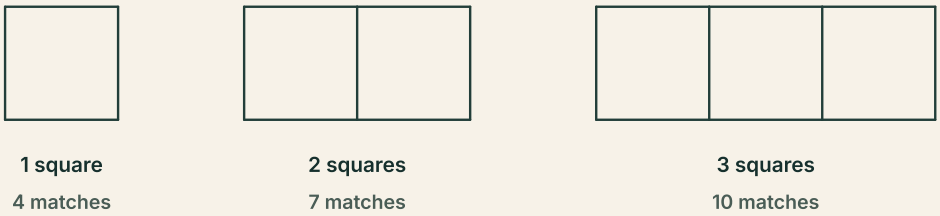


Figure 6.5 A chain of 1, 2 and 3 squares. Count the matches, including the shared sides only once.

MATCHES IN A CHAIN OF SQUARES

Number of squares n	1	2	3	4	5
Number of matches	4	7	10	13	16

1. Draw stage 4 and count its matches. Does your count agree with the table?
2. Find the first differences in the table. What does the constant difference tell you about the rule?
3. Explain, using the picture, why a chain of n squares uses $3n + 1$ matches rather than $4n$.
4. Predict the number of matches for 12 squares and for 50 squares. Show the substitution.
5. A student claims the rule is $4n$, because each square has 4 sides. Test the claim for $n = 2$ and explain why it fails.
6. Exactly 100 matches are available. Form and solve an equation to find how many squares can be made. Is there any match left over?
7. Two separate chains of 5 squares are made. How many matches are needed? Write an expression for two chains of n squares and simplify it.
8. Extension: a chain of equilateral triangles uses 3 matches for 1 triangle, 5 for 2 and 7 for 3. Find a rule, explain it with the shared side, and solve $2n + 1 = 41$.

Record each count in a table before writing a rule. A rule is a generalisation only when it explains why the count must be $3n + 1$ for every n, not merely when it fits the first few stages.

Real-life application · The courtyard pond

A school is building a square pond in the courtyard and surrounding it with one row of square paving slabs. The pond has side length n metres, and every slab is 1 m square. The gardener needs a rule for ordering slabs, and the business manager needs an equation for the budget.

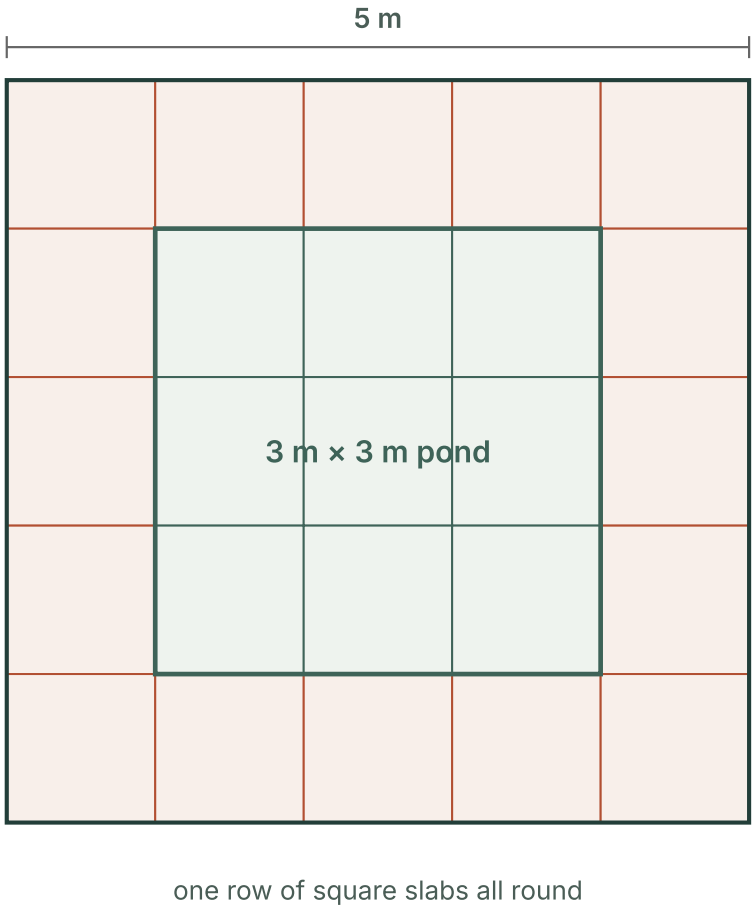


Figure 6.6 A 3 m × 3 m pond needs a 5 m × 5 m outer square. The border contains $25 - 9 = 16$ slabs.

POND SIDE AND BORDER SLABS

Pond side n (m)	1	2	3	4	5
Border slabs	8	12	16	20	24

Your task. Slabs cost £3.50 each, and delivery is a fixed £25 per order.

1. Explain why the outer square has side $n + 2$. Use the table to find a rule for the number of border slabs in terms of n.
2. How many slabs are needed for a pond of side 7 m? For 12 m?
3. A design uses 44 slabs. Form and solve an equation to find the pond side. Check that the pond and border fit together.
4. Write an expression for the total cost of a pond of side n. Expand it and use it to find the cost for a 7 m pond.
5. The budget is £200. Form an inequality or test values to find the largest whole-number pond side that can be paved. Explain why the next size is not possible.
6. Laying a slab takes 12 minutes, and preparation takes 2 hours. Find the total working time for the largest pond in question 5, in hours and minutes.

7. **Consider.** The school could enlarge the pond by 1 m instead of adding benches. Use your costs to explain the financial effect of that choice.

Reflect. Which representation — the diagram, the table or the expression — would you show the gardener first? Which would you show the business manager? Explain what each person needs from it.

Criterion-mapped mini-task

A · Knowing and understanding
C · Communicating
D · Applying mathematics

Context. The mathematics department is preparing a display for the school open evening. Younger pupils and parents will walk past a table of posters. Your poster must show one growing pattern clearly and explain the rule behind it, so that a reader can predict a far-away stage without counting every step.

Product. A single poster with a title, four labelled stages of your pattern, a table, a rule and a short explanation. It must be readable from about two metres away and must be honest about what the rule does and does not prove.

Requirements.

1. Choose a growing pattern that can be built from tiles, matches or counters. Draw the first four stages accurately and label them by position.
2. Record the stage number and the number of elements in a table. State the first differences and describe the term-to-term rule.
3. Write a position-to-term rule. Justify it from the structure of the pattern, not only from the numbers in the table.
4. Use the rule to predict stage 50, showing the substitution.
5. Add an inverse question: “Which stage uses exactly 100 elements?” Form an equation, solve it, and explain what the solution means.
6. Add a cost panel: each element costs £1.50 and there is a fixed £8 display charge. Write a cost expression and evaluate it for stage 20.
7. Include a “check the claim” box with one wrong rule and its correction. Explain the mistake in words a younger pupil would understand.
8. Test the poster on someone who has not studied the chapter. Record one misunderstanding and one improvement you made.

CRITERIA AND STRANDS A i, ii, iii: accurate pattern, table, rule and calculation; C i, ii, iii: clear layout, correct notation and explanation of the structure; D i, ii, iii, iv: apply the rule to a far stage, solve an inverse problem and cost the display.

BEFORE YOU SUBMIT

Check that the rule reproduces all four stages. Check that the inverse solution is a whole, positive stage number. Recalculate the cost of 20 elements, including the fixed charge. Ask a reader to find stage 50 using only your poster; if they cannot, the rule needs a clearer example, not more decoration.

Chapter review

SUMMARY

— **Sequences.** A term-to-term rule continues a list; a position-to-term rule calculates any term directly. A constant difference becomes the multiplier of n.

— **Finding a rule.** Compare the terms with the multiples of the difference, then adjust by a constant. Check the rule at the first and last known positions.

— **Expressions.** A variable changes, a coefficient multiplies it, and a constant does not change. Substitute with the order of operations and reinstate multiplication signs mentally.

— **Simplifying.** Collect only like terms. Expand $a(b + c)$ by multiplying both terms inside the bracket.

— **Equations.** Keep both sides balanced. Undo the operations in reverse order and check by substitution.

— **Formulas.** A formula is an equation for a context. Substitute known values, then solve for the unknown.

Position-to-term:
term $=$ difference $\times\, n +$ adjustment.

Distributive law: $a(b + c) = ab + ac$.

Perimeter of a rectangle: $P = 2(l + w)$.

Balance rule: do the same operation to both sides.

REVISION CHECKLIST

- ☐ I can continue a sequence and describe its term-to-term rule.
- ☐ I can find and use a position-to-term rule.
- ☐ I can write and evaluate expressions, including with negative values.
- ☐ I can collect like terms and expand a bracket.
- ☐ I can solve one- and two-step equations and check the answer.
- ☐ I can form an equation from a context and interpret the solution.

KEY VOCABULARY

sequence
An ordered list of numbers.

term
One number in a sequence.

position
The index of a term.

variable
A letter standing for a number.

coefficient
The number multiplying a variable.

constant
A fixed number in an expression.

expression
Numbers and variables combined without an equals sign.

equation
A statement that two expressions are equal.

solution
A value that makes an equation true.

like terms
Terms with the same variable and power.

expand
Multiply every term inside a bracket.

formula
An equation linking quantities in a context.

REFLECTION AND SELF-ASSESSMENT

Find a rule you use outside mathematics — a price per item, a score, a recipe or a travel time. Write it as an expression and explain what each letter and number means. Then describe one situation in which the rule stops working.

Rate your confidence on each success criterion, name one exercise to revisit, and explain in two sentences how you check whether two expressions are equivalent.

END-OF-CHAPTER PRACTICE

Mixed practice 6

Do not use a calculator. Show the rule, substitution or operation you used at each stage.

FLUENCY

Build accuracy

1. Write the next three terms: (a) 9, 14, 19, … (b) 80, 72, 64, … (c) 5, 10, 20, …
2. Find a position-to-term rule: (a) 6, 10, 14, 18, … (b) 2, 9, 16, 23, … (c) 1, 3, 5, 7, …
3. Find the 20th term of each rule: (a) $6n - 1$; (b) $4n + 3$.
4. Write an expression for: (a) 4 more than $3n$; (b) half of n plus 7; (c) 5 less than $2n$.
5. Substitute $x = 6$: (a) $4x - 5$; (b) $2x^2$; (c) $3(x + 2)$.
6. Simplify: (a) $8a + 3a - 2a$; (b) $5x + 4 - x$; (c) $7m + 2n - 3m + n$.
7. Expand: (a) $3(2x + 5)$; (b) $4(3 - y)$.
8. Expand and simplify: (a) $2(x + 4) + 3x$; (b) $5(2n - 1) + n$.
9. Solve: (a) $x + 12 = 20$; (b) $7x = 56$; (c) $\frac{x}{6} = 5$.
10. Solve: (a) $3x + 8 = 26$; (b) $4x - 5 = 27$; (c) $2(x + 3) = 18$.
11. I multiply a number by 5 and subtract 4. The result is 31. Form and solve an equation.
12. Use $P = 2(l + w)$ and $A = lw$ for a rectangle with $l = 9$ cm and $w = 4$ cm. Find P and A.

REASONING AND PROBLEM SOLVING

Connect ideas

13. A student writes $3(x + 4) = 3x + 4$ and $5x + 2 = 7x$. Explain both errors and give the correct results.
14. Compare $3n + 7$ and $4n - 1$. Which is larger when $n = 10$? Find the value of n for which the two expressions are equal.
15. A pattern uses $4n + 2$ matches at stage n. How many matches are used at stage 20? Which stage uses 102 matches?
16. A rectangle has length $2x + 3$ cm and width x cm. Its perimeter is 30 cm. Form and solve an equation, then state the dimensions.
17. Plan A costs £12 plus 10p per minute. Plan B costs £20 plus 5p per minute. Write an expression for each plan, compare them at 100 and 200 minutes, and find the number of minutes for which the costs are equal.
18. **The tiled stage (Criteria A, C, D).** A stage design uses $3n + 2$ tiles. Tiles cost £2 each and there is a fixed £10 delivery charge. (a) Write and simplify a cost expression. (b) Find the cost of stage 15. (c) A stage costs £68; form and solve an equation to find its number.
19. **The mosaic panel (Criteria A, C, D).** A mosaic grows by 4 tiles each stage, starting with 5 tiles. (a) Write the rule and find the 20th stage. (b) Tiles cost £1.50 each plus £30 for fixing. Write a cost expression. (c) The budget is £150. Find the largest whole stage number that can be afforded and the cost of the next stage. Explain your decision.
20. **Equivalent forms (Criteria C, D).** A trip costs £6 per student, less a £2 voucher for the group. One student writes $6n - 2$; another writes $2(3n - 1)$. Show that the expressions are equivalent using a table for $n = 1, 2, 3, 4$ and by expanding. Which form shows the £6 and the £2 more clearly? Explain.

Check your progress

ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

- **The difference is not the whole rule.** For 5, 8, 11, … the rule is $3n + 2$, not $3n$; always adjust to fit the first term.
- $3x$ **means** $3 \times x$**.** If $x = 5$, then $3x = 15$, not 8.
- **Only like terms combine.** $5a + 2b$ cannot become $7ab$.
- **Expand every term.** $3(x + 4) = 3x + 12$, because both x and 4 are multiplied by 3.
- **Keep the equation balanced.** Do the same operation to both sides, and check the solution by substitution.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can continue a sequence and describe its rule.			
I can find and use a position-to-term rule.			
I can write and evaluate expressions.			
I can collect like terms and expand brackets.			
I can solve equations and check my answer.			
I can form an equation from a context.			

WHERE THIS LEADS

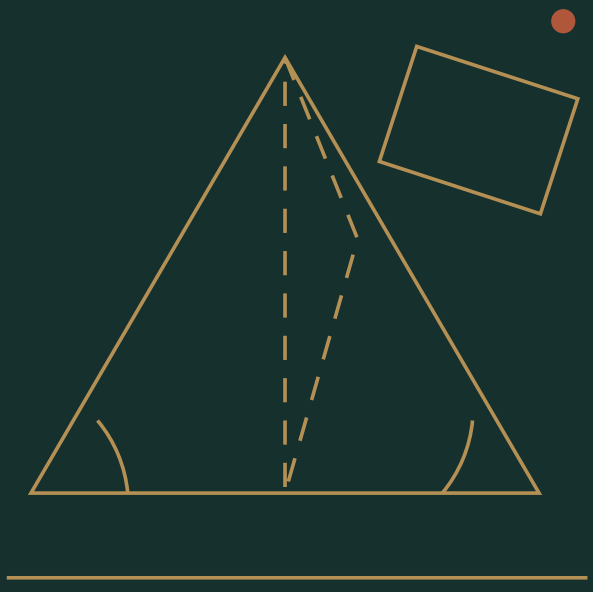
Chapter 7 uses these algebraic skills in geometry. Angle facts produce equations such as $2x + 30 = 180$, and a formula for the angle sum of a polygon grows from the same pattern reasoning as the tile and matchstick sequences in this chapter.

PART IV · SHAPE, STRUCTURE AND MEASURE

IV

Describing space and belonging

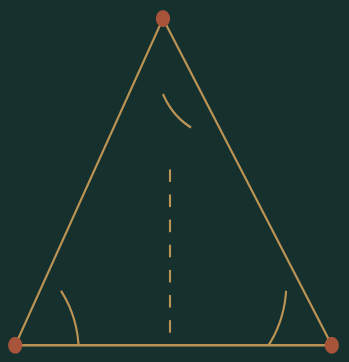
Geometry makes spatial relationships visible; measurement turns them into quantities; sets make classification explicit. These are different forms of structure, but they share a demand for precision. A diagram must label what is known, a measurement must carry its unit, and a set must state what belongs to it.



07 Lines, angles, triangles and constructions

08 Sets and classification

09 Perimeter, area, surface area and volume



7

REASONING WITH STRAIGHT LINES

Lines, angles, triangles and constructions

A single straight line contains 180°, and a full turn contains 360°. From those two facts, and one statement about parallel lines, an entire system of shape reasoning follows. Constructions turn that reasoning into a repeatable physical procedure.

KEY CONCEPT · RELATIONSHIPS | RELATED · SPACE | RELATED · MEASUREMENT | CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

A workshop needs a triangular roof truss with equal slopes and a horizontal collar beam. Which angles must the carpenter set, how can the shape be checked without measuring every angle, and how can a full-size template be reproduced exactly?

MYP UNIT PLANNER

Framing the inquiry

UNIT 7 · APPROX. 16 HOURS
ATL FOCUS · THINKING SKILLS

KEY AND RELATED CONCEPTS

Relationships. Angles are related by position: on a line, at a point, between parallel lines, in triangles and in polygons.

Space. Lines, angles and shapes describe how parts of a structure fit together. A construction locates a point using distance, not guesswork.

Measurement. Degrees measure turn. A construction enforces an exact relationship by repeating a radius; a measuring instrument only approximates.

GLOBAL CONTEXT AND EXPLORATION

Orientation in space and time. Exploration: how spatial relationships and precise construction allow structures to be designed, checked and reproduced.

A roof truss is a triangle of timbers. Equal base angles make the slopes symmetrical; a parallel collar creates equal angles; a perpendicular from the apex locates the centre.

STATEMENT OF INQUIRY

Spatial relationships obey angle and shape rules, and precise constructions let those rules be checked and reproduced in real structures.

FACTUAL

What is an angle, and how is it measured? Which angle facts hold on a line, at a point and between parallel lines?

CONCEPTUAL

Why is the angle sum of a triangle fixed even when the triangle changes? How does a construction guarantee a result that measurement can only approximate?

DEBATABLE

When is an exact construction more trustworthy than a careful measurement? Should a design follow the standard construction or adapt to the materials available?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Apply angle facts on lines, at points, between parallel lines, in triangles and in polygons; classify triangles; describe constructions accurately.
B · Investigating patterns	i, ii	Test sets of three lengths, record whether a triangle is possible, conjecture and justify the triangle inequality.
C · Communicating	i, ii, iii	Name angles and lines correctly, label diagrams, and write reasoned angle chains and construction steps.
D · Applying mathematics	i, ii, iii, iv	Design and check a roof truss, use scale and construction constraints, and evaluate a template.



ATL · THINKING SKILLS IN ACTION

One student measures the three angles of a triangle and obtains 179°. Another obtains 181°. Neither total is exactly 180°. Explain why the geometric result can still be exact while the measurements vary, and state one procedure that would reduce the variation.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** how angles are named and measured, and why angle facts on a line and at a point must be true.
- **Be able to** calculate angles between parallel lines, in triangles and in polygons, giving a reason at each step.
- **Be able to** classify triangles by sides and angles and use the exterior angle relationship.
- **Be able to** construct a perpendicular bisector, an angle bisector and a triangle from three side lengths.
- **Know** the triangle inequality and use it to decide whether a construction is possible.
- **Investigate** which triples of lengths form a triangle and justify the general condition.

SUCCESS CRITERIA

- [ ] I can name and classify an angle using correct notation.
- [ ] I can find missing angles on a straight line, at a point and between parallel lines, with reasons.
- [ ] I can use the triangle and polygon angle sums to solve problems.
- [ ] I can explain why a construction produces the required point or line.
- [ ] I can decide whether three lengths can form a triangle.
- [ ] I can produce a clear scale drawing and evaluate its accuracy.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Solve $x + 40 = 180$.
2. Calculate $180 - 65$.
3. Solve $3x = 180$.
4. Name a triangle with two equal sides.
5. How many degrees are in a quarter turn?
6. Calculate $2 \times (180 - 70)$.

Answers are in the Chapter 7 answer section. If more than two are incorrect, review angle measurement, solving one-step equations and the names of common triangles before continuing.

BEFORE YEAR 1
Recognising and measuring angles with a protractor; naming triangles and quadrilaterals; simple angle sums.

YEAR 1 · THIS CHAPTER
Angle facts with reasons; parallel lines; triangle and polygon sums; ruler-and-compass constructions; triangle inequality.

LATER
Congruence and similarity, Pythagoras' theorem, circle theorems, trigonometric ratios and geometric proof.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Roofs, ramps and frames are built from straight lines and triangles.	Which angles are forced once one angle is known, and when can a shape be built at all?	Relationships, space and measurement.	Which triples of lengths can make a triangle?	Labelled diagrams, angle chains with reasons, and construction instructions.	When does a drawing prove a relationship, and when does it only suggest one?

7.1 Lines, angles and angle facts

CRITERION A · I ◆ SPACE ◆ RELATIONSHIPS

An angle measures the turn between two rays that meet at a point. Its size is a relationship, not a property of either ray alone: the same two lines can form four different angles. Naming the position of an angle is therefore part of the mathematics.

A **point** marks a position. A **line** extends without end in two directions, while a **line segment** has two endpoints and a **ray** has one endpoint and continues in one direction. When two rays meet at a common endpoint, called the **vertex**, they form an angle. The angle is measured in degrees, where one full turn is $360°$.

Angles are named by three points: the middle letter is always the vertex. In Figure 7.1, the angle at O between OA and OB is written $\angle AOB$ or $\angle BOA$, but not $\angle OAB$, because that would name the angle at A. A single letter may be used when no other angle shares the vertex.

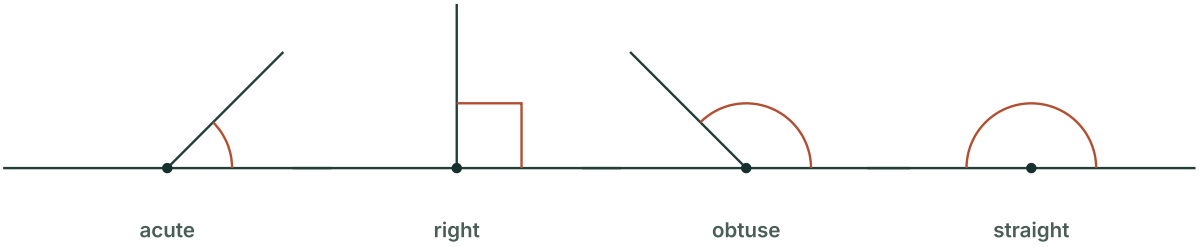


Figure 7.1 An acute angle is less than $90°$, a right angle is exactly $90°$, an obtuse angle is between $90°$ and $180°$, and a straight angle is $180°$. Angles between $180°$ and $360°$ are reflex.

DEFINITIONS

An **acute angle** is less than $90°$; a **right angle** is $90°$; an **obtuse angle** is between $90°$ and $180°$; a **straight angle** is $180°$; a **reflex angle** is between $180°$ and $360°$. Two angles are **complementary** if they sum to $90°$ and **supplementary** if they sum to $180°$. **Adjacent angles** share a vertex and a side but do not overlap. **Vertically opposite angles** are formed by two intersecting lines and lie opposite each other.

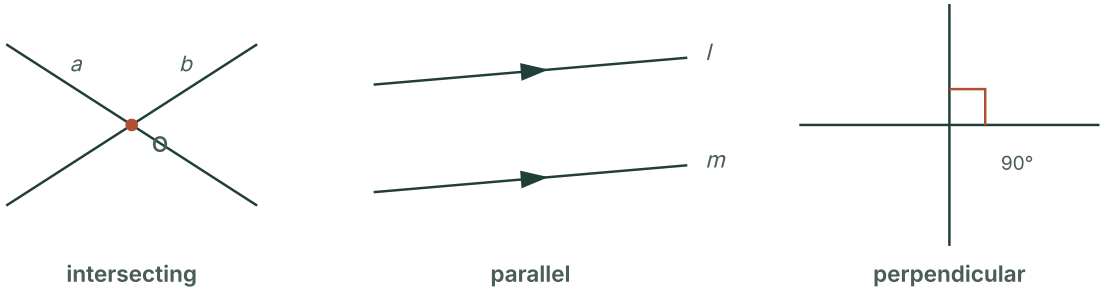


Figure 7.2 Intersecting lines cross at one point, parallel lines remain the same distance apart, and perpendicular lines meet at $90°$.

Three angle facts follow from the turn model. A half-turn along a straight line is $180°$, so adjacent angles on a straight line are supplementary. A full turn about a point is $360°$, so angles at a point sum to $360°$. When two straight lines cross, the two pairs of vertically opposite angles are equal, because each angle in a pair is supplementary to the same adjacent angle.

KEY IDEA

Angles on a straight line sum to $180°$. Angles at a point sum to $360°$. Vertically opposite angles are equal.

WORKED EXAMPLE 7.1.1 USING THE LINE AND POINT FACTS

PROBLEM

(a) Two adjacent angles on a straight line are $3x + 20°$ and $2x + 30°$. Find x and both angles. (b) Four angles at a point are $2x$, $3x$, $4x$ and $90°$. Find x and all four angles.

1. (a) The angles are supplementary: $(3x + 20) + (2x + 30) = 180$. Simplify: $5x + 50 = 180$.
2. Subtract 50: $5x = 130$. Divide by 5: $x = 26$.
3. The angles are $3(26) + 20 = 78 + 20 = 98°$ and $2(26) + 30 = 52 + 30 = 82°$. Check: $98 + 82 = 180$.
4. (b) The angles complete a full turn: $2x + 3x + 4x + 90 = 360$. Simplify: $9x + 90 = 360$, so $9x = 270$ and $x = 30$.
5. The angles are $60°$, $90°$, $120°$ and $90°$. Check: $60 + 90 + 120 + 90 = 360$.

ANSWER **(a)** $x = 26$; $98°$, $82°$ **(b)** $x = 30$; $60°$, $90°$, $120°$, $90°$.

Check. Substitute x back into each expression, not only into the simplified equation. A correct value of x must make every original angle sensible: none may be negative or greater than the total turn.

WORKED EXAMPLE 7.1.2 VERTICALLY OPPOSITE ANGLES

PROBLEM

Two straight lines cross. One angle is $5x - 10°$ and the vertically opposite angle is $3x + 30°$. Find x and the size of each angle. What is the size of the adjacent angle?

1. Vertically opposite angles are equal: $5x - 10 = 3x + 30$.
2. Subtract $3x$ from both sides: $2x - 10 = 30$. Add 10: $2x = 40$. Divide by 2: $x = 20$.
3. Each vertically opposite angle is $5(20) - 10 = 100 - 10 = 90°$. Check with the other expression: $3(20) + 30 = 90°$.
4. The adjacent angle lies on a straight line with $90°$, so it is $180 - 90 = 90°$. All four angles are right angles.

ANSWER $x = 20$; **each angle is** $90°$; **the adjacent angle is also** $90°$.

Reason. The two expressions must be equal because vertically opposite angles are equal. The adjacent angle is supplementary to either one, because it forms a straight line with it.

COMMON ERROR

Assuming that angles on a line add to $360°$ or that angles at a point add to $180°$. A straight line is half a turn, so its adjacent angles sum to $180°$; a complete turn about a point is $360°$. A second error is naming $\angle ABC$ when the vertex is B but reading the angle at A. The middle letter always names the vertex. A third is treating vertically opposite angles as supplementary: they are equal, and it is the adjacent pair that sums to $180°$.

GUIDED PRACTICE · COPY AND COMPLETE

Angles on a straight line: $118° + \square = 180°$, so the missing angle is $\square°$, because angles on a straight line sum to $\square°$. Angles at a point: $95° + 120° + 85° + \square = 360°$, so the missing angle is $\square°$. Vertically opposite: if one angle is $47°$, the angle opposite is $\square°$, and each adjacent angle is $180° - \square° = \square°$.

PRACTICE 7.1 · LINES AND ANGLE FACTS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

State the angle fact you used before giving each answer.

1. Classify each angle as acute, right, obtuse, straight or reflex: (a) $38°$; (b) $90°$; (c) $142°$; (d) $180°$; (e) $205°$; (f) $318°$. F
2. Two adjacent angles on a straight line are $47°$ and $68°$. Find the third angle. F
3. Four angles meet at a point: $85°$, $95°$, $120°$ and x. Find x. F
4. Two straight lines cross. One angle is $132°$. Write the sizes of the other three angles and name the relationship used for each. F
5. Two vertically opposite angles are $4x + 15°$ and $2x + 45°$. Find x and the size of each angle. P
6. Three adjacent angles on a straight line are $x + 30°$, $2x°$ and $x + 10°$. Find x and all three angles. P
7. A student says that two angles on a straight line must be equal because "a straight line is symmetrical". Explain why this is false, and give an example of a supplementary pair that is not equal. R
8. Extension: four angles at a point are in the ratio $1 : 2 : 3 : 4$. Find all four angles, and state whether any pair is complementary. E

7.2 Parallel lines and triangles

CRITERION A · I ◆ RELATIONSHIPS ◆ SPACE

A transversal is a line that crosses two or more other lines. When the crossed lines are parallel, the eight angles formed fall into equal or supplementary families. That single setup also explains why the angles of every triangle sum to $180°$.

In Figure 7.3, the line l is parallel to m, and a transversal cuts both. The eight angles are numbered. **Corresponding angles** sit in the same position at each intersection: 1 and 5, 2 and 6, 3 and 7, 4 and 8. They are equal. **Alternate angles** lie on opposite sides of the transversal between the parallel lines: 3 and 6, 4 and 5. They are equal. **Co-interior angles** lie on the same side of the transversal between the parallel lines: 3 and 5, 4 and 6. They are supplementary.

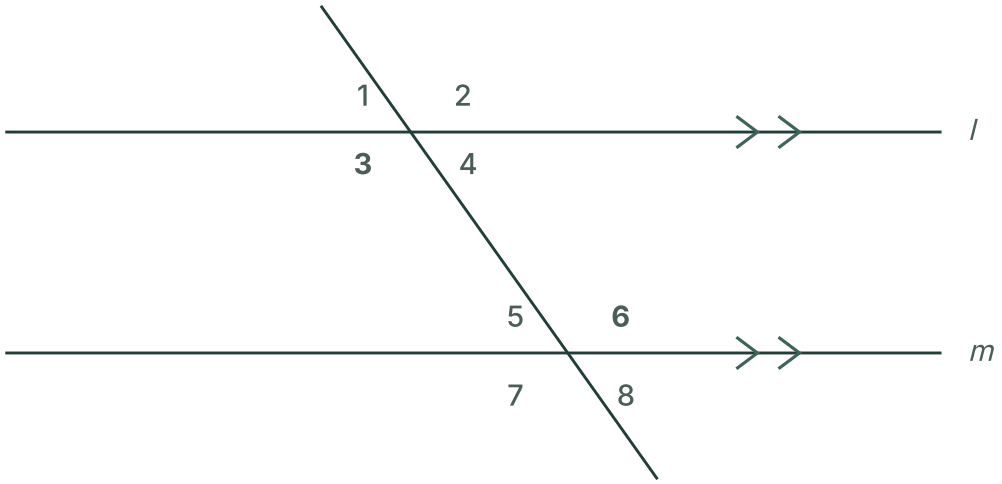


Figure 7.3 Two parallel lines cut by a transversal. Equal corresponding pairs are (1, 5), (2, 6), (3, 7) and (4, 8); equal alternate pairs are (3, 6) and (4, 5); co-interior pairs such as (3, 5) sum to $180°$.

KEY IDEA

When two parallel lines are cut by a transversal, corresponding angles are equal, alternate angles are equal, and co-interior angles are supplementary.

Now draw a triangle ABC and extend the base AB through B. The **interior angles** are at A, B and C. The angle between the extension of AB and the side BC is an **exterior angle**. Because the exterior angle and the interior angle at B lie on a straight line, they are supplementary. The two interior angles at A and C, which are opposite the exterior angle, add to the same amount: this is the **exterior angle theorem**.

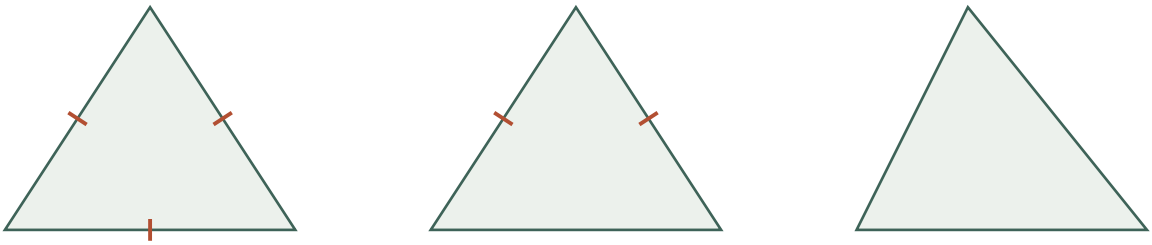
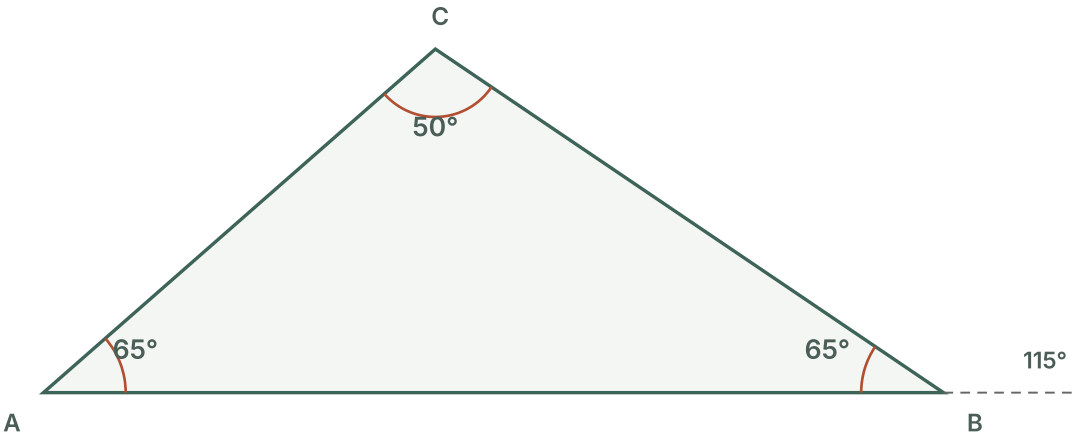


Figure 7.4 Triangles are classified by sides as equilateral, isosceles or scalene, and by angles as acute, right or obtuse. Tick marks show equal sides; the equilateral triangle also has three equal angles of $60°$.

Every triangle has an angle sum of $180°$. This is not a coincidence of drawing: it follows from the parallel-line facts. In an equilateral triangle the three equal angles must each be $60°$. In an isosceles triangle the two angles opposite the equal sides are equal; these are called the base angles. In a right triangle the two acute angles are complementary.



The exterior angle at B equals the sum of the two opposite interior angles: 65° + 50° = 115°.

Figure 7.5 In this isosceles triangle the equal base angles are $65°$, so the apex angle is $180 - 65 - 65 = 50°$. The exterior angle is $180 - 65 = 115°$, which also equals $65° + 50°$.

WORKED EXAMPLE 7.2.1 — ANGLES BETWEEN PARALLEL LINES

PROBLEM

In a diagram like Figure 7.3, angle 3 is $72°$. Find angles 6, 5 and 1, giving a reason for each.

1. Angle 6 is alternate to angle 3, so angle $6 = 72°$.
2. Angle 5 and angle 3 are co-interior, so angle $5 = 180 - 72 = 108°$.
3. Angle 1 and angle 5 are corresponding, so angle $1 = 108°$.
4. Check on the upper line: angle 1 and angle 3 are adjacent on a straight line, and $108 + 72 = 180$.

ANSWER $\angle 6 = 72°$ **(alternate),** $\angle 5 = 108°$ **(co-interior),** $\angle 1 = 108°$ **(corresponding).**

Alternative. Find angle 2 as vertically opposite to angle 3, so $\angle 2 = 72°$; then angle 6 is corresponding to angle 2, giving $72°$. Different routes must agree.

WORKED EXAMPLE 7.2.2 — TRIANGLE SUMS AND AN EXTERIOR ANGLE

PROBLEM

(a) The angles of a triangle are $2x$, $3x$ and $4x$. Find each angle. (b) In another triangle an exterior angle is $125°$ and one opposite interior angle is $68°$. Find the other opposite interior angle.

1. (a) The interior angles sum to $180°$: $2x + 3x + 4x = 180$.
2. $9x = 180$, so $x = 20$. The angles are $40°$, $60°$ and $80°$. Check: $40 + 60 + 80 = 180$.
3. (b) The exterior angle equals the sum of the two opposite interior angles: $125 = 68 + y$.
4. Subtract 68: $y = 57°$. Check: $68 + 57 = 125$, and the interior angle at that vertex is $180 - 125 = 55°$.

ANSWER **(a)** $40°$, $60°$, $80°$ **(b)** $57°$.

Check (b) another way. The third interior angle is $180 - 125 = 55°$. The triangle sum then gives $68 + 57 + 55 = 180$. Both methods use the same straight-line fact.

COMMON ERROR

Using the exterior angle as if it were supplementary to the sum of the opposite angles. The exterior angle is equal to the sum of the two opposite interior angles, not $180°$ minus that sum. A second error is assuming any two angles between parallel lines are equal; only corresponding and alternate pairs are equal, while co-interior pairs are supplementary.

GUIDED PRACTICE · COPY AND COMPLETE

A triangle has angles $48°$, $67°$ and x. Then $x = 180 - \square - \square = \square°$. An isosceles triangle has an apex angle of $40°$, so each base angle is $(180 - \square) \div 2 = \square°$. Between parallel lines, an angle alternate to $113°$ is $\square°$, and its co-interior partner is $180 - \square = \square°$.

PRACTICE 7.2 · PARALLEL LINES AND TRIANGLES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Give a reason for every angle you calculate.

1. In a parallel-line diagram one angle is $115°$. Find its corresponding angle, its alternate angle and its co-interior partner. F
2. Two angles of a triangle are $48°$ and $67°$. Find the third angle. F
3. An isosceles triangle has an apex angle of $40°$. Find each base angle. F
4. A right triangle has one acute angle of $32°$. Find the other acute angle. F
5. The angles of a triangle are $3x$, $4x$ and $5x$. Find x and classify the triangle by its angles. P
6. An exterior angle of a triangle is $118°$. One opposite interior angle is $54°$. Find the other opposite interior angle and the interior angle at the exterior vertex. P
7. A student claims that a triangle can have two obtuse angles, "because two angles can each be more than $90°$". Explain why this is impossible. R
8. Extension: in triangle ABC, angle B is twice angle A, and angle C is $20°$ more than angle A. Find all three angles. E

7.3 Polygons and their angles

CRITERION A · II ◆ PATTERN ◆ SPACE

A polygon is a closed plane shape made from straight sides. Its interior angles obey a rule that depends only on the number of sides, not on the particular shape. The rule can be discovered by cutting the polygon into triangles.

A **polygon** has straight sides that meet only at their endpoints. It is **regular** if every side is equal and every interior angle is equal. A quadrilateral has 4 sides, a pentagon 5, a hexagon 6, an octagon 8 and a decagon 10. The names matter because the angle rule is stated in terms of the number of sides n.

Choose one vertex of a polygon and draw diagonals to every non-adjacent vertex. The polygon is divided into triangles. A quadrilateral divides into 2 triangles, a pentagon into 3, a hexagon into 4. In general, an n-sided polygon divides into $n - 2$ triangles. Each triangle contributes $180°$, so the interior angle sum is $(n - 2) \times 180°$.

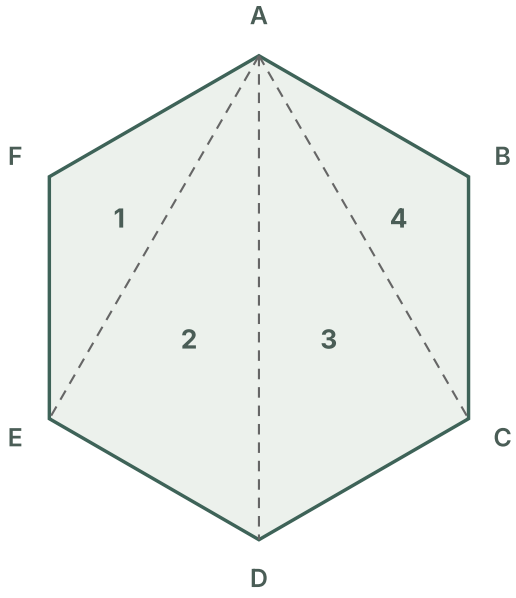


Figure 7.6 A hexagon divided from one vertex into $6 - 2 = 4$ triangles. The interior angle sum is $4 \times 180^\circ = 720^\circ$.

KEY IDEA

For a polygon with n sides, the interior angle sum is $(n - 2) \times 180^\circ$. The exterior angles of any convex polygon sum to 360°. In a regular polygon, each interior angle is $\frac{(n-2)\times 180^\circ}{n}$ and each exterior angle is $\frac{360^\circ}{n}$.

The exterior-angle result follows by imagining a walk around the polygon: at each vertex you turn through the exterior angle, and after one complete circuit you have turned through one full turn, 360°. This is also why an interior angle and its adjacent exterior angle are supplementary: together they form a straight line.

WORKED EXAMPLE 7.3.1 — A MISSING QUADRILATERAL ANGLE

PROBLEM

A quadrilateral has interior angles 85°, 95°, 110° and x. Find x.

1. A quadrilateral has $4 - 2 = 2$ triangles, so its interior angle sum is $2 \times 180 = 360^\circ$.
2. Add the known angles: $85 + 95 + 110 = 290^\circ$.
3. Subtract from 360: $x = 360 - 290 = 70^\circ$.
4. Check: $85 + 95 + 110 + 70 = 360$.

ANSWER $x = 70^\circ$.

Check with the exterior angles. The exterior angles are 95°, 85°, 70° and 110°; their sum is 360°, as required.

WORKED EXAMPLE 7.3.2 REGULAR POLYGONS

PROBLEM

(a) Find each interior angle of a regular pentagon. (b) A regular polygon has an interior angle of $156°$. How many sides does it have?

1. (a) A pentagon has 5 sides, so its interior angle sum is $(5-2)\times 180 = 3\times 180 = 540°$.
2. In a regular pentagon the five angles are equal: $540 \div 5 = 108°$.
3. (b) The exterior angle is $180 - 156 = 24°$, because interior and exterior angles are supplementary.
4. The exterior angles sum to $360°$, so the number of sides is $360 \div 24 = 15$.

ANSWER **(a)** $108°$ **(b) 15 sides.**

Check (b). A 15-sided regular polygon has each exterior angle $360 \div 15 = 24°$, so each interior angle is $180 - 24 = 156°$, as required.

COMMON ERROR

Using $n \times 180°$ for the interior sum instead of $(n-2)\times 180°$. A quadrilateral would then appear to have $720°$, which is the sum for a hexagon. A second error is dividing $360°$ by the number of interior angles to find each interior angle in an irregular polygon; that only works when the polygon is regular. A third is forgetting that an interior angle and its adjacent exterior angle are supplementary, not equal.

GUIDED PRACTICE · COPY AND COMPLETE

A pentagon has $\square - 2 = \square$ triangles, so its interior angle sum is $\square \times 180° = \square°$. In a regular hexagon, each exterior angle is $360° \div \square = \square°$, so each interior angle is $180° - \square° = \square°$. A regular polygon with exterior angle $20°$ has $360° \div 20° = \square$ sides.

PRACTICE 7.3 · POLYGONS AND THEIR ANGLES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

State the number of triangles or the exterior-angle fact you used.

1. Find the interior angle sum of: (a) a quadrilateral; (b) a pentagon; (c) an octagon; (d) a decagon. F
2. Find each interior angle of a regular: (a) pentagon; (b) hexagon; (c) octagon; (d) decagon. F
3. A quadrilateral has angles $100°$, $80°$, $95°$ and x. Find x. F
4. A pentagon has four angles $110°$, $120°$, $100°$ and $130°$. Find the fifth angle. P
5. Find the exterior angle of a regular 12-sided polygon, then its interior angle. P
6. A regular polygon has an interior angle of $156°$. Find the number of sides. P
7. Explain, using the idea of walking around the polygon, why the exterior angles of any convex polygon sum to $360°$. R
8. Extension: a regular polygon has an interior angle sum of $1800°$. Find the number of sides, the size of each exterior angle and the size of each interior angle. E

7.4 Constructions with ruler and compass

CRITERION A · II ◆ MEASUREMENT ◆ SPACE

A construction is a sequence of steps using only a straightedge and a pair of compasses. It does not rely on reading a scale. Instead, it uses equal radii to force a geometric relationship, so the result is exact even when the drawing is not.

The two instruments have different jobs. The straightedge draws a straight line through two points. The compasses draw a circle or an arc with a fixed radius from a chosen centre. Marking the same radius from two different centres locates points that are the same distance from both centres. That is the essential idea behind every construction in this chapter.

DEFINITIONS

The **perpendicular bisector** of a segment AB is the line that meets AB at its midpoint at $90°$. Every point on it is equidistant from A and B. The **bisector** of an angle divides it into two equal angles. A **construction** uses only a straightedge and compasses; a **scale drawing** uses measured lengths and angles.

WORKED EXAMPLE 7.4.1 PERPENDICULAR BISECTOR

PROBLEM

Construct the perpendicular bisector of a segment AB of length 6 cm. Describe each step and explain why the result is perpendicular.

1. Draw $AB = 6$ cm. Open the compasses to a radius greater than 3 cm, the half-length of AB.
2. With centre A, draw one arc above AB and one below it.
3. Without changing the radius, repeat from centre B. The two upper arcs meet at P; the two lower arcs meet at Q.
4. Draw the line through P and Q. It crosses AB at its midpoint M, and the angle at M is $90°$.
5. Why it works: P and Q are the same distance from A as from B. The line through two points equidistant from A and B is the perpendicular bisector.

ANSWER **$AM = MB = 3$ cm and $\angle PMB = 90°$. The arc radius must exceed 3 cm, or the arcs do not meet.**

Check. Choose any point on the constructed line and measure its distance to A and to B; the distances should be equal within drawing tolerance. This is a property of the construction, not a coincidence.

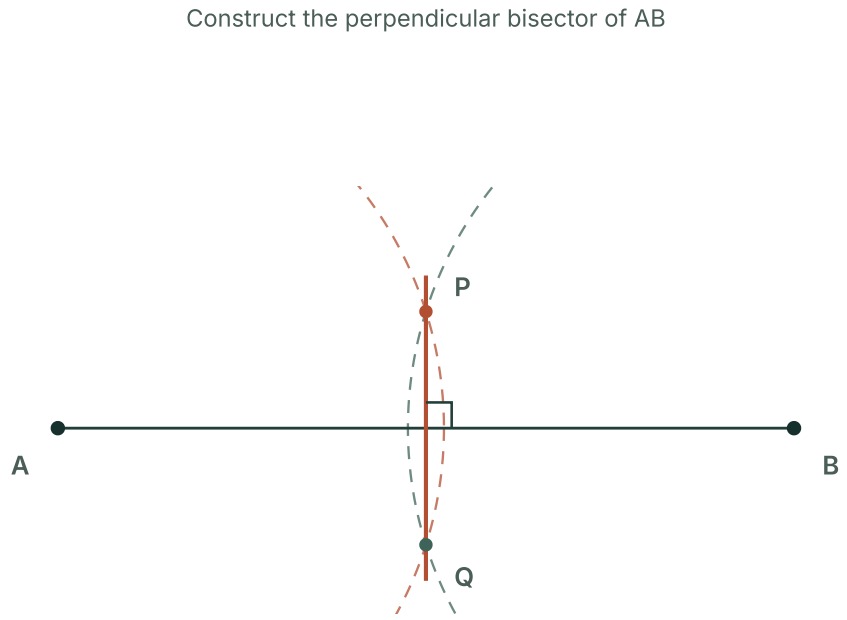


Figure 7.7 The perpendicular bisector of AB. The arcs from A and B use equal radii, so $PA = PB$ and $QA = QB$.

WORKED EXAMPLE 7.4.2 ANGLE BISECTOR AND A TRIANGLE

PROBLEM

(a) Construct the bisector of an angle of $70°$. (b) Construct a triangle with sides 5 cm, 6 cm and 7 cm.

1. (a) With centre at the vertex A, draw an arc that cuts both arms at P and Q.
2. With the same radius, draw arcs from P and Q that meet at R inside the angle. Draw AR. It bisects the angle, so each half is $35°$.
3. (b) Draw the longest side $AB = 7$ cm. (Any side may be drawn first, but the longest makes the arcs easiest to see.)
4. Set the compasses to 5 cm and draw an arc from A. Set them to 6 cm and draw an arc from B. The arcs meet at C.
5. Draw AC and BC. The triangle has sides 5 cm, 6 cm and 7 cm. Check the triangle inequality first: $5 + 6 = 11 > 7$, so the arcs meet.

ANSWER **(a) Two angles of $35°$ (b) a triangle with sides 5 cm, 6 cm and 7 cm.**

Check. In (a), measure both halves with a protractor; in (b), measure all three sides. A construction is checked by testing its defining property, not by trusting the appearance.

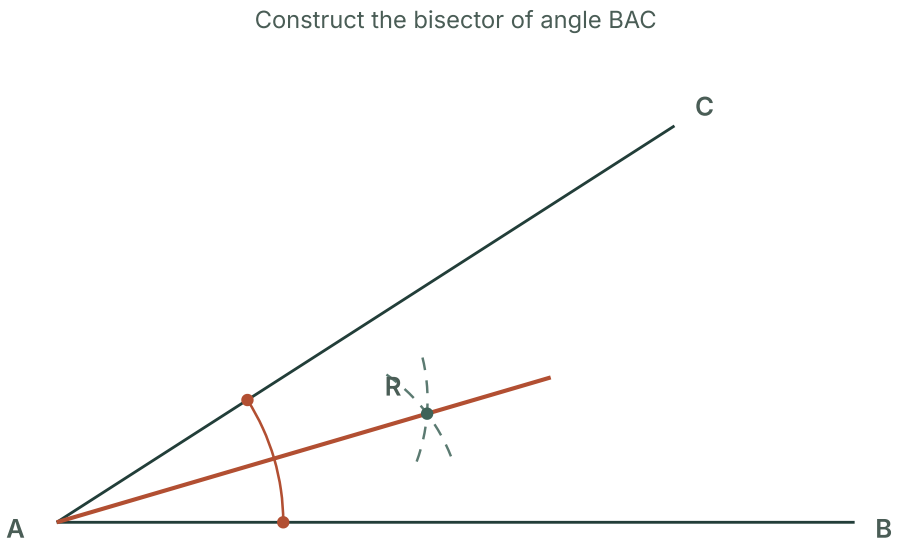


Figure 7.8 The angle bisector of $\angle BAC$. Equal arcs from P and Q locate R, and AR divides the angle into two equal parts.

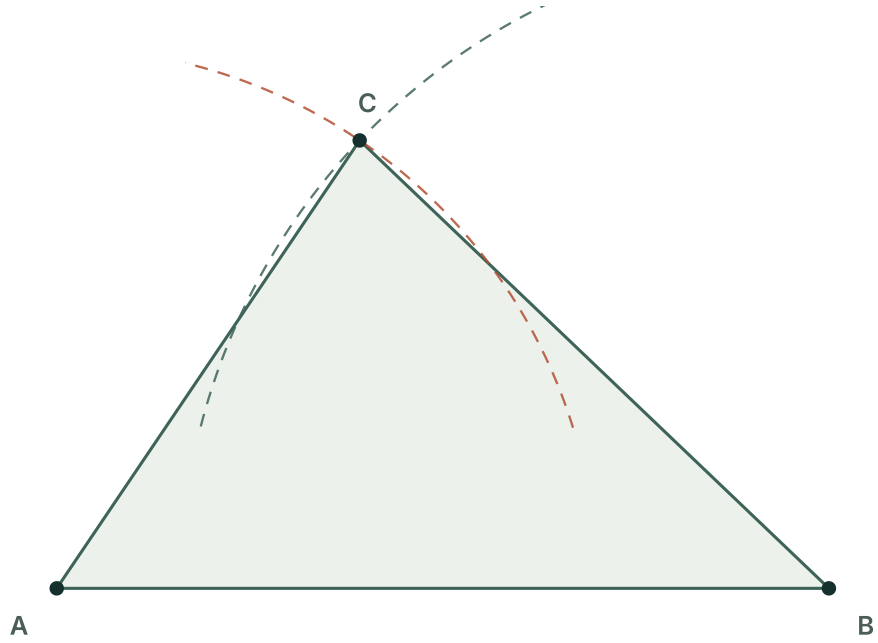


Figure 7.9 Constructing a triangle from three side lengths. The base is drawn first; the other two sides are transferred as arc radii from its endpoints.

COMMON ERROR

Changing the compass radius between the two arcs. For a perpendicular bisector or an angle bisector, the radii from the two centres must be equal. A second error is choosing a radius that is too short: for $AB = 6$ cm, a radius of 2 cm produces arcs that never meet. A third is erasing construction arcs; they are evidence of the method and should remain visible. Finally, do not use a protractor for a construction and call it a construction: measuring is a different method with a different kind of accuracy.

GUIDED PRACTICE · ORDER THE STEPS

Put these steps for a perpendicular bisector in order: (i) keep the same radius and draw arcs from B; (ii) draw segment AB; (iii) draw the line through the two intersection points; (iv) open the compasses to more than half of AB and draw arcs from A. Correct order: ____, ____, ____, ____. The line is perpendicular because both intersection points are the same distance from ____ and from ____. For $AB = 8$ cm, the radius must be greater than ____ cm.

PRACTICE 7.4 · CONSTRUCTIONS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Keep all construction arcs visible and label every point you create.

1. List the steps to construct the perpendicular bisector of AB. Include the condition on the compass radius. F
2. Construct $AB = 7.2$ cm and its perpendicular bisector. Measure each half. F
3. Construct an angle of 64° and bisect it. Measure both halves and state what they should be. F
4. Construct a triangle with sides 4 cm, 5 cm and 6 cm. Check that the arcs meet. P
5. A student uses a compass radius of 2 cm to bisect a segment of length 6 cm. The arcs do not meet. Explain why, and state a suitable radius. R
6. Construct an equilateral triangle of side 5 cm. Explain why every angle must be 60°. P
7. A path must pass through a point that is the same distance from two gates A and B. Which construction locates the path? Explain. P
8. Extension: construct an isosceles triangle with equal sides 6 cm and base 8 cm. Then construct the perpendicular bisector of the base. Explain why it passes through the apex. E

Construction clinic · Check before you draw

Every construction in this chapter depends on one decision: where to place the compass point and how wide to open it. The table fixes the setting for each method.

Construction	Compass setting	How to test the result
Perpendicular bisector of AB	One radius greater than $\frac{1}{2}AB$, used unchanged from A and from B.	The line crosses AB at its midpoint and makes 90°.
Bisector of $\angle BAC$	An arc from A cuts both arms; the same radius from those two points locates R.	$\angle BAR = \angle RAC$, so each half is half the original angle.
Triangle from three sides	Base equal to one side; radii equal to the other two sides, drawn from the base endpoints.	The arcs meet, and the two shorter sides together exceed the longest.
Equilateral triangle on AB	Radius AB from A, then the same radius from B; the arcs meet at C.	$AC = AB = BC$, so every angle is 60°.

Before you move on: never change the compass between a pair of arcs · keep construction arcs visible as evidence · label every new point · check the defining property, not the appearance of the drawing.

CONNECTIONS · DRAWING IS NOT PROVING

A construction is an argument made with instruments. Each arc records a fixed distance from a named point, so the finished drawing already contains evidence: equal radii, equal distances, equal angles. When you explain a construction, name that evidence instead of describing what the picture looks like.

Criterion B rewards the same habit: a correct method, a labelled diagram, and a reason that connects them. A neat drawing with no stated reason is only a sketch.

Investigation 7 · The triangle inequality

CRITERION B · I, II

Three strips joined end to end may or may not close a triangle. Figure 7.10 tests this with two compass arcs, without bending or guessing.

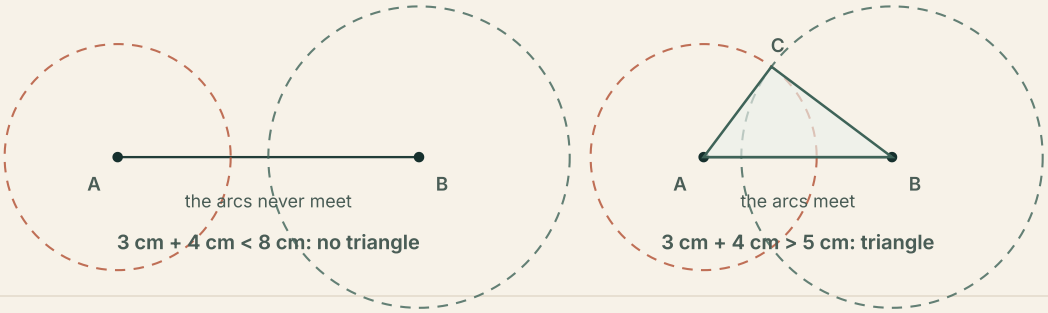


Figure 7.10 Arcs of radius 3 cm and 4 cm from the base ends: they miss for 8 cm, but meet for 5 cm.

Choose a longest side c and two shorter sides a and b. Set the base to c, then draw an arc of radius a from one endpoint and an arc of radius b from the other. A triangle exists exactly when the two arcs meet.

TESTING TRIPLES OF SIDE LENGTHS

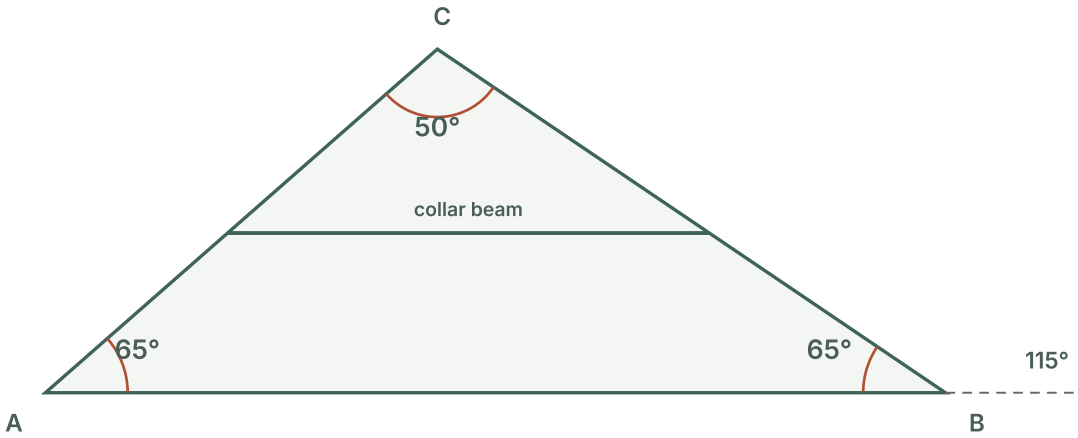
Sides a, b, c with c longest	3, 4, 5	3, 4, 8	4, 5, 9	6, 7, 10	2, 2, 3
$a + b$	7	7	9	13	4
Triangle possible?	yes	no	no (degenerate)	yes	yes

1. Complete the table, then explain what happens when the strips 4, 5 and 9 are joined.
2. Compare $a + b$ with c for each row. Write a rule that predicts whether a triangle can be made.
3. Test the rule on 5, 8 and 13; on 5, 8 and 12; and on 7, 7 and 14. Which cases are borderline, and what shape results in a borderline case?
4. Explain, using the two arcs in Figure 7.10, why the rule must hold. Your explanation should mention the distance between the endpoints of the base.
5. A student says “I only need to check the two shorter sides against the longest.” Is this sufficient? Explain why checking all three pairs gives the same conclusion.
6. Two sides of a triangle are 5 cm and 8 cm. The third side is a whole number of centimetres. Find every possible value. Justify the smallest and largest values.
7. Could a triangle have sides 4 cm, 4 cm and 9 cm? Answer using the rule, then describe how the arcs behave.
8. Extension: three sides of a triangle are x, $x + 1$ and $x + 2$. Find the smallest whole-number value of x for which a triangle is possible. Test the borderline value in a sketch.

A generalisation needs more than agreeing examples: explain why the arcs meet only when the two shorter sides together reach farther than the base, then test the borderline equality case.

Real-life application · The roof-truss workshop

A carpenter is making a triangular roof truss for a small garden shelter. The base is 6 m, and the two sloping rafters meet the base at equal angles of $65°$. A horizontal collar beam joins the rafters partway up, parallel to the base. The workshop first makes a 1:20 scale template, then checks the full-size frame.



The truss is isosceles: the two base angles are equal, so the apex angle is 50°.

Figure 7.11 The truss triangle. The exterior angle at the right-hand base is $115°$; the collar beam will be parallel to the base.

Your task. Use angle facts and a scale drawing to check the design before any timber is cut.

1. Explain why the apex angle must be $50°$. State the triangle fact you used.
2. What type of triangle is the truss? Which two sides are equal, and which angles are the base angles?
3. The collar beam is parallel to the base. At the left rafter, the angle between the collar and the rafter is $65°$. Find the angle on the other side of the collar at that joint, and explain your reasoning with a parallel-line fact.
4. Make a 1:20 scale drawing. The base becomes 30 cm. Use a protractor to draw base angles of $65°$, then measure the apex angle and both rafters. Record any difference from the calculated values.
5. A second design has a base of 8 m and two rafters of 2.5 m each. Use the triangle inequality to decide whether this truss can be built. Show the comparison.
6. The centre line from the apex to the base must be perpendicular to the base. Which construction locates the point where it meets the base? Explain why it also passes through the apex.
7. **Consider.** The workshop can use a measured template or a constructed one. Compare the two methods for accuracy, speed and the skill required. Recommend one, and state one situation in which the other is preferable.

Reflect. Which angle fact did you use most often in this task? Where did a construction replace a measurement, and what did that change about the reliability of the result?

Criterion-mapped mini-task

A · Knowing and understanding
C · Communicating
D · Applying mathematics

Context. A primary class is building models from card and needs a short geometry guide. The guide must explain the angle facts they will use, show one construction clearly, and justify why the angles of a triangle always add to 180°.

Product. A two-page guide with labelled diagrams, a step-by-step construction, a short explanation of the triangle-sum proof and a "check it yourself" section. It must be accurate enough for another teacher to use without asking you questions.

Requirements.

1. Explain, with a labelled diagram, the difference between an acute, right, obtuse and reflex angle. Use correct three-letter angle notation at least once.
2. State and illustrate the facts for angles on a straight line, angles at a point and vertically opposite angles.
3. Explain corresponding, alternate and co-interior angles using two parallel lines and a transversal. Label all eight angles.
4. Give a clear proof, in words and diagrams, that the interior angles of a triangle sum to 180°. Draw a line through one vertex parallel to the opposite side and use alternate angles.
5. Choose one construction: perpendicular bisector, angle bisector or a triangle from three sides. List each step, including the compass setting, and explain why the method works.
6. Include a worked example that finds a missing angle using an equation, with a reason for each step.
7. Add three "check it yourself" questions with answers hidden at the back of the guide. At least one question must involve parallel lines and one must involve a polygon.
8. Test the guide on a reader who has not studied the chapter. Record one unclear instruction and one improvement you made.

CRITERIA AND STRANDS A i, ii, iii: correct angle facts, triangle sum and construction steps; C i, ii, iii: precise notation, labelled diagrams and a logical explanation; D i, ii, iii, iv: apply the facts to a reader's task, choose an appropriate construction and evaluate the guide's clarity.

BEFORE YOU SUBMIT

Check every angle statement against a diagram. In the triangle proof, make sure the line through the vertex is parallel to the opposite side and that the three angles at the vertex form a straight angle. For the construction, a reader must be able to reproduce it with only your words. Remove any step that says "draw it carefully" without saying what to set the compasses to.

Chapter review

SUMMARY

- **Angle facts.** Angles on a straight line sum to $180°$, angles at a point sum to $360°$, and vertically opposite angles are equal.
- **Parallel lines.** Corresponding and alternate angles are equal; co-interior angles are supplementary.
- **Triangles.** Interior angles sum to $180°$. An exterior angle equals the sum of the two opposite interior angles. Isosceles triangles have equal base angles.
- **Polygons.** Interior sum $= (n-2) \times 180°$; exterior sum $= 360°$. Regular polygons divide each sum equally.
- **Constructions.** Equal compass radii locate points equidistant from two centres. This produces perpendicular bisectors, angle bisectors and triangles from three sides.
- **Triangle inequality.** Three lengths form a triangle only when the two shorter lengths together exceed the longest.

Straight line: $a + b = 180°$.

At a point: $a + b + c + \cdots = 360°$.

Triangle: $a + b + c = 180°$.

Polygon interior sum: $(n-2) \times 180°$.

Regular polygon: exterior $= \frac{360°}{n}$, interior $= 180° - \frac{360°}{n}$.

Triangle inequality: $a + b > c$ for every pairing of sides.

REVISION CHECKLIST

- ☐ I can name and classify angles correctly.
- ☐ I can use angle facts on a line, at a point and between parallel lines.
- ☐ I can calculate angles in triangles using the sum and exterior-angle facts.
- ☐ I can find interior and exterior angles of polygons.
- ☐ I can carry out and explain the three core constructions.
- ☐ I can test whether three lengths form a triangle.

KEY VOCABULARY

vertex
The point where two rays meet.

acute
An angle less than $90°$.

obtuse
An angle between $90°$ and $180°$.

reflex
An angle between $180°$ and $360°$.

supplementary
Two angles summing to $180°$.

complementary
Two angles summing to $90°$.

vertically opposite
Equal angles opposite each other at a crossing.

transversal
A line crossing two or more lines.

corresponding
Angles in matching positions at two intersections.

alternate
Equal angles on opposite sides of a transversal.

co-interior
Angles on the same side of a transversal, summing to $180°$.

exterior angle
An angle formed by a side and the extension of an adjacent side.

regular polygon
A polygon with equal sides and equal angles.

perpendicular bisector
The line at $90°$ through the midpoint of a segment.

REFLECTION AND SELF-ASSESSMENT

Find a structure near you — a window frame, a shelf bracket, a fence or a roof. Identify one triangle and one pair of parallel lines. Estimate, then calculate, one angle that the builder had to control. Explain why that angle matters.

Rate your confidence on each success criterion, name one construction that still needs practice, and explain in two sentences the difference between a measurement and a construction.

END-OF-CHAPTER PRACTICE

Mixed practice 7

Do not use a calculator. Draw a sketch, mark the known angles and give a reason for each step.

FLUENCY Build accuracy

1. Classify each angle: (a) 27°; (b) 90°; (c) 101°; (d) 199°.
2. Find the missing angle on a straight line: 63°, 79° and x.
3. Find the missing angle at a point: 120°, 75°, 85° and y.
4. Two straight lines cross. One angle is 64°. Find the other three angles.
5. In a parallel-line diagram, one angle is 128°. Find its alternate angle and its co-interior partner.
6. Two angles of a triangle are 39° and 88°. Find the third angle.
7. An isosceles triangle has base angles of 72°. Find the apex angle.
8. Find the interior angle sum of a heptagon (7 sides).
9. Find each interior angle of a regular nonagon (9 sides).
10. Find the exterior angle of a regular 18-sided polygon.
11. Which of these triples can form a triangle: (a) 4, 5, 9; (b) 4, 5, 8; (c) 2, 6, 7?
12. Construct $AB = 6.4$ cm and its perpendicular bisector. State the length of each half.

REASONING AND PROBLEM SOLVING Connect ideas

13. Two vertically opposite angles are $6x - 8^\circ$ and $4x + 22^\circ$. Find x and each angle.
14. Three angles on a straight line are $2x + 10^\circ$, $x + 25^\circ$ and $x - 15^\circ$. Find x and all three angles.
15. A triangle has angles x, $2x$ and $3x$. Find x and classify the triangle by its angles.
16. A quadrilateral has angles $3x$, $4x$, $5x$ and $6x$. Find x and all four angles.
17. A regular polygon has each exterior angle equal to 36°. Find the number of sides and each interior angle.
18. Two sides of a triangle are 7 cm and 11 cm. The third side is a whole number of centimetres. Find all possible values.
19. **The ramp frame (Criteria A, C, D).** A ramp frame is a right triangle. The angle between the ramp and the ground is 14°. Find the third angle. A straight handrail at the top continues the ramp direction; find the angle between the handrail and a vertical post.
20. **The kite design (Criteria A, C, D).** A kite is made from two congruent triangles. Each triangle has angles 40°, 70° and 70°. Find the angles at the left and right tips of the kite, explain your reasoning, and describe how a 1:10 scale drawing would be constructed.
21. **The truss check (Criteria A, C, D).** A truss has base 5 m and equal rafters 4 m. (a) Can it be built? Use the triangle inequality. (b) The base angles are 66°. Find the apex angle. (c) A collar parallel to the base makes a 114° co-interior angle with one rafter. Find the angle on the other side of the collar at that joint.

Check your progress

ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

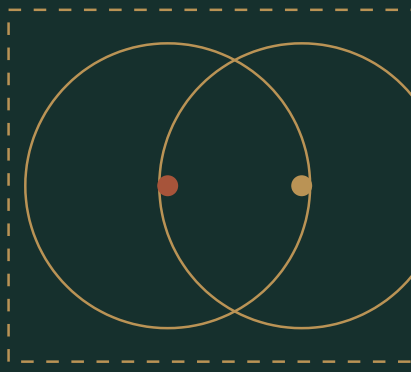
- **Match the fact to the position.** A line gives $180°$, a point gives $360°$, and co-interior angles are supplementary, not equal.
- **Use the correct pair.** Alternate and corresponding angles are equal only when the crossed lines are parallel.
- **The exterior angle is the sum of the two opposite interior angles.** It is not supplementary to that sum.
- **An n-gon has $n-2$ triangles, not n.** Use $(n-2) \times 180°$ for the interior sum.
- **A construction uses equal radii.** Keep the compass setting fixed between the two arcs and leave the arcs visible.
- **Check the triangle inequality before constructing.** If the two shorter sides do not reach, the arcs cannot meet.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can name and classify angles.			
I can use angle facts with reasons.			
I can calculate triangle and polygon angles.			
I can carry out and explain constructions.			
I can decide whether three lengths form a triangle.			
I can produce and evaluate a scale drawing.			

WHERE THIS LEADS

Chapter 8 turns classification into a precise language. Sets describe which objects share a property, and Venn diagrams show relationships of overlap and exclusion — the same kind of structural thinking used to sort triangles and angles here. **Chapter 9** then measures the shapes whose boundaries you have learned to reason about.

8



A ∪ B

THE MATHEMATICS OF BELONGING

Sets and classification

A set is a collection described by a rule. Once the rule is precise, belonging can be tested, overlap can be counted and a decision can be defended. Sets give classification a language and a diagram.

KEY CONCEPT · FORM | RELATED · REPRESENTATION | RELATED · SYSTEMS | CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

A community centre offers two activity programmes. Some members use only one, some use both and some use neither. How can the centre record this fairly, count each group without double-counting, and decide what a combined membership should include?

MYP UNIT PLANNER

Framing the inquiry

UNIT 8 · APPROX. 14 HOURS
ATL FOCUS · THINKING SKILLS

KEY AND RELATED CONCEPTS

Form. A set is defined by the property its elements share. The form of the definition determines which objects belong and which do not.

Representation. Roster notation, set-builder notation and Venn diagrams represent the same collection in different ways. Each makes a different feature visible.

Systems. Number sets form a nested system. Each larger set contains the smaller ones and adds new possibilities.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Exploration: how clear classification and accurate counting support fair access to shared resources.

A community centre must know how many members use each programme, how many use both and how many use neither. Counting the overlap twice would overstate demand; ignoring it would understate it.

STATEMENT OF INQUIRY

Precise classification reveals relationships between groups, and accurate set reasoning supports fair decisions about shared resources.

FACTUAL

What is a set, and how are its elements and subsets written? What do union, intersection and complement mean?

CONCEPTUAL

How can one collection be represented by a list, a rule and a diagram? Why does the overlap need special treatment when groups are combined?

DEBATABLE

When does a classification system help a decision, and when does it exclude people unfairly? Should a shared service be priced by use or by membership?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Use set notation, identify subsets, classify numbers, and calculate counts from Venn diagrams.
B · Investigating patterns	i, ii	Investigate how overlap constrains the possible counts in a survey, and generalise the addition rule.
C · Communicating	i, ii, iii	Represent sets by roster, rule and diagram; label a universal set; justify a count in words.
D · Applying mathematics	i, ii, iii, iv	Audit a membership scheme, interpret survey data transparently, and evaluate a classification guide.



ATL · THINKING SKILLS IN ACTION

Two students classify the same 30 classmates as “uses the library” or “uses the sports hall”. One counts 42 uses in total and concludes that 42 students were surveyed. The other counts 28. Explain how both can be working from the same data, and what information is missing from the first student’s total.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** sets as collections defined by a property, and use element and subset notation correctly.
- **Be able to** represent a set by roster, set-builder notation and a Venn diagram.
- **Be able to** find unions, intersections, complements and differences of sets.
- **Know** how the number sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are related.
- **Be able to** use the addition rule to count overlapping groups without double-counting.
- **Investigate** how the overlap between two groups is constrained by the total number surveyed.

SUCCESS CRITERIA

- ☐ I can list the elements of a set and state its cardinality.
- ☐ I can decide whether one set is a subset of another.
- ☐ I can read and draw a two-set Venn diagram with a universal set.
- ☐ I can calculate $n(A \cup B)$, $n(A \cap B)$ and $n(A')$.
- ☐ I can classify a number into the correct number sets.
- ☐ I can solve a two-group survey problem and explain each count.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. List the first five even numbers greater than zero.
2. Write 0.75 as a fraction in its simplest form.
3. Calculate $18 + 14 - 6$.
4. Which is greater: -4 or -9?
5. Write two multiples of 3 that are less than 10.
6. Calculate $10 - 3 - 2$.

Answers are in the Chapter 8 answer section. If more than two are incorrect, review factors and multiples, fraction equivalence and addition with compensation before continuing.

BEFORE YEAR 1
Sorting objects by one or two properties; using tables and simple lists; recognising odd, even, prime and square numbers.

YEAR 1 · THIS CHAPTER
Set notation and subsets; Venn diagrams; union, intersection and complement; number sets; two-group survey reasoning.

LATER
Three-set Venn diagrams, set algebra, logic and truth tables, probability using sets, and functions defined on number sets.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
A community centre records who uses which activity programme.	How can overlapping groups be counted once each, and what does “neither” mean?	Form, representation and systems.	How much overlap is possible when two totals and the overall total are known?	Roster lists, set-builder rules, labelled Venn diagrams and count statements.	When does a neat classification clarify a decision, and when does it hide an exception?

8.1 Sets, elements and subsets

CRITERION A · I ◆ FORM ◆ REPRESENTATION

A set is a well-defined collection of objects. "Well-defined" is the important part: given any object, there must be no doubt about whether it belongs. A vague description such as "some interesting numbers" is not a set.

The objects in a set are its **elements** or members. We write $x \in A$ when x belongs to A, and $x \notin A$ when it does not. A set can be written as a **roster**, with its elements inside braces, or by a **rule** in set-builder notation. The order of a roster does not matter, and repeated elements are written once.

DEFINITIONS

A **set** is a collection of distinct objects. The **empty set** $\varnothing$ contains no elements. The **universal set** ξ contains every object under discussion. The **cardinality** $n(A)$ is the number of elements in a finite set A. Set A is a **subset** of B, written $A \subseteq B$, if every element of A is also an element of B. It is a **proper subset**, written $A \subset B$, if in addition $A \neq B$.

For example, if $A = \{2, 4, 6, 8\}$, then $6 \in A$, $5 \notin A$ and $n(A) = 4$. The set of even numbers between 1 and 9 can also be written $\{x \in \mathbb{N} : x \text{ is even and } x < 9\}$. The colon means "such that". The rule and the roster describe exactly the same set.

The empty set $\varnothing$ is a subset of every set, because there is no element in it that could fail to belong. Every set is a subset of itself. A set with n elements has 2^n subsets, including $\varnothing$ and the set itself. This count comes from a choice at each element: include it or leave it out.

WORKED EXAMPLE 8.1.1 ROSTERS AND CARDINALITY

PROBLEM

Let A be the set of prime numbers less than 12, let $B = \{x \in \mathbb{N} : x \leq 12 \text{ and } x \text{ is a multiple of } 3\}$, and let C be the set of letters in the word "book". (a) List each set. (b) State $n(A)$, $n(B)$ and $n(C)$. (c) Decide whether $7 \in B$ and whether $C \subseteq \{b, o, k, e\}$.

1. (a) The primes less than 12 are 2, 3, 5, 7 and 11, so $A = \{2, 3, 5, 7, 11\}$.
2. The multiples of 3 from 1 to 12 are 3, 6, 9 and 12, so $B = \{3, 6, 9, 12\}$.
3. The word "book" contains b, o, o, k. The repeated o is listed once, so $C = \{b, o, k\}$.
4. (b) Counting the distinct elements: $n(A) = 5$, $n(B) = 4$ and $n(C) = 3$.
5. (c) 7 is prime but not a multiple of 3, so $7 \notin B$. Every element of C is in $\{b, o, k, e\}$, so $C \subseteq \{b, o, k, e\}$.

ANSWER **(a)** $A = \{2, 3, 5, 7, 11\}$, $B = \{3, 6, 9, 12\}$, $C = \{b, o, k\}$ **(b) 5, 4, 3 (c)** $7 \notin B$**; yes,** C **is a subset.**

Check. Re-read the rule for B. The phrase "less than 12" would exclude 12, but the symbol $\leq$ includes it. A single symbol can change the set.

WORKED EXAMPLE 8.1.2 RULES, SUBSETS AND THE POWER OF CHOICE

PROBLEM

(a) Write $D = \{1, 3, 5, 7, 9\}$ in set-builder notation using a property of odd numbers. (b) List every subset of $E = \{a, b, c\}$ and count them. (c) Explain why $\varnothing \subseteq D$.

1. (a) Each element is an odd natural number less than 10: $D = \{x \in \mathbb{N} : x < 10 \text{ and } x \text{ is odd}\}$.
2. (b) The subsets are $\varnothing$, $\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{a, c\}$, $\{b, c\}$ and $\{a, b, c\}$.
3. There are 8 subsets. Since E has 3 elements, this agrees with $2^3 = 8$.
4. (c) To show $\varnothing \not\subseteq D$, we would need an element of $\varnothing$ that is not in D. No such element exists, so $\varnothing \subseteq D$ is true.

ANSWER **(a) $D = \{x \in \mathbb{N} : x < 10,\ x \text{ odd}\}$ (b) 8 subsets, as listed (c) there is no element in $\varnothing$ to contradict the definition.**

Organise the list. Group subsets by size: 0 elements (1), 1 element (3), 2 elements (3), 3 elements (1). The row 1, 3, 3, 1 totals 8 and is a useful check that none has been missed.

COMMON ERROR

Writing $\{\varnothing\}$ for the empty set. These are different: $\varnothing$ has no elements, while $\{\varnothing\}$ has one element, namely the empty set. A second error is counting repeated items in a roster: $\{b, o, o, k\}$ has three elements, not four. A third is confusing $\in$ with $\subseteq$: $3 \in \{1, 3, 5\}$ but $\{3\} \subseteq \{1, 3, 5\}$.

GUIDED PRACTICE · COPY AND COMPLETE

For $S = \{2, 4, 6, 8, 10\}$: the element 8 $\square$ S, so we write $8 \square S$. The element 7 is not in S, so $7 \square S$. The cardinality is $n(S) = \square$. The set of even numbers from 2 to 10 inclusive can be written $\{x \in \mathbb{N} : x \leq \square,\ x \text{ is } \square\}$. A subset with three elements is $\{\square, \square, \square\}$. The empty set is written $\square$, and it is a subset of S because $\square$.

PRACTICE 8.1 · SETS AND SUBSETS

SUPPORT 1–3 · CORE 1–7 · EXTENSION 8–9

Use correct set notation throughout. A roster may be written in any order, but each element appears once.

1. List the elements of each set: (a) even numbers from 4 to 12; (b) factors of 18; (c) letters in "mathematics"; (d) prime numbers between 20 and 30. F
2. State the cardinality of each set in Question 1. F
3. Decide whether each statement is true: (a) $5 \in \{1, 3, 5, 7\}$; (b) $\{2\} \subseteq \{2, 4, 6\}$; (c) $\varnothing \subseteq \{0\}$; (d) $0 \in \varnothing$. F
4. Write each set in set-builder notation: (a) $\{3, 6, 9, 12\}$; (b) $\{1, 4, 9, 16, 25\}$; (c) $\{-4, -3, -2, -1, 0\}$. P
5. Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{2, 4, 6, 8\}$. List all elements that are in both sets, and all elements that are in at least one of the sets. P
6. List every subset of $\{x, y, z, w\}$ that contains exactly two elements. How many are there? P
7. A student writes the set of factors of 12 as $\{1, 2, 3, 4, 6, 12, 12\}$. Explain the error and state the correct cardinality. R
8. Extension: a set has 5 elements. How many subsets does it have? How many of these are proper subsets? Explain your reasoning. E
9. Extension: describe a universal set for the survey "Which form of transport do students use to travel to school?" and explain why the choice of universal set matters. E

8.2 Venn diagrams and set operations

CRITERION A · I, II ◆ REPRESENTATION ◆ SYSTEMS

A Venn diagram turns membership into position. Elements that belong to both sets sit in the overlap; elements outside both sit inside the rectangle but outside the circles. The rectangle is the universal set.

The **union** of A and B, written $A \cup B$, contains every element that belongs to A or to B or to both. The **intersection** $A \cap B$ contains the elements that belong to both. The **complement** A' contains the elements of the universal set ξ that are not in A. The **difference** $A \setminus B$ contains the elements of A that are not in B. Two sets are **disjoint** when $A \cap B = \varnothing$.

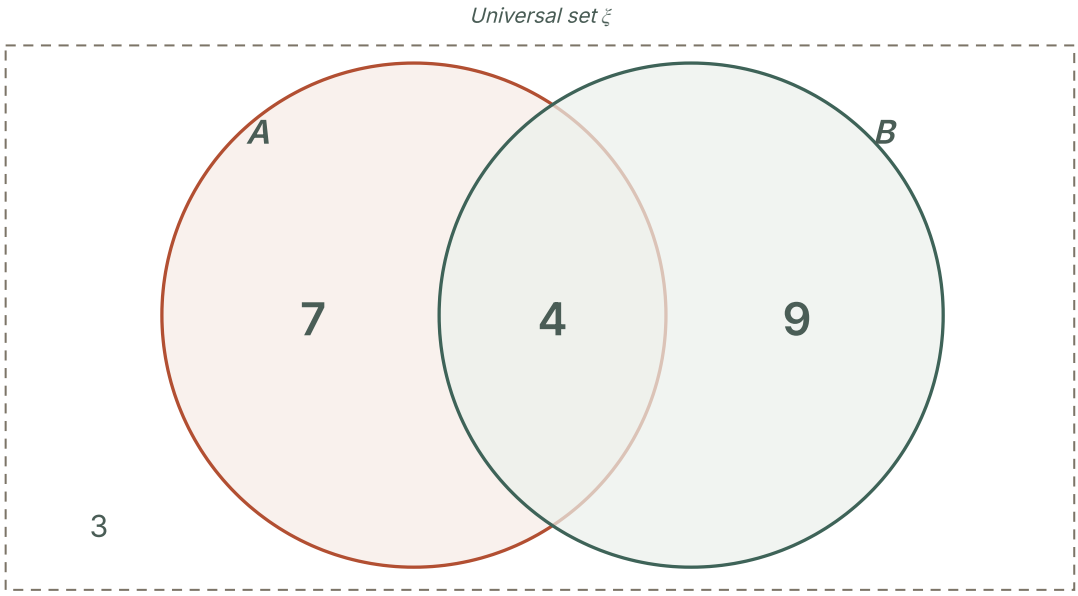


Figure 8.1 A two-set Venn diagram. The overlap contains the elements in $A \cap B$; the two crescents contain the elements in exactly one set; the region outside both circles contains the elements in neither.

In Figure 8.1, $n(A) = 7 + 4 = 11$ and $n(B) = 4 + 9 = 13$. The union has $7 + 4 + 9 = 20$ elements. Notice that $n(A) + n(B) = 11 + 13 = 24$, which counts the four elements in the overlap twice. Subtracting one copy gives the correct union count. This is the **addition rule** for two sets.

> KEY IDEA
>
> $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. In a two-set Venn diagram, $n(A)$ is the "only A" number plus the overlap, not the "only A" number alone.

When the universal set is finite and the diagram is complete, the four regions account for every element: only A, both, only B, and neither. Their sum equals $n(\xi)$. This gives a reliable check on any calculation.

WORKED EXAMPLE 8.2.1 OPERATIONS WITH A LISTED UNIVERSAL SET

PROBLEM

Let $\xi = \{1, 2, 3, \ldots, 12\}$, $A = \{1, 2, 3, 4, 6, 12\}$ (the factors of 12) and $B = \{2, 3, 5, 7, 11\}$ (the primes less than 12). Find (a) $A \cap B$, (b) $A \cup B$, (c) A', and (d) $A \setminus B$. Check the union count with the addition rule.

1. (a) Elements in both lists: $A \cap B = \{2, 3\}$.
2. (b) Combine the lists and remove duplicates: $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 11, 12\}$.
3. (c) The elements of ξ not in A are $5, 7, 8, 9, 10, 11$, so $A' = \{5, 7, 8, 9, 10, 11\}$.
4. (d) The elements of A that are not in B are $1, 4, 6, 12$, so $A \setminus B = \{1, 4, 6, 12\}$.
5. Check: $n(A) = 6$, $n(B) = 5$ and $n(A \cap B) = 2$, so $n(A \cup B) = 6 + 5 - 2 = 9$. The list in (b) has 9 elements.

ANSWER **(a)** $\{2, 3\}$ **(b)** $\{1, 2, 3, 4, 5, 6, 7, 11, 12\}$ **(c)** $\{5, 7, 8, 9, 10, 11\}$ **(d)** $\{1, 4, 6, 12\}$**; the union has 9 elements.**

Check the complement. $A \cup A' = \xi$, and $A \cap A' = \varnothing$. Here A has 6 elements and A' has 6, accounting for all 12 elements of ξ.

WORKED EXAMPLE 8.2.2 READING COUNTS FROM A SURVEY DIAGRAM

PROBLEM

In a group of 40 students, 23 study French and 18 study Spanish. Seven students study both languages. (a) Draw a Venn diagram and complete all four regions. (b) How many students study at least one language? (c) How many study neither? (d) What percentage study Spanish but not French?

1. (a) The overlap is 7. French only is $23 - 7 = 16$; Spanish only is $18 - 7 = 11$.
2. At least one language is $16 + 7 + 11 = 34$. This also agrees with $23 + 18 - 7 = 34$.
3. Neither is $40 - 34 = 6$. Check the regions: $16 + 7 + 11 + 6 = 40$.
4. (d) Spanish but not French is the "only Spanish" region: 11 students.
5. The percentage is $\frac{11}{40} \times 100 = 27.5\%$.

ANSWER **(a) 16 French only, 7 both, 11 Spanish only, 6 neither** **(b) 34** **(c) 6** **(d)** 27.5%**.**

Check the totals. French total $16 + 7 = 23$, Spanish total $7 + 11 = 18$, and the four regions total 40. Every given number is accounted for, so the diagram is consistent.

GUIDED PRACTICE · COPY AND COMPLETE

For sets $P = \{1, 2, 3, 4\}$ and $Q = \{3, 4, 5, 6\}$: $P \cap Q = \{\square, \square\}$, $P \cup Q = \{\square, \square, \square, \square, \square, \square\}$, and $n(P \cup Q) = \square$. If $\xi = \{1, 2, \ldots, 8\}$, then $P' = \{\square, \square, \square, \square\}$. The sets P and Q are not disjoint because $\square$. The addition rule gives $n(P \cup Q) = \square + \square - \square = \square$.

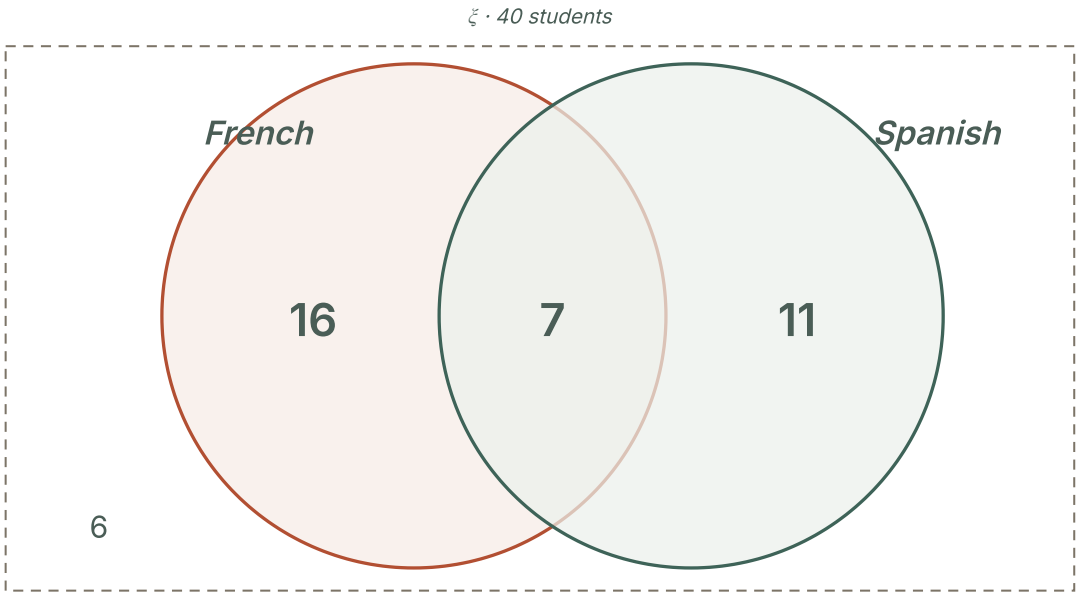


Figure 8.2 The completed diagram for Worked example 8.2.2; circle labels name whole sets.

> **COMMON ERROR**
>
> Placing the number of students who study French (23) in the "French only" region and then also placing 7 in the overlap. That counts the seven bilingual students twice and makes the circle totals too large. Always subtract the overlap before writing an "only" number. A second error is treating A' as "everything not in A" without reference to ξ; the complement depends on the universal set chosen.

PRACTICE 8.2 · VENN DIAGRAMS AND OPERATIONS

SUPPORT 1–3 · CORE 1–7 · EXTENSION 8–9

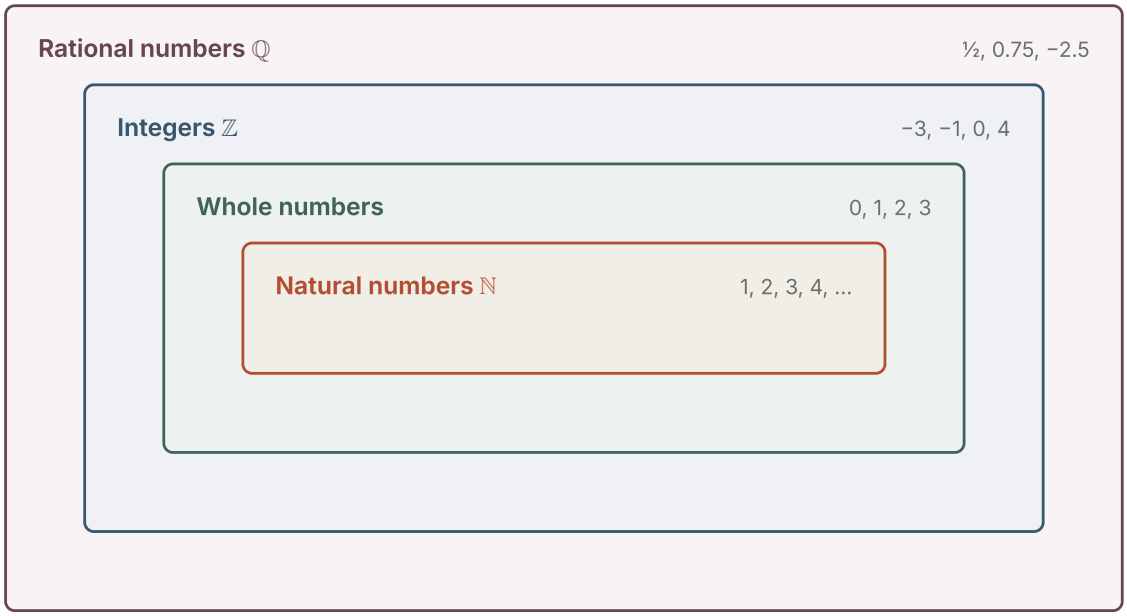
Copy each diagram. Say whether each answer is a set or a count.

1. $\xi = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3, 4\}$. Find (a) $A \cap B$; (b) $A \cup B$; (c) A'; (d) $B \setminus A$. F
2. A Venn diagram has 9 in A only, 5 in both, 8 in B only and 4 outside. Find $n(A)$, $n(B)$, $n(A \cup B)$, $n(A \cap B)$ and $n(\xi)$. F
3. In a class of 28, 16 play football, 12 play hockey and 5 play both. Draw a Venn diagram and find how many play neither. F
4. For $C = \{a, b, c, d\}$ and $D = \{c, d, e\}$, list the elements of $C \cup D$, $C \cap D$ and $C \setminus D$. P
5. If $n(A) = 14$, $n(B) = 11$ and $n(A \cap B) = 4$, find $n(A \cup B)$. P
6. A survey of 50 people finds 31 use a bus, 24 use a train and 12 use both. How many use at least one? How many use neither? P
7. A student says "$n(A \cup B) = n(A) + n(B)$ because union means add." Give a counter-example with a non-empty intersection and correct the statement. R
8. Extension: $n(A) = 20$, $n(B) = 17$ and $n(A \cup B) = 30$. Find $n(A \cap B)$ and explain why the overlap cannot be larger than 17. E
9. Extension: in a group of 35 students, 22 like tea, 19 like coffee and 8 like neither. Find how many like both. E

8.3 Number sets and the real line

CRITERION A · I ◆ SYSTEMS ◆ FORM

Number families are sets. Each is contained in the next, and the containment explains which operations are always available: subtraction needs integers, and division needs rational numbers.



Every natural number is whole, every whole number is an integer, and every integer is rational.

Figure 8.3 The natural numbers are a subset of the whole numbers, which are a subset of the integers, which are a subset of the rational numbers. The rational numbers are themselves a subset of the real numbers, which also contain the irrational numbers.

In this course, the **natural numbers** are the counting numbers $\mathbb{N} = \{1, 2, 3, \ldots\}$. The **whole numbers** add zero: $\{0, 1, 2, 3, \ldots\}$. The **integers** add the negatives: $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$. A **rational number** can be written as $\frac{a}{b}$, where a and b are integers and $b \neq 0$. The rational numbers are $\mathbb{Q}$.

Every integer is rational because it can be written with denominator 1. Terminating decimals and repeating decimals are also rational: $0.4 = \frac{2}{5}$ and $0.\overline{3} = \frac{1}{3}$. Numbers that cannot be written as a fraction of integers are **irrational**. Familiar examples are $\sqrt{2} = 1.414\ldots$, $\pi = 3.14159\ldots$ and $0.101001000\ldots$, where the decimal neither terminates nor repeats. Together, the rational and irrational numbers form the **real numbers** $\mathbb{R}$.

> KEY IDEA
>
> $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. An irrational number is a real number that is not rational. A number can belong to several sets at once, so "classify" means "name every set that contains it".

The symbol $\mathbb{Q}$ comes from "quotient", because a rational number is a quotient of integers. The symbol $\mathbb{R}$ denotes the complete number line. Between any two real numbers there is another real number, and every point on the line corresponds to exactly one real number. This is why the number line is a complete representation of $\mathbb{R}$.

WORKED EXAMPLE 8.3.1 CLASSIFYING WITH EVIDENCE

PROBLEM

Classify each number into every set it belongs to: (a) -7; (b) 0; (c) $\frac{3}{4}$; (d) $\sqrt{16}$; (e) $\sqrt{2}$; (f) 0.4; (g) π; (h) -2.5.

1. (a) -7 is an integer, so it is rational and real: $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$. It is not natural or whole.
2. (b) 0 is whole, an integer, rational and real: $\mathbb{Q}$ and $\mathbb{R}$, but not natural in this course's convention.
3. (c) $\frac{3}{4}$ is rational and real: $\mathbb{Q}, \mathbb{R}$. It is not an integer.
4. (d) $\sqrt{16} = 4$, which is natural, whole, an integer, rational and real.
5. (e) $\sqrt{2}$ is irrational and real: it cannot be written as a fraction of integers.
6. (f) $0.4 = \frac{2}{5}$, so it is rational and real. (g) π is irrational and real.
7. (h) $-2.5 = -\frac{5}{2}$, so it is rational and real, but not an integer.

ANSWER **(a)** $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ **(b) whole,** $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ **(c)** $\mathbb{Q}, \mathbb{R}$ **(d)** $\mathbb{N}$**, whole,** $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ **(e) irrational,** $\mathbb{R}$ **(f)** $\mathbb{Q}, \mathbb{R}$ **(g) irrational,** $\mathbb{R}$ **(h)** $\mathbb{Q}, \mathbb{R}$**.**

Test with a fraction. A number is rational if you can write it as $\frac{a}{b}$ with integers a and b. For a terminating decimal, read the digits after the point as the numerator over the matching power of 10, then simplify.

WORKED EXAMPLE 8.3.2 SUBSETS AND SET NOTATION

PROBLEM

(a) Is $\mathbb{N} \subseteq \mathbb{Z}$? (b) Is $\mathbb{Z} \subseteq \mathbb{Q}$? (c) Write the set of even integers between -4 and 4 inclusive in roster form. (d) Explain why $\sqrt{9} \in \mathbb{Q}$ but $\sqrt{10} \notin \mathbb{Q}$.

1. (a) Every natural number is an integer, so $\mathbb{N} \subseteq \mathbb{Z}$ is true.
2. (b) Every integer n can be written $\frac{n}{1}$, so every integer is rational. Thus $\mathbb{Z} \subseteq \mathbb{Q}$.
3. (c) The even integers are $-4, -2, 0, 2, 4$.
4. (d) $\sqrt{9} = 3$, and $3 = \frac{3}{1}$, so it is rational. There is no integer whose square is 10, and $\sqrt{10}$ cannot be written as a fraction of integers; it is irrational.

ANSWER **(a) yes** **(b) yes** **(c)** $\{-4, -2, 0, 2, 4\}$ **(d)** $\sqrt{9} = 3 \in \mathbb{Q}$**;** $\sqrt{10} \notin \mathbb{Q}$ **because it is irrational.**

Use the nested diagram. If a number lies inside a smaller box in Figure 8.3, it also lies inside every larger box around it. A rational number need not be an integer, so the arrows only point inward.

COMMON ERROR

Assuming that every decimal is rational. A decimal is rational only if it terminates or repeats; $0.101001000\ldots$ is a decimal but not rational. A second error is claiming $3.14 = \pi$. The decimal 3.14 is rational, while π is irrational; 3.14 is only an approximation. A third is writing $\sqrt{9}$ as irrational because it contains a square-root sign: evaluate the root first.

GUIDED PRACTICE · COPY AND COMPLETE

Complete the chain of subsets: $\mathbb{N} \subset \square \subset \mathbb{Q} \subset \square$. The number -6 belongs to $\square$, $\square$ and $\square$. The number $\frac{7}{2}$ belongs to $\square$ and $\square$, but not to $\square$. The decimal 0.125 is rational because $0.125 = \frac{\square}{\square}$. The number $\sqrt{25}$ is actually the integer $\square$, so it belongs to $\square$.

PRACTICE 8.3 · NUMBER SETS

SUPPORT 1–3 · CORE 1–7 · EXTENSION 8–9

Name every set that contains each number. Use $\mathbb{N}$, whole, $\mathbb{Z}$, $\mathbb{Q}$, irrational and $\mathbb{R}$.

1. Classify: (a) 12; (b) -5; (c) 0.75; (d) $\frac{9}{4}$. F
2. Classify: (a) $\sqrt{36}$; (b) $\sqrt{7}$; (c) π; (d) $0.333\ldots$ F
3. Decide whether each statement is true: (a) $-3 \in \mathbb{N}$; (b) $0 \in \mathbb{Z}$; (c) $\frac{5}{2} \in \mathbb{Q}$; (d) $\sqrt{2} \in \mathbb{Q}$; (e) $\mathbb{Z} \subset \mathbb{R}$. F
4. Write each number as a fraction of integers: (a) 0.6; (b) 1.25; (c) $0.\overline{7}$; (d) -3.5. P
5. Sort these numbers into rational and irrational: $\sqrt{81}$, π, $-\frac{2}{3}$, $0.101001000\ldots$, $\sqrt{15}$, 4.5. P
6. Explain why every integer is rational, using the denominator 1. Then explain why not every rational number is an integer. R
7. A student says "$\pi = \frac{22}{7}$, so π is rational." Explain the error, and state whether $\frac{22}{7}$ is greater or less than π. R
8. Extension: the number $0.121212\ldots$ repeats. Show that it equals $\frac{4}{33}$ by writing it as a fraction. E
9. Extension: explain why $\sqrt{2}$ cannot be a terminating decimal. Use the fact that squaring a terminating decimal cannot produce exactly 2. E

8.4 Sets in surveys and decisions

CRITERION A · II, III ◆ SYSTEMS ◆ FAIRNESS AND DEVELOPMENT

Survey data often describes overlapping groups. The overlap is the most important region: it determines how many people are counted twice if the groups are simply added, and it changes what a "combined" total actually means.

A two-group survey can be summarised by four counts: only A, both, only B, and neither. If any three of the four are known, the fourth follows from the total. The addition rule connects the group totals to the union. In a decision, the union measures how many people use at least one service, while the intersection measures how many use both.

THE FOUR REGIONS OF A TWO-SET SURVEY

Region	Set notation	Meaning in a survey
Only A	$A \setminus B$	Belongs to A but not to B.
Both	$A \cap B$	Belongs to A and to B.
Only B	$B \setminus A$	Belongs to B but not to A.
Neither	$(A \cup B)'$	Belongs to neither group.

WORKED EXAMPLE 8.4.1 — A CLUB SURVEY

PROBLEM

Of 32 students, 19 belong to the drama club and 15 belong to the music club. Eight students belong to both. (a) Complete a Venn diagram. (b) How many belong to exactly one club? (c) How many belong to neither? (d) What percentage belong to at least one club?

1. (a) Drama only: $19 - 8 = 11$. Music only: $15 - 8 = 7$. Both: 8. Neither: $32 - (11 + 8 + 7) = 6$.
2. (b) Exactly one club is $11 + 7 = 18$ students.
3. (c) Neither is 6 students. Check: $11 + 8 + 7 + 6 = 32$.
4. (d) At least one is $11 + 8 + 7 = 26$, or $19 + 15 - 8 = 26$.
5. The percentage is $\frac{26}{32} \times 100 = 81.25\%$.

ANSWER **(a) 11 drama only, 8 both, 7 music only, 6 neither (b) 18 (c) 6 (d) 81.25%.**

Use the addition rule first. $n(D \cup M) = 19 + 15 - 8 = 26$. Then neither is $32 - 26 = 6$, and the "only" counts follow by subtracting the overlap from each club total.

WORKED EXAMPLE 8.4.2 — FROM A DESCRIPTION TO A DIAGRAM

PROBLEM

A leisure centre has 60 members. 27 use the pool, 22 use the gym and 9 use both. (a) Draw a Venn diagram. (b) How many use exactly one facility? (c) What percentage use neither? (d) The centre claims "two thirds of members use at least one facility." Is the claim accurate?

1. (a) Pool only: $27 - 9 = 18$. Gym only: $22 - 9 = 13$. Both: 9. Neither: $60 - (18 + 9 + 13) = 20$.
2. (b) Exactly one is $18 + 13 = 31$ members.
3. (c) Neither is 20, so the percentage is $\frac{20}{60} \times 100 = 33.3\%$ (to one decimal place).
4. (d) At least one is $18 + 9 + 13 = 40$, and $\frac{40}{60} = \frac{2}{3}$. The claim is accurate.

ANSWER **(a) 18 pool only, 9 both, 13 gym only, 20 neither (b) 31 (c) 33.3% (d) yes, exactly two thirds.**

Check with the rule. $n(P \cup G) = 27 + 22 - 9 = 40$, so neither is $60 - 40 = 20$. The four regions total 60, confirming the diagram.

TWO DIFFERENT QUESTIONS

"How many use at least one facility?" counts the union, overlap included. "How many use exactly one?" counts the two outer regions and excludes the overlap. The leisure centre answers are 40 and 31, so one diagram supports two different totals; name the region before calculating.

The "neither" count is the complement of the union: $n((A \cup B)') = n(\xi) - n(A \cup B)$. It is the region outside both circles, and it is the one most often omitted when a diagram is copied in a hurry.

MAKING THE UNIVERSAL SET VISIBLE

Before drawing, decide what ξ represents. In a survey it is the group questioned, not the whole population; changing ξ changes every complement and every "neither" count. Write $n(\xi)$ beside the rectangle so the check is visible to a reader.

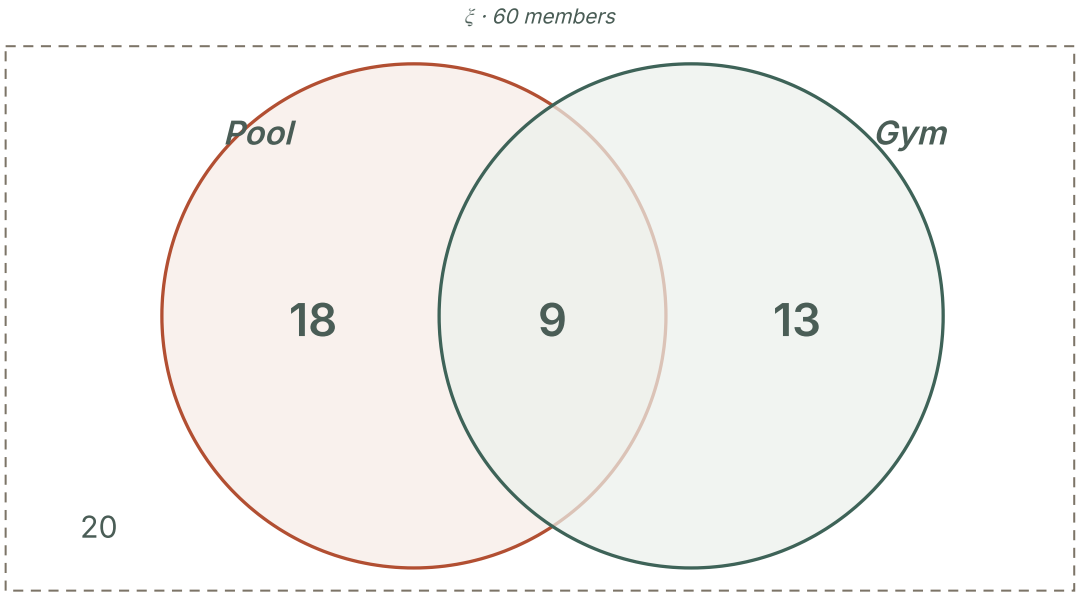


Figure 8.4 The leisure-centre data from Worked example 8.4.2; 40 members use at least one facility.

COMMON ERROR

Answering "how many use at least one facility?" with $18 + 13 = 31$, which is the "exactly one" count and leaves out the 9 who use both. The union must include every region inside the circles. A second error is reporting the intersection as a percentage of the union instead of the whole group; always state the denominator.

GUIDED PRACTICE · COPY AND COMPLETE

A survey of 40 people finds 25 read newspapers, 18 read magazines and 10 read both. Newspaper only is $25 - \square = \square$; magazine only is $18 - \square = \square$; at least one is $\square + \square + \square = \square$; neither is $40 - \square = \square$. The percentage who read neither is $\frac{\square}{40} \times 100 = \square\%$.

PRACTICE 8.4 · SURVEYS AND DECISIONS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Draw a Venn diagram for each two-group problem, then write a sentence explaining what each region means in the context.

1. Of 45 students, 28 walk to school, 21 cycle and 12 do both. Find the number who walk only, cycle only, do at least one and do neither. F
2. In a survey of 80 households, 52 have a garden, 37 have a balcony and 24 have both. Find the number with neither. F
3. A club has 36 members: 20 play tennis, 17 play badminton and 8 play neither. Find the number who play both. F
4. In a year group of 120 students, 74 study a language, 61 study a humanity and 38 study both. How many study at least one? How many study neither? P
5. A leisure centre has 150 members: 96 use the pool, 64 use the gym and 45 use both. The manager claims that more than half the members use exactly one facility. Test the claim with counts and a percentage. P
6. A student writes "neither = total − only A − only B". Explain why this is wrong, and write the correct expression. R

7. Extension: 50 people were asked about two flavours. 34 like vanilla, 29 like chocolate and 7 like neither. Find how many like both, and how many like vanilla but not chocolate. E
8. Extension: a survey reports only that 30 people use service A and 25 use service B. The total number surveyed is 60. Find the smallest and largest possible number who use both, and explain your reasoning. E

Investigation 8 · How much overlap is possible?

CRITERION B · I, II

A committee has 30 students. Seventeen help with the summer fair, 14 help with the concert and 6 help with both. The overlap is stated, so the diagram is determined. The more interesting question is what happens when the overlap is not stated: the two totals and the overall total may still restrict it.

COMMITTEE MEMBERSHIP

Group	Number of students
Help with the summer fair (F)	17
Help with the concert (C)	14
Help with both ($F \cap C$)	6
Total in the committee (ξ)	30

1. Draw a Venn diagram for the stated data and complete all four regions. Check that the four numbers sum to 30.
2. How many students help with at least one event? How many help with exactly one event?
3. A student calculates $17 + 14 = 31$ and says "there must be one more student than we thought." Explain what the student has counted twice and why the correct union is smaller than 31.
4. Now suppose only the two totals 17 and 14 and the total 30 are known, but the overlap is not stated. Let $x = n(F \cap C)$. Write an expression for $n(F \cup C)$ in terms of x.
5. Use $n(F \cup C) \leq 30$ to find the smallest possible value of x. Explain why $x = 0$ is impossible.
6. Find the largest possible value of x. Explain why it cannot exceed 14, and describe the Venn diagram when x takes this value.
7. The committee needs 20 helpers for a combined event. Using the stated overlap of 6, decide whether there are enough helpers. Then decide whether the answer could change if the overlap were different but still possible.
8. Extension: the committee adds a third event with 11 helpers. Explain why a two-set diagram is no longer sufficient, and list one extra piece of information you would need before drawing a three-set diagram.

A useful generalisation states the allowed range of the overlap, not only one value. Justify each endpoint with an inequality or a diagram, then test your range on a second pair of totals.

Real-life application · The centre audit

A community centre has 80 members: 44 use the pool, 37 use the gym and 18 use both. A pool pass costs $4, a gym pass costs $3, and a combined pass costs $6. The centre needs accurate counts before it sets next year's budget.

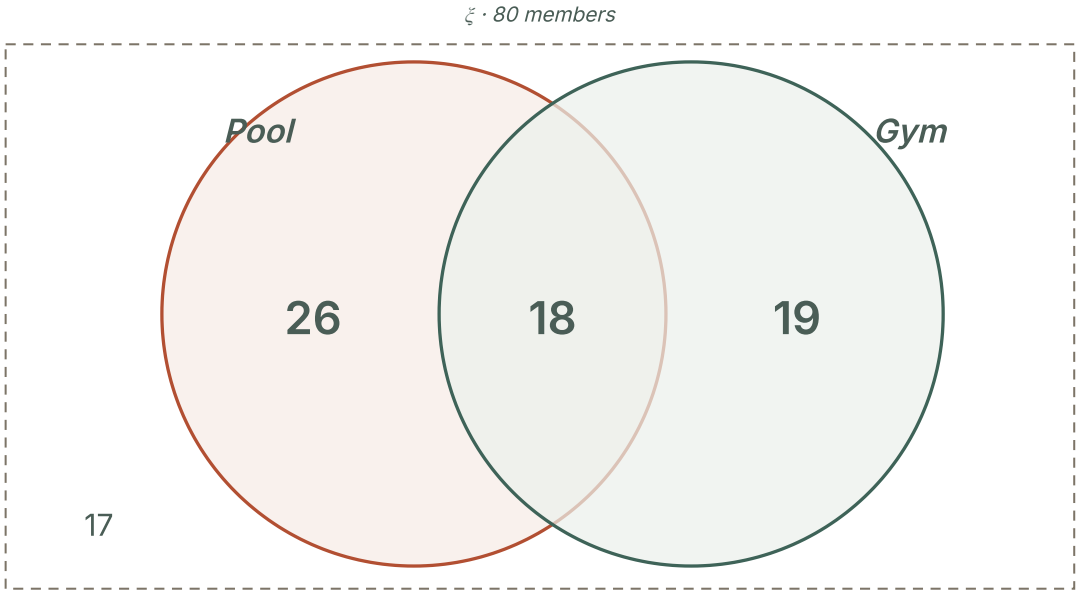


Figure 8.5 The audit data; the overlap is the group for whom a combined pass is relevant.

Your task. Use the diagram and set reasoning to advise the centre. Show every count and state the region you are using.

1. Find the number of members who use the pool only, the gym only, at least one facility and neither.
2. Check your four regions by adding them. Explain why the sum must equal 80.
3. Calculate the income from single-visit passes if every member who uses exactly one facility buys the pass for that facility, and every member who uses both buys the combined pass.
4. What percentage of members use at least one facility? What percentage use neither? Round sensibly and state the denominator.
5. The centre claims "more than a third of our members use the gym." Is the claim true if "use the gym" includes the members who use both? Is it true if it means "gym only"? Explain the difference.
6. **Consider.** A member who uses both facilities pays $6, while two single passes would cost $7. Is the combined pass fair? Give one argument for the member and one for the centre, then recommend a pricing rule.
7. **Evaluate.** The centre wants to increase income without raising prices. Suggest one change to the membership categories and one piece of additional data it should collect before deciding.

Reflect. Where did a Venn diagram make a count easier than a list? Where did the diagram still require a written explanation to avoid ambiguity?

Criterion-mapped mini-task

A · Knowing and understanding
C · Communicating
D · Applying mathematics

Context. A youth centre runs art and coding workshops. It has 48 registered students: 31 are in the art workshop, 27 are in the coding workshop and 14 are in both. The centre needs a one-page guide that explains who can attend each workshop, who can attend both and who is in neither.

Product. A one-page access guide containing a labelled Venn diagram, a set-notation summary, three worked counts and a short "how to check" section. It must be usable by a receptionist who has not studied set theory.

Requirements.

1. Define the universal set and the two workshop sets. State $n(\xi)$, $n(A)$, $n(C)$ and $n(A \cap C)$.
2. Draw a Venn diagram with all four regions completed. Check that the regions total 48.
3. Explain, in one sentence for each, what $A \cup C$, $A \cap C$, A' and $A \setminus C$ mean in this context.
4. Calculate the number of students in exactly one workshop and the percentage in both. Round the percentage to one decimal place.
5. Write a rule for a student who wants to join both workshops, including the condition on places if each workshop has a capacity of 30.
6. Add two "check it yourself" questions with answers at the foot of the guide. One must ask for a complement; the other must ask for an "only" count.
7. Test the guide on a reader. Record one place where the reader hesitated and the change you made.
8. **Evaluate.** The centre wants to classify students by year group as well as workshop. Explain why a two-set diagram would then be insufficient, and describe what extra representation would be needed.

CRITERIA AND STRANDS A i, ii, iii: correct set notation, counts and classification; C i, ii, iii: labelled diagram, precise vocabulary and a clear rule; D i, ii, iii, iv: apply set reasoning to access, check consistency and evaluate the guide.

BEFORE YOU SUBMIT

Check that the circle labels name whole sets, not "only" regions. Verify that $17 + 14 + 13 + 4 = 48$. Every percentage must state its denominator. A reader should be able to answer "Can I attend both?" without interpreting set symbols, so include a plain-language sentence beside each symbol.

Chapter review

SUMMARY

— **Sets and elements.** A set is a well-defined collection. $x \in A$ means x belongs; $n(A)$ counts the elements; $\varnothing$ is the empty set.

— **Subsets.** $A \subseteq B$ when every element of A is in B. A set with n elements has 2^n subsets.

— **Operations.** $A \cup B$ is "either or both"; $A \cap B$ is "both"; A' is "not in A" relative to ξ; $A \setminus B$ is "in A but not B".

— **Counting.** $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. The overlap is counted twice if the group totals are added directly.

— **Number sets.** $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. Irrational numbers are real but not rational.

— **Surveys.** The four regions of a two-set diagram are only A, both, only B and neither; they sum to $n(\xi)$.

Belonging: $x \in A$, $x \notin A$.

Subset: $A \subseteq B$; proper subset $A \subset B$.

Union and intersection: $A \cup B$, $A \cap B$.

Complement: $A' = \{x \in \xi : x \notin A\}$.

Addition rule:
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Subset count: a set with n elements has 2^n subsets.

REVISION CHECKLIST

- ☐ I can use $\in$, $\notin$, $\subseteq$ and $\varnothing$ correctly.
- ☐ I can list elements from a rule and write a rule from a list.
- ☐ I can complete a two-set Venn diagram.
- ☐ I can calculate union, intersection and complement counts.
- ☐ I can classify numbers into $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$.
- ☐ I can solve and explain a two-group survey problem.

KEY VOCABULARY

set
A well-defined collection of distinct objects.

element
An object that belongs to a set.

roster
A set written as a list inside braces.

set-builder
A set written by a rule, using a colon for "such that".

cardinality
The number of elements in a finite set.

empty set
The set with no elements, $\varnothing$.

universal set
The set of all objects under discussion, ξ.

subset
A set whose elements all belong to another set.

union
Elements in either set or both.

intersection
Elements in both sets.

complement
Elements of the universal set not in the set.

disjoint
Sets with no common element.

rational
A number that can be written as a fraction of integers.

irrational
A real number that is not rational.

REFLECTION AND SELF-ASSESSMENT

Find a club, team, class or online community that you belong to. Identify another group that overlaps with it. Describe one person in each of the four regions — only your group, both, only the other group, neither — without using anyone's name.

Rate your confidence on each success criterion, name one set operation that still needs practice, and explain in two sentences why an overlap must be subtracted when two group totals are combined.

END-OF-CHAPTER PRACTICE

Mixed practice 8

Use set notation where it makes the reasoning clearer. Do not use a calculator unless a question asks for a percentage.

FLUENCY Build accuracy

1. List the elements of $\{x \in \mathbb{N} : x < 12,\ x \text{ is a multiple of } 4\}$.
2. State $n(\{a, b, c, d, e\})$ and list two of its subsets with exactly three elements.
3. For $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 3, 5, 8\}$, find $A \cap B$ and $A \cup B$.
4. If $\xi = \{1, 2, \ldots, 10\}$ and $A = \{2, 4, 6, 8, 10\}$, find A'.
5. Classify $-\frac{7}{2}$ and $\sqrt{49}$ into every number set that contains them.
6. Which of these are irrational: π, $\sqrt{25}$, $0.\overline{6}$, $\sqrt{3}$, -8?
7. In a group of 24, 15 like swimming, 12 like cycling and 7 like both. Find the number who like neither.
8. If $n(A) = 18$, $n(B) = 13$ and $n(A \cup B) = 25$, find $n(A \cap B)$.
9. Write 0.45 as a fraction in simplest form and state whether it is rational.
10. Write the set of integers between -3 and 3 inclusive that are not natural numbers.
11. How many subsets does a set with 4 elements have? How many are proper?
12. Explain why $\varnothing \subset \{0\}$ is true but $0 \in \varnothing$ is false.

REASONING AND PROBLEM SOLVING Connect ideas

13. Two sets are disjoint. $n(A) = 12$, $n(B) = 9$ and 6 elements are in neither. Find $n(\xi)$.
14. In a survey of 90 students, 48 study history, 41 study geography and 22 study both. Find the number who study exactly one of the two subjects.
15. A student claims that if $n(A) = n(B)$, then $A = B$. Use a counter-example to show the claim is false, then state what can be concluded.
16. **The library audit (Criteria A, C, D).** A library has 200 members. 124 borrow fiction, 98 borrow non-fiction and 57 borrow both. (a) Draw a Venn diagram. (b) How many borrow neither? (c) What percentage borrow fiction but not non-fiction? (d) The librarian wants to order more of the less-used category. Which count should inform the decision, and why?
17. **The timetable overlap (Criteria A, C, D).** A school has 36 students in the orchestra and 29 in the choir. At least 8 students are in both. (a) What is the largest possible number in the orchestra or choir? (b) What is the smallest? (c) If the school has 55 students in these year groups, is it possible that every student is in at least one group? Justify your answer.
18. **The number-line sort (Criteria A, C).** A set contains $\sqrt{4}$, $\sqrt{5}$, $\frac{7}{3}$, 2.5, -1, 0 and π. Sort the numbers into rational and irrational, then explain why the rational numbers are not all integers.
19. **The fair-access proposal (Criteria C, D).** A centre proposes a rule: “Members who use both facilities pay the combined pass; everyone else pays a single pass.” Write one advantage and one disadvantage of the rule, using counts or percentages, and propose one modification that would make it fairer.
20. **The three-group question (Criteria B, C).** A survey has three groups A, B and C. Explain why a two-set Venn diagram cannot represent all possible overlaps, and name two pieces of information that a three-set diagram would need that a two-set diagram does not.

Check your progress

ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

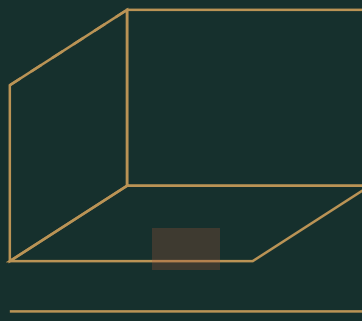
- $\in$ **and** $\subseteq$ **are different.** An element belongs to a set; a set is contained in another set.
- **The empty set is not the set containing zero.** $\varnothing$ has no elements; $\{0\}$ has one.
- **Rosters ignore order and repetition.** List each distinct element once.
- **Subtract the overlap before writing "only".** A circle total includes the overlap; an "only" region does not.
- **Complement depends on the universal set.** Always identify ξ before finding A'.
- **Not every decimal is rational.** Only terminating or repeating decimals can be written as fractions of integers.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can use set notation and list elements.			
I can decide subset relationships.			
I can complete and read a Venn diagram.			
I can classify numbers into number sets.			
I can solve two-group survey problems.			
I can explain a count with set reasoning.			

WHERE THIS LEADS

Chapter 9 measures the shapes you have classified. Perimeter, area and volume are quantities, so each must carry a unit; the set language of this chapter helps you separate the boundary, the surface and the space inside. **Chapter 10** returns to groups and categories through data, where clear classification is the first defence against a misleading graph.

Review the answer key only after attempting each question.



9

MEASURING THE SPACE WE USE

Perimeter, area, surface area and volume

A boundary has length, a surface has area and a solid encloses space. Each quantity needs its own formula, its own unit and its own check. Measuring well is therefore as much about choosing the right question as about calculating accurately.

KEY CONCEPT · FORM

RELATED · MEASUREMENT

RELATED · SPACE

CONTEXT · GLOBALIZATION AND SUSTAINABILITY

INQUIRY

A community centre is redesigning its courtyard. It must buy turf for the lawn, paving for a semicircular stage, fencing for the boundary, soil for three raised beds and paint for their outsides. Which measurements does each purchase require, and how can the centre avoid paying for material it does not need?

MYP UNIT PLANNER

Framing the inquiry

UNIT 9 · APPROX. 16 HOURS
ATL FOCUS · RESEARCH SKILLS

KEY AND RELATED CONCEPTS

Form. The shape of an object determines which measurements are needed and how they combine. A rectangle, a prism and a composite courtyard each have a different form.

Measurement. Length, area and volume are different quantities; the unit records which one has been measured.

Space. Decomposing a shape into familiar parts turns an unfamiliar space into a calculation that can be checked.

GLOBAL CONTEXT AND EXPLORATION

Globalization and sustainability. Exploration: how measurement supports responsible decisions about materials, packaging, land and water.

A courtyard, a carton and a water tank raise the same question: how much material is needed for the surface, and how much space is available inside? Accurate measurement reduces waste.

STATEMENT OF INQUIRY

Form determines how length, area and volume are measured, and measuring accurately supports responsible use of materials and space.

FACTUAL

What is the difference between perimeter, area, surface area and volume? Which units are appropriate for each?

CONCEPTUAL

Why does doubling every dimension multiply volume by eight but surface area by only four? How does decomposition simplify an unfamiliar shape?

DEBATABLE

Should packaging use the least material even when the result is less attractive? Who pays for unnecessary packaging — buyer, seller or environment?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Select and use measurement formulas; convert units correctly; classify measurements by dimension.
B · Investigating patterns	i, ii	Compare the surface areas of cuboids with the same volume, record the pattern and justify a conjecture.
C · Communicating	i, ii, iii	Label diagrams with dimensions and units, show the decomposition of composite shapes, and explain each step of a calculation.
D · Applying mathematics	i, ii, iii, iv	Cost a courtyard project, decide quantities of turf, soil and paint, and evaluate a storage-crate design.

ATL · RESEARCH SKILLS IN ACTION

Two paint tins sit on a shelf. Tin A states “coverage 12 m^2 per litre”; tin B states “6 m^2 per litre, two coats required” and costs less per litre. A student says B is better value because it is cheaper. Explain what the student has not checked, and list three pieces of information needed before the comparison can be made fairly.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** why perimeter, area, surface area and volume are different quantities with different units.
- **Be able to** calculate the perimeter and area of rectangles, triangles, parallelograms, trapezoids and circles.
- **Be able to** break a composite shape into familiar parts and combine the results.
- **Be able to** draw and interpret nets, and calculate the surface area of cuboids and triangular prisms.
- **Be able to** calculate the volume of prisms and convert between cubic units, millilitres and litres.
- **Investigate** how the shape of a cuboid affects its surface area when the volume is fixed.

SUCCESS CRITERIA

- ☐ I can name the quantity being measured and state its unit.
- ☐ I can choose the correct formula and substitute the perpendicular dimensions.
- ☐ I can decompose a composite shape and account for every boundary edge.
- ☐ I can calculate surface area from a net or from a formula.
- ☐ I can calculate volume and convert capacity units accurately.
- ☐ I can evaluate a measurement-based design against a budget and a constraint.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate 14×6.
2. Convert 2.4 m to centimetres.
3. Calculate $0.5 \times 10 \times 7$.
4. Calculate 3.14×4.
5. Calculate $3^2 + 4^2$.
6. Calculate $1000 \div 8$.

Answers are in the Chapter 9 answer section. If more than two are incorrect, review multiplication of decimals, metric length conversions and the meaning of square units before continuing.

BEFORE YEAR 1

Perimeter and area of rectangles and squares; naming common 2D and 3D shapes; simple metric conversions.

YEAR 1 · THIS CHAPTER

Area of triangles, parallelograms, trapezoids and circles; composite shapes; nets and surface area of prisms; volume and capacity.

LATER

Arc length and sector area, cylinders, cones and spheres, Pythagoras' theorem, similarity and scaling, and optimisation.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Gardens, packaging and water tanks are sized before they are built or bought.	Which measurement does each decision require, and which unit makes it meaningful?	Form, measurement and space.	Which cuboid uses the least material for a fixed volume?	Labelled nets, dimensioned diagrams and calculations that carry units.	When is an estimate sufficient, and when must a measurement be exact?

9.1 Perimeter and area of triangles and quadrilaterals

CRITERION A · I, II ◆ MEASUREMENT ◆ FORM

Perimeter measures the distance around a boundary. Area measures the surface enclosed by that boundary. Both are calculated from lengths, but they answer different questions, and only area is measured in square units.

A length is measured in centimetres, metres or kilometres. A surface is measured by counting how many unit squares cover it, so its unit is a square: one square centimetre is the area of a square with side 1 cm, written $1\ \text{cm}^2$. A square metre is $100\ \text{cm} \times 100\ \text{cm} = 10\,000\ \text{cm}^2$, not $100\ \text{cm}^2$.

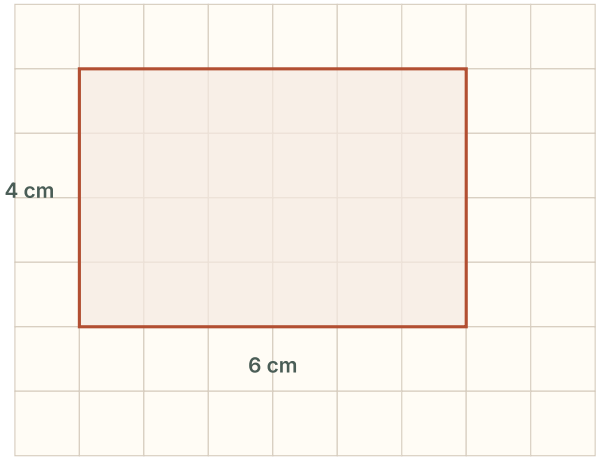


Figure 9.1 Counting squares gives the area of a rectangle. The perimeter counts the outer edges, not the grid lines inside.

For a rectangle of length l and width w, the area is $A = lw$. The perimeter is the total length of the four sides, so $P = 2(l + w)$. A square is a special rectangle with $l = w = s$, giving $A = s^2$ and $P = 4s$. The formulas work in any unit, provided every length is converted to the same unit first.

A triangle does not fill a rectangle completely. Two identical triangles can be joined along a diagonal to form a parallelogram, and that parallelogram can be rearranged into a rectangle with the same base and height. One triangle is therefore half of the rectangle: $A = \frac{1}{2}bh$. The height must be the **perpendicular height**, measured at right angles to the base, not the length of a sloping side.

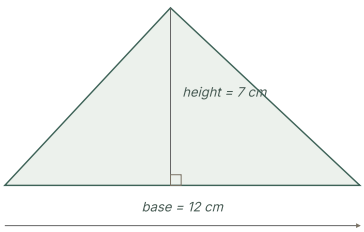


Figure 9.2 The perpendicular height is measured at $90°$ to the base. A sloping side is longer and cannot replace it in the area formula.

The same reasoning gives the area of a parallelogram. Cut a triangular piece from one end and move it to the other end; the shape becomes a rectangle with the same base and perpendicular height. Hence $A = bh$. Its perimeter, however, uses the sloping side as well as the base, so area and perimeter require different measurements from the same diagram.

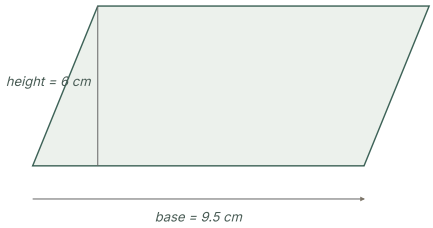


Figure 9.3 The area uses the base and the perpendicular height. The sloping side is needed only for the perimeter.

DEFINITIONS AND FORMULAS

Perimeter is the total length of the boundary of a shape, measured in linear units. **Area** is the amount of surface enclosed, measured in square units. **Perpendicular height** is the shortest distance from a vertex to the line containing the opposite side, meeting it at 90°. For a rectangle, $P = 2(l + w)$ and $A = lw$; for a triangle, $A = \frac{1}{2}bh$; for a parallelogram, $A = bh$.

WORKED EXAMPLE 9.1.1 A RECTANGULAR ALLOTMENT

PROBLEM

An allotment is a rectangle 8.4 m long and 3.6 m wide. (a) Find its perimeter. (b) Find its area. (c) Edging is sold in whole metres. How many metres of edging are needed to go once around the plot?

1. (a) $P = 2(l + w) = 2(8.4 + 3.6) = 2 \times 12 = 24$ m.
2. (b) $A = lw = 8.4 \times 3.6 = 30.24$ m^2.
3. (c) The perimeter is already 24 m, a whole number, so 24 m of edging is needed.
4. Check the units: perimeter is in metres, area in square metres. The numerical values are different because they measure different quantities.

ANSWER **(a)** 24 **m** **(b)** 30.24 m^2 **(c)** 24 **m.**

Check by splitting both lengths. $8.4 \times 3.6 = (8 + 0.4)(3 + 0.6) = 24 + 4.8 + 1.2 + 0.24 = 30.24$. The estimate $8 \times 4 = 32$ confirms that the answer is the right size.

WORKED EXAMPLE 9.1.2 TRIANGLE AND PARALLELOGRAM

PROBLEM

(a) A triangle has base 12 cm, perpendicular height 7 cm and other sides 8 cm and 9 cm. Find its area and perimeter. (b) A parallelogram has base 9.5 cm, perpendicular height 6 cm and sloping side 7 cm. Find its area and perimeter.

1. (a) Area $= \frac{1}{2}bh = \frac{1}{2} \times 12 \times 7 = 42$ cm^2.
2. Perimeter is the sum of the three sides: $12 + 8 + 9 = 29$ cm.
3. (b) Area $= bh = 9.5 \times 6 = 57$ cm^2.
4. Perimeter $= 2(9.5 + 7) = 2 \times 16.5 = 33$ cm. The height 6 cm is not a side and is not used in the perimeter.

ANSWER **(a)** 42 cm^2, 29 **cm** **(b)** 57 cm^2, 33 **cm.**

Check the triangle. A 12×7 rectangle has area 84 cm^2; the triangle is half of it, so 42 cm^2. The perimeter cannot be found from the area alone, because different triangles can have the same base and height but different sloping sides.

COMMON ERROR

Using a sloping side as the height of a triangle or parallelogram. The area formula needs the perpendicular height, which is shorter than any sloping side. A second error is adding the height to the perimeter: the height is an internal measurement unless it is also an edge. A third is writing cm instead of cm^2 for area; the unit must match the quantity.

GUIDED PRACTICE · COPY AND COMPLETE

A rectangle is 7 cm by 4 cm. Its perimeter is $2(7 + \square) = \square$ cm and its area is $7 \times \square = \square$ cm^2. A triangle has base 10 cm and perpendicular height 6 cm, so its area is $\frac{1}{2} \times \square \times \square = \square$ cm^2. A parallelogram has base 8 cm, height 5 cm and sloping side 6 cm; its area is $\square \times \square = \square$ cm^2 and its perimeter is $2(8 + \square) = \square$ cm.

PRACTICE 9.1 · PERIMETER AND AREA

SUPPORT 1–3 · CORE 1–5 · EXTENSION 6–8

Draw a labelled sketch before substituting. Write the unit with every answer and check that area answers use square units.

1. A rectangle is 15 cm long and 8 cm wide. Find its perimeter and area. F
2. A triangle has base 14 m and perpendicular height 9 m. Find its area. F
3. A parallelogram has base 11 cm, perpendicular height 7 cm and sloping side 9 cm. Find its area and perimeter. F
4. A square has perimeter 36 cm. Find its side length and area. P
5. A rectangle has area 96 cm^2 and length 12 cm. Find its width and perimeter. P
6. A student calculates the area of a triangle as $\frac{1}{2} \times 8 \times 6$, where 8 cm is the base and 6 cm is a sloping side. The perpendicular height is 4 cm. Explain the error and give the correct area. R
7. Extension: a rectangle is twice as long as it is wide and has perimeter 36 cm. Find its dimensions and area. E
8. Extension: two different triangles both have base 12 cm and height 9 cm. Explain why their areas are equal even though their perimeters may differ. E

9.2 Circles and composite shapes

CRITERION A · II, III ◆ FORM ◆ SPACE

A circle has no straight sides, but its boundary and its interior are still measurable. Both depend on one length — the radius — through the constant π. Composite shapes extend the same idea: split an unfamiliar outline into familiar parts.

The **radius** r is the distance from the centre to the circle. The **diameter** d passes through the centre and is twice the radius. The **circumference** is the distance around the circle. For every circle, the ratio of circumference to diameter is the same number, $\pi \approx 3.14$ (or $\frac{22}{7}$ when a question asks for an exact-looking estimate).

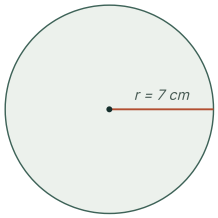


Figure 9.4 The radius starts at the centre. The diameter is twice the radius, and the circumference is π times the diameter.

> **CIRCLE AND TRAPEZOID FORMULAS**
>
> Circumference $C = \pi d = 2\pi r$. Area $A = \pi r^2$. For a standalone semicircle, the curved edge has length πr and the straight edge has length $2r$, so its perimeter is $\pi r + 2r$; its area is $\frac{1}{2}\pi r^2$. A trapezoid with parallel sides a and b and perpendicular height h has area $A = \frac{1}{2}(a + b)h$.

The area formula is often confused with the circumference formula because both contain π and r. The difference is dimensional: $2\pi r$ is a length, while πr^2 is an area. Squaring the radius is what turns the measurement from one dimension into two. A common check is to compare the numerical size: for $r = 7$, the circumference is about 44 cm and the area is about 154 cm^2. They are not interchangeable.

WORKED EXAMPLE 9.2.1 A CIRCULAR TABLETOP

PROBLEM

A circular tabletop has radius 7 cm in a scale model. Using $\pi = 3.14$, find (a) its circumference and (b) its area. (c) Show that the circumference divided by the diameter is approximately π.

1. (a) $C = 2\pi r = 2 \times 3.14 \times 7 = 43.96$ cm.
2. (b) $A = \pi r^2 = 3.14 \times 49 = 153.86$ cm^2.
3. (c) The diameter is 14 cm, and $43.96 \div 14 = 3.14$.
4. The answers are in different units because one measures a boundary and the other measures a surface.

ANSWER **(a)** 43.96 **cm** **(b)** 153.86 cm^2 **(c)** 3.14**.**

Use the diameter directly. $C = \pi d = 3.14 \times 14 = 43.96$ cm. For the area, halve the diameter first: $r = 7$ cm, then square it. Squaring the diameter by mistake would give an area four times too large.

WORKED EXAMPLE 9.2.2 A COURTYARD WITH A CURVED END

PROBLEM

A courtyard is a rectangle 12 m by 8 m with a semicircular stage attached to one 8 m end. The stage has diameter 8 m. Find (a) the total area and (b) the length of the outer boundary. Use $\pi = 3.14$.

1. (a) Rectangle area $= 12 \times 8 = 96$ m^2. Stage radius $= 4$ m, so stage area $= \frac{1}{2} \times 3.14 \times 16 = 25.12$ m^2.
2. Total area $= 96 + 25.12 = 121.12$ m^2.
3. (b) The outer boundary has two 12 m sides, one 8 m side and the semicircular arc. Arc length $= \pi r = 3.14 \times 4 = 12.56$ m.
4. Boundary $= 12 + 8 + 12 + 12.56 = 44.56$ m. The diameter is not part of the outer boundary because the stage is attached there.

ANSWER **(a)** 121.12 m^2 **(b)** 44.56 **m.**

Check the boundary. Trace the outline with a finger: top edge, left end, bottom edge, then the arc. The 8 m diameter is an internal join. For a standalone semicircle the diameter would be included, which is why the context matters.

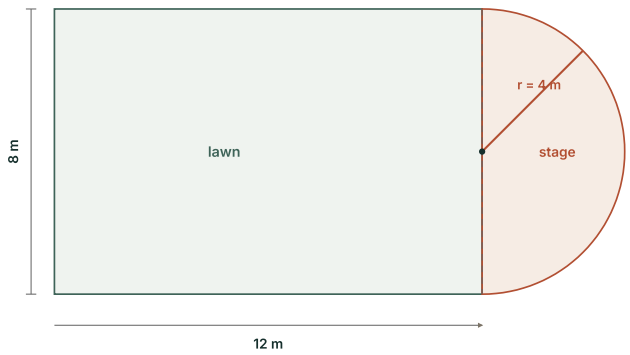


Figure 9.5 The courtyard from Worked example 9.2.2. Only the arc belongs to the outer boundary; the dashed diameter is where the two parts meet.

A trapezoid can be split into a rectangle and one or two triangles, or two copies can be rotated to form a parallelogram with base $a + b$. Both arguments give $A = \frac{1}{2}(a + b)h$. The formula averages the two parallel sides and multiplies by the perpendicular distance between them.

> **COMMON ERROR**
>
> Using the diameter in $A = \pi r^2$. The radius is half the diameter, so square the radius, not the diameter. For a standalone semicircle, forgetting the straight edge gives only the arc length, not the perimeter. Rounding π to 3 at the start can also change the final answer noticeably; keep 3.14 (or the calculator value) until the final step.

> **GUIDED PRACTICE · COPY AND COMPLETE**
>
> A circle has diameter 10 cm, so $r = \square$ cm, $C = 3.14 \times \square = \square$ cm and $A = 3.14 \times \square^2 = \square$ cm^2. A standalone semicircle has radius 6 cm: its area is $\frac{1}{2} \times 3.14 \times \square = \square$ cm^2, and its perimeter is $3.14 \times 6 + \square = \square$ cm. A trapezoid has parallel sides 5 cm and 9 cm and height 4 cm, so its area is $\frac{1}{2}(\square + \square) \times \square = \square$ cm^2.

PRACTICE 9.2 · CIRCLES AND COMPOSITE SHAPES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Use $\pi = 3.14$. Sketch each shape, mark the radius or perpendicular height, and state the unit of every answer.

1. A circle has radius 5 cm. Find its circumference and area. F
2. A circle has diameter 18 m. Find its circumference and area. F
3. A standalone semicircle has radius 3 cm. Find its area and perimeter. F
4. A trapezoid has parallel sides 7 cm and 13 cm and perpendicular height 6 cm. Find its area. P
5. A shape is made from a rectangle 10 m by 6 m with a semicircle of diameter 6 m attached to one short side. Find its total area and outer boundary length. P
6. A circle has area 78.5 cm^2. Using $\pi = 3.14$, find its radius and diameter. P
7. A student writes "area of a semicircle $= \pi r^2$". Explain the mistake and write the correct formula. R
8. Extension: a circular path surrounds a circular pond. The pond has radius 5 m and the outside of the path has radius 8 m. Find the area of the path. E

9.3 Nets and surface area

CRITERION A · I, II ◆ SPACE ◆ FORM

A solid has a surface, and that surface can be cut open and laid flat. The resulting net shows every face without overlap, so the surface area is simply the sum of the face areas. Folding the net back up checks that no face has been missed.

Surface area is the total area of all the faces of a solid, measured in square units. A cuboid has six rectangular faces in three identical pairs: top and bottom, front and back, and two ends. Cutting along some edges and unfolding produces its net.

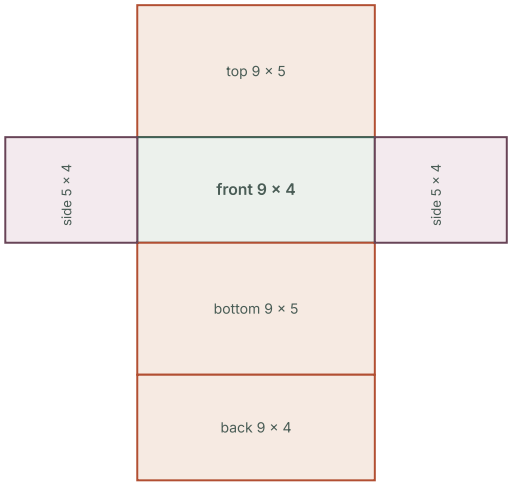


Figure 9.6 The net of a $9 \times 5 \times 4$ cuboid. Opposite faces are equal, so the six rectangles form three pairs.

Adding the six faces gives $SA = 2(lw + lh + wh)$. The factor 2 accounts for each pair. For a cube of side s, all six faces are squares, so $SA = 6s^2$. For a triangular prism, the net has two congruent triangular ends and three rectangles whose widths are the three sides of the triangle.

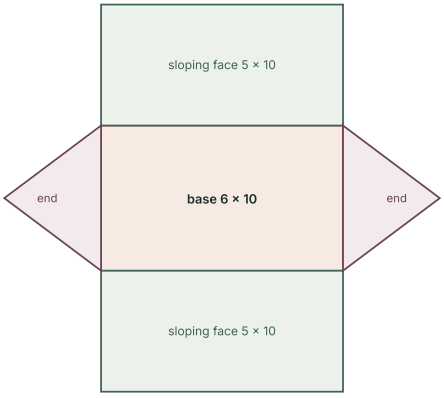


Figure 9.7 A triangular-prism net: two triangular ends plus three rectangular faces. The rectangles' widths are the sides of the triangle.

WORKED EXAMPLE 9.3.1 SURFACE AREA OF A CUBOID

PROBLEM

A box is a cuboid 9 cm long, 5 cm wide and 4 cm high. Find its surface area.

1. The three different faces have areas $9 \times 5 = 45$, $9 \times 4 = 36$ and $5 \times 4 = 20$ cm^2.
2. Each face appears twice, so $SA = 2(45 + 36 + 20)$.
3. $45 + 36 + 20 = 101$, and $2 \times 101 = 202$ cm^2.
4. Check against the net: two 45 cm^2 faces, two 36 cm^2 faces and two 20 cm^2 faces, six faces in total.

ANSWER 202 cm^2.

Add the net directly. $45 + 45 + 36 + 36 + 20 + 20 = 202$ cm^2. This is slower but makes it obvious why the formula doubles each product.

WORKED EXAMPLE 9.3.2 SURFACE AREA OF A TRIANGULAR PRISM

PROBLEM

A triangular prism has a cross-section with base 6 cm and perpendicular height 4 cm. The two equal sloping sides are 5 cm each, and the prism is 10 cm long. Find its surface area.

1. Area of one triangular end $= \frac{1}{2} \times 6 \times 4 = 12$ cm^2. There are two ends, so 24 cm^2.
2. The three rectangular faces have areas $6 \times 10 = 60$, $5 \times 10 = 50$ and $5 \times 10 = 50$ cm^2.
3. Total $= 24 + 60 + 50 + 50 = 184$ cm^2.
4. Check the net: two triangles and three rectangles, with no overlap and no missing face.

ANSWER 184 cm^2.

Use the perimeter of the cross-section. The three rectangles together have area $(6 + 5 + 5) \times 10 = 160$ cm^2. Adding the two ends gives $160 + 24 = 184$ cm^2.

COMMON ERROR

Calculating only three faces of a cuboid, or only one triangular end of a prism. A closed solid has every face on its surface, and opposite faces usually come in pairs. Another error is using the volume formula for surface area: lwh gives cubic units and describes the space inside, not the material around it.

GUIDED PRACTICE · COPY AND COMPLETE

A cuboid is 7 cm by 4 cm by 3 cm. The three face areas are ☐, ☐ and ☐ cm^2. Its surface area is $2(\square + \square + \square) = \square$ cm^2. A triangular prism has triangular ends with base 8 cm and height 3 cm, sloping sides 5 cm, and length 12 cm. Each end has area ☐ cm^2; the rectangles total $(\square + \square + \square) \times 12 = \square$ cm^2; the surface area is ☐ cm^2.

PRACTICE 9.3 · NETS AND SURFACE AREA

SUPPORT 1–3 · CORE 1–5 · EXTENSION 6–8

Sketch the net or list the faces before calculating. Every surface-area answer needs square units.

1. A cuboid is 6 cm by 4 cm by 2 cm. Find its surface area. F
2. A cube has side 5 cm. Find its surface area. F
3. A cuboid is 5 cm by 3 cm by h cm and has surface area 94 cm^2. Find h. P
4. A triangular prism has a cross-section with base 10 cm, height 12 cm and equal sides 13 cm. Its length is 8 cm. Find its surface area. P
5. A rectangular tank is 2 m long, 1.5 m wide and 1 m high. One litre of paint covers 5 m^2. How many whole litres are needed to paint all six faces? P
6. A student calculates the surface area of a cuboid as $2(lw + lh)$. Explain what has been left out and correct the formula. R
7. Extension: a box has no lid. Its base is 30 cm by 20 cm and its height is 15 cm. Find the area of cardboard needed, counting only the five faces. E
8. Extension: a cube has surface area 150 cm^2. Find its side length and its volume. E

9.4 Volume of prisms and capacity

CRITERION A · I, III ◆ MEASUREMENT ◆ SPACE

Volume measures the space inside a solid. For a prism, that space is a constant cross-section carried along a length, so one multiplication extends an area into three dimensions.

A **prism** has the same cross-section along its whole length. A cuboid is a prism whose cross-section is a rectangle; a triangular prism has a triangle as its cross-section. The volume of any prism is $V = A_{\text{cross-section}} \times l$, where l is the length between the two parallel faces. For a cuboid this becomes $V = lwh$, and for a cube $V = s^3$.

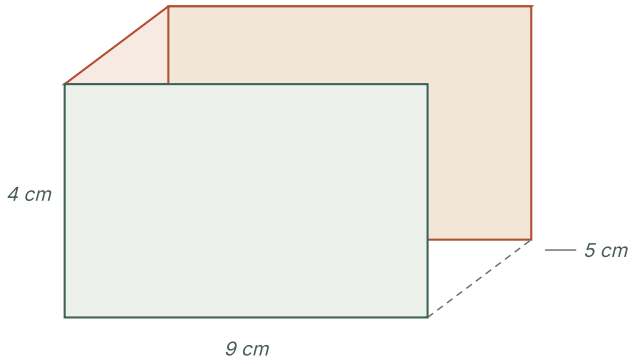


Figure 9.8 A $9 \times 5 \times 4$ cuboid. Its volume is the area of the 9×5 base multiplied by the height 4 cm.

Volume is measured in cubic units. A cube with side 1 cm has volume $1\ \text{cm}^3$, and $1\ \text{cm}^3$ holds exactly 1 millilitre of liquid. Capacity is the volume a container can hold, often quoted in millilitres or litres: $1\ \text{L} = 1000\ \text{mL} = 1000\ \text{cm}^3$. A cubic metre contains $100 \times 100 \times 100 = 1\,000\,000\ \text{cm}^3$, which is 1000 litres, not 100.

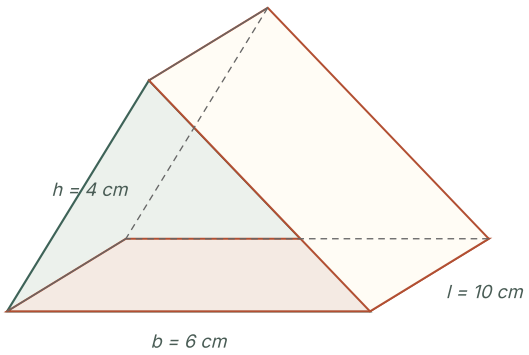


Figure 9.9 The triangular end is the cross-section. Its area is multiplied by the length of the prism.

WORKED EXAMPLE 9.4.1 — VOLUME AND CAPACITY OF A BOX

PROBLEM

A box is a cuboid 9 cm long, 5 cm wide and 4 cm high. (a) Find its volume. (b) Find its capacity in litres.

1. (a) $V = lwh = 9 \times 5 \times 4 = 180\ \text{cm}^3$.
2. (b) $180\ \text{cm}^3 = 180\ \text{mL}$, because $1\ \text{cm}^3 = 1\ \text{mL}$.
3. $180\ \text{mL} = 0.18\ \text{L}$, because $1000\ \text{mL} = 1\ \text{L}$.
4. The box holds 0.18 litres, a little less than a small glass.

ANSWER **(a)** $180\ \text{cm}^3$ **(b)** 0.18 **L.**

Check with the base. The base area is $9 \times 5 = 45\ \text{cm}^2$, and the height is 4 cm, so $V = 45 \times 4 = 180\ \text{cm}^3$. This is the same calculation read as a prism.

WORKED EXAMPLE 9.4.2 — A TRIANGULAR WATER TROUGH

PROBLEM

A trough is a triangular prism with cross-section base 6 cm and perpendicular height 4 cm; its length is 10 cm. (a) Find its volume. (b) Find its capacity in millilitres. (c) How much water is in it when it is 75% full?

1. (a) Cross-section area $= \frac{1}{2} \times 6 \times 4 = 12\ \text{cm}^2$, so $V = 12 \times 10 = 120\ \text{cm}^3$.
2. (b) $120\ \text{cm}^3 = 120\ \text{mL} = 0.12\ \text{L}$.
3. (c) 75% of 120 mL is $0.75 \times 120 = 90\ \text{mL}$.
4. The sloping sides are 5 cm, but they are not needed for the volume; only the cross-section area and the length matter.

ANSWER **(a)** $120\ \text{cm}^3$ **(b)** 120 **mL** **(c)** 90 **mL.**

Check by halves. The triangular face is half of a 6×4 rectangle. A rectangular trough with the same base and height would hold $6 \times 4 \times 10 = 240\ \text{cm}^3$; half of that is $120\ \text{cm}^3$.

COMMON ERROR

Using a sloping side as the height of a triangular cross-section. A second error is mixing the dimensions of area and volume: a surface is in cm^2, a space in cm^3. A third is the conversion $1\ \text{m}^3 = 100\ \text{L}$; the correct value is 1000 L. When every length is doubled, surface area is multiplied by 4 but volume is multiplied by 8, because three dimensions have changed.

GUIDED PRACTICE · COPY AND COMPLETE

A cuboid is 7 cm by 4 cm by 3 cm, so $V = 7 \times \square \times \square = \square\ \text{cm}^3$. A triangular prism has cross-section base 8 cm, height 3 cm and length 12 cm; the cross-section area is $\square\ \text{cm}^2$, and $V = \square \times 12 = \square\ \text{cm}^3$. Since $1000\ \text{cm}^3 = 1\ \text{L}$, this capacity is $\square$ L.

PRACTICE 9.4 · VOLUME AND CAPACITY

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Write the formula first, then substitute. Use cm^3 or m^3 for volume and mL or L for capacity.

1. A cuboid is 12 cm by 5 cm by 4 cm. Find its volume. F

2. A cube has side 6 cm. Find its volume. F
3. A triangular prism has cross-section base 8 cm, height 5 cm and length 14 cm. Find its volume. F
4. A tank is 50 cm long, 30 cm wide and 40 cm high. Find its capacity in litres. P
5. Convert: (a) 2.5 L to cm^3; (b) 450 mL to cm^3; (c) $0.8\ \text{m}^3$ to litres. P
6. A cuboid 20 cm by 10 cm by 5 cm has a $10 \times 5 \times 5$ block removed from one corner. Find the volume remaining. P
7. A student says $1\ \text{m}^3 = 100$ L "because there are 100 cm in a metre." Explain the error and give the correct conversion. R
8. Extension: a prism has volume $360\ \text{cm}^3$ and cross-section area $30\ \text{cm}^2$. Find its length. Explain what happens to the volume if every dimension is doubled. E

Investigation 9 · The packaging problem

CRITERION B · I, II

A box manufacturer wants to use as little cardboard as possible. The volume is fixed at $24\ \text{cm}^3$, and every edge must be a whole number of centimetres. Different cuboids with the same volume may need very different amounts of cardboard.

CUBOIDS WITH VOLUME $24\ \text{CM}^3$

Length (cm)	Width (cm)	Height (cm)	Volume (cm^3)	Surface area (cm^2)
1	1	24	24	98
1	2	12	24	
1	3	8	24	
1	4	6	24	
2	2	6	24	
2	3	4	24	

1. Complete the surface-area column. Check the first row: $2(1 \times 1 + 1 \times 24 + 1 \times 24) = 98\ \text{cm}^2$.
2. Which cuboid uses the least cardboard? Which uses the most?
3. Describe the pattern in your results. How do the dimensions of the most efficient box compare with those of the least efficient box?
4. Explain why a long, thin box such as $1 \times 1 \times 24$ needs so much more cardboard than a compact box with the same volume.
5. Test the pattern with a volume of $27\ \text{cm}^3$. Compare $3 \times 3 \times 3$ with $1 \times 3 \times 9$ and $1 \times 1 \times 27$. Does the cube use the least cardboard?
6. Now design a box with volume $36\ \text{cm}^3$ and whole-centimetre edges. Find two possible designs, calculate their surface areas, and recommend one.
7. A manufacturer might not choose the box with the smallest surface area. Suggest two practical reasons, other than material cost, that could affect the design.
8. Extension: write a short recommendation to the manufacturer. State the pattern, support it with at least two calculations, and explain one limitation of your conclusion.

A conjecture needs more than one example. Show that the pattern holds for a second volume, then explain why bringing the dimensions closer together reduces the surface area.

Real-life application · The courtyard project

A community centre has a rectangular courtyard 18 m long and 12 m wide, with a semicircular stage of diameter 12 m on one short end. The lawn needs turf, the stage needs paving, the whole boundary needs fencing, and three raised beds need soil and paint. The budget is $6,000.

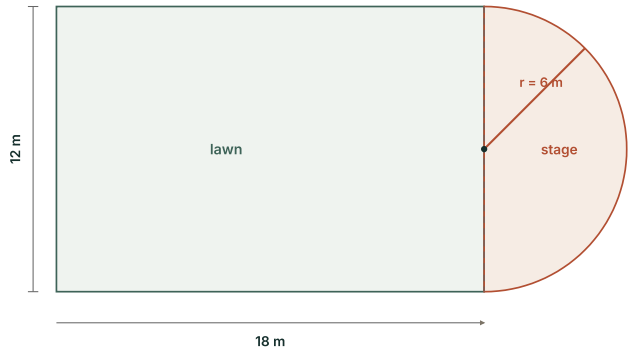


Figure 9.10 The courtyard plan. Turf covers the rectangle; paving covers the semicircle; fencing follows the outer boundary.

PRICES AND QUANTITIES

Item	Price	Coverage or size
Turf	$7.50 per m^2	Lawn only
Paving	$42 per m^2	Stage only
Fencing	$28 per m	Outer boundary
Soil	$9.50 per 40 L bag	Each bed filled to 80%
Paint	$22 per 1 L tin	1 L covers 9 m^2

Your task. Use $\pi = 3.14$ and show every calculation with its unit. Each raised bed is a cuboid 1.2 m long, 0.6 m wide and 0.5 m high, open at the top.

1. Find the area of the lawn, the area of the stage and the total area of the courtyard.
2. Find the length of fencing needed for the outer boundary. Explain why the 12 m diameter is not included.
3. Calculate the cost of turf, paving and fencing.
4. Find the volume and capacity of one raised bed. At 80% full, how many litres of soil does one bed need? How many 40 L bags are needed for three beds?
5. Find the outside surface area of one open-topped bed (five faces). How many litres of paint are needed for all three beds if one litre covers 9 m^2?
6. Add the five costs, including soil and paint. Is the project within the $6,000 budget? By how much is it over or under?
7. **Evaluate.** The centre wants a 10% contingency for cutting waste and breakages. Recalculate the total with 10% added. Suggest one change that would bring the project within budget without reducing the lawn area.

Reflect. Which purchase depended on a perimeter, which on an area and which on a volume? How did choosing the right quantity prevent an expensive mistake?

Criterion-mapped mini-task

A · Knowing and understanding
C · Communicating
D · Applying mathematics

Context. A workshop needs 12 open-topped storage crates. Each crate must hold 36 litres. Plywood costs $24 per square metre, and the workshop wants the design that uses the least plywood while remaining stable and easy to stack.

Product. A design proposal containing two possible crate designs, a net for the recommended design, surface-area and cost calculations, and a short evaluation.

Requirements.

1. State the required volume in cm^3 and in litres.
2. **Design A** has a base 40 cm by 30 cm. Find the height needed for a volume of 36 litres. Check your answer.
3. Draw a net for Design A. It has five faces: one base and four sides.
4. Calculate the surface area of Design A and the plywood cost for 12 crates.
5. **Design B** has a base 60 cm by 20 cm and the same height. Show that its volume is also 36 litres, then calculate its surface area and cost for 12 crates.
6. Which design uses less plywood? By how much, in square centimetres and in dollars, across 12 crates?
7. If a crate is filled to 90% of its capacity, how many litres does it hold?
8. **Evaluate.** Recommend one design. Use the surface-area evidence, one practical consideration such as stability or stacking, and one limitation of your recommendation.

CRITERIA AND STRANDS A i, ii, iii: correct volume, capacity and surface-area calculations with units; C i, ii, iii: labelled net, clear comparison and precise language; D i, ii, iii, iv: apply measurement to a real constraint, compare designs and evaluate the recommendation.

BEFORE YOU SUBMIT

Check that each design has volume $36\,000\ cm^3$. Count five faces, not six. Convert the plywood area to square metres before costing, and keep the dollar value to two decimal places. State clearly whether “least plywood” and “best design” are the same claim.

Chapter review

SUMMARY

- **Perimeter and area.** Perimeter is a length around a boundary; area is the surface inside, in square units. $P = 2(l + w)$ and $A = lw$ for a rectangle.
- **Triangles and parallelograms.** $A = \frac{1}{2}bh$ for a triangle and $A = bh$ for a parallelogram. The height is perpendicular to the base.
- **Circles.** $C = \pi d = 2\pi r$ and $A = \pi r^2$. A semicircle has area $\frac{1}{2}\pi r^2$; a standalone semicircle has perimeter $\pi r + 2r$.
- **Composite shapes.** Split into familiar parts, add or subtract areas, and trace only the outer boundary for perimeter.
- **Surface area.** Unfold to a net, or use $SA = 2(lw + lh + wh)$ for a cuboid. A prism has two ends plus rectangles.
- **Volume and capacity.** $V = A_{\text{cross-section}} \times l$. Convert with $1\ \text{cm}^3 = 1\ \text{mL}$, $1000\ \text{cm}^3 = 1\ \text{L}$ and $1\ \text{m}^3 = 1000\ \text{L}$.

Rectangle: $P = 2(l + w)$, $A = lw$.

Triangle: $A = \frac{1}{2}bh$.

Parallelogram: $A = bh$.

Trapezoid: $A = \frac{1}{2}(a + b)h$.

Circle: $C = 2\pi r$, $A = \pi r^2$.

Cuboid: $SA = 2(lw + lh + wh)$, $V = lwh$.

Prism: $SA = 2 \times \text{end} + (\text{perimeter of end}) \times l$, $V = \text{end area} \times l$.

Capacity: $1\ \text{cm}^3 = 1\ \text{mL}$, $1000\ \text{cm}^3 = 1\ \text{L}$, $1\ \text{m}^3 = 1000\ \text{L}$.

REVISION CHECKLIST

- ☐ I can name the quantity and choose its unit.
- ☐ I can use perpendicular heights correctly.
- ☐ I can calculate circumference and area of a circle.
- ☐ I can decompose a composite shape and trace its boundary.
- ☐ I can calculate the surface area of a cuboid and a triangular prism.
- ☐ I can calculate volume and convert capacity units.

KEY VOCABULARY

perimeter
The total length of a boundary.

area
The amount of surface enclosed, in square units.

perpendicular height
The shortest distance from a vertex to the opposite side.

radius
The distance from the centre of a circle to its edge.

diameter
The distance across a circle through its centre.

circumference
The distance around a circle.

composite shape
A shape made from simpler shapes.

net
A flat pattern that folds into a solid.

surface area
The total area of all faces of a solid.

prism
A solid with a constant cross-section along its length.

cross-section
The shape seen when a solid is cut parallel to its end.

volume
The space inside a solid, in cubic units.

capacity
The volume a container can hold, often in mL or L.

REFLECTION AND SELF-ASSESSMENT

Find a box, carton or container at home. Estimate its length, width and height, then calculate its volume and surface area. Check your estimate with a ruler.

Rate your confidence on each success criterion, name one formula you still need to check, and explain why every measurement needs a unit.

END-OF-CHAPTER PRACTICE

Mixed practice 9

Use $\pi = 3.14$. Draw a sketch, name the quantity, and carry units through every line of working.

FLUENCY Build accuracy

1. Find the perimeter and area of a rectangle 18 m by 7 m.
2. Find the area of a triangle with base 16 cm and height 9 cm.
3. Find the area of a parallelogram with base 13 cm and height 6 cm.
4. Find the area of a trapezoid with parallel sides 8 cm and 14 cm and height 5 cm.
5. Find the circumference and area of a circle with radius 4 cm.
6. Find the surface area of a cuboid 8 cm by 5 cm by 3 cm.
7. Find the volume of a cube of side 7 cm.
8. Find the volume of a triangular prism with cross-section base 9 cm, height 4 cm and length 15 cm.
9. Convert 3.2 L to cm^3 and 750 mL to L.
10. Find the area of a semicircle of radius 10 m.
11. A rectangle has area $84\ \text{cm}^2$ and width 7 cm. Find its length and perimeter.
12. Find the surface area of a triangular prism whose ends have area $18\ \text{cm}^2$ each and whose rectangular faces total $96\ \text{cm}^2$.

REASONING AND PROBLEM SOLVING Connect ideas

13. A path 1 m wide runs around a rectangular lawn 20 m by 12 m. Find the area of the path and explain how you decomposed the shape.
14. A circular pizza has diameter 30 cm. A square box has side 32 cm. Which area is larger, and by how much? Is the box large enough for the pizza? Explain.
15. **The paint estimate (Criteria A, C, D).** A wall is 8 m long and 2.5 m high, with a rectangular window 2 m by 1.2 m. One litre of paint covers 10 m^2 and two coats are needed. How many whole litres must be bought?
16. **The water tank (Criteria A, C, D).** A tank is a cuboid 1.2 m long, 0.8 m wide and 0.6 m high. (a) Find its capacity in litres. (b) It is 65% full. How many litres does it contain? (c) A family uses 150 L per day. How many full days will the water last?
17. **The packaging claim (Criteria C, D).** A company says a new box "uses 20% less cardboard" than an old box. The old box is 30 cm by 20 cm by 10 cm; the new box is 25 cm by 20 cm by 12 cm. Calculate both surface areas and test the claim.
18. **The garden design (Criteria B, C, D).** A garden is a rectangle 15 m by 9 m with a semicircular flower bed of diameter 9 m on one short side. (a) Find the lawn area. (b) Find the boundary length. (c) Turf costs \$8 per m^2. Estimate the turf cost and explain any assumption.
19. **The doubling effect (Criteria A, C).** A cuboid has dimensions 3 cm, 4 cm and 5 cm. Calculate its surface area and volume. Then double every dimension and calculate again. Describe how each result has changed and explain why.
20. **The open question (Criteria B, C).** Design two different prisms with a volume of $60\ \text{cm}^3$. One must have a rectangular cross-section and one a triangular cross-section. Calculate the surface area of each and explain which design uses less material.

Check your progress

ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

- **Area uses perpendicular height.** A sloping side is not a height.
- **Perimeter and area need different measurements.** The height may not be an edge.
- **Square the radius, not the diameter.** If the diameter is given, halve it first.
- **A semicircle perimeter includes the straight edge.** A composite boundary may not.
- **Surface area is not volume.** cm^2 measures a surface; cm^3 measures a space.
- **Convert before comparing.** 1 m^3 is 1000 L, not 100 L.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can identify the quantity and its unit.			
I can calculate perimeter and area.			
I can work with circles and composite shapes.			
I can use nets to find surface area.			
I can calculate volume and capacity.			
I can evaluate a design against a budget.			

WHERE THIS LEADS

Chapter 10 turns measurement into evidence. Tables, graphs and averages describe data, and the same care with units and scales prevents a misleading conclusion. Beyond Year 1, these measurement ideas extend to circles and sectors, cylinders, cones and spheres, and to similarity, where doubling a length has predictable effects on area and volume.

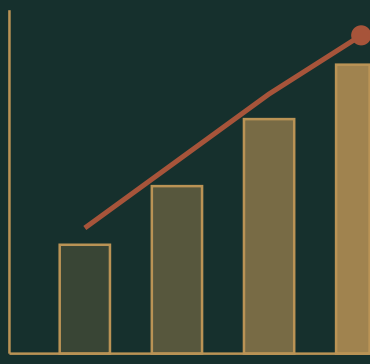
PART V · DATA AND DECISIONS

V

Reading the story in data

A graph can clarify a pattern or exaggerate a difference. A summary can describe a group or conceal its variation. This part develops the habits needed to read data critically: choose an appropriate display, calculate useful summaries, and ask how the design of a graph shapes the claim a reader is likely to make.





10

THE STORY BEHIND THE NUMBERS

data → claim

Data, tables and graphs

A table can organise a crowd of responses into a pattern. A graph can make that pattern visible in a glance. But every display is a set of choices, and those choices decide what a reader notices, what they overlook and what they are willing to believe.

KEY CONCEPT · RELATIONSHIPS

RELATED · REPRESENTATION

RELATED · PATTERN

CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

A school surveys 200 students about where they eat lunch. The newsletter reports that “most students buy school lunch”, while a second article claims that “off-site eating is out of control”. Both use the same survey. Which tables and graphs would let you check each claim, and what could a careless display hide?

MYP UNIT PLANNER

Framing the inquiry

UNIT 10 · APPROX. 16 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Relationships. A data set records how individuals vary. Tables, graphs and averages are ways of showing relationships between a variable and its frequency, or between two variables.

Representation. Every table or graph models the data and can reveal or distort a pattern through its scale and labels.

Pattern. Trends, clusters and outliers are patterns; describing them needs evidence, not first impressions.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Exploration: how data displays inform decisions about shared resources, and how a misleading display can make a claim seem stronger than the evidence.

Survey results about meals, travel or screen time justify budgets and rules. A table that hides a subgroup, or a graph that exaggerates a difference, can shift decisions away from the people affected.

STATEMENT OF INQUIRY

Data becomes evidence when it is represented accurately; the choice of table, graph or average shapes the conclusion a reader can draw.

FACTUAL

What are the main types of data? How is a frequency table built, and how is a sector angle calculated?

CONCEPTUAL

How does the choice of representation reveal or conceal a pattern? When does one average represent a group fairly?

DEBATABLE

Is a graph that is technically accurate but visually exaggerated misleading? Who is responsible for the interpretation — the creator or the reader?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii	Classify data, construct frequency and two-way tables, calculate averages and sector angles accurately.
B · Investigating patterns	i, ii	Compare how different class intervals change a histogram, describe the pattern and justify a choice.
C · Communicating	i, ii, iii	Label axes and units, choose an appropriate display, and explain a trend or a misleading feature precisely.
D · Applying mathematics	i, ii, iii, iv	Analyse survey data, cost a canteen decision, and evaluate a public claim against the evidence behind it.



ATL · CRITICAL THINKING IN ACTION

An advertisement states that a new phone has “twice the battery life” of the previous model and shows a bar twice as tall. List three questions you would ask before accepting the claim, and explain which feature of the graph could make the difference look larger than it is.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** the difference between categorical and numerical data, and between discrete and continuous measurements.
- **Be able to** organise data in frequency tables and two-way tables.
- **Be able to** construct and interpret bar charts, pie charts, dot plots, histograms and time-series graphs.
- **Be able to** calculate and compare the mean, median, mode and range.
- **Be able to** describe a trend and calculate a percentage change.
- **Investigate** how the choice of class interval changes the picture a histogram gives.

SUCCESS CRITERIA

- ☐ I can classify a variable and justify the classification.
- ☐ I can build a table with correct totals and relative frequencies.
- ☐ I can calculate sector angles and draw a labelled pie chart.
- ☐ I can choose an average and explain why it suits the data.
- ☐ I can describe a trend using values from a graph.
- ☐ I can identify how an axis, interval or missing context misleads.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate 25% of 80.
2. Write $\frac{3}{8}$ as a decimal and a percentage.
3. Calculate $360 \div 8$.
4. Find the mean of 4, 7 and 13.
5. List these numbers in order: 12, 5, 9, 5, 14.
6. Calculate the difference between 86 and 104.

Answers are in the Chapter 10 answer section. If more than two are incorrect, review fractions, percentages and the idea of an average before continuing.

BEFORE YEAR 1

Tally charts, simple bar charts, reading tables, and finding the mean of a small set of numbers.

YEAR 1 · THIS CHAPTER

Data types; frequency and two-way tables; bar charts, pie charts, dot plots and histograms; averages and range; trends and misleading graphs.

LATER

Scatter graphs and correlation, sampling and bias, probability from data, cumulative frequency and statistical inference.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Schools and news outlets use surveys to make claims about groups of people.	Which display communicates each claim honestly, and which display exaggerates it?	Relationships, representation and pattern.	How does the width of a class interval change the shape of a histogram?	Labelled axes, stated totals and a caption that names the pattern.	When is an average a fair summary, and when does it hide the people behind the number?

10.1 Data, variables and tables

CRITERION A · I, II ◆ RELATIONSHIPS ◆ REPRESENTATION

Data are not just numbers. A data set records a characteristic of each individual, and the type of that characteristic decides which table, graph and average are appropriate. Classifying the data correctly is the first step in any honest analysis.

A **data set** is a collection of observations. Each person or object described is an **individual**; each characteristic recorded is a **variable**. In a survey of 30 students, the individuals are the students and the variables might be year group, favourite fruit and journey time. A single row of a table describes one individual; a single column describes one variable.

Variables fall into two broad families. **Categorical** data place individuals in groups. **Numerical** data record a quantity. Categorical data may be **nominal**, with no natural order, such as transport mode, or **ordinal**, with a meaningful order, such as a satisfaction rating from "very dissatisfied" to "very satisfied". Numerical data may be **discrete**, taking separate countable values such as the number of siblings, or **continuous**, taking any value in a range such as height or journey time.

TYPES OF DATA WITH EXAMPLES

Type	Subtype	Example	Why
Categorical	Nominal	Preferred transport	Groups with no order.
Categorical	Ordinal	T-shirt size S, M, L	Groups with a clear order.
Numerical	Discrete	Number of pets	Countable whole-number values.
Numerical	Continuous	Mass of a school bag	Any value in a range, limited only by the measuring instrument.

The classification is about the variable, not the way it is written. A rating of 1 to 5 is often treated as ordinal categorical data because the numbers label ordered categories. A journey time recorded as "15 minutes" is still continuous data, even though the response has been rounded to a whole number. What matters is what the values represent.

WORKED EXAMPLE 10.1.1 CLASSIFYING VARIABLES

PROBLEM

Classify each variable and explain your reasoning. (a) Number of siblings. (b) Journey time to school. (c) Preferred mode of transport. (d) T-shirt size. (e) Height. (f) Number of books read last month.

1. (a) Numerical, discrete: the values are whole-number counts.
2. (b) Numerical, continuous: time can take any value in a range, even if it is recorded to the nearest minute.
3. (c) Categorical, nominal: the categories have no natural order.
4. (d) Categorical, ordinal: S, M and L have a clear order, but the gaps are not equal numerical amounts.
5. (e) Numerical, continuous: height can take any value within the range measured.
6. (f) Numerical, discrete: a count of whole books.

ANSWER **Discrete numerical: (a), (f). Continuous numerical: (b), (e). Nominal categorical: (c). Ordinal categorical: (d).**

Test with a question. Ask "Can the answer be counted, or only measured?" A count gives discrete data; a measurement gives continuous data. Ask "Is there a natural order?" If not, the categorical data are nominal.

A **frequency table** counts how often each response occurs. The **frequency** is the count for one category or value. The **relative frequency** is the frequency divided by the total number of observations; it can be written as a fraction, a decimal or a percentage. Relative frequencies make groups of different sizes comparable, because they measure a share of the whole rather than a raw count.

WORKED EXAMPLE 10.1.2 RELATIVE FREQUENCIES

PROBLEM

Thirty students were asked to name their favourite fruit. The results are shown in the table. (a) Complete the relative-frequency column as a fraction, a decimal and a percentage. (b) Check that the percentages total 100%. (c) What percentage chose apple or banana?

Fruit	Frequency	Relative frequency
Apple	12	
Banana	9	
Orange	6	
Mango	3	
Total	**30**	**1 · 100%**

1. Apple: $12 \div 30 = \frac{2}{5} = 0.4 = 40\%$.
2. Banana: $9 \div 30 = \frac{3}{10} = 0.3 = 30\%$.
3. Orange: $6 \div 30 = \frac{1}{5} = 0.2 = 20\%$. Mango: $3 \div 30 = \frac{1}{10} = 0.1 = 10\%$.
4. Check: $40 + 30 + 20 + 10 = 100\%$.
5. (c) Apple and banana together: $40\% + 30\% = 70\%$, or $(12 + 9) \div 30 = 21 \div 30 = 0.7$.

ANSWER **(a)** 40%, 30%, 20%, 10% **(b)** 100% **(c)** 70%.

Use the total row. Once three percentages are known, the fourth is $100\% - 40\% - 30\% - 20\% = 10\%$. This check catches an arithmetic slip before it reaches a graph.

When two categorical variables are recorded for the same individuals, a **two-way table** shows both at once. Each cell is a count for one combination of categories, and the row and column totals are called **marginal totals**. A two-way table is more informative than two separate tables because it allows comparisons within rows and within columns.

WORKED EXAMPLE 10.1.3 A TWO-WAY TABLE

PROBLEM

A survey of 100 students recorded year group and usual way of travelling to school. Year 7 (52 students): 18 by bus, 20 walk, 14 by car. Year 8 (48 students): 26 by bus, 10 walk, 12 by car. (a) Copy and complete the two-way table. (b) How many Year 8 students walk? (c) What percentage of all students travel by bus? (d) Compare walking in the two year groups. (e) A newsletter says "walking is equally common in both years". Test the claim.

	Bus	Walk	Car	Total
Year 7	18	20	14	52
Year 8	26	10	12	48
Total	44	30	26	100

1. (a) Bus total $= 18 + 26 = 44$; walk total $= 20 + 10 = 30$; car total $= 14 + 12 = 26$. The grand total is 100, which matches the survey size.
2. (b) Year 8 students who walk: 10.
3. (c) Bus travellers: $44 \div 100 = 0.44 = 44\%$.
4. (d) Year 7 walkers: $20 \div 52 \approx 0.385 = 38.5\%$. Year 8 walkers: $10 \div 48 \approx 0.208 = 20.8\%$.
5. (e) The claim is false. Walking is more common in Year 7 by about 17.7 percentage points, even though 30 students walk in total.

ANSWER **(a) totals 44, 30, 26, 100 (b) 10 (c)** 44% **(d)** 38.5% **vs** 20.8% **(e) false.**

Compare shares, not counts. Ten walkers is not directly comparable with a year group of 52 and a year group of 48. Dividing by each year-group total converts both counts into shares of the same kind.

COMMON ERROR

Confusing frequency with relative frequency. A frequency of 12 is not "12%" unless the total is 100; here it is 40%. A second error is comparing raw counts across groups of different sizes. A third is adding percentages from different groups as if they came from the same total. Always state the denominator behind a percentage.

GUIDED PRACTICE · COPY AND COMPLETE

A survey of 40 students records journey time. The frequencies for walk, bus, car and bike are 14, 11, 9 and 6. The total is ☐. The relative frequency for walk is $14 \div \square = \square = \square\%$. For bike it is $6 \div 40 = \square = \square\%$. The four percentages should add to ☐%. A two-way table of year group and journey time has row totals 22 and 18; if 9 Year 7 students walk and 5 Year 8 students walk, the total number of walkers is ☐.

PRACTICE 10.1 · DATA AND TABLES

SUPPORT 1–3 · CORE 1–5 · EXTENSION 6–8

For each variable, state whether it is categorical or numerical, and give the subtype. For tables, check every total.

1. Classify each variable: (a) number of pets; (b) favourite colour; (c) mass of a school bag; (d) number of goals scored; (e) month of birth; (f) a satisfaction rating from 1 to 5. F
2. A die is rolled 20 times: 3, 5, 2, 6, 3, 1, 4, 3, 5, 2, 6, 3, 4, 1, 5, 3, 2, 6, 4, 3. Make a frequency table and calculate each relative frequency as a percentage. F

3. In a survey of 25 students, 8 walk, 11 take the bus, 4 come by car and 2 cycle. Make a frequency table with relative frequencies. F
4. A two-way table records whether 60 students own a bicycle. There are 32 boys, of whom 24 own a bicycle; 15 of the 28 girls own one. Complete the table. What percentage of all students own a bicycle? P
5. Explain why "number of pages in a book" is discrete but "time spent reading" is continuous. P
6. A student adds the frequencies in a table and gets 39, but the survey recorded 42 responses. Explain what the student should check before changing any number in the table. R
7. A survey of 60 responses has 25% choosing "bus" and one fifth choosing "walk". How many chose each? How many chose neither? P
8. Extension: design a survey question about how students travel to school. Give four options that do not overlap and cover the likely answers. Explain why "walk or bike" and "bike" cannot both be options. E

10.2 Bar charts and pie charts

CRITERION A · I; C · I, II ◆ REPRESENTATION ◆ RELATIONSHIPS

A bar chart compares categories; a pie chart shows parts of one whole. Both use length or angle to represent a count, and both can mislead if their scale or labels are chosen carelessly. Choosing the right display begins with the question you want to answer.

In a **bar chart**, each category has a bar whose height is proportional to its frequency. The bars are separated by gaps because the categories are distinct, not points on a continuous scale. The vertical axis must start at zero: starting at another value makes a small difference look dramatic. A horizontal bar chart works in the same way, with the categories down the side; it is useful when category names are long.

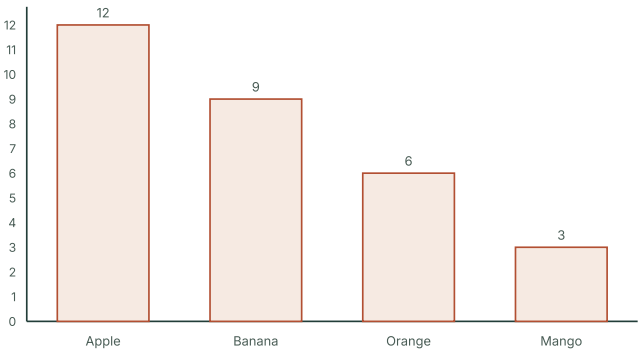


Figure 10.1 A bar chart of the 30 fruit choices from Worked example 10.1.2. The gaps separate the categories; the heights compare the counts.

To read a bar chart, first identify what each axis represents and check the scale. Then compare the bar heights and use the numbers on the frequency axis, not the area of the bars. A **clustered bar chart** places two or more bars side by side for each category, which allows two groups to be compared; a **stacked bar chart** places them on top of one another and is better for showing a total, though the lower segments are easier to compare than the upper ones.

A **pie chart** divides a circle into sectors whose angles are proportional to the frequencies. Since a full turn is $360°$, the angle for a category is $\frac{\text{frequency}}{\text{total}} \times 360°$. The same fraction multiplied by 100 gives

the percentage. A pie chart is most useful when the categories are parts of a single whole and there are only a few of them; with many similar-sized categories, the sectors become hard to compare.

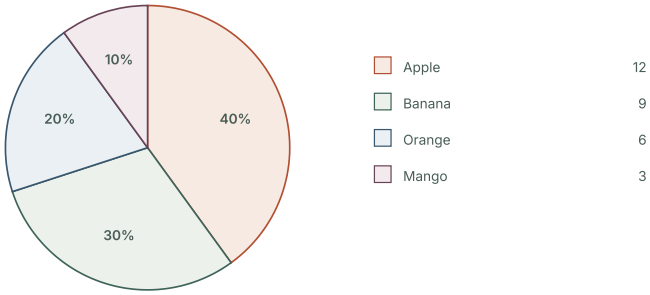


Figure 10.2 The same fruit data as a pie chart. Each sector shows a share of the 30 students; the legend gives the count for each category.

WORKED EXAMPLE 10.2.1 READING A BAR CHART

PROBLEM

Use Figure 10.1. (a) How many students chose apple? (b) Which fruit was chosen least often, and how many chose it? (c) How many more chose apple than mango? (d) What percentage chose banana? (e) Why are there gaps between the bars?

1. (a) The apple bar reaches 12, so 12 students chose apple.
2. (b) The mango bar is the shortest: 3 students.
3. (c) $12 - 3 = 9$ more students chose apple than mango.
4. (d) $9 \div 30 = 0.3 = 30\%$.
5. (e) The categories are separate groups, not values on a continuous scale; the gaps show that no category lies between them.

ANSWER **(a) 12 (b) mango, 3 (c) 9 (d)** 30% **(e) the categories are distinct.**

Check the total. $12 + 9 + 6 + 3 = 30$, which matches the survey size. A bar chart that does not match the stated total signals a counting or copying error.

WORKED EXAMPLE 10.2.2 SECTOR ANGLES

PROBLEM

The 30 fruit choices are to be shown in a pie chart. (a) Calculate the sector angle for each fruit. (b) Check that the angles total $360°$. (c) The same survey is repeated with 60 students, and every frequency doubles. What happens to the angles?

1. One student corresponds to $360° \div 30 = 12°$ of the circle.
2. Apple: $12 \times 12° = 144°$. Banana: $9 \times 12° = 108°$.
3. Orange: $6 \times 12° = 72°$. Mango: $3 \times 12° = 36°$.
4. Check: $144 + 108 + 72 + 36 = 360°$.
5. With 60 students, each angle is calculated from the same share: $24/60 = 40\%$, so apple is still 40% of $360° = 144°$. The angles are unchanged because only the total count has changed.

ANSWER **(a)** $144°, 108°, 72°, 36°$ **(b)** $360°$ **(c) they stay the same.**

Use percentages. Apple is 40% of the total, and $0.4 \times 360° = 144°$. This method is quicker when the percentages are already known and gives the same angle.

WHICH DISPLAY?

Use a bar chart to compare categories, a pie chart to show parts of one whole, a dot plot for a small set of discrete values, a histogram for grouped continuous data, and a line graph for change over time. The display should make the comparison easy without changing the data.

COMMON ERROR

Calculating a pie sector from the largest frequency instead of the total: every angle must use the same denominator. A second error is labelling a pie chart with counts but no total, so a reader cannot tell what the whole represents. A third is starting a bar-chart axis at a value other than zero; even a technically correct bar then exaggerates the differences.

GUIDED PRACTICE · COPY AND COMPLETE

A pie chart represents 40 students. The angle for "bus" is $90°$, so the fraction is $90 \div 360 = \square$ and the number of bus users is $\square \times 40 = \square$. If 8 students walk, their sector angle is $8 \div 40 \times 360° = \square°$. A bar chart of the same data must have a vertical axis starting at $\square$.

PRACTICE 10.2 · BAR CHARTS AND PIE CHARTS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Label every axis and state the total. When drawing a pie chart, show the angle calculation for at least one sector.

1. A bar chart shows 8, 12, 5 and 3 for categories A, B, C and D. Find the total, the most common category and the difference between the largest and smallest frequencies. F
2. Favourite sports in a class of 21: football 9, netball 6, swimming 4, athletics 2. Draw a bar chart and calculate the four pie-chart angles. F
3. A pie chart represents 20 students: 8 choose art, 5 music, 4 drama and 3 coding. Calculate each sector angle. F
4. A pie chart represents 40 people. One sector is 25% and another is $90°$. How many people are in each? How many are left for the remaining sectors? P

5. A sector represents 45% of a pie chart. What is its angle? P
6. A student draws a bar chart with the vertical axis starting at 5. The values are 10 and 5. Explain why the chart is misleading and how to redraw it. R
7. A pie chart represents 240 people and one sector has angle 60°. How many people does the sector represent, and what percentage is it? P
8. Extension: a pie chart must show five categories for 200 people. One category is 50%, and the other four are equal. Find the angle and count for each category. Explain why a bar chart might communicate the four small categories better. E

10.3 Dot plots, histograms and averages

CRITERION A · I, II ◆ PATTERN ◆ QUANTITY

Numerical data can be displayed value by value or grouped into intervals. A display shows the shape of the data; an average summarises it in one number. Neither replaces the other, because a single average can hide a cluster, a gap or an outlier.

A **dot plot** uses one dot for each observation, stacked above its value on a number line. It is practical for small sets of discrete data because every value remains visible. Clusters show where most values lie, gaps show missing ranges and an isolated dot far from the rest is a possible **outlier**. The same information can be recorded in a frequency table, with one row for each value.

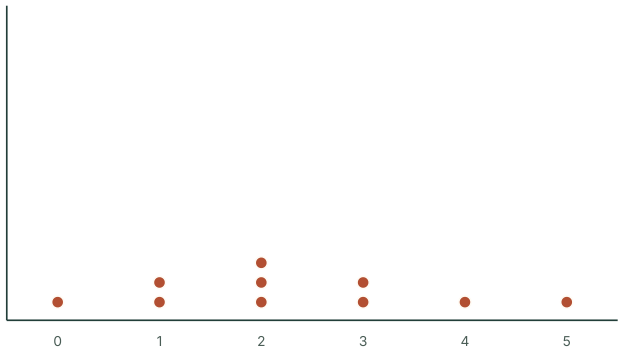


Figure 10.3 Goals scored in ten matches. The mode is 2, but the values spread from 0 to 5, so a single average would not describe every match.

When data are continuous, or when there are many different values, a **histogram** groups the data into **class intervals**. The bars touch because the intervals continue without gaps, and the height of each bar is the frequency of that interval. The intervals must have equal width; otherwise the areas of the bars cannot be compared fairly. A histogram shows the overall shape of the distribution but loses the individual values inside each interval.

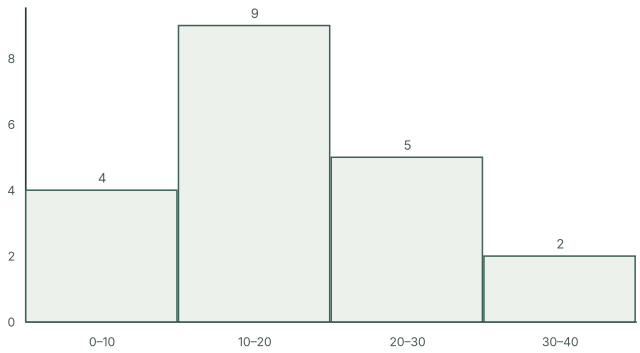


Figure 10.4 Journey times for 20 students. The bars touch because the time intervals are continuous; the tallest bar is the 10–20 minute interval.

An average is a single value used to summarise a set. The **mean** is the sum of the values divided by how many there are. The **median** is the middle value when the data are ordered; with an even number of values, it is the mean of the two middle values. The **mode** is the most frequent value, and the **range** is the largest value minus the smallest. The mean uses every value, the median is resistant to outliers, and the mode is the only average that works for categorical data.

WORKED EXAMPLE 10.3.1 FOUR SUMMARIES

PROBLEM

Eight students scored 6, 7, 7, 8, 9, 9, 9 and 10 in a quiz. Find (a) the mean, (b) the median, (c) the mode and (d) the range.

1. (a) Sum $= 6 + 7 + 7 + 8 + 9 + 9 + 9 + 10 = 65$. Mean $= 65 \div 8 = 8.125$.
2. (b) With eight values, the median is the mean of the 4th and 5th: $(8 + 9) \div 2 = 8.5$.
3. (c) 9 occurs three times, more than any other value, so the mode is 9.
4. (d) Range $= 10 - 6 = 4$.
5. The mean is not one of the scores. That is normal: an average need not be a value that actually occurs.

ANSWER **(a)** 8.125 **(b)** 8.5 **(c)** 9 **(d)** 4.

Balance the mean. Start from 8: the deviations are $-2, -1, -1, 0, +1, +1, +1, +2$, which total $+1$. Spreading that over eight values adds 0.125, giving 8.125. The positive and negative deviations nearly cancel, which is why the mean sits close to the middle.

WORKED EXAMPLE 10.3.2 READING A HISTOGRAM

PROBLEM

Use Figure 10.4, which shows the journey times of 20 students. (a) How many students take less than 20 minutes? (b) What percentage is that? (c) Which is the modal class? (d) Estimate the mean journey time using the midpoint of each interval. (e) Explain why the estimate may differ from the true mean.

1. (a) The first two bars contain $4 + 9 = 13$ students.
2. (b) $13 \div 20 = 0.65 = 65\%$.
3. (c) The tallest bar is 10–20 minutes, with frequency 9.
4. (d) Midpoints are 5, 15, 25 and 35 minutes. Estimated total $= 4 \times 5 + 9 \times 15 + 5 \times 25 + 2 \times 35 = 20 + 135 + 125 + 70 = 350$. Estimated mean $= 350 \div 20 = 17.5$ minutes.
5. (e) The midpoint assumes every student in an interval has the same journey time, which is not true. The histogram keeps the interval totals but loses the individual values.

ANSWER **(a) 13 (b)** 65% **(c) 10–20 minutes (d)** 17.5 **minutes (e) midpoints are approximations.**

Check the total frequency. $4 + 9 + 5 + 2 = 20$, matching the 20 students. If the frequencies do not add to the stated total, the histogram or the total is wrong before any average is calculated.

COMMON ERROR

Finding the median of an even-numbered set by choosing one of the two middle values instead of averaging them. A second error is using the mean with an outlier and calling it typical: one very large value can pull the mean far from most of the data. A third is leaving gaps between histogram bars or using unequal interval widths, which changes the visual comparison.

GUIDED PRACTICE · COPY AND COMPLETE

The values 3, 4, 6, 7, 9 and 12 have sum ☐, so the mean is ☐. With six values, the median is the mean of the ☐rd and ☐th values, which is ☐. The range is ☐ − ☐ = ☐. A histogram has frequencies 3, 7, 6 and 4, so the total is ☐; the modal class is the one with frequency ☐.

PRACTICE 10.3 · NUMERICAL DATA

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Order the values before finding a median. State whether each average is exact or an estimate.

1. For the data in Figure 10.3, find the mean, median, mode and range. F
2. Find the mean, median, mode and range of 5, 8, 8, 10 and 14. F
3. Find the median and mean of 3, 4, 6, 7, 9 and 10. Explain why the two answers are equal. F
4. The values 4, 5, 5, 6 and 30 include an outlier. Calculate the mean and the median. Which is the fairer summary, and why? P
5. A histogram has frequencies 3, 7, 6 and 4 for intervals 0–10, 10–20, 20–30 and 30–40. Find the total, the modal class, and the number of values below 20. P
6. A student says the median of 3, 5, 7, 9, 11 and 13 is 7. Explain the error and give the correct median. R

7. Which average would you choose for each purpose: (a) the most common shoe size in a shop; (b) the typical house price in a street with one very expensive house; (c) the total score of a team divided among its players? Explain each choice. P

8. Extension: the mean of five numbers is 10. Four of them are 6, 8, 11 and 13. Find the fifth number. Then add a sixth number so that the new mean is 11. E

10.4 Trends and misleading graphs

CRITERION C · I, II; D · I, II ◆ PATTERN ◆ CHANGE

A time-series graph shows how a quantity changes over time. The direction of change is the trend, but a trend is only as reliable as the points behind it. A careless axis, a short time window or a missing denominator can turn a small change into a dramatic story.

A **time-series graph** plots a numerical value against time. The points are joined because the order of the observations matters. A **trend** is the overall direction over the period: increasing, decreasing or roughly steady. Short-term rises and falls are **fluctuations**; a repeated pattern at the same time each year is **seasonal**. To describe a trend precisely, quote values and dates rather than saying only that a line “goes up”.

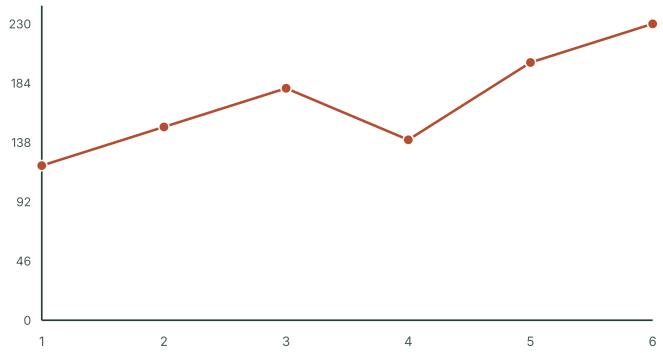


Figure 10.5 Library visitors in the first six months of a year. The overall trend is upward, with a dip in April; the six points do not prove what will happen in July.

The **percentage change** from an old value to a new value is $\frac{\text{new} - \text{old}}{\text{old}} \times 100\%$. A rise from 120 to 230 is a change of 110, which is $110 \div 120 \approx 91.7\%$, not 110%. Percentage change always compares the change with the starting value, so the same absolute change can be a different percentage depending on where it starts.

A graph can be accurate in every plotted number and still mislead through its design. The most common devices are a **truncated vertical axis** that does not start at zero, **unequal intervals** on an axis, a **broken axis** marked with a zigzag, pictures whose **area or volume** grows faster than the quantity, a **cherry-picked time window** that begins or ends at a convenient point, and a comparison of raw counts between groups of very different sizes.

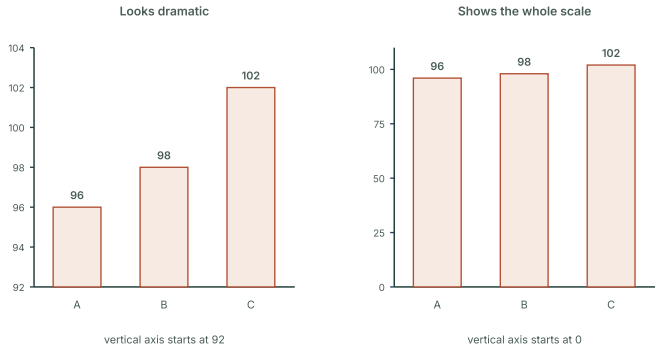


Figure 10.6 The same three values drawn twice. The truncated axis on the left makes C look far larger than A; the honest axis on the right shows a small difference.

WORKED EXAMPLE 10.4.1 DESCRIBING A TREND

PROBLEM

Use Figure 10.5. (a) Describe the overall trend from January to June. (b) Identify the month with the lowest number of visitors after January and suggest a reason. (c) Calculate the percentage increase from January to June. (d) Find the mean number of visitors over the six months. (e) Why should the graph not be used to predict July with confidence?

1. (a) Visitors rise overall from 120 in January to 230 in June, with a dip in April.
2. (b) April has 140 visitors, the lowest after January. A school holiday or exam period could explain the dip, but the graph alone cannot confirm the cause.
3. (c) Increase $= 230 - 120 = 110$. Percentage increase $= 110 \div 120 \times 100\% \approx 91.7\%$.
4. (d) Total $= 120 + 150 + 180 + 140 + 200 + 230 = 1020$. Mean $= 1020 \div 6 = 170$ visitors.
5. (e) Six monthly points show a pattern but not a rule. July may follow the trend or break it, and the graph gives no information about the causes of the changes.

ANSWER **(a) upward, 120 to 230 (b) April, 140 (c)** 91.7% **(d) 170 (e) limited evidence.**

Check the percentage with a multiplier. $230 \div 120 = 1.916\ldots$, so the new value is about 191.7% of the old value. That is a 91.7% increase, not a 191.7% increase.

WORKED EXAMPLE 10.4.2 TESTING A DRAMATIC CLAIM

PROBLEM

A company newsletter shows Figure 10.6 and claims that "product C sells 50% more than product A". (a) Read the three values. (b) Calculate the true percentage difference between A and C. (c) Explain how the left-hand chart creates the impression of a large difference. (d) Rewrite the claim accurately.

1. (a) The values are A 96, B 98 and C 102.
2. (b) Difference $= 102 - 96 = 6$. Percentage difference $= 6 \div 96 \times 100\% = 6.25\%$.
3. (c) The left axis starts at 92, not 0. A difference of 6 then occupies a large part of the visible axis, so the bars differ dramatically in height even though the values differ by only 6.25%.
4. (d) A fair claim is "product C sells about 6% more than product A", supported by a chart whose axis starts at zero.
5. The claim is not supported by the data; the graph exaggerates a real but small difference.

ANSWER **(a) 96, 98, 102 (b)** 6.25% **(c) truncated axis (d) "about** 6% **more".**

Compare ratios. $102 \div 96 = 1.0625$, so C is 106.25% of A, a 6.25% increase. If C really were 50% more, it would be $1.5 \times 96 = 144$, far above the plotted value.

A READER'S CHECKLIST

Before accepting a graph, check five things: Does the vertical axis start at zero? Are the intervals equal? Is the time window long enough to support the claim? Are counts compared with the right denominator? And does the caption state the source and the sample size? A graph that fails one of these may still be accurate, but it is not yet evidence for the claim beside it.

COMMON ERROR

Reading the height of a bar as a percentage change. A bar twice as tall shows twice the value only when the axis starts at zero; the percentage increase from 50 to 100 is 100%, not 50%. A second error is confusing "percentage points" with "per cent": a rise from 20% to 25% is 5 percentage points but a 25% increase. A third is choosing a start date that makes a short-term rise look like a long-term trend.

GUIDED PRACTICE · COPY AND COMPLETE

A value rises from 80 to 92. The change is $\square$, and the percentage increase is $\square \div \square \times 100\% = \square\%$. A graph of the values 70, 72 and 75 has a vertical axis starting at 60. The apparent height of the first bar is $\square$ units instead of 70, so the differences are exaggerated by a factor of $\square$.

PRACTICE 10.4 · TRENDS AND MISLEADING GRAPHS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Quote values from the graph, calculate the percentage change, and name the misleading feature precisely.

1. Monthly sales are 40, 45, 42, 55, 60 and 58. Describe the trend, identify the dip, and calculate the overall percentage increase from the first to the last value. F
2. A price rises from 80 to 92. Calculate the percentage increase. F

3. Three values 70, 72 and 75 are plotted with a vertical axis starting at 60. State two features that make the graph misleading. F
4. Redraw the three values from Question 3 with an axis starting at zero. Describe how the visual impression changes. P
5. A headline says "sales doubled" when sales rose from 50 to 90. Calculate the true percentage increase and rewrite the headline accurately. P
6. A pictogram uses a picture twice as tall and twice as wide for a value twice as large. Explain why the picture exaggerates the change. R
7. Choose the better display for each purpose: (a) change in temperature over a week; (b) share of a budget spent on four items; (c) comparison of five categories. Explain each choice. P
8. Extension: a news report says "crime doubled" after incidents rose from 2 to 4. Explain why the percentage is correct but the headline may still mislead. Suggest what additional information a reader should ask for. E

Investigation 10 · How should data be grouped?

CRITERION B · I, II

A class recorded the daily screen time, in minutes, of 30 students. The values are already ordered from smallest to largest. A histogram needs class intervals, but the width of those intervals is a choice. This investigation asks how that choice changes the picture.

DAILY SCREEN TIME OF 30 STUDENTS (MINUTES)

45	60	60	75	80	90	90	90	105	110
120	120	120	120	135	140	150	150	165	180
180	195	210	210	240	240	270	300	330	360

1. Make a frequency table using intervals of 60 minutes: 0–60, 60–120, 120–180, 180–240, 240–300 and 300–360. Decide which interval contains a value exactly on a boundary, and state your rule.
2. Draw a histogram from your table. Label both axes and use equal interval widths.
3. Describe the shape: where is the cluster, and is there a long tail of high values?
4. Regroup the same data into intervals of 120 minutes: 0–120, 120–240 and 240–360. Draw a second histogram.
5. Compare the two histograms. Which grouping shows the cluster more clearly, and which loses more detail?
6. Calculate the mean, median and mode of the 30 values. Which average best represents a "typical" student, and why?
7. A report claims that "the typical student spends four hours a day on screens". Test the claim using your summaries. How many students actually spend 240 minutes or more?
8. Write a recommendation for how the class should display the data in a newsletter aimed at families. Justify your choice of interval and average, and state one limitation of the display.

Changing the interval width changes the number of bars, not the data. A good grouping reveals the cluster without hiding the shape of the tails; a poor one can make a smooth distribution look like a single spike.

Real-life application · The canteen survey

A school surveys all 200 students about where they eat lunch. The canteen manager uses the results to plan staffing and prices, while the student council uses the same results to argue for more outdoor seating. The survey recorded year group and lunch choice.

LUNCH CHOICE BY YEAR GROUP

	School lunch	Packed lunch	Off-site	Total
Year 7	38	24	8	70
Year 8	31	25	9	65
Year 9	27	25	13	65
Total	96	74	30	200

The canteen charges $4.20 for a school lunch; a packed lunch costs about $3.10 in ingredients, and an off-site meal costs about $5.50. School-lunch uptake over the last five years was 88, 72, 80, 90 and 96 students.

Your task. Use the tables and show every percentage with its denominator. When you make a claim, quote the numbers that support it.

1. How many students buy school lunch? What percentage of all students is that?
2. Calculate the percentage of Year 9 students who eat off-site, and the percentage of Year 7 students who do. Compare the two year groups fairly.
3. Calculate the sector angle for each lunch choice in a pie chart of all 200 students. Check that the angles total 360°.
4. Draw a bar chart or clustered bar chart that compares the three lunch choices across the three year groups. Explain why a single pie chart would hide the year-group comparison.
5. Describe the trend in school-lunch uptake over the five years. Identify the year with the largest fall and calculate the percentage fall from the previous year.
6. Calculate the daily income from each lunch choice and the total daily income. If 24 students switch from packed lunch to school lunch, how much extra income does the canteen receive per day?
7. **Evaluate.** The newsletter claims that “off-site eating is out of control”. Test the claim using at least two calculations. Then write a more accurate headline of no more than twelve words. Explain one way a graph could make the off-site share look larger than it is.

Reflect. Which claim was supported by the data, and which depended on a misleading comparison? How did stating the denominator change the meaning of a percentage?

Criterion-mapped mini-task

A · Knowing and understanding
C · Communicating
D · Applying mathematics

Context. A factory reports its annual water use in thousands of litres: 2019, 420; 2020, 415; 2021, 408; 2022, 402; 2023, 396. A local newsletter publishes a bar chart with the vertical axis starting at 390 and the headline “Water use plummets by 80%”.

Product. A short data report containing a correct table, a fair graph, the true percentage change, and an evaluation of the headline.

Requirements.

1. Copy the data into a table with a column for the change from the previous year.
2. Describe the trend from 2019 to 2023, quoting values.
3. Calculate the total decrease and the percentage decrease over the five years.
4. Calculate the mean annual water use. Explain whether the mean is a useful summary here.
5. Explain how an axis starting at 390 changes the visual impression of the bars.
6. Redraw the bar chart with a vertical axis from 0 to 450, using equal intervals.
7. Test the “80%” claim. Calculate what the 2023 value would be if water use had fallen by 80% from 2019.
8. **Evaluate.** Write a fair headline and a two-sentence caption for your graph. State one limitation of the data and one question you would ask the factory.

CRITERIA AND STRANDS A i, ii: accurate table, percentage change and mean; C i, ii, iii: correctly labelled axes, precise headline and caption; D i, ii, iii, iv: interpret the data in context, evaluate the claim and communicate a recommendation.

BEFORE YOU SUBMIT

Check that the percentage denominator is the 2019 value, not the 2023 value. An 80% fall from 420 would leave 84, not 396. Make sure your redrawn graph starts at zero and that the caption names the period and the unit.

Chapter review

SUMMARY

— **Data types.** Categorical data name groups; numerical data record quantities. Numerical data are discrete if counted and continuous if measured.

— **Frequency tables.** Frequency counts responses; relative frequency divides by the total and can be written as a fraction, decimal or percentage.

— **Two-way tables.** Cells show combinations of two variables; marginal totals sit at the edges. Compare shares when group sizes differ.

— **Bar charts and pie charts.** Bars compare categories and must start at zero. Pie sectors use $\frac{\text{frequency}}{\text{total}} \times 360°$.

— **Dot plots and histograms.** Dot plots keep every value visible; histograms group continuous data into equal intervals with touching bars.

— **Averages.** Mean = total ÷ count; median = middle ordered value; mode = most frequent; range = largest − smallest.

— **Trends.** Describe direction, quote values and calculate percentage change from the starting value.

— **Misleading graphs.** Check the axis, intervals, time window, denominator and sample size before accepting a claim.

Relative frequency: $\frac{\text{frequency}}{\text{total}}$.

Sector angle: $\frac{\text{frequency}}{\text{total}} \times 360°$.

Mean: $\frac{\text{sum of values}}{\text{number of values}}$.

Median: middle value of the ordered data.

Range: largest − smallest.

Percentage change: $\frac{\text{new}-\text{old}}{\text{old}} \times 100\%$.

REVISION CHECKLIST

- ☐ I can classify a variable and justify it.
- ☐ I can complete and check a frequency table.
- ☐ I can compare groups using percentages.
- ☐ I can calculate sector angles and draw pie charts.
- ☐ I can find mean, median, mode and range.
- ☐ I can describe a trend and find percentage change.
- ☐ I can spot a misleading axis or comparison.

KEY VOCABULARY

data set A collection of observations.

variable A recorded characteristic.

categorical data Groups with no order.

numerical data A counted or measured quantity.

discrete Countable separate values.

continuous Any value in a range.

frequency The count for a value.

relative frequency Frequency ÷ total.

two-way table Two variables at once.

sector A slice of a pie chart.

histogram A graph of grouped data.

class interval A group of values.

mean Total ÷ count.

median Middle ordered value.

mode Most frequent value.

range Largest − smallest.

trend Overall direction over time.

outlier A value far from the rest.

percentage change Change ÷ starting value.

truncated axis An axis not starting at zero.

REFLECTION AND SELF-ASSESSMENT

Find one graph in a news article, advertisement or school notice. Identify the claim it makes and check the axis, the time window and the denominator behind the numbers.

Rate your confidence on each success criterion, name one display you would now choose differently, and explain in two sentences how a graph can be accurate yet still mislead.

END-OF-CHAPTER PRACTICE

Mixed practice 10

Show the table, formula or calculation that supports each answer. For a claim, state whether the evidence is sufficient.

FLUENCY Build accuracy

1. Classify “number of pages in a book” as categorical or numerical, and discrete or continuous.
2. A survey of 40 responses gives 14 A, 10 B, 9 C and 7 D. Calculate the percentage for each response.
3. A pie chart represents 120 people. A sector has angle 90°. How many people does it represent?
4. Find the mean of 12, 15, 9, 14 and 10.
5. Find the median of 3, 8, 5, 9, 4 and 7.
6. Find the range of 23, 17, 30, 19 and 25.
7. A histogram has frequencies 5, 8, 4 and 3. Find the total and the modal class.
8. A value rises from 60 to 75. Calculate the percentage increase.
9. A two-way table has 26 boys and 24 girls; 18 boys and 10 girls play chess. What percentage of girls play chess?
10. A bar chart’s vertical axis starts at 90 and the bars are 96, 100 and 104. Explain why the chart is misleading.

REASONING AND PROBLEM SOLVING Connect ideas

11. Explain why a pie chart is a poor choice for comparing twelve categories with similar frequencies. Suggest a better display.
12. A data set has one very large outlier. Compare the mean and the median and explain which is the fairer summary.
13. A student writes that 12 out of 30 is “12%”. Identify the error and give the correct percentage.
14. A shop’s sales rise from 40 to 70. A poster says “sales doubled”. Calculate the true percentage increase and rewrite the claim accurately.
15. Choose the most appropriate display for (a) change in a town’s population over ten years; (b) the share of a budget spent on four items; (c) the distribution of 40 journey times. Explain each choice.
16. A survey question asks: “Don’t you agree that the school day should start later?” Explain two ways the wording could bias the responses and write a fairer question.
17. **The screen-time claim (Criteria A, C, D).** The mean daily screen time of 30 students is 158 minutes and the median is 137.5 minutes. A report says “the average student spends over two and a half hours on screens”. Is the statement supported? Explain which average you would report and why.
18. **The misleading chart (Criteria C, D).** A graph of values 95, 97 and 100 starts its axis at 94. Describe the visual effect, calculate the true percentage difference between 95 and 100, and redraw the graph honestly in words.

Check your progress

ERRORS WORTH CATCHING BEFORE THE NEXT CHAPTER

- **Frequency is not a percentage.** Divide by the total before writing a percentage.
- **Compare shares, not raw counts.** Groups of different sizes need a common denominator.
- **Pie sectors use the whole total.** Every angle must share the same denominator.
- **Bars must start at zero.** A truncated axis exaggerates differences.
- **The median of an even set is an average of two values.** Do not choose one of them.
- **Percentage change uses the starting value.** A rise from 40 to 70 is 75%, not 30%.
- **A trend needs enough points.** A short series can suggest a pattern that the data cannot support.

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: QUESTION OR TASK I CAN SHOW	NEXT STEP
I can classify data and build tables.			
I can calculate relative frequencies and angles.			
I can choose and read an appropriate display.			
I can calculate and compare averages.			
I can describe a trend and percentage change.			
I can identify and explain a misleading graph.			

WHERE THIS LEADS

Beyond Year 1 these ideas extend to scatter graphs and correlation, sampling and bias, probability calculated from data, and displays that compare several distributions at once. The habit that matters most is the one practised here: ask what the display is measuring, what it leaves out and whether the claim is proportionate to the evidence.

Review the answer key only after attempting each question.

Answers and worked solutions

COMPLETE SOLUTIONS FOR EVERY EXERCISE

An answer ends a calculation; a solution records a decision. The entries below show the reasoning behind each result, so a difference between your work and mine becomes a precise question.

HOW EACH ENTRY IS ARRANGED

Exercise numbers match the chapter. Part letters follow the number, for example **3c**. Working is shown where it helps you locate a disagreement; routine arithmetic is compressed. Units and rounding are stated with each result.

Some questions invite more than one valid route. The key gives one efficient method and, where useful, a note beginning *Alternatively*. A different correct method is not an error. Validity, accuracy and a conclusion that answers the question are what count.

INVESTIGATIONS AND CRITERION TASKS

Open tasks receive *sample responses*, not a single script: representative data, a conjecture, a verification and a limitation. Your data may differ. Use the sample to judge the completeness of your table, the quality of your reasoning and the care of your conclusion.

Academic honesty. These solutions support your attempt; they do not replace it. If you use a sample response, rewrite the reasoning in your own words and explain the choice of method. That is the evidence a teacher can assess.

ACCURACY AND NOTATION

Unless a question states otherwise, exact answers are given for integers, fractions and terminating decimals. Measurements are rounded to a sensible degree of accuracy, generally three significant figures. Never round intermediate values; rounding early is a common source of "correct method, wrong answer".

- **Fractions** in simplest form; **percentages** with the % symbol and the denominator stated where it matters.
- **Sets** in roster or set-builder form; elements listed once, without repetition.
- **Angles** with the degree symbol; lengths, areas and volumes with units.
- **Money** to two decimal places where the context uses currency.
- **Data summaries** with the unit and number of values.

Equivalent fractions, ratios and expressions can look unlike each other. Test numerical equivalence before assuming an error.

Criterion tasks use course-specific rubrics that paraphrase the MYP criteria; they are not official IB descriptors, and sample responses show one level of performance.

Chapter 1 · Number systems and place value

PREREQUISITE CHECK

1 4,060.
2 0.5, since $0.50 > 0.45$.
3 56.
4 3,050.
5 0.7.
6 $-5, -2, 0, 3$.

GUIDED PRACTICE

1 Tenths, 0.4; thousandths, 0.007; $\times 100$ gives 2,840.7.

2 Hundreds digit 6, next digit 8, so round up to 700: 4,700. Nearest ten: 4,680. The rounding unit differs.

PRACTICE 1.1 · NUMBER FAMILIES AND ORDER

1 (a) natural, whole, integer, rational; (b) whole, integer, rational; (c) integer, rational; (d) rational; (e) rational.
2 $-3, -2, -1, 0, 1, 2$.
3 (a) $<$; (b) $>$; (c) $<$.
4 $-7, -2, 0, 3, 5$ °C.
5 No. Natural numbers here are $1, 2, 3, \ldots$; 0 is whole but not natural. "Counting zero items" is colloquial, not the definition.
6 $\frac{5}{4} = 1.25 > 1$; a fraction records a division and may exceed 1.
7 $1 \to -2$: -3 (down 3); $-2 \to 3$: $+5$ (up 5); final floor is 2 above the start.
8 2.5 lies halfway between 2 and 3; its fractional part makes it non-integer. Mark -4, 0 and 2.5 between 2 and 3.

PRACTICE 1.2 · PLACE VALUE AND ZERO

1 (a) 700; (b) 0.5; (c) 0.2; (d) 6.
2 4,632.
3 (a) 362; (b) 548; (c) 92; (d) 0.073.
4 The zero holds the tenths place, so 6 means 0.06. Removing it gives 2.6, an increase of 0.54.
5 3.5 is larger: tenths 5 vs 0. More digits does not mean a larger decimal.
6 $24 \times 100 = 2,400$ pencils; the 2 is in the thousands place (2,000).
7 Works for whole numbers (4→40), but $0.4 \to 0.40$ and $4.50 \to 4.500$ keep their value. Corrected rule: appending 0 after a whole number multiplies by 10; a trailing decimal zero changes only precision.
8 Greatest 8,520; least 2,058. A four-digit number cannot start with 0, so 0 must be in the hundreds place.

PRACTICE 1.3 · EXPANDED FORM

1 (a) $2 \times 1000 + 5 \times 100 + 0 \times 10 + 6$; (b) $3 \times 10 + 0 + 7 \times 0.1 + 8 \times 0.01$; (c) $4 \times 0.1 + 0 \times 0.01 + 9 \times 0.001$.
2 (a) 496; (b) 6.58; (c) 2,070.003.
3 (a) 700; (b) 0.009.
4 $0.2 + 0.05 = 0.25$; shifting both digits one place right gives $0.02 + 0.005 = 0.025$. Position fixes value.
5 B shows the tens place with 0×10; both equal 407.3. The zero term records an empty place without changing value.
6 $50 + 3 + 0.4 = 53.4$ cm: 5 tens, 3 ones, 4 tenths.
7 Example 1,605; swapping 6 and 0 gives 1,065, which is 540 less because the 6 moves from hundreds to tens.
8 1,000 has only one non-zero term; its chart still needs four columns, three of them empty.

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

1 (a) $=$; (b) $>$; (c) $>$.
2 0.07, 0.7, 0.707, 0.77.
3 (a) 8,600; (b) 8,650.
4 3.75; 0.9.
5 60.0: 9 tenths rounds up and carries into the ones, changing the whole part.
6 $4.56 \to 4.6$ to the nearest tenth; both signs agree if 4.6 is labelled as rounded.
7 E.g. 6.46, 6.50, 6.53. The interval is $6.45 \le x < 6.55$, so infinitely many values are possible.
8 Reasonable to the nearest thousand; difference £52. Ask for the audited total, the purpose and the rounding convention.

PRACTICE 1.5 · ESTIMATE AND JUDGE ACCURACY

1 $700 + 3,100 = 3,800$; exact 3,833.
2 $60 \times 30 = 1,800$; exact 1,798.
3 $10 \times 4 = 40$; exact 41.16, so the estimate is low (both factors rounded down).
4 $600 \div 30 = 20$; exact 20.83 (2 d.p.), so the estimate is low.
5 $400 + 400 + 300 = 1,100$; exact 1,098.
6 Total 20.03 km; "about 20 km" is reasonable to the nearest km or 10 km. State which.
7 Exact 1,247,936 is over-precise for a noticeboard. Use "about 1.25 million" (nearest 10,000) or "about 1.2 million" (nearest 100,000) and state the accuracy.
8 Inputs to the nearest centimetre justify 3 cm; the calculator's extra digits suggest false precision.

INVESTIGATION 1 · COUNTING WITH DIGITS

1 $9 \times 10^3 = 9,000$. For 2 digits, 10 to 99 gives $99 - 10 + 1 = 90$.
2 Each extra digit multiplies the count by 10.
3 $9 \times 10^{n-1}$; for $n = 5$, 90,000.
4 First digit: 1–9 (9 choices); later digits: 0–9 (10 choices).
5 Base 5: $4 \times 5^{n-1}$, giving 4, 20, 100, 500. **Evaluation:** tabulate, state the rule, test an unused case and explain why independent choices multiply.

APPLICATION · COMMUNITY NOTICEBOARD

1 Population about 130,000 (nearest 10,000); visitors about 1 million (nearest 100,000); recycling 3,680 tonnes exact; year 1937 exact.
2 Round both comparison figures to the same precision, e.g. 0.13 million and 1.05 million visits.
3 Define "visitor": visits or unique people, and the time period.
4 Recycling must stay exact for the grant; the year is historical. Visitors are most sensitive in absolute size: rounding to 1 million hides about 47,000 visits.

CRITERION TASK · MUSEUM LABEL

1 48,672: 4 in ten-thousands (40,000); 6,509: 6 in thousands (6,000); 214,830: 2 in hundred-thousands (200,000); 7,408.5: 7 in thousands (7,000).
2 $6,000 + 500 + 0 + 9$.
3 Visitors 214,830 → 215,000 (nearest thousand, readable); area 7,408.5 → 7,400 m² (nearest hundred, avoids false precision); keep 48,672 exact for credibility.
4 Rounding the collection to 49,000 suggests "nearly 50,000"; the exact figure signals a catalogued collection. Rounded visitors suggest scale without claiming an exact count.
5 Exact figures may be uncertain (loans, uncatalogued objects); heavy rounding can hide a difference that matters to an audit. State source and rounding rule.

> **Sample label (67 words).** "The Hall of Numbers opened in 1894. Its collection holds 48,672 objects — 40,000 + 8,000 + 600 + 70 + 2 — and 6,509 are on display; more than seven times as many are held as displayed. About 215,000 people visit yearly, and the galleries cover about 7,400 m²."

MIXED PRACTICE 1

1 (a) 800; (b) 0.8; (c) 0.08.
2 $5 \times 1000 + 0 \times 100 + 4 \times 10 + 0 \times 1 + 2 \times 0.1 + 0 \times 0.01 + 7 \times 0.001$.
3 7,030.06.
4 65,000; 64,800; 64,780.
5 0.50; 13.0.
6 (a) 64; (b) 0.82; (c) 3,500.
7 $-2.5, -0.5, 0.25, 2.05$.
8 (a) $<$; (b) $=$; (c) $<$.

9 E.g. 4,651, 4,700, 4,749; range $4,650 \le x < 4,750$.
10 6.09 < 6.1: align as 6.09 and 6.10; tenths 0 < 1.
11 -3 °C; 7 degrees colder (7 units apart).
12 Underestimate by 480; "about 12,500" (nearest hundred) is more informative.
13 500 vs 50; the zero shifts the 5 ten times left.
14 $4,000 + 4,000 + 3,000 = 11,000$; exact 10,997; estimate is 3 high.
15 $640 + d$, strictly between 646 and 650: only 648 is even, so $d = 8$.
16 Same value; 2.60 records hundredths, 2.6 only tenths.
17 E.g. 3.501 and 3.567; 3.501 is closer to 3.5.
18 $h + 3d = 9$: 3,900; 3,621; 3,342; 3,063.
19 "About 2,500" for the bid, 2,495 for the audited report; difference 5. Add visiting groups or repeat visits.
20 -12 basement, 0 boundary/ground, 45 new wing; 0 is the reference point, not "no boxes".

Common errors. More digits do not make a decimal larger (0.08 < 0.8). A zero inside a number may hold a place (3.05 ≠ 3.5). Name the rounding unit. An estimate needs a direction: $10 \times 4 = 40$ is below 9.8×4.2 because both factors were rounded down.

Chapter 2 · Powers, expanded form and order of operations

PREREQUISITE CHECK

1 42 and 81.
2 4,536.
3 $12 + 12 = 24$.
4 $3 \times 3 \times 3 = 27$.
5 64 cm².
6 $(2 + 4) \times 3 = 18$.

GUIDED PRACTICE

1 Base 8, exponent 3; $8 \times 8 \times 8$; value 512; with exponent 1 the value is 8. The exponent counts factors of 8, not 8×3.
2 $13^2 = 169$, so $\sqrt{169} = 13$; $6^3 = 216$, so $\sqrt[3]{216} = 6$; side 14 cm. The side is the root whose square equals the area.
3 For $x = 5$: $2x^2 = 2 \times 25 = 50$, while $(2x)^2 = 10^2 = 100$. For $x = 2$: $x^2 + 3^2 = 4 + 9 = 13$, while $(x + 3)^2 = 5^2 = 25$. Brackets change the base: $2x^2$ squares only x, whereas $(2x)^2$ squares the whole product $2x$.

PRACTICE 2.1 · POWERS AND INDEX NOTATION

1 (a) 4^3; (b) 7^2; (c) 10^5; (d) 2^7.
2 (a) 8; (b) 9; (c) 125; (d) 1,000; (e) 64; (f) 64.
3 (a) 4; (b) 6; (c) 3; (d) 3.
4 $4^2 = 16$, but $4 \times 2 = 8$. The exponent counts equal factors.
5 $9^2 = 81$ cm².
6 $2^4 = 4^2 = 16$, but $2^3 = 8 \neq 9 = 3^2$. Swapping base and exponent can change the value, so the claim is false.
7 Round 4: $3^4 = 81$ new people. Rounds 1–4: $3 + 9 + 27 + 81 = 120$.
8 $2^{10} = (2^5)^2 = 32^2 = 1,024 > 10^3 = 1,000$. The estimate via $32 > 30$ avoids the full calculation; exact check confirms 1,024.

PRACTICE 2.2 · POWERS OF TEN AND EXPANDED FORM

1 (a) 100; (b) 10,000; (c) 1,000,000; (d) 1.
2 $8 \times 10^3 + 3 \times 10^2 + 0 \times 10^1 + 4 \times 10^0$.
3 4,706.
4 $9 \times 10^4 + 0 \times 10^3 + 0 \times 10^2 + 5 \times 10^1 + 0 \times 10^0$; the 5 is in the tens place, value 50.
5 $20,000 + 50 = 20,050$; the zeros hold the thousands, hundreds and ones places, so their values are not omitted.
6 $1 \times 10^4 + 2 \times 10^3 + 4 \times 10^2 + 8 \times 10^1 + 0 \times 10^0$. The 1 contributes 10,000, the largest share.
7 3,000 is 10 times 300; the digit has moved one place left, where each place is ten times the one to its right.
8 Example 12,690: $1 \times 10^4 + 2 \times 10^3 + 6 \times 10^2 + 9 \times 10^1 + 0 \times 10^0$. Digits sum to 18 and exactly one term is zero. Many answers exist: choose four non-zero digits summing to 18 and place the zero in any non-leading position.
9 $10^3 \to 10^2 \to 10^1 \to 10^0$ by dividing by 10 each time: $1,000 \to 100 \to 10 \to 1$. Since $10^1 = 10$, one more division by 10 gives $10^0 = 1$, the ones place.

PRACTICE 2.3 · SQUARES, CUBES AND ROOTS

1 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.
2 (a) 169; (b) 225; (c) 216; (d) 1,000.
3 (a) 7; (b) 10; (c) 13; (d) 2; (e) 3; (f) 5.
4 $\sqrt{144} = 12$ tiles.
5 (a) 8; (b) 6; (c) 15.
6 $9 + 4 = 13$, but $\sqrt{97} \approx 9.85$; not equal. A root of a sum is not the sum of the roots.
7 Between 7 and 8, since $49 < 50 < 64$; closer to 7 (distance 1, against 14 from 8).
8 Edge 7 cm. Doubled edge 14 cm gives $14^3 = 2,744$ cm³ $= 8 \times 343$; the factor is $2^3 = 8$.

PRACTICE 2.4 · ORDER OF OPERATIONS

1 $8 + 10 = 18$.
2 $20 - 12 = 8$.
3 (a) 22; (b) 100; (c) 17.
4 (a) $6 \times 2 = 12$; (b) $24 \div 8 = 3$; (c) 9.
5 $5 \times 8 - 9 = 40 - 9 = 31$.
6 (a) $(3 + 5) \times 2$; (b) $(12 - 4) \times 2$; (c) $18 \div (3 + 6)$.
7 Division precedes addition: $6 \div 3 = 2$, then $12 + 2 = 14$. The student added first to get 18, then divided.
8 Examples: $2 + 3 \times 4 = 14$; $(2 + 3) \times 4 = 20$; $2^3 + 4 = 12$; $4^2 - 3 = 13$; $2 \times 3 + 4 = 10$; $(4 - 2)^3 = 8$.

PRACTICE 2.5 · SUBSTITUTION AND POWERS

1 (a) 18; (b) 50; (c) 200.
2 (a) 36; (b) 100; (c) 400. Each is double the matching result in question 1, because $(2n)^2 = 4n^2 = 2(2n^2)$.
3 $9 + 16 = 25$, but $(3 + 4)^2 = 49$; not equal.
4 $12^2 = 144$ cm².
5 (a) 27 cm³; (b) 125 cm³.
6 (a) 20 m; (b) 80 m. Doubling t multiplies t^2 by 4, so d quadruples.
7 (a) $(2x)^2 = 36$; (b) $(a + b)^2 = 49$. Brackets put the whole product or sum in the base.
8 Side 9 cm. Tripled side 27 cm gives 729 cm² $= 9 \times 81$; the factor is 3^2.

APPLICATION · THE SCHOOL DATA STORE

1 $10^4 \times (1 - 0.15) \div (5 + 1) = \lfloor 8,500 \div 6 \rfloor = 1,416$ projects, with 4 GB unused.

2 The 15% reserve is 1,500 GB, leaving 8,500 GB; each project needs 6 GB. A 1,417th project needs 8,502 GB, so round down.
3 10 TB $= 10^{13}$ bytes; 1 TiB $= 2^{40}$ bytes, so $10^{13} \div 2^{40} \approx 9.09$ TiB. No storage is lost; the units differ (powers of 10 against powers of 2).
4 Quote 10 TB to a technician ordering a replacement (drives are sold in decimal TB); for a family poster, show 10 TB with "about 9.09 TiB reported by a computer". Honest means stating the unit, not choosing the larger number.

INVESTIGATION 2 · THE DOUBLING SHEET

1 $n = 3 : 0.8$ mm; $n = 6 : 6.4$ mm; $n = 9 : 51.2$ mm (each double the row above).
2 $102.4 \div 0.1 = 1,024 = 2^{10}$, so 1,024 times thicker.
3 $T = 0.1 \times 2^n$ mm.
4 $n = 13 : 819.2$ mm; $n = 14 : 1,638.4$ mm. The least is 14 folds.
5 The room is 3,000 mm high. $n = 14 : 1,638.4$ mm; $n = 15 : 3,276.8$ mm; first 15 folds. One extra fold doubles the stack again.
6 The table does not disprove the physical claim; it models perfect doubling. Real paper cannot be folded indefinitely because thickness, creasing and sheet size make further folds impossible. Separate mathematical prediction from physical feasibility.
7 Assumptions: perfect folds and constant thickness. Improve by measuring a stack of many sheets with a micrometer and averaging.
8 $T = 0.1 \times 3^n$: 0.1, 0.3, 0.9, 2.7, 8.1, 24.3 mm. At $n = 5$ tripling gives 24.3 mm against 3.2 mm. After 10 folds the ratio is $3^{10} \div 2^{10} = 57,665 \div 1,024 \approx 57.7$. **Evaluation:** a complete table, rule, comparison and limitation of the model.

CRITERION TASK · POWER PRIMER

1 5^3: base 5, exponent 3, meaning $5 \times 5 \times 5 = 125$.
2 $5^3 = 125, 5^2 = 25, 5^1 = 5, 5^0 = 1$: each step divides by 5. $5^0 = 1$, not 0.
3 A 4×4 square contains 16 unit squares; a $4 \times 4 \times 4$ cube contains 64 unit cubes.
4 $4 + 3 \times (8 - 2)^2 \div 6$: brackets $= 6$; index $= 36$; multiply $= 108$; divide $= 18$; add $= 22$. Rules: B (brackets), I (indices), M/D left to right, A/S left to right.
5 When $x = 2$, $3x^2 = 12$, but $(3x)^2 = 36$.
6 500×2^h; at $h = 5$, $500 \times 32 = 16,000$. It is not $500 \times 5 = 2,500$ because the culture doubles each hour rather than increasing by a fixed amount.
7 Test evaluation: $2 + 5 \times 3^2 = 2 + 5 \times 9 = 47$. If the test reader multiplied 2 by 5 first, revise the card to label the operation order with arrows.
8 Sample evaluation: strength — one worked line per step; limitation — too many symbols for one metre; change — add a large "indices before multiplication" banner and fewer examples. **Before submitting:** check every number, test a reader and revise for clarity rather than adding words.

MIXED PRACTICE 2

1 (a) 81; (b) 128; (c) 1,000,000; (d) 216; (e) 121; (f) 256.
2 (a) 9^3; (b) 12^2; (c) 10^5; (d) 5^7.
3 $6 \times 10^4 + 0 \times 10^3 + 3 \times 10^2 + 0 \times 10^1 + 4 \times 10^0$.
4 80,205.
5 (a) 9; (b) 14; (c) 3; (d) 10.
6 (a) 16; (b) 40; (c) 26; (d) 6; (e) 9; (f) 25.
7 (a) 13; (b) 25; (c) 54; (d) 36.
8 (a) $(2 + 3) \times 4$; (b) $(24 - 8) \div 4$; (c) $(3 \times 2)^2$; (d) $(10 - 4)^2$.
9 $2^5 = 32$ uses five factors of 2; $5^2 = 25$ uses two factors of 5. 2^5 is larger.
10 $7^2 = 49$, not $7 \times 2 = 14$; the exponent counts factors.
11 (a) 6; (b) 5; (c) 4; (d) 5.
12 Divide each power by 10: $1,000 \to 100 \to 10 \to 1$, so $10^0 = 1$.
13 1,111; each place receives a 1, creating one more digit in the answer.
14 Side 13 cm, perimeter 52 cm. Doubling gives area 676 cm^2 (factor 4) and perimeter 104 cm (factor 2).
15 8 cm^3 and 64 cm^3; 8 times larger, since $2^3 = 8$.
16 (a) $4 \times 3 + 2 = 14$; (b) $4 \times (3 + 2) = 20$.
17 (a) 81; (b) $1 + 3 + 9 + 27 + 81 = 121$; (c) $3^5 = 243$ would add a fifth round; the total stops at round 4.
18 $\sqrt{100} = 10$ against $6 + 8 = 14$. The example disproves $\sqrt{a + b} = \sqrt{a} + \sqrt{b}$.
19 (a) 400×2^h; (b) 3,200, 12,800 and 102,400; (c) $400 \times 3 = 1,200$ adds 400 each hour, but the culture doubles; (d) 102,400 rounds to 100,000 (difference 2,400, about 2.3%), so the approximation is sensible if the lab states the rounding; (e) no — the second is always twice the first while both double.
20 (a) 2×10^{12} bytes; (b) $2 \times 10^{12} \div 2^{40} \approx 1.82$ TiB; (c) no storage is missing, the units differ; (d) $250 \div 2,000 = 12.5\%$; (e) "2 TB (decimal) ≈ 1.82 TiB (binary). Same drive; no storage is missing."

> **Common errors.** $5^2 = 25$, not 10. Brackets change the base: $3x^2$ squares only x. Multiplication and division share rank and run left to right: $24 \div 4 \times 2 = 12$. Roots do not distribute: $\sqrt{36 + 64} = 10$, not 14.

Chapter 3 · Factors, multiples and primes

PREREQUISITE CHECK

1 $7 \times 8 = 56$; $9 \times 6 = 54$.
2 12.
3 4, 8, 12, 16, 20.
4 $100 = 10 \times 10$.
5 $3 \times 3 \times 3 = 27$ is larger than $4 \times 4 = 16$.
6 Yes: $56 = 7 \times 8$, so 56 is in the 7 times table.

GUIDED PRACTICE

1 $1 \times 40, 2 \times 20, 4 \times 10, 5 \times 8$; 8 factors. 40 is a multiple of 8 because $40 = 8 \times 5$.
2 Last digit 4, so divisible by 2. Digit sum 12, so divisible by 3 but not by 9. Last two digits 14, not a multiple of 4, so not divisible by 4. Therefore not divisible by 6.
3 $48 = 4 \times 12 = 2 \times 2 \times 3 \times 2 \times 2 = 2^4 \times 3$. Check: $16 \times 3 = 48$.
4 HCF $= 2 \times 3 = 6$; LCM $= 2 \times 3^2 \times 5 = 90$. $18 \times 30 = 540$ and $6 \times 90 = 540$, so HCF × LCM = product of the two numbers.

PRACTICE 3.1 · FACTORS AND MULTIPLES

1 (a) 1, 2, 4, 5, 10, 20: 6 factors. (b) 1, 2, 4, 7, 14, 28: 6 factors. (c) 1, 3, 5, 9, 15, 45: 6 factors.
2 (a) $1 \times 30, 2 \times 15, 3 \times 10, 5 \times 6$. (b) $1 \times 42, 2 \times 21, 3 \times 14, 6 \times 7$. (c) $1 \times 64, 2 \times 32, 4 \times 16, 8 \times 8$.
3 (a) 8, 16, 24, 32, 40. (b) 13, 26, 39, 52, 65.
4 (a) True. (b) False: 42 is a multiple of 6, not a factor of 6. (c) True. (d) True.

5 Cartons of 6 give 6 cartons; cartons of 4 give 9 cartons. Cartons of 10 leave 6 eggs, so they cannot be used.
6 $8 \times 9 = 72$. Factors: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72 (12 factors).
7 $6 \times 6 = 36$, so 6 is a factor and 36 is a multiple of 6. $36 \div 5$ leaves remainder 1, so 5 is not a factor.
8 21, 22, 26 and 27. Each is a product of two primes (21, 22, 26) or a prime cube (27), which gives four factors; primes, prime squares and numbers with three prime factors do not qualify. Listing factor pairs for each number confirms the counts.

PRACTICE 3.2 · DIVISIBILITY TESTS

1 (a) 2, 3. (b) 5. (c) 2, 3, 5, 10. (d) 2, 3, 9. (e) 3, 5, 9.
2 (a) 4 yes, 8 yes. (b) 4 yes, 8 no (516 leaves 4). (c) 4 yes, 8 yes. (d) neither (10 and 310 are not multiples of 4 or 8).
3 (a) 15: 3 yes, 9 no. (b) 18: 3 and 9. (c) 9: 3 and 9. (d) 18: 3 and 9.
4 (a) 2, 5, 8. (b) 2. (c) 1, 3, 5, 7, 9. (d) 2, 5, 8.
5 Ending in 0 gives factors 10 and $10 = 2 \times 5$, so both 2 and 5 divide it. The last two digits need not be a multiple of 4: 30 is divisible by 2 and 5 but not 4; 40 is divisible by all three.
6 The test for 6 needs divisibility by 2 as well as by 3. 513 is odd, so it is not divisible by 6, even though its digit sum is 9. It is composite: $513 = 3 \times 171$.
7 Divisible by 3 and 4 implies divisible by 12 because 3 and 4 share no factor. 36 and 48 both confirm this. Divisible by 8 and 6 does not imply 48: 24 is divisible by both 8 and 6 but not by 48.
8 564. The last two digits, 64, are divisible by 4. Of the other arrangements, 654 and 546 end in 54 and 46, which fail; 456 works but is smaller; 465 and 645 are odd. So 564 is the largest valid number.

PRACTICE 3.3 · PRIMES AND COMPOSITES

1 23, 29, 31, 37.
2 $33 = 3 \times 11$; $57 = 3 \times 19$; $87 = 3 \times 29$. 43 and 61 are prime.
3 (a) $121 = 11^2$, composite. (b) 127: tested 2, 3, 5, 7, 11; none divides, prime. (c) 131: tested 2, 3, 5, 7, 11; none divides, prime. (d) $133 = 7 \times 19$, composite.
4 2 has exactly two factors, 1 and 2; being even does not add a third factor. 1 has only one factor (itself), so it is neither prime nor composite.
5 The student applied the test for 3 incorrectly: $8 + 7 = 15$, which is a multiple of 3. So $87 = 3 \times 29$ is composite.
6 59 and 61; the sum is 120.
7 (a) $28 = 5 + 23$ (also $11 + 17$); $40 = 3 + 37$ (also $11 + 29$); $100 = 3 + 97$ (also $11 + 89$, $17 + 83$, $29 + 71$, $41 + 59$, $47 + 53$). (b) Three examples support the conjecture but cannot prove it for every even number; a proof or a counterexample is needed.
8 $5 = 0 \times 6 + 5$, $7 = 1 \times 6 + 1$, $11 = 1 \times 6 + 5$, $13 = 2 \times 6 + 1$, $17 = 2 \times 6 + 5$, $19 = 3 \times 6 + 1$. Remainders 0, 2, 3 and 4 are impossible: 0 gives a multiple of 6; 2 and 4 give even numbers greater than 2; 3 gives a multiple of 3 greater than 3.

PRACTICE 3.4 · PRIME FACTORISATION

1 (a) $2 \times 2 \times 3 \times 5$. (b) $2 \times 3 \times 3 \times 5$. (c) $2 \times 3 \times 3 \times 7$.
2 (a) $2^2 \times 3^2$. (b) 2×3^3. (c) $2^3 \times 3^2$. (d) $2^5 \times 3$. (e) $2 \times 3 \times 5^2$. (f) $3^2 \times 5^2$.
3 (a) $2^3 \times 3^2 \times 5 = 360$. (b) $3^2 \times 5 \times 7 = 315$. (c) $2 \times 5^3 \times 11 = 2{,}750$.
4 (a) 28. (b) 135. (c) 350. (d) 144.
5 4 and 21 are composite, so the factorisation is not finished. Splitting them gives $84 = 2^2 \times 3 \times 7$.
6 Not a square: the exponents 3 and 2 are not all even. Multiply by 2 to make the exponent of 2 even: $72 \times 2 = 144 = 12^2$, so the smallest multiplier is 2.
7 $1{,}000{,}000 = 10^6 = (2 \times 5)^6 = 2^6 \times 5^6$. Each factor of 10 contributes one 2 and one 5, and there are six such factors.
8 60 and 18; 60 is larger. Common prime factors are 2 and 3; the HCF is $2 \times 3 = 6$.

PRACTICE 3.5 · HCF AND LCM

1 (a) 6. (b) 10. (c) 8. (d) 15.
2 (a) 24. (b) 30. (c) 36. (d) 36.
3 $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$: HCF 12, LCM 144. $12 \times 144 = 1{,}728 = 36 \times 48$.
4 $18 = 2 \times 3^2$, $30 = 2 \times 3 \times 5$: HCF 6, LCM 90.
5 HCF 2, LCM 72. 72 is a multiple of 8, 12 and 18, and 2 divides each number.
6 HCF of 48 and 60 is 12, so pieces are 12 cm long; $4 + 5 = 9$ pieces.
7 LCM of 15 and 20 is 60, so they next leave together at 9:00. Coincidences occur at common multiples; the HCF 5 would only give the largest interval dividing both schedules, not the meeting times.
8 A common factor cannot exceed either number, so the HCF cannot. Each number is a multiple of itself, so the LCM is at least either number. Pairs with HCF 6 and LCM 36: (6, 36) and (12, 18).

APPLICATION · THE FAIRNESS RELAY

1 Sample judgement: Plan A is fairer if "fair" means every team has the same size and every team mixes year groups; its disadvantage is that only 12 teams exist, so fewer students are active at once. Plan B is fairer if year groups should compete separately; its disadvantage is that teams are not mixed. State the criterion before choosing.
2 Both stations change together every LCM(20, 30) = 60 minutes. From 9:00, this happens at 10:00, 11:00 and 12:00 before lunch; 9:00 is the starting coincidence. Each 60-minute gap is a common multiple of 20 and 30, and 60 is the least.
3 Equal team size is guaranteed by using the HCF, 12. Equal proportions from each year are also guaranteed, since every team receives $7/84 = 5/60 = 4/48 = 1/12$ of each year group. Equal raw numbers from each year are impossible because the year groups have different totals. The arithmetic constrains the organiser to promises based on common factors.
4 Smaller teams are fairer when the priority is more simultaneous participation or mixing across more groups; they require a smaller common team size, which may not divide all three year groups exactly, so the organiser must check the factor first.

INVESTIGATION 3 · THE FACTOR TABLE

1 Counts 21–30: 4, 4, 2, 8, 3, 4, 4, 6, 2, 8.
2 2, 3, 5, 7, 11, 13, 17, 19, 23, 29: the primes.
3 1, 4, 9, 16, 25: the square numbers. 36 has 9 factors (1, 2, 3, 4, 6, 9, 12, 18, 36) and 49 has 3 (1, 7, 49), both odd.
4 Factors occur in pairs $a \times b = n$. The pair is two different factors unless $a = b$, which happens only for a square, where the middle factor is counted once. Hence non-squares have an even count and squares an odd count.
5 p^2 has 3 factors ($1, p, p^2$); p^3 has 4 ($1, p, p^2, p^3$); $p \times q$ has 4 ($1, p, q, pq$). Checks: 25 has 3, 27 has 4, 15 has 4.
6 24 has 8 factors: 1, 2, 3, 4, 6, 8, 12, 24, i.e. 1×24, 2×12, 3×8, 4×6.
7 No. 24 has 8 factors but 25 has 3; 28 has 6 but 29 has 2. The number of factors depends on the prime factorisation, not on size alone.
8 Exactly three factors must be 1, p and p^2, so the number is p^2. Below 100: 4, 9, 25, 49.

CRITERION TASK · THE NUMBER MUSEUM

1 Sample number 84. Factor pairs: 1×84, 2×42, 3×28, 4×21, 6×14, 7×12; 12 factors.

2 Divisible by 2 (even), 3 (digit sum 12), 4 ($84 = 4 \times 21$) and 6 (2 and 3); not divisible by 5 or 10 (does not end in 0 or 5) or 9 (digit sum 12).

3 $84 = 4 \times 21 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$. Check: $4 \times 3 \times 7 = 84$.

4 Sample second exhibit 60. $84 = 2^2 \times 3 \times 7$, $60 = 2^2 \times 3 \times 5$: HCF 12, LCM 420. The HCF answers "largest equal groups" because it is the largest number dividing both exhibits' counts.

5 Sample context: 84 chairs in equal rows. Possible arrangements include 7 rows of 12, 12 rows of 7, 6 rows of 14 and 14 rows of 6, since each row number must be a factor of 84.

6 Surprising fact: 84 is abundant. Its proper factors sum to $1 + 2 + 3 + 4 + 6 + 7 + 12 + 14 + 21 + 28 + 42 = 140 > 84$, so it is smaller than the sum of its proper factors.

7 Sample questions: "What is the smallest prime factor?" 2. "Write the prime factorisation." $2^2 \times 3 \times 7$. "How many factor pairs are there?" Six.

8 Sample evaluation: the visitor understood factor pairs but misread 2^2 as $2 \times 2 = 4$ without connecting it to the tree. Revision: label the exponent as "two factors of 2" beside the index. Accuracy, audience fit and one tested revision complete the exhibit.

MIXED PRACTICE 3

1 1, 2, 3, 6, 9, 18, 27, 54.

2 9, 18, 27, 36, 45, 54.

3 1×48, 2×24, 3×16, 4×12, 6×8.

4 2 yes (even); 3 yes (digit sum 18); 4 no (42 not a multiple of 4); 5 no; 9 yes (digit sum 18); 10 no.

5 41, 43, 47, 53, 59.

6 $87 = 3 \times 29$, composite; 89 prime (no factor 2, 3, 5 or 7); $117 = 9 \times 13$, composite; 149 prime; $221 = 13 \times 17$, composite.

7 (a) $2^2 \times 3 \times 7$. (b) $2^3 \times 3 \times 5$. (c) $2^2 \times 7^2$. (d) $2 \times 3^2 \times 5^2$.

8 (a) 72. (b) 245. (c) 300. (d) 675.

9 HCF 14, LCM 84.

10 HCF 15, LCM 225.

11 HCF 6, LCM 72.

12 HCF 4, LCM 240.

13 180 is a common multiple, but 12 and 15 share the factor 3, so it is not the lowest: LCM = 60. The product equals the LCM only when the numbers are coprime, for example 8 and 9.

14 1 has one factor; 2 has exactly two, 1 and 2, so 2 is prime. A prime must have exactly two distinct factors.

15 (8, 48) and (16, 24). Checks: both pairs have HCF 8 and LCM 48.

16 The odd exponents are 1 (for 2) and 1 (for 5), so multiply by $2 \times 5 = 10$: $90 \times 10 = 900 = 30^2$.

17 HCF(240, 360) = 120 cm tiles; $2 \times 3 = 6$ tiles.

18 LCM(8, 12, 18) = 72 seconds, so the next simultaneous flash is at 10:01:12.

19 (a) HCF(96, 72, 60) = 12 bags. (b) Each bag has 8 apples, 6 bananas and 5 oranges. (c) HCF(186, 72, 60) = 6 bags, each with 31 apples, 12 bananas and 10 oranges. (d) $186 = 2 \times 3 \times 31$ lacks the factor 4 shared by 96 and 60, so the common factor drops from 12 to 6; changing one quantity can change the entire common-factor structure.

20 (a) LCM(45, 60) = 180 s = 3 min. (b) 07:03, 07:06, 07:09. (c) 21 times (07:00 through 08:00 inclusive). (d) LCM(45, 60, 90) is still 180 s because $90 = 2 \times 3^2 \times 5$ divides $2^2 \times 3^2 \times 5 = 180$, so the coincidence times do not change.

Common errors. Factors and multiples run in opposite directions. Stop factor lists at the square root. 1 is neither prime nor composite. A prime factorisation contains only primes. HCF answers "largest equal groups"; LCM answers "when do events coincide?"

Chapter 4 · Fractions and decimals

PREREQUISITE CHECK

1 Shade 6 of the 8 equal parts.

2 $0.5 = \frac{1}{2}$; $0.25 = \frac{1}{4}$.

3 15.

4 $\frac{1}{2} = \frac{4}{8} > \frac{3}{8}$.

5 $\frac{1}{2}$.

6 $0.7 = \frac{7}{10}$.

GUIDED PRACTICE

1 HCF 12; $\frac{24}{36} = \frac{2}{3}$. $\frac{5}{8} = \frac{25}{40}$ (denominator multiplied by 5). $\frac{3}{7} = \frac{15}{21}$ (numerator multiplied by 5).

2 Common denominator 63: $\frac{4}{7} = \frac{36}{63}$, $\frac{5}{9} = \frac{35}{63}$; $36 > 35$, so $\frac{4}{7}$ is larger.

3 $\frac{2}{3} + \frac{1}{4} = \frac{8}{12} + \frac{3}{12} = \frac{11}{12}$. $3\frac{1}{2} = 3\frac{5}{10}$, $1\frac{2}{5} = 1\frac{4}{10}$, difference $2\frac{1}{10}$.

4 $\frac{3}{5} \times \frac{5}{9} = \frac{1}{3}$. $\frac{5}{8} \div \frac{5}{12} = \frac{5}{8} \times \frac{12}{5} = \frac{3}{2} = 1\frac{1}{2}$.

5 0.450 is smallest and 0.504 largest. Rounding 6.347 to two decimal places gives 6.35. $1.2 \times 0.5 = 0.6$.

6 $\frac{9}{20} = \frac{45}{100} = 0.45$. $0.45 = \frac{45}{100} = \frac{9}{20}$. $\frac{2}{3} = 0.\overline{6}$.

PRACTICE 4.1 · EQUIVALENT FRACTIONS

1 (a) $\frac{4}{10}$, $\frac{6}{15}$. (b) $\frac{6}{8}$, $\frac{9}{12}$. (c) $\frac{14}{20}$, $\frac{21}{30}$. Other correct multiples are accepted.

2 (a) 15. (b) 30. (c) 3. (d) 2.

3 (a) $\frac{3}{4}$. (b) $\frac{5}{8}$. (c) $\frac{2}{3}$. (d) $\frac{3}{4}$.

4 $\frac{9}{15}$, $\frac{15}{25}$ and $\frac{18}{30}$ are equivalent to $\frac{3}{5}$; $\frac{12}{18} = \frac{2}{3}$. Check by scaling both terms by the same number or by cross-multiplying.

5 $\frac{18}{27} = \frac{2}{3}$. Two of every three students walk to school.

6 Adding the same number to both terms changes the value: $\frac{12}{18} = \frac{2}{3}$, while $\frac{13}{19} \approx 0.684$. Correct simplest form: $\frac{2}{3}$. Check: $12 \times 19 = 228 \neq 234 = 13 \times 18$.

7 Cut each of the 4 quarters into 3 equal pieces. There are now $4 \times 3 = 12$ pieces and $3 \times 3 = 9$ are shaded. The number of pieces is multiplied by 3 and each piece is one third as large, so the shaded amount is unchanged.

8 Denominator 70: yes, $70 = 7 \times 10$, so $\frac{3}{7} = \frac{30}{70}$. Denominator 50: no, because 50 is not a multiple of 7. $\frac{4}{5} = \frac{100}{125}$.

PRACTICE 4.2 · COMPARING AND ORDERING

1 (a) $>$ since $\frac{36}{60} > \frac{35}{60}$. (b) $<$ since $\frac{15}{24} < \frac{16}{24}$. (c) $<$ since $\frac{21}{30} < \frac{22}{30}$. (d) $<$ since $\frac{9}{12} < \frac{10}{12}$.

2 $\frac{7}{12}, \frac{3}{5}, \frac{5}{8}, \frac{2}{3}$ (70, 72, 75, 80 hundred-twentieths).

3 $\frac{11}{12}, \frac{7}{8}, \frac{5}{6}, \frac{3}{4}$ (22, 21, 20, 18 twenty-fourths).

4 0.25, 0.625, 0.667, 0.7: $\frac{1}{4}$ lies below $\frac{1}{2}$; $\frac{5}{8}$, $\frac{2}{3}$, $\frac{7}{10}$ lie above it, in that order.

5 $\frac{9}{20}$. Using twentieths, $\frac{2}{5} = \frac{8}{20}$ and $\frac{1}{2} = \frac{10}{20}$; $\frac{9}{20}$ lies between. Check: 0.45 is between 0.4 and 0.5.

6 The denominators name part size, not the number of parts. With common denominator 63: $\frac{5}{9} = \frac{35}{63}$ and $\frac{4}{7} = \frac{36}{63}$, so $\frac{5}{9} < \frac{4}{7}$ is justified by the numerators.

7 Common denominator 48: $\frac{7}{12} = \frac{28}{48} > \frac{27}{48} = \frac{9}{16}$. Decimal check: $0.583 > 0.5625$. The common-denominator method is efficient here because 48 is the LCM and the numerators are close.

8 $\frac{100}{101}$ is closer to 1. Its distance from 1 is $\frac{1}{101}$, while the distance of $\frac{99}{100}$ is $\frac{1}{100}$; since $\frac{1}{101} < \frac{1}{100}$, $\frac{100}{101}$ is closer and larger.

PRACTICE 4.3 · ADDITION AND SUBTRACTION

1 (a) $\frac{4}{5}$. (b) $\frac{5}{7}$. (c) $\frac{3}{4}$. (d) 1.

2 (a) $\frac{5}{6}$. (b) $\frac{11}{12}$. (c) $\frac{29}{24} = 1\frac{5}{24}$. (d) $\frac{11}{10} = 1\frac{1}{10}$.

3 (a) $\frac{1}{2}$. (b) $\frac{1}{3}$. (c) $\frac{7}{20}$. (d) $\frac{4}{9}$.

4 (a) $\frac{43}{12} = 3\frac{7}{12}$. (b) $\frac{7}{3} = 2\frac{1}{3}$. (c) $\frac{31}{12} = 2\frac{7}{12}$. (d) $\frac{11}{2} = 5\frac{1}{2}$.

5 $\frac{2}{3} + \frac{3}{4} = \frac{8}{12} + \frac{9}{12} = \frac{17}{12} = 1\frac{5}{12}$ cups.

6 The student added numerators and denominators. Rewrite in fifteenths: $\frac{3}{5} = \frac{9}{15}$, $\frac{2}{3} = \frac{10}{15}$, sum $\frac{19}{15} = 1\frac{4}{15}$. The answer exceeds 1 because the two fractions are greater than $\frac{1}{2}$ each.

7 (a) $\frac{1}{3}$. (b) $\frac{1}{2}$.

8 Used $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$, so $\frac{5}{12}$ remains. Check: $\frac{4}{12} + \frac{3}{12} + \frac{5}{12} = 1$.

PRACTICE 4.4 · MULTIPLICATION AND DIVISION

1 (a) $\frac{3}{10}$. (b) $\frac{1}{2}$. (c) $\frac{1}{4}$. (d) $\frac{1}{2}$.

2 (a) 2. (b) 2. (c) $\frac{5}{4} = 1\frac{1}{4}$. (d) $\frac{1}{10}$.

3 (a) 1. (b) 3. (c) 4. (d) 2.

4 (a) 30. (b) 24 m. (c) $\frac{5}{6} \times 2 = \frac{5}{3} = 1\frac{2}{3}$ hours = 1 hour 40 minutes.

5 The product is smaller than $\frac{3}{4}$ because the second factor is less than 1; the quotient is larger because the divisor is less than 1. Values: $\frac{5}{8}$ and $\frac{9}{10}$.

6 The student multiplied instead of using the reciprocal. Correct: $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4} = \frac{5}{6}$. The reciprocal of $\frac{4}{5}$ is $\frac{5}{4}$; division asks how many $\frac{4}{5}$ are in the dividend.

7 (a) $4\frac{1}{2} \div \frac{3}{4} = \frac{9}{2} \times \frac{4}{3} = 6$ whole batches. (b) $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ kg.

8 $\frac{3}{4} = \frac{6}{8}$, so there are 6 groups of $\frac{1}{8}$; the quotient 6 is larger than $\frac{3}{4}$. Then $\frac{2}{3} \div \frac{4}{9} = \frac{2}{3} \times \frac{9}{4} = \frac{3}{2}$, and $\frac{3}{2} \times \frac{3}{8} = \frac{9}{16}$.

PRACTICE 4.5 · DECIMALS

1 (a) 0.007. (b) 0.2. (c) 0.7. (d) 0.05.

2 (a) $4 + 0.3 + 0.06$. (b) $0.7 + 0.008$. (c) $10 + 2 + 0.05$.

3 0.56, 0.6, 0.605, 0.65 (0.560, 0.600, 0.605, 0.650).

4 (a) 3.85 and 3.8. (b) 1.00. (c) 12.3.

5 (a) 8.35. (b) 4.55. (c) 0.42. (d) 9.

6 0.45 is 45 hundredths; 0.5 is 50 hundredths, so 0.5 is larger. $0.3 \times 0.4 = 0.12$, not 1.2, because tenths multiplied by tenths give hundredths.

7 2.38 m, 2.4 m, 2.405 m. 2.405 m is the most precise: it is recorded to thousandths, so its rounding unit is 0.001 m rather than 0.01 m or 0.1 m.

8 0.50 has 5 tenths and 0 hundredths, equal to 5 tenths. 0.05 has 0 tenths and 5 hundredths, which is 5 hundredths, ten times smaller than 0.5.

PRACTICE 4.6 · FRACTIONS AND DECIMALS

1 (a) 0.6. (b) 0.35. (c) 1.125. (d) 0.3125.

2 (a) $\frac{7}{10}$. (b) $\frac{9}{25}$. (c) $\frac{1}{8}$. (d) $\frac{12}{5} = 2\frac{2}{5}$.

3 (a) Terminates: denominator 2^3. (b) Repeats: factor 3. (c) Terminates: $40 = 2^3 \times 5$. (d) Repeats: factor 3.

4 $\frac{2}{3}, 0.67, 0.7, \frac{3}{4}$ (0.666..., 0.67, 0.70, 0.75).

5 $\frac{1}{3} = 0.333\ldots$, not 0.33 exactly; 0.33 is a rounded value. $0.3 = \frac{3}{10}$, not $\frac{1}{3}$. Three copies of 0.33 make 0.99, not 1.

6 $\frac{3}{8} = 0.375 < 0.4$, so the 0.4 inch bit is wider by 0.025 inch.

7 $\frac{1}{3}$ has denominator 3, which is neither 2 nor 5, so it repeats. $\frac{3}{8}$ has denominator 2^3, so it terminates at 0.375. Another repeating example: $\frac{1}{6} = 0.1\overline{6}$; another terminating example: $\frac{1}{20} = 0.05$.

8 $\frac{1}{3} = 0.333\ldots$, so multiplying by 3 gives $1 = 0.999\ldots$. The two decimals name the same number: their difference is smaller than every positive number, so it is zero. Algebraically, if $x = 0.999\ldots$, then $10x = 9.999\ldots$, and subtracting gives $9x = 9$, so $x = 1$.

APPLICATION · THE SCALING KITCHEN

1 The closest single measure to $\frac{4}{9}$ cup is $\frac{1}{2}$ cup. Error: $\frac{1}{2} - \frac{4}{9} = \frac{1}{18}$ cup $= \frac{240}{18} \approx 13.3$ mL. This exceeds a 10 mL tolerance, so it is not acceptable under that rule; a combination of cups or a marked jug is needed.

2 $5\frac{5}{8} = 5.625$; the rounded record 5.6 differs by 0.025 cup = 6 mL, which is within 10 mL. The supplier may prefer 5.6 because it is a single-decimal record that is easy to enter and read, and the difference is below the agreed tolerance.

3 Multiplying by $\frac{5}{2}$ is the same as doubling and adding half: $2x + \frac{1}{2}x = \frac{5}{2}x$. For sugar, $\frac{5}{2} \times \frac{1}{2} = \frac{5}{4} = 1\frac{1}{4}$ cups; doubling gives 1 cup and half the original is $\frac{1}{4}$ cup, totalling $1\frac{1}{4}$ cups. The single multiplication is less likely to introduce rounding error because it needs one measurement.

4 The fraction is more useful to the cook: it records the exact ratio and can be scaled again without losing the relationship. The decimal is more useful to the supplier: it fits measuring equipment and forms. The fraction hides how the quantity reads on a scale; the decimal hides that it may be a rounded value.

5 A measured decimal becomes the more honest record when the measuring instrument itself reports decimal places and no exact fractional relation exists — for example, a digital scale reading 0.44 kg. The decimal then states the precision actually available.

INVESTIGATION 4 · WHICH FRACTIONS STOP?

1 Terminating: 2, 4, 5, 8, 10. Repeating: 3, 6, 7, 9, 11, 12.

2 $2 = 2$, $4 = 2^2$, $5 = 5$, $8 = 2^3$, $10 = 2 \times 5$: only the primes 2 and 5 appear.

3 $3 = 3$, $6 = 2 \times 3$, $7 = 7$, $9 = 3^2$, $11 = 11$, $12 = 2^2 \times 3$: each contains a prime other than 2 or 5.

4 In simplest form, a fraction terminates exactly when its denominator has no prime factor except 2 and 5. Tests: $\frac{1}{16} = 0.0625$, $\frac{1}{20} = 0.05$, $\frac{1}{25} = 0.04$, $\frac{1}{40} = 0.025$ terminate; $\frac{1}{30} = 0.0\overline{3}$ repeats.

5 The 2 alone is not enough: $\frac{1}{6}$ also has the prime factor 3, and no power of 10 is divisible by 3.

6 $\frac{3}{12} = \frac{1}{4} = 0.25$, which terminates. The unsimplified denominator 12 contains 3 only because numerator and denominator share the factor 3; simplifying removes it.

7 Example terminating: $\frac{1}{16} = 0.0625$, denominator 2^4. Example repeating: $\frac{1}{13} = 0.\overline{076923}$, denominator 13.

8 $\frac{1}{27} = 0.\overline{037}$, whose repeating block has three digits; $\frac{1}{37} = 0.\overline{027}$ also works. Search by long division until a remainder repeats.

CRITERION TASK · THE CONVERSION CARD

1 Sample quantities: $\frac{5}{6}$, $1\frac{3}{4}$, 0.36, 1.125 and 0.7. Two are fractions in simplest form and two are decimals.

2 $\frac{5}{6} = 0.8\overline{3}$, $1\frac{3}{4} = 1.75$, $0.36 = \frac{9}{25}$, $1.125 = 1\frac{1}{8} = \frac{9}{8}$, $0.7 = \frac{7}{10}$. A mixed number and a thousandths decimal are included.

3 As decimals: 0.36, 0.7, 0.833..., 1.125, 1.75. On a number line they sit in that order between 0 and 2.

4 To two decimal places: 0.36, 0.70, 0.83, 1.13, 1.75. The first, second and last are exact; $\frac{5}{6}$ and 1.125 lose information because their exact decimals continue or end in a thousandths digit.

5 Addition: $\frac{5}{6} + 0.36 = \frac{5}{6} + \frac{9}{25} = \frac{125}{150} + \frac{54}{150} = \frac{179}{150} \approx 1.19$. Subtraction: $1.75 - 0.7 = 1.05 = \frac{21}{20}$.

6 $\frac{5}{6}$ repeats because 6 has the prime factor 3. $1\frac{3}{4}$, 0.36, 1.125 and 0.7 terminate because their simplest denominators contain only 2s and 5s.

7 Use the fraction when scaling a recipe or preserving an exact ratio; use the decimal when entering a measurement on a digital instrument. Risk of the wrong form: a rounded decimal scaled again can multiply the rounding error, while a fraction may suggest a precision the measuring tool cannot deliver.

8 Sample evaluation: a classmate converted 1.125 correctly but marked $\frac{5}{6}$ at 0.83 exactly. The card was at fault for not marking 0.83 as rounded; revise the number line to show $0.8\overline{3}$ and state the rounding. Accuracy, audience fit and a tested revision complete the card.

MIXED PRACTICE 4

1 (a) $\frac{2}{3}$. (b) $\frac{2}{3}$. (c) $\frac{2}{3}$. (d) $\frac{3}{4}$.

2 (a) 10. (b) 64. (c) 4. (d) 81.

3 (a) $\frac{15}{40} < \frac{16}{40}$, so $<$. (b) $\frac{49}{84} > \frac{48}{84}$, so $>$. (c) $\frac{15}{18} < \frac{16}{18}$, so $<$. (d) $\frac{54}{60} < \frac{55}{60}$, so $<$.

4 $\frac{7}{12}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$ (7, 8, 9, 10 twelfths).

5 (a) $\frac{29}{24} = 1\frac{5}{24}$. (b) $\frac{10}{9} = 1\frac{1}{9}$. (c) $\frac{6}{5} = 1\frac{1}{5}$.

6 (a) $\frac{1}{12}$. (b) $\frac{1}{3}$. (c) $\frac{8}{15}$.

7 (a) $\frac{103}{20} = 5\frac{3}{20}$. (b) $\frac{4}{3} = 1\frac{1}{3}$.

8 (a) $\frac{1}{6}$. (b) $\frac{3}{4}$. (c) 1.

9 (a) $\frac{2}{3}$. (b) $\frac{2}{7}$. (c) 4.

10 (a) 0.875. (b) 0.55. (c) 0.3125. (d) $1.\overline{3}$.

11 (a) $\frac{9}{20}$. (b) $\frac{3}{8}$. (c) $\frac{8}{5} = 1\frac{3}{5}$. (d) $\frac{2}{25}$.

12 (a) 5.28 and 5.3. (b) 1.00. (c) 17.5 (the digit after the tenths is 5).

13 (a) 9.15. (b) 6.4. (c) 0.56. (d) 5.

14 The student added numerators and denominators. Correct: $\frac{2}{3} + \frac{1}{5} = \frac{10}{15} + \frac{3}{15} = \frac{13}{15}$. This is greater than $\frac{2}{3}$ because a positive amount $\frac{1}{5}$ was added.

15 $\frac{3}{3}$ is a whole, so multiplying by it leaves $\frac{4}{5}$ unchanged: cutting a whole into 3 equal groups and taking all 3 gives the whole. $\frac{0}{0}$ is undefined: no number multiplied by 0 gives a non-zero numerator, and every number multiplied by 0 gives 0, so the result is not unique.

16 Dividing by $\frac{4}{5}$ is multiplying by its reciprocal $\frac{5}{4}$: $\frac{3}{4} \times \frac{5}{4} = \frac{15}{16}$. The answer is larger than $\frac{3}{4}$ because the divisor is less than 1.

17 $\frac{5}{7} = \frac{50}{70} > \frac{49}{70} = \frac{7}{10}$. Decimal check: $0.714 > 0.7$. Under time pressure, comparing with $\frac{7}{10}$ as a benchmark or using decimals is quick; the common-denominator method is the most reliable written check.

18 The first bottle: $\frac{5}{8} = 0.625$ L > 0.6 L. Difference $\frac{5}{8} - \frac{3}{5} = \frac{25}{40} - \frac{24}{40} = \frac{1}{40}$ L $= 0.025$ L.

19 (a) $\frac{3}{4} - \frac{2}{5} = \frac{15}{20} - \frac{8}{20} = \frac{7}{20}$. (b) 0.35. (c) $0.35 \times 2{,}400 = 840$ L. (d) Yes: starting half full gives $\frac{1}{2} - \frac{2}{5} = \frac{1}{10}$, so the remaining fraction changes.

20 (a) 0.35 m, 0.375 m, 0.4 m. (b) 1.125 m. (c) The longest strip is 0.4 m; each piece is 0.125 m, so 3 pieces are made and 0.025 m remains. (d) Rounding each to one decimal place gives $0.4 + 0.4 + 0.4 = 1.2$ m, which overstates the total by 0.075 m (7.5 cm); for fitting or ordering material this may matter and should be stated as a tolerance.

> **Common errors.** Equivalent fractions use multiplication or division, never addition. Denominators are part sizes, not quantities to add. Division by a fraction inverts the divisor. A longer decimal is not larger; align place values. Termination depends on the simplest denominator.

Chapter 5 · Percentages and applications

PREREQUISITE CHECK

1 0.75.

2 8.

3 30.

4 $\frac{9}{20}$.

5 60.

6 0.6.

GUIDED PRACTICE

1 $62\% = \frac{62}{100} = \frac{31}{50} = 0.62$. $4\% = \frac{1}{25} = 0.04$. $12.5\% = \frac{1}{8} = 0.125$. $0.09 = 9\%$.

2 $10\% = 6$, $30\% = 18$, $5\% = 3$, so 35% is 21. $1\% = 4$, so 2.5% is 10. $100\% = 25$ and $40\% = 10$, so 140% is 35.

3 Increase £15, new amount £75. Multiplier 0.9, result 81. $60 \div 0.75 = £80$.

4 Discount £24, sale price £56. Service charge £3, total £63. Amount spent £540, left £660, which is 55% of the budget.

PRACTICE 5.1 · PERCENT, FRACTIONS AND DECIMALS

1 (a) $\frac{1}{4}$, 0.25. (b) $\frac{3}{5}$, 0.6. (c) $\frac{2}{25}$, 0.08. (d) $\frac{3}{2}$, 1.5.

2 (a) 25%. (b) 35%. (c) 112.5%. (d) 60%.

3 (a) 70%. (b) 9%. (c) 135%. (d) 0.5%.

4 (a) 35, 35. (b) 0.375, 37.5. (c) 6, 6. (d) 250, 2.5.

5 (a) $0.4 = 40\% > 39\%$. (b) $\frac{5}{8} = 62.5\% > 62\%$. (c) $1.2 = 120\% > 115\%$.

6 0.5% means half of one hundredth, so it is 0.005, not 0.5. As a fraction, $0.5\% = \frac{5}{1000} = \frac{1}{200}$. The decimal 0.5 is 50%.

7 $85\% = \frac{85}{100} = \frac{17}{20} = 0.85$; $0.85 \times 40 = 34$ marks.

8 No. £50 increased by 100% becomes £100; a second 100% increase doubles £100 to £200. The overall multiplier is $2 \times 2 = 4$, so the final price is 400% of the original, not 200%.

PRACTICE 5.2 · A PERCENTAGE OF A QUANTITY

1 (a) 9. (b) 15. (c) 3.5. (d) 4.5.

2 (a) 12. (b) 21. (c) £15.60. (d) 54.

3 (a) 10. (b) 4.5. (c) 35. (d) 30.

4 $0.15 \times 80 = £12$.

5 $0.48 \times 650 = 312$ students.

6 30% of 80 is 24; 25% of 100 is 25. The second is larger by 1.

7 The student misplaced the decimal: $0.065 \times 240 = 15.6$, not 156. An estimate confirms this: 10% of 240 is 24, so 6.5% must be less than 24; 156 is far too large.

8 Both equal $0.15 \times 80 = 0.8 \times 15 = 12$, because multiplication is commutative: $\frac{15}{100} \times 80 = \frac{80}{100} \times 15$. Another pair: 25% of 40 = 10 = 40% of 25.

PRACTICE 5.3 · INCREASE AND DECREASE

1 (a) 44. (b) 84. (c) 260. (d) 27.

2 (a) 60. (b) 98. (c) 56. (d) 495.

3 $240 \times 1.15 = £276$.

4 $1{,}200 \times 0.92 = 1{,}104$ g.

5 Decrease $= 2{,}250$; $\frac{2{,}250}{45{,}000} = 0.05 = 5\%$.

6 $54 \div 1.2 = £45$.

7 £100 becomes £120 after the increase, then £96 after the decrease. The second 20% is taken of £120, not £100, so the changes do not cancel. Multipliers: $1.2 \times 0.8 = 0.96$.

8 The final price is 99% of the original, so it is 1% less. Multipliers $1.1 \times 0.9 = 0.99$. Reversing the order gives $0.9 \times 1.1 = 0.99$; multiplication is commutative, so the order does not change the final amount, although the intermediate amounts differ.

PRACTICE 5.4 · DISCOUNTS, TAX AND BUDGETS

1 (a) Discount £20, sale £60. (b) Discount £18, sale £132. (c) Discount £1.80, sale £34.20.

2 (a) £72. (b) £252. (c) £22.20.

3 (a) Rent £720, transport £360, food £600. (b) 30% remains, which is £720.

4 No. If £15 is 20% of the original, the original is $15 \div 0.2 = £75$, not £90. A 20% saving on £90 would be £18. The shopper treated the saving as a percentage of the sale price.

5 Shop A: £60 − 25% = £45. Shop B: £60 − £12 = £48. Shop A is cheaper by £3.

6 Service charge $0.125 \times 48 = £6$, total £54.

7 40% of £250 is £100, but £120 was spent. That is $\frac{120}{250} = 48\%$ of the whole budget, or 120% of the planned amount; the student is £20 over the materials budget.

8 No. The sale price is $0.8 \times 80 = £64$; adding 20% tax gives $1.2 \times 64 = £76.80$. The overall multiplier is $0.8 \times 1.2 = 0.96$, so the final price is 96% of the original.

9 $\frac{450}{1{,}500} = 30\%$. At 25%, rent would be £375, so the household would save £75 per month.

10 Two items of price p cost $2p$ at full price; "buy one, get one half price" charges $1.5p$. Saving $= 0.5p$, which is $\frac{0.5p}{2p} = 25\%$ of the pair's full price. It is not 50% because the reduction applies to only one of the two items.

APPLICATION · THE COMMUNITY MARKET STALL

1 Honey: sale £8.10, profit £2.10. Soap: sale £3.00, profit £0.50. Jam: sale £5.00, profit £1.80.

2 Revenue $= 20 \times 8.10 + 36 \times 3.00 + 25 \times 5.00 = £395$. Cost of goods $= 20 \times 6 + 36 \times 2.50 + 25 \times 3.20 = £290$. Gross profit £105. Fixed costs £63, so net profit £42.

3 Honey $162 \div 395 \approx 41.0\%$; soap $108 \div 395 \approx 27.3\%$; jam $125 \div 395 \approx 31.6\%$. The rounded shares sum to 99.9%; the 0.1% difference is rounding error. The exact shares sum to 100%.

4 15% more: honey 23, soap 41.4 → 42, jam 28.75 → 29. New cost $= 23 \times 6 + 42 \times 2.50 + 29 \times 3.20 = 138 + 105 + 92.80 = £335.80$.

5 New fixed costs $= 63 \times 1.08 = £68.04$. With original revenue and cost of goods, net profit $= 395 - 290 - 68.04 = £36.96$.

6 Offer price per jar $= 6 \times 0.90 = £5.40$; 30 jars cost £162. The 23 jars needed at normal cost £138. The offer costs £24 more and supplies 7 extra jars; it is only worth accepting if the group can sell those extras. Risk: unsold stock reduces net profit and ties up money.

7 A blanket 15% discount at the same quantities gives revenue $20 \times 7.65 + 36 \times 3.40 + 25 \times 4.25 = 153 + 122.40 + 106.25 = £381.65$, £13.35 less than £395. The group prefers selective discounts because jam has no discount and a higher margin, while the blanket offer cuts its price too; discounts can be targeted where they increase sales most.

8 The group needs net profit to cover costs; gross profit is useful for comparing products but ignores the £63 fixed cost. A customer cares about the final price, not the percentage alone: a 10% discount on £9 saves £0.90, while 25% on £4 saves £1.00. State both percentage and amount.

INVESTIGATION 5 · DO TWO DISCOUNTS ADD UP?

1 Offer A: £250 → £200 → £180, total reduction £70. Offer B: £250 → £175, reduction £75. Offer B is cheaper by £5.

2 10% first: £250 → £225; then 20% of £225 is £45, giving £180. The final price is unchanged. Multipliers $0.9 \times 0.8 = 0.72$ and $0.8 \times 0.9 = 0.72$.

3 $0.8 \times 0.9 = 0.72$, equivalent to a single 28% discount, not 30%.

4 £40: A gives £28.80, B gives £28, difference £0.80. £80: A gives £57.60, B gives £56, difference £1.60. The difference is not the same amount; it is always 2% of the original price (£0.80 is 2% of £40, £1.60 is 2% of £80).

5 Multipliers $1.2 \times 0.8 = 0.96$; £100 becomes £96. The decrease is taken of £120, so it removes £24 while the increase added only £20.

6 Three 10% discounts: $0.9^3 = 0.729$, so £200 becomes £145.80. A single 30% discount gives £140. The single 30% is cheaper by £5.80. The three reductions are equivalent to 27.1% off, not 30%.

7 Overall multiplier $(1 - \frac{a}{100})(1 - \frac{b}{100})$; the equivalent single discount is $100\left[1 - (1 - \frac{a}{100})(1 - \frac{b}{100})\right]$ %. For 20% and 10%: 28%. For 25% and 20%: $1 - 0.75 \times 0.8 = 0.40$, so 40%.

8 Successive percentage changes multiply, and multiplication is commutative, so the order of the multipliers does not change the final amount. Adding £10 then taking 10% gives $0.9(x + 10) = 0.9x + 9$, whereas taking 10% then adding £10 gives $0.9x + 10$. The operations do not commute.

9 25% and 4%: $0.75 \times 0.96 = 0.72$, equivalent to 28% off. Check on £100: £100 → £75 → £72.

CRITERION TASK · THE FAIR-PRICE CARD

1 Sample item marked at £80. Every percentage is stated as a percentage of the marked price unless another base is named.

2 20% discount gives £64. A fixed £15 off gives £65. The percentage offer is cheaper by £1, which is 1.25% of the marked price.

3 20% tax on £64 is £12.80, total £76.80. The total is 1.2×64, i.e. 120% of the pre-tax price.

4 Increase: $80 \times 1.15 = £92$, estimated as $80 + 8 + 4$. Decrease: $80 \times 0.85 = £68$, estimated as $80 - 12$.

5 Sample budget: item £64 = 3.2%; rent £700 = 35%; food £400 = 20%; transport £200 = 10%; savings £300 = 15%; remaining £336 = 16.8%. The percentages sum to 100%.

6 A 25% discount on £90 gives £67.50, while a 20% discount on £80 gives £64. The smaller percentage is the better deal because its base is lower; the percentage alone does not compare the final prices.

7 Error 1: "0.5% = 0.5." Correct: 0.5% = 0.005 = $\frac{1}{200}$. Error 2: "+20% then −20% returns to the start." Correct: $1.2 \times 0.8 = 0.96$, so the result is 96% of the original.

8 Sample evaluation: the reader correctly found the sale price but added 20% tax to the marked price rather than the sale price. The card was ambiguous about the tax base; revise it to label "20% of the reduced price" beside the calculation. Accuracy, clarity of bases and a tested revision complete the card.

MIXED PRACTICE 5

1 (a) $\frac{7}{20}$, 0.35. (b) $\frac{2}{25}$, 0.08. (c) $\frac{3}{2}$, 1.5. (d) $\frac{1}{200}$, 0.005.

2 (a) 60%. (b) 28%. (c) 110%. (d) 62.5%.

3 (a) 42%. (b) 8%. (c) 160%. (d) 12.5%.

4 30.

5 36.

6 30.

7 78.

8 $150 \times 1.12 = 168$.

9 $84 \times 0.75 = 63$.

10 $360 \div 1.2 = 300$.

11 $140 \times 0.85 = 119$.

12 $85 \times 1.2 = 102$.

13 Spent £1,300; left £700.

14 $0.15 \times 80 = 12$ and $0.80 \times 15 = 12$. They agree because $\frac{15}{100} \times 80 = \frac{80}{100} \times 15$; multiplication is commutative.

15 Both equal 45. Estimate: 18% of 250 is 20% of 250 minus 2% of 250, i.e. $50 - 5 = 45$; 25% of 180 is a quarter of 180, also 45.

16 No. Multipliers $0.9 \times 1.1 = 0.99$, so the final price is 99% of the original, 1% lower. The increase is taken of the reduced amount.

17 Shop A: £64. Shop B: £61.20. Shop B is cheaper by £2.80.

18 (a) $800 \div 2{,}500 = 32\%$. (b) New share $900 \div 2{,}500 = 36\%$; rent itself increased by $100 \div 800 = 12.5\%$. The share of income rose by 4 percentage points.

19 (a) Fee £96, transport £48, materials £660, profit £396. (b) New takings $= 1{,}200 \times 0.9 \times 1.25 = £1{,}350$; fee £108, transport £54, materials $660 \times 1.25 = £825$; profit $= 1{,}350 - 108 - 54 - 825 = £363$. (c) No: profit fell from £396 to £363 despite higher takings, because materials and percentage fees rose. The discount increased sales but reduced profit.

20 (a) $180 \times 0.75 \times 0.9 = £121.50$. (b) $0.75 \times 0.9 = 0.675$, so the single equivalent discount is 32.5%. (c) 35% would give £117, so the customer overstates the discount by 2.5 percentage points, or £4.50 on a £180 jacket.

Common errors. Per cent means per hundred: 7.5% = 0.075. A percentage needs a base. An increase of 15% uses the multiplier 1.15, not "add £15". To reverse a change, divide by the multiplier. Successive percentages multiply, so 20% then 10% is 28% off, not 30%.

Chapter 6 · Patterns, expressions and equations

PREREQUISITE CHECK

1 42.

2 8.

3 5×4.

4 14, 17.

5 14.

6 9.

GUIDED PRACTICE

1 Difference 4, multiplier 4; each term is 1 more; rule $4n + 1$. 10th term $4(10) + 1 = 41$. $4n + 1 = 45$ gives $4n = 44$, $n = 11$.

2 $4x$: 8, 12, 20. $x + 7$: 9, 10, 12. $3x - 2$: 4, 7, 13. x^2: 4, 9, 25.

3 $8x$; $5y$; $3a + 15$; $8n - 2$; $7p - 3$; $7x + 8$.

4 $x = 7$; $x = 8$; $2x = 16$, $x = 8$; $x + 1 = 6$, $x = 5$; check $3(5 + 1) = 18$.

PRACTICE 6.1 · SEQUENCES AND GENERAL RULES

1 (a) 26, 31, 36; add 5. (b) 24, 18, 12; subtract 6. (c) 48, 96, 192; multiply by 2. (d) 25, 36, 49; add successive odd numbers 3, 5, 7, ...

2 6, 10, 14, 18, 22; 10th term $4(10) + 2 = 42$.

3 Stage 1 has 4, so the rule is $3n + 1$. Stage 12: $3(12) + 1 = 37$ tiles.

4 Difference 3, so matches $= 3n + 1$. 15 squares: 46 matches. $3n + 1 = 61$ gives $n = 20$.

5 $3(1) + 5 = 8$, $3(2) + 5 = 11$, $3(3) + 5 = 14$; 10th term $3(10) + 5 = 35$.

6 The multiplier is the constant difference 3, not the first term 2; and $2n$ gives 2, 4, 6, 8, which fails from the second term. Correct rule $3n - 1$.

7 $3 + 2d$. 12 km: £27. 25 km: £53.

8 $5n - 3 = 72$ gives $5n = 75$, $n = 15$. For 100: $5n = 103$, $n = 20.6$, not a whole number, so 100 is not a term; the neighbouring terms are 97 and 102.

PRACTICE 6.2 · EXPRESSIONS AND SUBSTITUTION

1 (a) $x + 6$. (b) $x - 8$. (c) $5x$. (d) $\frac{x}{4}$. (e) $2x + 3$.

2 (a) 2 terms; coefficient 5; constant 2. (b) 2 terms; coefficient -3; constant 7. (c) 3 terms; coefficients 4 and 2; constant -1.

3 $x = 4$: (a) 12. (b) 13. (c) 18. (d) 16. $x = 10$: (e) 22. (f) 8.

4 (a) $6 + 5 = 11$. (b) $3 \times 3 \times 5 = 45$. (c) $25 - 9 = 16$. (d) $2(8) = 16$.

5 (a) 5. (b) -8. (c) $3 + 4 = 7$. (d) 4.

6 $2x$ means $2 \times x$, not $2 + x$. When $x = 5$, $2x + 1 = 10 + 1 = 11$, not 8.

7 $8 + 0.10m$. 60 minutes: £14. 200 minutes: £28.

8 $P = 2((2x + 1) + x) = 2(3x + 1) = 6x + 2$. When $x = 4$, $P = 26$ cm.

PRACTICE 6.3 · SIMPLIFY AND EXPAND

1 (a) $7a$. (b) $7x$. (c) $7m$. (d) 0.

2 (a) $8x + 2$. (b) $5a + 4$. (c) $2n + 3$. (d) $5p + 3q$.

3 (a) $2x + 10$. (b) $5y - 15$. (c) $6m + 3$. (d) $12 - 4n$.

4 (a) $3x + 6 + 4x = 7x + 6$. (b) $6n - 2 + 5 = 6n + 3$. (c) $10a + 5 + 2a = 12a + 5$.

5 $4x + 8$ and $8 + 4x$ are equivalent; $4x + 2$ is not. Both terms inside the bracket are multiplied by 4, and addition is commutative.

6 Only the first term was multiplied; the 4 must also be multiplied: $3(x + 4) = 3x + 12$. Also $5a$ and $2b$ are unlike terms and cannot be combined: $5a + 2b$ stays as it is.

7 Perimeter $= 2((x + 5) + 3) = 2x + 16$ cm; area $= 3(x + 5) = 3x + 15$ cm^2. When $x = 4$: perimeter 24 cm, area 27 cm^2.

8 $6x + 2 + 3x + 6 = 9x + 8$. When $x = 5$, $9(5) + 8 = 53$. Original: $2(16) + 3(7) = 32 + 21 = 53$.

PRACTICE 6.4 · EQUATIONS AND FORMULAS

1 (a) $x = 7$. (b) $y = 13$. (c) $m = 7$. (d) $n = 30$.

2 (a) $x = 5$. (b) $x = 6$. (c) $y = 7$. (d) $n = 6$.

3 (a) $x = 5$. (b) $x = 7$. (c) $x = 4$.

4 $3x + 7 = 25$, so $3x = 18$ and $x = 6$.

5 $2(12 + w) = 40$, $12 + w = 20$, $w = 8$ cm.

6 Adding 4 destroys the balance; 4 must be subtracted. $3x = 15$, so $x = 5$. Check $3(5) + 4 = 19$.

7 $3 + 2d = 23$, $2d = 20$, $d = 10$ km.

8 $2((x + 3) + x) = 4x + 6 = 26$, so $4x = 20$, $x = 5$. The rectangle is 8 cm by 5 cm; check $2(8 + 5) = 26$.

INVESTIGATION 6 · THE MATCHSTICK CHAIN

1 Stage 4 uses 13 matches, agreeing with the table.

2 Differences $7 - 4 = 3$, $10 - 7 = 3$, $13 - 10 = 3$: constant, so the rule is linear, of the form $3n + c$.

3 The first square uses 4 matches. Each extra square shares one side with the previous square, so it adds only 3 matches. For n squares: $4 + 3(n - 1) = 3n + 1$.

4 12 squares: $3(12) + 1 = 37$. 50 squares: $3(50) + 1 = 151$.

5 At $n = 2$, $4n = 8$, but the chain uses 7. The shared side is counted twice by "4 per square"; there are $n - 1$ shared sides, so $4n - (n - 1) = 3n + 1$.

6 $3n + 1 = 100$, $3n = 99$, $n = 33$. Exactly 33 squares can be made with no match left over.

7 Two separate chains of 5 use $16 + 16 = 32$ matches. Two chains of n: $2(3n + 1) = 6n + 2$.

8 First triangle 3 matches; each extra triangle adds 2 because one side is shared, so $3 + 2(n - 1) = 2n + 1$. $2n + 1 = 41$ gives $n = 20$ triangles.

APPLICATION · THE COURTYARD POND

1 A 1 m slab is added on every side, so the outer square has side $n + 2$. Border $= (n + 2)^2 - n^2 = 4n + 4$; the table increases by 4 each time.

2 $n = 7$: $4(7) + 4 = 32$ slabs. $n = 12$: $4(12) + 4 = 52$ slabs.

3 $4n + 4 = 44$, so $4n = 40$ and $n = 10$ m. Outer square $12 \times 12 = 144$, pond $10 \times 10 = 100$, border $144 - 100 = 44$.

4 Cost $= 3.50(4n + 4) + 25 = 14n + 14 + 25 = 14n + 39$. For $n = 7$: £98 + £39 = £137.

5 $14n + 39 \le 200$, so $14n \le 161$ and $n \le 11.5$. Largest whole side 11 m, costing £193. A 12 m pond costs $14(12) + 39 =$ £207, £7 over budget.

6 A side of 11 m needs 48 slabs. Laying $48 \times 12 = 576$ min $= 9$ h 36 min; with 2 h preparation, total 11 h 36 min.

7 Enlarging to 12 m adds 4 slabs, costing £14, and 48 minutes of laying. The cost rises from £193 to £207, so the project is £7 over budget with £0 left for benches. The school must raise at least £7 or save £14 elsewhere; the rule makes both the cost and the overrun predictable before work starts.

CRITERION TASK · THE PATTERN EXPLAINER

Sample response. A square frame of counters is built around a growing square hole. Stage 1 has 8 counters, stage 2 has 12, stage 3 has 16 and stage 4 has 20.

1–2 Labelled stages 1–4; table: $1 \to 8$, $2 \to 12$, $3 \to 16$, $4 \to 20$. First differences are all 4, so the term-to-term rule is "add 4".

3 Rule $4n + 4$. Top and bottom rows each hold $n + 2$ counters; the two side columns each hold n more, because the corners are already counted: $2(n + 2) + 2n = 4n + 4$. The rule follows from the construction for every n, not only from four values.

4 Stage 50: $4(50) + 4 = 204$ counters.

5 $4n + 4 = 100$, so $4n = 96$ and $n = 24$. Stage 24 uses exactly 100 counters.

6 Cost $= 1.50(4n + 4) + 8 = 6n + 6 + 8 = 6n + 14$. Stage 20: $6(20) + 14 =$ £134.

7 Wrong claim: $4n$, "four sides of n". At stage 1 it gives 4, but the frame has 8; it counts one counter per side and forgets the four corners. Correction: $4n + 4$. A younger pupil can walk the frame: count the top and bottom rows, then the two side columns, and do not count a corner twice.

8 Sample test record: a reader used $4n$ and predicted 20 counters at stage 5 instead of 24. Improvement: colour the four corner counters and label the top, bottom and side groups on the poster.

Marking guide. Criterion A: accurate stages, table and rule, with correct substitution. Criterion C: labelled layout, correct notation, and a structural justification that distinguishes evidence from proof. Criterion D: correct stage 50, valid inverse solution, correct cost with the fixed charge, and a tested, revised poster.

MIXED PRACTICE 6

1 (a) 24, 29, 34. (b) 56, 48, 40. (c) 40, 80, 160.

2 (a) $4n + 2$. (b) $7n - 5$. (c) $2n - 1$.

3 (a) 119. (b) 83.

4 (a) $3n + 4$. (b) $\frac{n}{2} + 7$. (c) $2n - 5$.

5 (a) 19. (b) 72. (c) 24.

6 (a) $9a$. (b) $4x + 4$. (c) $4m + 3n$.

7 (a) $6x + 15$. (b) $12 - 4y$.

8 (a) $5x + 8$. (b) $11n - 5$.

9 (a) 8. (b) 8. (c) 30.

10 (a) 6. (b) 8. (c) 6.

11 $5x - 4 = 31$, $5x = 35$, $x = 7$.

12 $P = 2(9 + 4) = 26$ cm; $A = 9 \times 4 = 36$ cm^2.

13 $3(x + 4) = 3x + 12$: both terms are multiplied by 3. $5x + 2$ cannot become $7x$: 2 is a constant, not an x-term. Correct results $3x + 12$ and $5x + 2$.

14 When $n = 10$: $3n + 7 = 37$, $4n - 1 = 39$, so $4n - 1$ is larger by 2. Equal when $3n + 7 = 4n - 1$, so $n = 8$; both equal 31.

15 Stage 20: $4(20) + 2 = 82$. $4n + 2 = 102$ gives $4n = 100$, $n = 25$.

16 $2((2x + 3) + x) = 6x + 6 = 30$, so $6x = 24$, $x = 4$. Length 11 cm, width 4 cm.

17 A: $12 + 0.10m$; B: $20 + 0.05m$. At 100 min: A £22, B £25 (A cheaper by £3). At 200 min: A £32, B £30 (B cheaper by £2). Equal when $12 + 0.10m = 20 + 0.05m$, so $0.05m = 8$, $m = 160$ min; both cost £28.

18 (a) $2(3n + 2) + 10 = 6n + 4 + 10 = 6n + 14$. (b) $6(15) + 14 =$ £104. (c) $6n + 14 = 68$, $6n = 54$, $n = 9$.

19 (a) Rule $4n + 1$; 20th stage $4(20) + 1 = 81$ tiles. (b) $1.50(4n + 1) + 30 = 6n + 31.50$. (c) $6n + 31.50 \le 150$, so $6n \le 118.50$ and $n \le 19.75$. Largest whole stage 19, costing £114 + £31.50 = £145.50, with £4.50 left. Stage 20 costs £120 + £31.50 = £151.50, £1.50 over budget, so choose stage 19.

20 Table: $6n - 2$ gives 4, 10, 16, 22; $2(3n - 1)$ gives $2(2) = 4$, $2(5) = 10$, $2(8) = 16$, $2(11) = 22$. Expanding gives $2(3n - 1) = 6n - 2$. The form $6n - 2$ shows the £6 per student and the £2 voucher separately; $2(3n - 1)$ is compact for calculation but hides the two contextual amounts.

Common errors. The constant difference is the multiplier, but the rule still needs an adjustment to fit the first term. A coefficient multiplies; it does not add. Only like terms combine. Expanding multiplies every term inside the bracket. Equations stay balanced, and every solution should be checked by substitution.

Chapter 7 · Lines, angles, triangles and constructions

PREREQUISITE CHECK

1 $x = 140$.

2 115.

3 $x = 60$.

4 Isosceles.

5 90°.

6 220.

GUIDED PRACTICE

1 $118 + 62 = 180$, so the missing angle is $62°$. At a point: $95 + 120 + 85 = 300$, so the fourth angle is $60°$. Vertically opposite $47°$; each adjacent angle is $180 - 47 = 133°$.

2 $x = 180 - 48 - 67 = 65°$. Isosceles with apex $40°$: each base angle is $(180 - 40) \div 2 = 70°$. Alternate to $113°$ is $113°$; co-interior partner $180 - 113 = 67°$.

3 Pentagon: $5 - 2 = 3$ triangles, sum $3 \times 180 = 540°$. Regular hexagon: exterior $360 \div 6 = 60°$, interior $120°$. Exterior $20°$: $360 \div 20 = 18$ sides.

4 Order ii, iv, i, iii. Both intersection points are the same distance from A and from B. For $AB = 8$ cm, the radius must exceed 4 cm.

PRACTICE 7.1 · LINES AND ANGLE FACTS

1 (a) Acute. (b) Right. (c) Obtuse. (d) Straight. (e) Reflex. (f) Reflex.

2 $180 - 47 - 68 = 65°$, angles on a straight line.

3 $360 - (85 + 95 + 120) = 60°$, angles at a point.

4 The vertically opposite angle is $132°$; the two adjacent angles are $48°$ each, because they are supplementary to $132°$.

5 $4x + 15 = 2x + 45$, so $2x = 30$ and $x = 15$. Each angle is $4(15) + 15 = 75°$. Check: $2(15) + 45 = 75$.

6 $x + 30 + 2x + x + 10 = 180$, so $4x + 40 = 180$, $x = 35$. The angles are $65°$, $70°$ and $45°$.

7 The statement is false: supplementary angles sum to $180°$, but they need not be equal. For example, $60°$ and $120°$ lie on a straight line and are not equal. A straight line is not a line of symmetry for an arbitrary pair of adjacent angles.

8 $1 + 2 + 3 + 4 = 10$ parts, so one part is $360 \div 10 = 36°$. The angles are $36°$, $72°$, $108°$ and $144°$. No pair is complementary: the smallest sum is $36 + 72 = 108°$, already above $90°$.

PRACTICE 7.2 · PARALLEL LINES AND TRIANGLES

1 Corresponding $115°$; alternate $115°$; co-interior $180 - 115 = 65°$.

2 $180 - 48 - 67 = 65°$.

3 $(180 - 40) \div 2 = 70°$ for each base angle.

4 $90 - 32 = 58°$; the acute angles of a right triangle are complementary.

5 $3x + 4x + 5x = 180$, so $12x = 180$ and $x = 15$. The angles are $45°$, $60°$ and $75°$, so the triangle is acute.

6 Other opposite interior angle: $118 - 54 = 64°$. Interior angle at the exterior vertex: $180 - 118 = 62°$. Check: $54 + 64 + 62 = 180$.

7 Two obtuse angles already sum to more than $180°$, leaving no positive amount for the third angle. For example, $100° + 100° = 200°$, which exceeds the triangle sum.

8 Let $A = x$; then $B = 2x$ and $C = x + 20$. So $4x + 20 = 180$, $x = 40$. The angles are $40°$, $80°$ and $60°$.

PRACTICE 7.3 · POLYGONS AND THEIR ANGLES

1 (a) $360°$. (b) $540°$. (c) $1080°$. (d) $1440°$.

2 (a) $108°$. (b) $120°$. (c) $135°$. (d) $144°$.

3 $360 - (100 + 80 + 95) = 85°$.

4 Pentagon sum $540°$; the known angles total $460°$, so the fifth angle is $80°$.

5 Exterior angle $360 \div 12 = 30°$; interior angle $180 - 30 = 150°$.

6 Exterior angle $180 - 156 = 24°$, so the number of sides is $360 \div 24 = 15$.

7 Walking once around the polygon, at each vertex you turn through the exterior angle. After a complete circuit you face the original direction, so the total turn is one full turn, $360°$.

8 $(n - 2) \times 180 = 1800$, so $n - 2 = 10$ and $n = 12$. Each exterior angle is $360 \div 12 = 30°$; each interior angle is $150°$.

PRACTICE 7.4 · CONSTRUCTIONS

1 Draw AB. Open the compasses to more than half AB. With centre A, draw arcs above and below. With the same radius and centre B, draw arcs meeting the first pair at P and Q. Draw line PQ; it bisects AB at $90°$. The radius must exceed $\frac{1}{2}AB$.

2 Each half is $7.2 \div 2 = 3.6$ cm. Measurement should agree within drawing tolerance.

3 Each half should be $64 \div 2 = 32°$.

4 The arcs meet because $4 + 5 = 9 > 6$: the two shorter sides together exceed the longest. The construction produces the required triangle.

5 A radius of 2 cm is less than half of 6 cm, so the arcs from A and B cannot reach each other. Use any radius greater than 3 cm, for example 4 cm.

6 Use radius 5 cm from both ends of a 5 cm base. The arcs meet at C, and $AC = AB = BC = 5$ cm. Equal sides give equal opposite angles; their sum is $180°$, so each is $60°$.

7 The perpendicular bisector of AB locates every point equidistant from the two gates. Draw the perpendicular bisector; the path follows it.

8 Draw the base $AB = 8$ cm. With radius 6 cm from A and from B, the arcs meet at the apex C. The apex is equidistant from A and B, so it lies on the perpendicular bisector of AB; that is why the bisector passes through it. The triangle is isosceles by construction.

INVESTIGATION 7 · THE TRIANGLE INEQUALITY

1 The completed table agrees with the printed entries. For 4, 5 and 9, $4 + 5 = 9$, so the two shorter strips just reach across the longest. They lie flat in a straight line; the triangle is degenerate, with zero height.

2 A triangle is possible exactly when $a + b > c$, where c is the longest side. Equality gives a degenerate straight-line "triangle"; a smaller sum leaves a gap.

3 5, 8, 13: $5 + 8 = 13$, degenerate. 5, 8, 12: $13 > 12$, a triangle. 7, 7, 14: $14 = 14$, degenerate. The borderline cases satisfy equality and collapse to a straight line.

4 The base endpoints are c apart. An arc of radius a from one endpoint and an arc of radius b from the other meet only if the two radii can span the distance between the centres: $a + b \geq c$. For a genuine triangle the meeting points must be off the base, so $a + b > c$.

5 Yes. If the two shorter sides sum to more than the longest, then the longest plus either shorter side certainly exceeds the remaining side, because the longest is at least as large as that side. Checking the two shorter sides against the longest is therefore sufficient.

6 \(8-58\); the largest, 12, satisfies $5 + 8 > 12$. Values 3 and 13 would be degenerate.

7 No: $4 + 4 = 8 < 9$. The arcs of radius 4 cm from the ends of a 9 cm base do not meet, leaving a gap of 1 cm.

8 $x + (x + 1) > x + 2$, so $2x + 1 > x + 2$ and $x > 1$. The smallest whole number is $x = 2$, giving sides 2, 3 and 4; indeed $2 + 3 > 4$.

REAL-LIFE APPLICATION · THE ROOF-TRUSS WORKSHOP

1 The interior angles of a triangle sum to $180°$, so the apex is $180 - 65 - 65 = 50°$.

2 The truss is isosceles. The two sloping rafters are equal, and the equal base angles are the $65°$ angles at the base.

3 The collar is parallel to the base, so the angle on the other side of the collar at the same joint is $180 - 65 = 115°$. The $65°$ and $115°$ angles are co-interior (or supplementary along the rafter).

4 At 1:20, the 6 m base becomes 30 cm. The apex should measure $50°$. Each full-size rafter is $3 \div \cos 65° \approx 7.10$ m, so the drawing should show about 35.5 cm. Measurements within a few millimetres and $1°$ are acceptable.

5 $2.5 + 2.5 = 5 < 8$. The two rafters cannot span the 8 m base, so the design cannot be built as stated.

6 Construct the perpendicular bisector of the base. The apex is equidistant from the two base endpoints because the rafters are equal, so it lies on that bisector. The centre line is therefore perpendicular to the base and meets it at the midpoint.

7 A constructed centre line is exact in principle but slower; a measured template is quick to reproduce but accumulates measurement error. Recommend the construction for the master template and measurement for repeated copies. Measurement is preferable when speed matters and a small tolerance is acceptable.

CRITERION TASK · THE GEOMETRY GUIDE

Sample response. A complete guide uses labelled diagrams rather than paragraphs alone. The entries below show the required content and one consistent set of check questions; a reader's own examples are acceptable if the angle facts and reasons are correct.

1 Acute $< 90°$, right $= 90°$, obtuse between $90°$ and $180°$, reflex between $180°$ and $360°$. Notation: $\angle ABC$ names the angle at B between BA and BC; the vertex letter sits in the middle.

2 Straight line: adjacent angles sum to $180°$. At a point: angles sum to $360°$. Vertically opposite angles are equal. Each fact is illustrated with a labelled diagram and one numerical example.

3 With two parallel lines and a transversal, the eight angles form four pairs of vertically opposite equal angles. Corresponding pairs are equal; alternate pairs are equal; co-interior pairs sum to $180°$. The guide labels all eight positions and gives one pair of each type.

4 Draw line DE through C, parallel to AB. Then $\angle DCA = \angle A$ and $\angle ECB = \angle B$ (alternate angles). The three angles at C form a straight angle, so $\angle A + \angle B + \angle C = 180°$.

5 Sample: perpendicular bisector of AB. Draw AB; set compasses to more than half AB; draw equal-radius arcs from A and from B, meeting at P and Q; draw PQ. The equal radii make P and Q equidistant from A and B, so PQ is the perpendicular bisector.

6 Worked example: the angles of a triangle are $2x$, $3x$ and $4x$. $2x + 3x + 4x = 180$ (triangle sum), so $9x = 180$, $x = 20$. The angles are $40°$, $60°$ and $80°$. Each line states the fact used.

7 Check questions: (i) parallel lines with one angle $73°$ — find the alternate angle and the co-interior angle ($73°$, $107°$); (ii) interior angle sum of a pentagon ($540°$); (iii) triangle angles in the ratio $2 : 3 : 4$ ($40°, 60°, 80°$). Answers are hidden at the back.

8 Test record: for example, the reader hesitated at "draw arcs from both ends" because the radius was not stated; the guide was changed to "set the compasses to 5 cm and keep that setting for both arcs".

Marking guide. Criterion A: all angle facts and construction steps correct, with the triangle-sum proof valid. Criterion C: labelled diagrams, correct notation and instructions a non-specialist can follow. Criterion D: the guide is usable, the check questions are consistent with the explanations, and the test record leads to a specific improvement.

MIXED PRACTICE 7

1 (a) Acute. (b) Right. (c) Obtuse. (d) Reflex.

2 $180 - 63 - 79 = 38°$.

3 $360 - (120 + 75 + 85) = 80°$.

4 The other angles are $64°$, $116°$ and $116°$: one vertically opposite and two supplementary.

5 Alternate $128°$; co-interior $180 - 128 = 52°$.

6 $180 - 39 - 88 = 53°$.

7 Apex $= 180 - 72 - 72 = 36°$.

8 $(7 - 2) \times 180 = 900°$.

9 $(9 - 2) \times 180 \div 9 = 140°$.

10 $360 \div 18 = 20°$.

11 (a) No: $4 + 5 = 9$, degenerate. (b) Yes: $4 + 5 > 8$. (c) Yes: $2 + 6 > 7$.

12 Each half is $6.4 \div 2 = 3.2$ cm.

13 $6x - 8 = 4x + 22$, so $2x = 30$, $x = 15$. Each angle is $6(15) - 8 = 82°$.

14 $2x + 10 + x + 25 + x - 15 = 180$, so $4x + 20 = 180$, $x = 40$. The angles are $90°$, $65°$ and $25°$.

15 $x + 2x + 3x = 180$, so $x = 30$. The angles are $30°$, $60°$ and $90°$: a right triangle.

16 $3x + 4x + 5x + 6x = 360$, so $18x = 360$, $x = 20$. The angles are $60°$, $80°$, $100°$ and $120°$.

17 $n = 360 \div 36 = 10$; each interior angle is $180 - 36 = 144°$.

18 \(11-7

19 Third angle $= 180 - 90 - 14 = 76°$. The handrail continues the $14°$ ramp direction; the angle between it and a vertical post is $90 - 14 = 76°$.

20 The left and right tips are each $70 + 70 = 140°$; the top and bottom tips are the $40°$ angles. The two triangles share the side between their $70°$ angles, so each pair of $70°$ angles meets at a tip. A 1:10 drawing uses a 40 mm shared diagonal and $70°$ angles at each end, with the $40°$ vertices on opposite sides; check that the two outer sides are equal.

21 (a) $4 + 4 = 8 > 5$, so it can be built. (b) Apex $= 180 - 66 - 66 = 48°$. (c) The angle on the other side of the collar is $180 - 114 = 66°$, supplementary along the collar.

Common errors. A line gives $180°$, a point $360°$; co-interior angles are supplementary, not equal. The exterior angle equals the sum of the two opposite interior angles, not $180°$ minus that sum. An n-gon contains $n - 2$ triangles. A construction keeps the compass radius fixed and leaves the arcs visible; the triangle inequality must be checked before drawing.

Chapter 8 · Sets and classification

PREREQUISITE CHECK

1 2, 4, 6, 8, 10.

2 $\frac{3}{4}$.

3 26.

4 -4 is greater.

5 For example 3 and 6 (or 3 and 9, or 6 and 9).

6 5.

GUIDED PRACTICE

1 $8 \in S$; $7 \notin S$; $n(S) = 5$; $\{x \in \mathbb{N} : x \le 10,\ x \text{ is even}\}$; a subset such as $\{2, 4, 6\}$; $\varnothing$ is a subset because it has no element that is outside S.

2 $P \cap Q = \{3, 4\}$; $P \cup Q = \{1, 2, 3, 4, 5, 6\}$; $n(P \cup Q) = 6$; $P' = \{5, 6, 7, 8\}$. They are not disjoint because their intersection is non-empty. Addition rule: $4 + 4 - 2 = 6$.

3 $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. $-6 \in \mathbb{Z}, \mathbb{Q}, \mathbb{R}$. $\frac{7}{2} \in \mathbb{Q}, \mathbb{R}$ but not $\mathbb{Z}$. $0.125 = \frac{125}{1000} = \frac{1}{8}$. $\sqrt{25} = 5$, so it belongs to $\mathbb{N}$, the whole numbers, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$.

4 Newspaper only $25 - 10 = 15$; magazine only $18 - 10 = 8$; at least one $15 + 10 + 8 = 33$; neither $40 - 33 = 7$. Percentage neither: $\frac{7}{40} \times 100 = 17.5\%$.

PRACTICE 8.1 · SETS AND SUBSETS

1 (a) $\{4, 6, 8, 10, 12\}$. (b) $\{1, 2, 3, 6, 9, 18\}$. (c) $\{m, a, t, h, e, i, c, s\}$. (d) $\{23, 29\}$.

2 (a) 6. (b) 6. (c) 8. (d) 2.

3 (a) True. (b) True. (c) True. (d) False: the empty set has no elements.

4 (a) $\{3n : n = 1, 2, 3, 4\}$. (b) $\{n^2 : n = 1, 2, 3, 4, 5\}$. (c) $\{x \in \mathbb{Z} : -4 \le x \le 0\}$.

5 Both: $A \cap B = \{2, 4, 6\}$. At least one: $A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$.

6 $\{x, y\}$, $\{x, z\}$, $\{x, w\}$, $\{y, z\}$, $\{y, w\}$, $\{z, w\}$: six subsets.

7 A set lists each distinct element once, so the repeated 12 is not counted twice. The correct set is $\{1, 2, 3, 4, 6, 12\}$, with cardinality 6.

8 $2^5 = 32$ subsets. Proper subsets exclude the set itself, so there are 31.

9 A suitable universal set is all students at the school (or all students surveyed). It matters because the complement and the "neither" count are defined relative to it; changing the universal set changes the answer.

PRACTICE 8.2 · VENN DIAGRAMS AND OPERATIONS

1 (a) $\{2, 4\}$. (b) $\{1, 2, 3, 4, 6, 8\}$. (c) $\{1, 3, 5, 7\}$. (d) $\{1, 3\}$.

2 $n(A) = 14$, $n(B) = 13$, $n(A \cup B) = 22$, $n(A \cap B) = 5$, $n(\xi) = 26$.

3 Football only $16 - 5 = 11$; hockey only $12 - 5 = 7$; neither $28 - (11 + 5 + 7) = 5$.

4 $C \cup D = \{a, b, c, d, e\}$; $C \cap D = \{c, d\}$; $C \setminus D = \{a, b\}$.

5 $n(A \cup B) = 14 + 11 - 4 = 21$.

6 At least one $31 + 24 - 12 = 43$; neither $50 - 43 = 7$.

7 Counter-example: $A = \{1, 2\}$, $B = \{2, 3\}$. Then $n(A \cup B) = 3$, while $n(A) + n(B) = 4$. The correct statement is $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

8 $n(A \cap B) = 20 + 17 - 30 = 7$. The overlap cannot exceed 17 because it is contained in B, and $n(B) = 17$.

9 At least one $= 35 - 8 = 27$; both $= 22 + 19 - 27 = 14$.

PRACTICE 8.3 · NUMBER SETS

1 (a) $\mathbb{N}$, whole, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$. (b) $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$. (c) $\mathbb{Q}$, $\mathbb{R}$. (d) $\mathbb{Q}$, $\mathbb{R}$.

2 (a) $\sqrt{36} = 6$: $\mathbb{N}$, whole, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$. (b) Irrational, $\mathbb{R}$. (c) Irrational, $\mathbb{R}$. (d) $\frac{1}{3}$: $\mathbb{Q}$, $\mathbb{R}$.

3 (a) False. (b) True. (c) True. (d) False. (e) True.

4 (a) $\frac{3}{5}$. (b) $\frac{5}{4}$. (c) $\frac{7}{9}$. (d) $-\frac{7}{2}$.

5 Rational: $\sqrt{81} = 9$, $-\frac{2}{3}$, 4.5. Irrational: π, $0.101001000\ldots$, $\sqrt{15}$.

6 Every integer n equals $\frac{n}{1}$, a quotient of integers, so it is rational. Not every rational is an integer: $\frac{3}{4}$ is rational but lies between integers.

7 $\frac{22}{7}$ is only an approximation to π, not equal to it. Since $\frac{22}{7} = 3.\overline{142857} \approx 3.142857$ and $\pi \approx 3.141593$, $\frac{22}{7}$ is greater than π.

8 Let $x = 0.\overline{12}$. Then $100x = 12.\overline{12}$. Subtracting, $99x = 12$, so $x = \frac{12}{99} = \frac{4}{33}$.

9 If x is a terminating decimal, $x = \frac{a}{10^n}$. Then $x^2 = \frac{a^2}{10^{2n}}$. For $x^2 = 2$ we would need $a^2 = 2 \times 10^{2n} = 2^{2n+1}5^{2n}$. The right side has an odd exponent on 2, while a square has even exponents in its prime factorisation. Contradiction, so $\sqrt{2}$ cannot be terminating.

PRACTICE 8.4 · SURVEYS AND DECISIONS

1 Walk only $28 - 12 = 16$; cycle only $21 - 12 = 9$; at least one $16 + 12 + 9 = 37$; neither $45 - 37 = 8$.

2 Garden only $52 - 24 = 28$; balcony only $37 - 24 = 13$; neither $80 - (28 + 24 + 13) = 15$.

3 At least one $= 36 - 8 = 28$; both $= 20 + 17 - 28 = 9$.

4 At least one $= 74 + 61 - 38 = 97$; neither $= 120 - 97 = 23$.

5 Pool only $96 - 45 = 51$; gym only $64 - 45 = 19$; exactly one $51 + 19 = 70$. $\frac{70}{150} = 46.7\%$, which is below half; the claim is false.

6 The expression omits the "both" region. Correctly, neither $=$ total $-$ only $A-$ both $-$ only B, or neither $= \xi - n(A \cup B)$.

7 At least one $= 50 - 7 = 43$; both $= 34 + 29 - 43 = 20$; vanilla only $= 34 - 20 = 14$.

8 Let x be the overlap. Then $n(A \cup B) = 55 - x \le 60$, which gives $x \ge -5$; since $x \ge 0$, the smallest is 0. The largest is 25, because the overlap cannot exceed the smaller group B. Every value from 0 to 25 is possible.

INVESTIGATION 8 · HOW MUCH OVERLAP IS POSSIBLE?

1 F only $17 - 6 = 11$; both 6; C only $14 - 6 = 8$; neither $30 - (11 + 6 + 8) = 5$. The regions total 30.

2 At least one $= 11 + 6 + 8 = 25$; exactly one $= 11 + 8 = 19$.

3 The six students who help with both have been counted in both 17 and 14, so $17 + 14$ counts them twice. Removing one copy gives $31 - 6 = 25$, the correct union.

4 $n(F \cup C) = 17 + 14 - x = 31 - x$.

5 $31 - x \le 30$ gives $x \ge 1$. If $x = 0$, the union would be 31, exceeding the 30 students available, so at least one student must be in both.

6 $x \le 14$, because the overlap is contained in C, which has 14 members. At $x = 14$, all concert helpers also help at the fair: $C \subset F$, with F only 3, both 14 and neither 13.

7 With $x = 6$, 25 helpers are available, enough for 20. But the union is $31 - x$; if $x > 11$, the union is below 20. So the answer could change for overlaps 12, 13 or 14, which give 19, 18 or 17 helpers.

8 A third event creates overlaps of three groups, which a two-circle diagram cannot show. A three-set diagram needs, for example, $n(A \cap B \cap C)$ and the three pairwise-only overlaps $n(A \cap B \cap C')$, $n(A \cap C \cap B')$ and $n(B \cap C \cap A')$.

REAL-LIFE APPLICATION · THE CENTRE AUDIT

1 Pool only $44 - 18 = 26$; gym only $37 - 18 = 19$; at least one $26 + 18 + 19 = 63$; neither $80 - 63 = 17$.

2 $26 + 18 + 19 + 17 = 80$. The four regions are disjoint and cover every member, so their sum is the whole group.

3 $26 \times \$4 = \$104$; $19 \times \$3 = \$57$; $18 \times \$6 = \$108$. Total $= \$269$.

4 At least one: $\frac{63}{80} = 78.75\%$. Neither: $\frac{17}{80} = 21.25\%$. The denominator is all 80 members.

5 Including both, gym users $= 37$, so $\frac{37}{80} = 46.25\%$, which is more than a third: true. Gym only $= 19$, so $\frac{19}{80} = 23.75\%$, which is less than a third: false. The difference is the 18 dual users.

6 For the member, the combined pass saves \$1 compared with two single passes. For the centre, the combined pass earns \$1 less than two singles from a dual user, although more than a single pass. A fairer rule might price the combined pass at \$6.50, or offer it only off-peak. Either argument must use the counts 18, 26 and 19.

7 A change could add an off-peak or family category, or bundle ten visits. Before deciding, collect frequency of visits and peak times, because the present data record only whether a member uses a facility, not how often.

CRITERION TASK · THE ACCESS GUIDE

Sample response. The guide is a communication task, so the values below are a consistency check rather than the only acceptable wording.

1 ξ = the 48 registered students. A = students in the art workshop, $n(A) = 31$. C = students in the coding workshop, $n(C) = 27$. $n(A \cap C) = 14$.

2 Art only $31 - 14 = 17$; coding only $27 - 14 = 13$; both 14; neither $48 - (17 + 14 + 13) = 4$. Check: $17 + 14 + 13 + 4 = 48$.

3 $A \cup C$: students in at least one workshop (44). $A \cap C$: students in both (14). A': registered students not in art (17). $A \setminus C$: students in art but not coding (17).

4 Exactly one $= 17 + 13 = 30$. Both $= \frac{14}{48} \times 100 = 29.2\%$ (one decimal place).

5 A student may join both only if there is a free place in each workshop. Coding has $30 - 27 = 3$ places, but art already has 31 registered against a capacity of 30, so no new dual enrolment is possible until the art number falls to 29 or fewer.

6 Check questions: "How many registered students are not in the art workshop?" Answer 17 (the complement of A). "How many are in coding only?" Answer 13 (the "only" region).

7 Test record: for example, the reader asked whether "both" meant both workshops or both days; the guide was changed to "both workshops" throughout and the symbols were paired with plain-language labels.

8 A two-set diagram cannot show year group as well, because each student would need a year label inside every region and the overlaps between year group and workshop would not be visible. A three-set diagram or a two-way table of year group by workshop is needed.

Marking guide. Criterion A: correct universal set, four regions and operations. Criterion C: labels name whole sets, percentages state the denominator, and the rule is understandable without set symbols. Criterion D: the guide is internally consistent, the capacity condition is handled correctly, and the evaluation identifies why a third variable requires a richer representation.

MIXED PRACTICE 8

1 $\{4, 8\}$.

2 $n = 5$; for example $\{a, b, c\}$ and $\{c, d, e\}$.

3 $A \cap B = \{3, 5\}$; $A \cup B = \{1, 2, 3, 5, 7, 8, 9\}$.

4 $A' = \{1, 3, 5, 7, 9\}$.

5 $-\frac{7}{2} \in \mathbb{Q}, \mathbb{R}$. $\sqrt{49} = 7 \in \mathbb{N}$, whole, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.

6 Irrational: π and $\sqrt{3}$. The others are rational.

7 At least one $= 15 + 12 - 7 = 20$; neither $= 24 - 20 = 4$.

8 $n(A \cap B) = 18 + 13 - 25 = 6$.

9 $0.45 = \frac{45}{100} = \frac{9}{20}$; rational.

10 $\{-3, -2, -1, 0\}$.

11 $2^4 = 16$ subsets; 15 proper subsets.

12 Every element of $\varnothing$ belongs to $\{0\}$ (there are none), so $\varnothing \subset \{0\}$. But 0 is not an element of $\varnothing$, because the empty set has no elements.

13 Disjoint sets have empty intersection, so the union is $12 + 9 = 21$. With 6 outside, $n(\xi) = 27$.

14 Exactly one $= (48 - 22) + (41 - 22) = 26 + 19 = 45$.

15 Counter-example: $A = \{1, 2\}$, $B = \{3, 4\}$. Then $n(A) = n(B) = 2$ but $A \neq B$. Equal cardinality implies only that the sets have the same number of elements.

16 Fiction only $124 - 57 = 67$; non-fiction only $98 - 57 = 41$; neither $200 - (67 + 57 + 41) = 35$. Fiction but not non-fiction: $\frac{67}{200} = 33.5\%$. The librarian should use the "only" counts: non-fiction only is 41, below fiction only 67, so non-fiction is the less-used category and the one to promote or expand.

17 (a) Largest union occurs at the smallest overlap 8: $36 + 29 - 8 = 57$. (b) Smallest occurs when the choir is entirely inside the orchestra (overlap 29): $36 + 29 - 29 = 36$. (c) Yes: an overlap of 10 gives union $36 + 29 - 10 = 55$, and 10 is within the allowed range 8 to 29.

18 Rational: $\sqrt{4} = 2$, $\frac{7}{3}$, 2.5, -1, 0. Irrational: $\sqrt{5}$, π. The rationals include non-integers such as $\frac{7}{3}$ and 2.5, so "rational" is broader than "integer".

19 Advantage: the 18 dual users save \$1 each compared with two singles, and the rule is simple. Disadvantage: the 26 pool-only and 19 gym-only members receive no discount, and the centre forgoes \$1 from each dual user. A fairer modification might discount the second single pass, or offer the combined pass at \$6.50 with off-peak access.

20 Two circles have four regions, but three groups can overlap in pairs and all together; the triple overlap and the three pairwise-only regions cannot be shown. A three-set diagram needs the triple intersection and the pairwise intersections (or the corresponding "only" counts), which a two-set diagram never requires.

Common errors. $\in$ relates an element to a set; $\subseteq$ relates two sets. $\varnothing$ is not $\{0\}$. Rosters ignore order and repetition. Subtract the overlap before writing an "only" count, and state the universal set before using a complement. A decimal is rational only if it terminates or repeats; 3.14 is not π.

Chapter 9 · Perimeter, area, surface area and volume

PREREQUISITE CHECK

1 84.

2 240 cm.

3 35.

4 12.56.

5 $9 + 16 = 25$.

6 125.

GUIDED PRACTICE

1 $2(7 + 4) = 22$ cm; $7 \times 4 = 28$ cm^2. Triangle: $\frac{1}{2} \times 10 \times 6 = 30$ cm^2. Parallelogram: base 8, height 5, so 40 cm^2; perimeter with sides 8 and 6 is 28 cm.

2 Diameter 10, so $r = 5$ cm; $C = 3.14 \times 10 = 31.4$ cm; $A = 3.14 \times 5^2 = 78.5$ cm^2. Semicircle $r = 6$: area 56.52 cm^2; perimeter $3.14 \times 6 + 12 = 30.84$ cm. Trapezoid: $\frac{1}{2}(5 + 9) \times 4 = 28$ cm^2.

3 Face areas $7 \times 4 = 28$, $7 \times 3 = 21$, $4 \times 3 = 12$. $SA = 2(28 + 21 + 12) = 122\ \text{cm}^2$. Triangular prism: each end $\frac{1}{2} \times 8 \times 3 = 12\ \text{cm}^2$; rectangles $(8 + 5 + 5) \times 12 = 216\ \text{cm}^2$; total $240\ \text{cm}^2$.

4 $V = 7 \times 4 \times 3 = 84\ \text{cm}^3$. Triangular prism: end area $\frac{1}{2} \times 8 \times 3 = 12\ \text{cm}^2$; $V = 12 \times 12 = 144\ \text{cm}^3$; capacity 0.144 L.

PRACTICE 9.1 · PERIMETER AND AREA

1 $P = 2(15 + 8) = 46$ cm; $A = 15 \times 8 = 120\ \text{cm}^2$.

2 $A = \frac{1}{2} \times 14 \times 9 = 63\ \text{m}^2$.

3 Area $= 11 \times 7 = 77\ \text{cm}^2$; perimeter $= 2(11 + 9) = 40$ cm.

4 Side $= 36 \div 4 = 9$ cm; area $= 81\ \text{cm}^2$.

5 Width $= 96 \div 12 = 8$ cm; perimeter $= 2(12 + 8) = 40$ cm.

6 The 6 cm side is sloping, not perpendicular to the base, so it cannot be used as the height. Area $= \frac{1}{2} \times 8 \times 4 = 16\ \text{cm}^2$.

7 Let width $= w$, length $= 2w$. Then $2(2w + w) = 6w = 36$, so $w = 6$ cm and the length is 12 cm. Area $= 72\ \text{cm}^2$.

8 Both areas are $\frac{1}{2} \times 12 \times 9 = 54\ \text{cm}^2$, because the area depends only on the base and the perpendicular height. The side lengths can differ, so the perimeters can differ while the areas stay equal.

PRACTICE 9.2 · CIRCLES AND COMPOSITE SHAPES

1 $C = 2 \times 3.14 \times 5 = 31.4$ cm; $A = 3.14 \times 25 = 78.5\ \text{cm}^2$.

2 $d = 18$ m, so $r = 9$ m. $C = 3.14 \times 18 = 56.52$ m; $A = 3.14 \times 81 = 254.34\ \text{m}^2$.

3 Area $= \frac{1}{2} \times 3.14 \times 9 = 14.13\ \text{cm}^2$; perimeter $= 3.14 \times 3 + 6 = 15.42$ cm.

4 $A = \frac{1}{2}(7 + 13) \times 6 = 60\ \text{cm}^2$.

5 Rectangle area $60\ \text{m}^2$; semicircle radius 3 m, area $14.13\ \text{m}^2$. Total $74.13\ \text{m}^2$. Boundary: two 10 m sides, one 6 m side and the arc $3.14 \times 3 = 9.42$ m; total 35.42 m.

6 $r^2 = 78.5 \div 3.14 = 25$, so $r = 5$ cm and $d = 10$ cm.

7 A semicircle is half a circle, so its area is $\frac{1}{2}\pi r^2$, not πr^2. The formula πr^2 gives the whole circle.

8 Path area $= 3.14(8^2 - 5^2) = 3.14(64 - 25) = 3.14 \times 39 = 122.46\ \text{m}^2$.

PRACTICE 9.3 · NETS AND SURFACE AREA

1 $SA = 2(24 + 12 + 8) = 88\ \text{cm}^2$.

2 $SA = 6 \times 5^2 = 150\ \text{cm}^2$.

3 $2(15 + 5h + 3h) = 94$, so $30 + 16h = 94$, $16h = 64$, $h = 4$ cm.

4 Ends: $2 \times \frac{1}{2} \times 10 \times 12 = 120\ \text{cm}^2$. Rectangles: $(10 + 13 + 13) \times 8 = 288\ \text{cm}^2$. Total $408\ \text{cm}^2$.

5 $SA = 2(3 + 2 + 1.5) = 13\ \text{m}^2$. At 5 m^2 per litre, $13 \div 5 = 2.6$, so 3 whole litres are needed.

6 The formula $2(lw + lh)$ counts the top and bottom and the front and back but omits the two end faces $2wh$. Correct: $SA = 2(lw + lh + wh)$.

7 Base $30 \times 20 = 600\ \text{cm}^2$; two long sides $2 \times 30 \times 15 = 900$; two short sides $2 \times 20 \times 15 = 600$. Total $2100\ \text{cm}^2$.

8 $6s^2 = 150$, so $s^2 = 25$ and $s = 5$ cm. Volume $= 5^3 = 125\ \text{cm}^3$.

PRACTICE 9.4 · VOLUME AND CAPACITY

1 $V = 12 \times 5 \times 4 = 240\ \text{cm}^3$.

2 $V = 6^3 = 216\ \text{cm}^3$.

3 End area $= \frac{1}{2} \times 8 \times 5 = 20\ \text{cm}^2$; $V = 20 \times 14 = 280\ \text{cm}^3$.

4 $V = 50 \times 30 \times 40 = 60\,000\ \text{cm}^3 = 60$ L.

5 (a) $2500\ \text{cm}^3$. (b) $450\ \text{cm}^3$. (c) 800 L.

6 Original volume $1000\ \text{cm}^3$; removed block $250\ \text{cm}^3$; remaining $750\ \text{cm}^3$.

7 A cubic metre is $100 \times 100 \times 100 = 1\,000\,000\ \text{cm}^3$. Since $1000\ \text{cm}^3 = 1$ L, this is 1000 L, not 100 L. The error is treating a linear conversion as a volume conversion.

8 $l = 360 \div 30 = 12$ cm. Doubling every dimension multiplies the volume by $2^3 = 8$, so the new volume is $2880\ \text{cm}^3$.

INVESTIGATION 9 · THE PACKAGING PROBLEM

1 Surface areas, in order: 98, 76, 70, 68, 56 and $52\ \text{cm}^2$.

2 Least cardboard: $2 \times 3 \times 4$, with $52\ \text{cm}^2$. Most: $1 \times 1 \times 24$, with $98\ \text{cm}^2$.

3 As the dimensions become closer together, the surface area falls. The most efficient box is compact, with dimensions 2, 3 and 4; the least efficient is long and thin, with two dimensions 1.

4 A long, thin box has a large length and therefore large side faces, while its small end faces contribute little. The same volume is spread over a shape with more exposed surface relative to its volume.

5 $3 \times 3 \times 3$: $SA = 54\ \text{cm}^2$. $1 \times 3 \times 9$: $SA = 78\ \text{cm}^2$. $1 \times 1 \times 27$: $SA = 110\ \text{cm}^2$. The cube uses the least cardboard.

6 For example $3 \times 3 \times 4$: $SA = 2(9 + 12 + 12) = 66\ \text{cm}^2$. $2 \times 3 \times 6$: $SA = 2(6 + 12 + 18) = 72\ \text{cm}^2$. Recommend $3 \times 3 \times 4$, which uses less cardboard and is more compact; $1 \times 6 \times 6$ would need $96\ \text{cm}^2$.

7 Practical reasons include stacking stability, a shape that fits a shelf or pallet, ease of carrying, room for a label, protection of the contents, and manufacturing constraints such as cutting and folding.

8 Recommendation: bring the dimensions as close together as the product allows. Evidence: $2 \times 3 \times 4$ uses $52\ \text{cm}^2$ against $98\ \text{cm}^2$ for $1 \times 1 \times 24$ at the same volume, and the cube $3 \times 3 \times 3$ uses $54\ \text{cm}^2$ against $110\ \text{cm}^2$ for $1 \times 1 \times 27$. Limitation: the test uses whole-centimetre edges and a closed box; product shape, stacking and strength may outweigh the material saving.

REAL-LIFE APPLICATION · THE COURTYARD PROJECT

1 Lawn $= 18 \times 12 = 216\ \text{m}^2$. Stage radius 6 m, so stage $= \frac{1}{2} \times 3.14 \times 36 = 56.52\ \text{m}^2$. Total $= 272.52\ \text{m}^2$.

2 Boundary $= 18 + 12 + 18 + 3.14 \times 6 = 66.84$ m. The 12 m diameter is the internal join between lawn and stage, not part of the outer boundary.

3 Turf $= 216 \times \$7.50 = \$1620$. Paving $= 56.52 \times \$42 = \$2373.84$. Fencing $= 66.84 \times \$28 = \$1871.52$.

4 Bed volume $= 1.2 \times 0.6 \times 0.5 = 0.36\ \text{m}^3 = 360$ L. At 80%, one bed needs 288 L. Three beds need 864 L, so $864 \div 40 = 21.6$, meaning 22 bags.

5 Open bed surface area: base $1.2 \times 0.6 = 0.72$; long sides $2 \times 1.2 \times 0.5 = 1.20$; short sides $2 \times 0.6 \times 0.5 = 0.60$. Total $2.52\ \text{m}^2$ per bed, $7.56\ \text{m}^2$ for three. Paint needed $= 7.56 \div 9 = 0.84$ L, so one 1 L tin.

6 Soil $= 22 \times \$9.50 = \$209$; paint $= \$22$. Total $= 1620 + 2373.84 + 1871.52 + 209 + 22 = \$6096.36$. The project is over budget by $\$96.36$.

7 With 10% contingency: $6096.36 \times 1.10 = \$6706.00$ (to the nearest cent), over by $\$706.00$. One change is to use paving at $\$29$ per m^2: saving $56.52 \times \$13 = \$734.76$, which brings the total to $\$5361.60$ before contingency and $\$5897.76$ with it, within budget. Reducing the stage to a quarter-circle would also save, but changes the design rather than the lawn.

CRITERION TASK · THE STORAGE CRATE

Sample response. Both designs are open-topped, so each has five faces. Costs are for plywood only and assume no waste.

1 $36\ \text{L} = 36\,000\ \text{cm}^3$.

2 Base $40 \times 30 = 1200\ \text{cm}^2$, so $h = 36\,000 \div 1200 = 30\ \text{cm}$.
Check: $40 \times 30 \times 30 = 36\,000\ \text{cm}^3$.

3 Net: one 40×30 base, two 40×30 sides and two 30×30 sides. The five faces fold up to an open box 40 cm by 30 cm by 30 cm.

4 $SA = 1200 + 2(40 \times 30) + 2(30 \times 30) = 1200 + 2400 + 1800 = 5400\ \text{cm}^2 = 0.54\ \text{m}^2$.
Cost per crate $= 0.54 \times \$24 = \$12.96$; for 12 crates, \$155.52.

5 Base $60 \times 20 = 1200\ \text{cm}^2$; height 30 cm; volume $60 \times 20 \times 30 = 36\,000\ \text{cm}^3 = 36\ \text{L}$.
$SA = 1200 + 2(60 \times 30) + 2(20 \times 30) = 1200 + 3600 + 1200 = 6000\ \text{cm}^2 = 0.60\ \text{m}^2$.
Cost per crate $= \$14.40$; for 12, \$172.80.

6 Design A uses less plywood: $6000 - 5400 = 600\ \text{cm}^2$ per crate, or $7200\ \text{cm}^2 = 0.72\ \text{m}^2$ across 12 crates. Saving $= \$17.28$.

7 90% of $36\ \text{L} = 32.4\ \text{L}$.

8 Recommend Design A: it uses 600 cm^2 less plywood per crate and is more compact (40 by 30) than the long, narrow Design B (60 by 20), so it should be more stable and easier to stack. Limitation: the comparison counts only plywood area; it ignores waste, joint strength, handles, a lid, and whether the 40 cm width fits the storage shelf.

Marking guide. Criterion A: correct litre–cm^3 conversion, height, five-face surface area and costs with units. Criterion C: labelled net, clear comparison and stated assumptions. Criterion D: valid recommendation that uses the surface-area evidence, considers stability or stacking, and names a genuine limitation.

MIXED PRACTICE 9

1 $P = 2(18 + 7) = 50\ \text{m}$; $A = 18 \times 7 = 126\ \text{m}^2$.

2 $\frac{1}{2} \times 16 \times 9 = 72\ \text{cm}^2$.

3 $13 \times 6 = 78\ \text{cm}^2$.

4 $\frac{1}{2}(8 + 14) \times 5 = 55\ \text{cm}^2$.

5 $C = 2 \times 3.14 \times 4 = 25.12\ \text{cm}$; $A = 3.14 \times 16 = 50.24\ \text{cm}^2$.

6 $SA = 2(40 + 24 + 15) = 158\ \text{cm}^2$.

7 $V = 7^3 = 343\ \text{cm}^3$.

8 End area $= \frac{1}{2} \times 9 \times 4 = 18\ \text{cm}^2$; $V = 18 \times 15 = 270\ \text{cm}^3$.

9 $3.2\ \text{L} = 3200\ \text{cm}^3$; $750\ \text{mL} = 0.75\ \text{L}$.

10 $\frac{1}{2} \times 3.14 \times 10^2 = 157\ \text{m}^2$.

11 Length $= 84 \div 7 = 12\ \text{cm}$; perimeter $= 2(12 + 7) = 38\ \text{cm}$.

12 $2 \times 18 + 96 = 132\ \text{cm}^2$.

13 Outer rectangle $22 \times 14 = 308\ \text{m}^2$; inner lawn $20 \times 12 = 240\ \text{m}^2$. Path $= 308 - 240 = 68\ \text{m}^2$. It is the outer rectangle minus the inner rectangle; alternatively, two strips 22×1 and two strips 12×1, taking care not to double-count the corners.

14 Pizza area $= 3.14 \times 15^2 = 706.5\ \text{cm}^2$. Box area $= 32^2 = 1024\ \text{cm}^2$, larger by $317.5\ \text{cm}^2$. The box can hold the pizza because its side 32 cm exceeds the 30 cm diameter; the area comparison alone is not the test of fit.

15 Wall area $= 8 \times 2.5 = 20\ \text{m}^2$; subtract the window $2 \times 1.2 = 2.4\ \text{m}^2$, leaving $17.6\ \text{m}^2$. Two coats cover $35.2\ \text{m}^2$. At $10\ \text{m}^2$ per litre, 3.52 L are needed, so 4 whole litres must be bought.

16 $V = 1.2 \times 0.8 \times 0.6 = 0.576\ \text{m}^3 = 576\ \text{L}$. At 65% full: $0.65 \times 576 = 374.4\ \text{L}$. At 150 L per day, $374.4 \div 150 = 2.496$, so 2 full days.

17 Old: $SA = 2(600 + 300 + 200) = 2200\ \text{cm}^2$. New: $SA = 2(500 + 300 + 240) = 2080\ \text{cm}^2$. The reduction is $120\ \text{cm}^2$, which is $120 \div 2200 \times 100 = 5.45\%$, not 20%. The claim is false.

18 Lawn area $= 15 \times 9 = 135\ \text{m}^2$ (the semicircular bed is additional). Boundary $= 15 + 9 + 15 + 3.14 \times 4.5 = 53.13$ m. Turf cost $= 135 \times \$8 = \$1080$. This assumes the turf covers the rectangular lawn only and that the quoted price already includes waste; if cutting waste is extra, the cost rises.

19 $3 \times 4 \times 5$: $SA = 2(12 + 15 + 20) = 94\ \text{cm}^2$; $V = 60\ \text{cm}^3$. Doubled $6 \times 8 \times 10$: $SA = 2(48 + 60 + 80) = 376\ \text{cm}^2$; $V = 480\ \text{cm}^3$. Surface area is multiplied by 4 and volume by 8, because area has two length dimensions and volume has three.

20 Example rectangular prism: $3 \times 4 \times 5$, $V = 60\ \text{cm}^3$, $SA = 94\ \text{cm}^2$. Example triangular prism: right-triangle cross-section with legs 3 and 4 (area $6\ \text{cm}^2$) and hypotenuse 5, length 10 cm, so $V = 60\ \text{cm}^3$; $SA = 2(6) + (3 + 4 + 5) \times 10 = 132\ \text{cm}^2$. The rectangular prism uses less material for this pair. The comparison is design-dependent; the triangular prism is not automatically more efficient.

Common errors. Area uses the perpendicular height, never a sloping side. Halve the diameter before squaring. A standalone semicircle's perimeter includes the straight edge, but an internal join does not belong to an outer boundary. Surface area is in square units and volume in cubic units. $1\ \text{m}^3 = 1000$ L. Doubling every length multiplies area by 4 and volume by 8.

Chapter 10 · Data, tables and graphs

PREREQUISITE CHECK

1 20.

2 $0.375 = 37.5\%$.

3 45.

4 $(4 + 7 + 13) \div 3 = 8$.

5 5, 5, 9, 12, 14.

6 $104 - 86 = 18$.

GUIDED PRACTICE

1 Total 40. Walk: $14 \div 40 = 0.35 = 35\%$. Bike: $6 \div 40 = 0.15 = 15\%$. The four percentages total 100%. Walkers: $9 + 5 = 14$.

2 $90 \div 360 = \frac{1}{4}$; bus users $= \frac{1}{4} \times 40 = 10$. Walkers: $8 \div 40 \times 360^\circ = 72^\circ$. The vertical axis must start at 0.

3 Sum 41, mean $41 \div 6 \approx 6.83$. Median is the mean of the 3rd and 4th values: $(6 + 7) \div 2 = 6.5$. Range $= 12 - 3 = 9$. Total frequency 20; modal frequency 7.

4 Change $= 12$; $12 \div 80 \times 100\% = 15\%$. The first bar appears 10 units tall instead of 70, so heights are exaggerated by a factor of 7.

PRACTICE 10.1 · DATA AND TABLES

1 (a) Numerical, discrete. (b) Categorical, nominal. (c) Numerical, continuous. (d) Numerical, discrete. (e) Categorical, ordinal: months follow a calendar order. (f) Categorical, ordinal.

2 Frequencies: $1 \to 2$, $2 \to 3$, $3 \to 6$, $4 \to 3$, $5 \to 3$, $6 \to 3$; total 20. Percentages: $10\%, 15\%, 30\%, 15\%, 15\%, 15\%$.

3 Walk $8/25 = 32\%$; bus $11/25 = 44\%$; car $4/25 = 16\%$; cycle $2/25 = 8\%$; total 100%.

4 Boys who do not own: $32 - 24 = 8$; girls who do not own: $28 - 15 = 13$. Owners $= 24 + 15 = 39$, so $39 \div 60 = 65\%$.

5 Pages are counted in whole units, so the values are separate and discrete. Reading time is measured and can take any value in a range, so it is continuous even when rounded to the nearest minute.

6 Check that each response appears exactly once, that every response fits one category, that no value is miscopied or omitted, and that the category frequencies add to the recorded 42 before changing any entry.

7 Bus $= 25\% \times 60 = 15$; walk $= \frac{1}{5} \times 60 = 12$; neither $= 60 - 15 - 12 = 33$.

8 For example: walk, cycle, bus, car, other. The options must not overlap: "walk or bike" already contains every cyclist, so a separate "bike" option would count those students twice and make the total exceed the number surveyed.

PRACTICE 10.2 · BAR CHARTS AND PIE CHARTS

1 Total $= 28$; most common B (12); difference $= 12 - 3 = 9$.

2 Angles: football $9/21 \times 360^\circ \approx 154.3^\circ$; netball $6/21 \times 360^\circ \approx 102.9^\circ$; swimming $4/21 \times 360^\circ \approx 68.6^\circ$; athletics $2/21 \times 360^\circ \approx 34.3^\circ$. They total 360° apart from rounding.

3 Art $8/20 \times 360^\circ = 144^\circ$; music $5/20 \times 360^\circ = 90^\circ$; drama $4/20 \times 360^\circ = 72^\circ$; coding $3/20 \times 360^\circ = 54^\circ$.

4 25% of 40 is 10 people; 90° is one quarter, also 10 people. The remaining 20 people fill the other sectors.

5 $45\% \times 360^\circ = 162^\circ$.

6 The bar for 5 has no visible height, and the bar for 10 appears twice as tall although the values are in the ratio $2 : 1$ only because the scale begins at 5. Redraw with the vertical axis starting at 0 and equal intervals.

7 $60/360 = 1/6$, so the sector represents $240 \div 6 = 40$ people; $40/240 = 16.7\%$.

8 The 50% category has 100 people and angle 180°. Each of the other four has 25 people and angle 45°. A bar chart compares the four equal small categories more clearly than four thin, similar pie sectors.

PRACTICE 10.3 · NUMERICAL DATA

1 Figure 10.3: sum $= 23$, mean $= 2.3$; median $= 2$; mode $= 2$; range $= 5 - 0 = 5$.

2 Sum $= 45$, mean $= 9$; median $= 8$; mode $= 8$; range $= 14 - 5 = 9$.

3 Sum $= 39$, mean $= 6.5$; median $= (6 + 7) \div 2 = 6.5$. The data are symmetric about 6.5: $3 + 10 = 4 + 9 = 6 + 7 = 13$, so the balancing point and the middle value coincide.

4 Sum $= 50$, mean $= 10$; median $= 5$. The median is fairer because the outlier 30 pulls the mean above every typical value; 5 represents the cluster.

5 Total $= 3 + 7 + 6 + 4 = 20$; modal class 10–20 (frequency 7); below $20 = 3 + 7 = 10$.

6 With six values the median is the mean of the 3rd and 4th: $(7 + 9) \div 2 = 8$. Choosing 7 uses only one of the two middle values.

7 (a) Mode, because shoe sizes are categories and the shop needs the most frequent. (b) Median, because one very expensive house would inflate the mean. (c) Mean, because the total score is shared equally among all players.

8 Total of five $= 5 \times 10 = 50$. The four known values sum to 38, so the fifth is 12. For a new mean of 11 over six values, the total must be 66, so the sixth number is $66 - 50 = 16$.

PRACTICE 10.4 · TRENDS AND MISLEADING GRAPHS

1 Sales rise overall from 40 to 58, with a dip in month 3 at 42. Overall increase $= 18$, so $18 \div 40 \times 100\% = 45\%$.

2 $12 \div 80 \times 100\% = 15\%$.

3 The vertical axis starts at 60 rather than 0, and the visible heights 10, 12 and 15 make a 5-unit change look like half the first bar. The differences are exaggerated even though the plotted values are correct.

4 With an axis from 0, the bars are 70, 72 and 75 and look almost equal; the impression of a dramatic rise disappears.

5 Increase $= 40$, so $40 \div 50 \times 100\% = 80\%$. An accurate headline is "Sales rise 80%", not "Sales doubled".

6 Doubling both height and width multiplies the area by $2 \times 2 = 4$, while the value only doubles. The eye compares area, so the picture overstates the change by a factor of 2.

7 (a) Line graph, because it shows change over time. (b) Pie chart, because the four items are parts of one budget. (c) Bar chart, because it compares separate categories.

8 The percentage is correct: $2 \to 4$ is a 100% increase. But two incidents becoming four is a change of only two, so "crime doubled" sounds far more serious than the counts support. Ask for the actual counts, the population, the period and the definition of "crime"; a small base makes a large percentage unreliable.

INVESTIGATION 10 · HOW SHOULD DATA BE GROUPED?

Guidance. Use one boundary rule throughout, for example "include the lower bound, exclude the upper bound, except the final interval which includes 360". A different but consistent rule is acceptable; the counts below use that convention.

1 Frequencies for 0–60, 60–120, 120–180, 180–240, 240–300, 300–360: 1, 9, 9, 5, 3, 3. The boundary values 60, 120, 180, 240 and 300 are placed in the interval beginning at that value.

2 Bars touch, widths are equal and heights are 1, 9, 9, 5, 3, 3; the horizontal axis is screen time in minutes and the vertical axis is frequency.

3 The cluster is from 60 to 180 minutes, containing $9 + 9 = 18$ of the 30 students. There is a long tail of high values up to 360 minutes.

4 For 0–120, 120–240 and 240–360 the frequencies are 10, 14 and 6.

5 The 60-minute grouping shows the cluster and the separate peaks at 60–120 and 120–180. The 120-minute grouping merges those peaks into one broad bar and loses detail, although it makes the overall shape simpler.

6 Sum $= 4740$, mean $= 158$ min; median $= (135 + 140) \div 2 = 137.5$ min; mode $= 120$ min. The median best represents a typical student because the high values pull the mean upward.

7 The claim is not supported. Four hours is 240 minutes; only 6 of 30 students (20%) reach 240 minutes, and the mean is 158 minutes, about 2 h 38 min. The median is 137.5 minutes, about 2 h 18 min.

8 A histogram with 60-minute intervals, a caption stating the sample size, and the median quoted as the typical value. Limitation: grouping hides individual values and the sample is one class of 30, so it may not represent all students. The display should also note that screen time is self-reported.

REAL-LIFE APPLICATION · THE CANTEEN SURVEY

1 96 students buy school lunch; $96/200 = 48\%$.

2 Year 9 off-site: $13/65 = 20\%$. Year 7 off-site: $8/70 \approx 11.4\%$. Year 9 is higher by about 8.6 percentage points; compare percentages because the year-group totals differ.

3 School lunch $96/200 \times 360^\circ = 172.8^\circ$; packed $74/200 \times 360^\circ = 133.2^\circ$; off-site $30/200 \times 360^\circ = 54^\circ$. Total 360°.

4 A clustered bar chart with three bars for each year group. A single pie chart shows only the overall shares and hides how the choices differ between Year 7, 8 and 9.

5 Uptake falls from 88 to 72, then rises to 96. The largest fall is $88 - 72 = 16$ students, $16/88 \times 100\% \approx 18.2\%$.

6 Canteen income from school lunch $= 96 \times \$4.20 = \$403.20$. Total spending across all choices $= 403.20 + 74 \times \$3.10 + 30 \times \$5.50 = 403.20 + 229.40 + 165.00 = \$797.60$; only the school-lunch amount is canteen income. If 24 switch, extra canteen income $= 24 \times \$4.20 = \$100.80$ per day.

7 Off-site is $30/200 = 15\%$, so 85% eat on-site; Year 9 is highest at 20%, Year 7 lowest at about 11.4%. "Out of control" is not supported: 15% is a minority and the rise is concentrated in one year group. Fair headline (10 words): "One in five Year 9 students eats lunch off-site." A graph could make the share look larger by using a truncated axis for the Year 9 counts, or by showing only the off-site sector without the 200-student total.

CRITERION TASK · THE WATER-USE HEADLINE

Sample response. Values are in thousands of litres; percentage changes use the starting value unless stated.

1 Changes: 2020 -5, 2021 -7, 2022 -6, 2023 -6.

2 Water use falls every year, from 420 in 2019 to 396 in 2023; the total decrease is 24 thousand litres.

3 $24/420 \times 100\% \approx 5.7\%$.

4 Mean $= 2041/5 = 408.2$ thousand litres. It is a valid average, but it hides the steady downward trend; the endpoints and yearly changes describe the data better.

5 With the axis starting at 390, the visible bars are 30, 25, 18, 12 and 6 units. The 2023 bar looks one-fifth as tall as the 2019 bar, although the true fall is only 5.7%.

6 On an axis from 0 to 450 with intervals of 50, the five bars are 420, 415, 408, 402 and 396 and look almost equal; the decline is visible but modest.

7 An 80% fall from 420 would leave $420 \times 0.20 = 84$, not 396. The claim is false; the true fall is 24 thousand litres, or 5.7%.

8 Fair headline: "Factory water use falls 5.7% over four years." Caption: "Annual water use fell from 420,000 L in 2019 to 396,000 L in 2023, a decrease of 24,000 L (5.7%)." Limitation: the figures are self-reported and cover only five years. A useful question is whether production changed over the same period, because lower output could explain lower water use.

Marking guide. Criterion A: correct changes, total, percentage and mean with units. Criterion C: an appropriately scaled graph, a headline of at most twelve words and a caption naming period and unit. Criterion D: the 80% claim is tested with the correct base value and rejected; the evaluation distinguishes a real but small decline from a dramatic one and names a genuine limitation.

MIXED PRACTICE 10

1 Numerical, discrete.

2 35%, 25%, 22.5%, 17.5%.

3 $90/360 = 1/4$, so $120 \div 4 = 30$ people.

4 Sum 60, mean 12.

5 Ordered: 3, 4, 5, 7, 8, 9. Median $= (5 + 7) \div 2 = 6$.

6 $30 - 17 = 13$.

7 Total 20; modal class is the second, with frequency 8.

8 $15/60 \times 100\% = 25\%$.

9 $10/24 \times 100\% \approx 41.7\%$.

10 The axis starts at 90, so the visible heights are 6, 10 and 14. The value 104 is only 8.3% above 96, but its bar appears more than twice as tall.

11 Twelve similar sectors are hard to distinguish and their labels crowd the circle. A horizontal bar chart compares the categories more accurately.

12 The mean is pulled toward the outlier; the median stays near the cluster. The median is fairer when the outlier is not typical of the group.

13 12 is a count, not a percentage. $12/30 \times 100\% = 40\%$.

14 Increase $= 30$, so $30/40 \times 100\% = 75\%$, not 100%. A fair poster says "Sales rise 75%".

15 (a) Line graph, for change over time. (b) Pie chart, for shares of one budget. (c) Histogram, for the distribution of continuous journey times.

16 "Don't you agree...?" presupposes that the reader agrees and offers no neutral option, so it pressures a yes response; the word "don't" also frames disagreement as negative. A fairer question: "Should the school day start later, stay the same, or start earlier?"

17 The mean is 158 minutes, which is over 150 minutes (two and a half hours), so the statement is true only if "average" means the mean. The median is 137.5 minutes, below two and a half hours, and is more representative because high values pull the mean up. Report the median (about 2 h 18 min) and note that 6 of 30 students reach 4 hours.

18 The visible bar heights are 1, 3 and 6, so 100 looks six times 95. The true difference is $5/95 \times 100\% \approx 5.3\%$. An honest graph starts the axis at 0, where the three bars look almost equal.

Common errors. A frequency is not a percentage until it is divided by the total. Compare shares, not raw counts, when group sizes differ. Every pie sector uses the same denominator. A bar chart must start at zero. The median of an even set is the mean of the two middle values. Percentage change uses the starting value, and a claim based on a very small count needs the counts as well as the percentage.

REFERENCE · FORMULA BANK

Formula bank

A compact reminder of the relationships used in this course. A formula is a tool, not a substitute for reasoning: identify the quantities, check the units, and estimate whether the result is sensible.

NUMBER AND CALCULATION

Place value	digit value = digit × place value
Expanded form	$352 = 3 \times 10^2 + 5 \times 10 + 2$
Powers	$a^n = a \times a \times \cdots \times a$ (n factors)
Zero and one	$a^0 = 1$ $(a \neq 0)$, $a^1 = a$
Squares and cubes	$a^2 = a \times a$, $a^3 = a \times a \times a$
Order of operations	brackets, indices, × and ÷, then + and −
Rounding	look at the next digit; 5 or more rounds up

FRACTIONS, DECIMALS AND PERCENTAGES

Equivalent fractions	$\frac{a}{b} = \frac{ka}{kb}$ for $k \neq 0$
Add or subtract	$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$
Multiply	$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$
Divide	$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$
Fraction to decimal	divide numerator by denominator
Percentage	$p\% = \frac{p}{100}$
Percentage of a quantity	$\frac{p}{100} \times Q$
Percentage change	$\frac{\text{new} - \text{old}}{\text{old}} \times 100\%$
Increase by $p\%$	multiply by $\left(1 + \frac{p}{100}\right)$
Decrease by $p\%$	multiply by $\left(1 - \frac{p}{100}\right)$

FACTORS, MULTIPLES AND ALGEBRA

Divisibility by 3	digit sum is a multiple of 3
Divisibility by 4	last two digits form a multiple of 4
Divisibility by 9	digit sum is a multiple of 9
HCF and LCM	compare prime factorisations
Linear sequence	$u_n = dn + c$, d common difference
Distributive law	$a(b + c) = ab + ac$
Linear equation	$ax + b = c \Rightarrow x = \frac{c - b}{a}$
Check a solution	substitute and compare both sides

ANGLES AND PLANE SHAPES

Angles on a line	sum = 180°
Angles at a point	sum = 360°
Triangle	angle sum = 180°
Exterior angle	equals the sum of the two opposite interior angles
Polygon interior sum	$(n - 2) \times 180°$
Regular polygon	exterior = $\frac{360°}{n}$
Rectangle	$P = 2(l + w)$, $A = lw$
Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Trapezoid	$A = \frac{1}{2}(a + b)h$
Circle	$C = 2\pi r$, $A = \pi r^2$
Triangle inequality	$a + b > c$ for every pairing

SOLIDS AND MEASUREMENT

Cuboid surface area	$SA = 2(lw + lh + wh)$
Cuboid volume	$V = lwh$
Prism surface area	$SA = 2B + Ph$
Prism volume	$V = Bh$
Cube	$SA = 6s^2$, $V = s^3$
Capacity	$1\ \text{cm}^3 = 1\ \text{mL}$, $1000\ \text{cm}^3 = 1\ \text{L}$
Cubic metre	$1\ \text{m}^3 = 1000\ \text{L}$

SETS AND DATA

Union count	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Complement count	$n(A') = n(\xi) - n(A)$
Number of subsets	2^n for a set with n elements
Relative frequency	$\frac{\text{frequency}}{\text{total}}$
Sector angle	$\frac{\text{frequency}}{\text{total}} \times 360°$
Mean	$\frac{\text{sum of values}}{\text{number of values}}$
Median	middle value of the ordered data
Range	largest − smallest

Notation. l length · w width · h height · r radius · b base · B area of a base · P perimeter · n number of sides, terms or elements · p percentage · Q quantity · A, B sets · ξ universal set · $\varnothing$ empty set. Units are part of the answer: cm, cm^2, cm^3, mL, L, m, m^2, m^3.

REFERENCE · VOCABULARY

Glossary

Definitions are written for this course. Where a word has a technical meaning, the definition names the quantities involved and the context in which the term is used.

absolute value — the distance of a number from zero; never negative.

acute angle — an angle less than $90°$.

alternate angles — equal angles on opposite sides of a transversal between parallel lines.

area — the surface enclosed by a two-dimensional shape, in square units.

balance method — solve an equation by applying the same operation to both sides.

bias — a systematic tendency for a survey or display to favour some responses or groups.

capacity — the volume a container can hold, often measured in millilitres or litres.

categorical data — data that place individuals in groups rather than measuring a quantity.

circumference — the distance around a circle.

class interval — a group of values used in a frequency table or histogram.

coefficient — the number multiplying a variable in a term.

common difference — the constant change between consecutive terms of a linear sequence.

common factor — a number that divides two or more given numbers exactly.

common multiple — a number that is a multiple of two or more given numbers.

complement — the elements of the universal set that are not in a given set.

composite number — a whole number greater than 1 with more than two factors.

composite shape — a shape made by joining or removing simpler shapes.

constant — a quantity with a fixed value.

continuous data — numerical data that can take any value in a range, limited only by measurement.

corresponding angles — equal angles in matching positions at two parallel lines cut by a transversal.

cross-section — the shape seen when a solid is cut parallel to its end.

cube number — a number formed by multiplying an integer by itself twice, such as $27 = 3^3$.

cuboid — a solid with six rectangular faces.

data set — a collection of observations.

decimal — a number written using place value to the right of the decimal point.

denominator — the number below the line in a fraction; the number of equal parts in the whole.

diameter — the distance across a circle through its centre.

discrete data — numerical data that take separate, countable values.

distributive law — $a(b + c) = ab + ac$; multiplication is shared over addition.

divisibility — the property of dividing exactly, leaving no remainder.

element — an object that belongs to a set.

empty set — the set with no elements, written $\varnothing$.

equation — a statement that two expressions are equal.

equivalent fractions — fractions that represent the same value.

equivalent expressions — expressions equal for every permitted variable value.

evaluate — substitute values and calculate a result.

expanded form — a number written as the sum of its place-value parts.

expression — numbers, variables and operations without an equals sign.

exterior angle — an angle formed by a side of a polygon and the extension of an adjacent side.

factor — a number that divides another exactly.

factor pair — two numbers whose product is a given number.

frequency — the number of times a value or category occurs.

frequency table — a table showing each value or category and its frequency.

generalisation — a statement describing a pattern for all its terms.

highest common factor — the largest number that divides two or more given numbers exactly.

histogram — a graph of grouped numerical data with touching bars of equal width.

improper fraction — a fraction whose numerator is greater than or equal to its denominator.

index — the power to which a base is raised, as in 5^3.

integer — a whole number, positive, negative or zero.

intersection — the elements belonging to both sets, written $A \cap B$.

irrational number — a real number that cannot be written as a fraction of integers.

like terms — terms with identical variables and powers.

linear sequence — a sequence with a constant difference between consecutive terms.

lowest common multiple — the smallest positive number that is a multiple of two or more given numbers.

mean — the sum of the values divided by their number.

median — the middle value of ordered data; for an even count, the mean of the two middle values.

mixed number — a whole number and a fraction written together.

mode — the most frequent value.

multiple — the result of multiplying a number by an integer.

net — a flat pattern that folds into a solid.

nth term — an expression giving any term of a sequence from its position n.

numerator — the number above the line in a fraction; the number of parts taken.

order of operations — brackets, indices, multiplication and division, then addition and subtraction.

ordinal data — categorical data whose categories have a meaningful order.

outlier — a value far outside the rest of the data.

parallel lines — lines in the same plane that never meet.

parallelogram — a quadrilateral with two pairs of parallel sides.

percentage — a proportion expressed out of 100.

perimeter — the distance around a two-dimensional shape.

perpendicular — meeting at a right angle.

pi — circumference divided by diameter of a circle, approximately 3.14159.

place value — the value of a digit determined by its position in a number.

polygon — a closed plane figure made of straight sides.

prime factorisation — a number written as a product of prime factors.

prime number — a whole number greater than 1 with exactly two factors, 1 and itself.

prism — a solid with a constant cross-section along its length.

radius — the distance from a circle's centre to its circumference.

range — the maximum value minus the minimum value.

regular polygon — a polygon with equal sides and equal angles.

relative frequency — frequency divided by the total number of observations.

rounding — replacing a number by a nearby number with fewer significant digits.

sector — the region of a circle between two radii and an arc.

sequence — an ordered list of numbers formed by a rule.

set — a well-defined collection of distinct objects.

simplest form — a fraction whose numerator and denominator share no common factor other than 1.

square number — a number formed by multiplying an integer by itself, such as $16 = 4^2$.

square root — a number that produces a given number when squared.

subset — a set whose elements all belong to another set.

substitution — replacing a variable with a value.

surface area — the total area of all faces of a solid.

term — a number or product in an expression; an element of a sequence.

term-to-term rule — a rule for finding the next term from the previous one.

trapezoid — a quadrilateral with exactly one pair of parallel sides (also called a trapezium).

triangle — a polygon with three sides; its angles sum to $180°$.

universal set — the set of all objects under discussion, written ξ.

variable — a symbol for a value that can change.

Venn diagram — a diagram using overlapping circles to show sets and their relationships.

vertex — a point where two or more sides or rays meet.

vertically opposite angles — equal angles opposite each other where two straight lines cross.

volume — the space occupied by a solid, in cubic units.

whole number — a non-negative integer: 0, 1, 2, 3, ...

Notation at a glance

$=$	is equal to	π	pi, about 3.14159
$\neq$	is not equal to	$\in$	is an element of
$\approx$	is approximately equal to	$\notin$	is not an element of
$<, >$	is less than, is greater than	$\subseteq$	is a subset of
$\frac{a}{b}$	a divided by b	$\varnothing$	the empty set
$a^2,\ a^3$	a squared, a cubed	ξ	the universal set
$\sqrt{a}$	the square root of a	$A \cup B$	union: in A or B or both
u_n	the nth term of a sequence	$A \cap B$	intersection: in both A and B
$\angle ABC$	the angle at vertex B	A'	the complement of A
$45°$	degrees	$A \setminus B$	elements of A not in B
$\parallel$	is parallel to	$\bar{x}$	the mean of a data set
$\perp$	is perpendicular to	cm, cm^2, cm^3	length, area and volume units

Units matter. Write the unit with every measurement and read it as part of the answer. A number without its unit is often incomplete.

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